

~Supplementary Manual~

WuKong: An LLM-Enhanced Framework for Adaptable Proteomics Analysis

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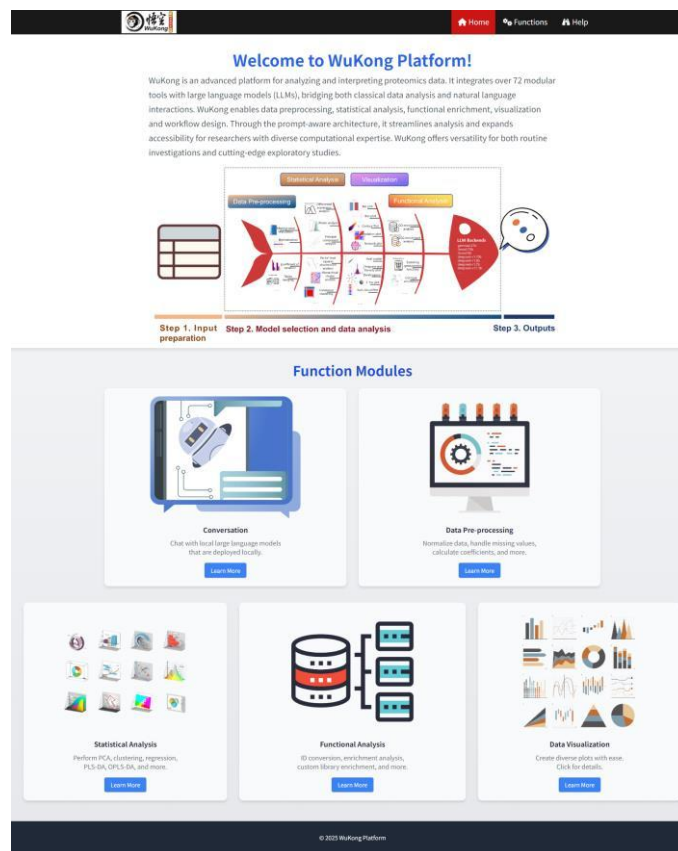
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Supplementary Manual

Abstract

WuKong is a powerful platform designed to elevate the way researchers approach proteomics data analysis. It integrates over 72 well-established analysis modules that cover the entire range of data preprocessing, statistical analysis, functional annotation, visualization, and workflow design. On the one hand, these modules retain the classic analysis frameworks, allowing users to adjust parameters within each module to directly generate results tailored to their specific research needs. This flexibility ensures that both beginners and experienced bioinformaticians can efficiently conduct complex analyses without being constrained by rigid workflows. On the other hand, WuKong incorporates large language models (LLMs) to enhance user experience by facilitating conversational interactions with the platform. Researchers can select from a variety of locally installed LLMs, enabling them to conduct analyses through an intuitive chat interface. This approach not only simplifies the analytical process but also provides real-time guidance and support, making it easier for users to navigate complex datasets and analytical procedures.

The whole source codes: <https://github.com/wangshisheng/WuKong>. The tool homepage is shown like this:



1. How to run this tool locally?

WuKong is an open source software for non-commercial use and all codes can be obtained on our GitHub: <https://github.com/wangshisheng/WuKong>. If users want to run *WuKong* on their own computer, they should operate as below:

As this tool was developed with R, you may:

- Install R. You can download R from here: <https://www.r-project.org/>.
- Install RStudio. (Recommendatory but not necessary). You can download RStudio from here: <https://www.rstudio.com/>.
- Check packages. After installing R and RStudio, you should check whether you have installed these packages (shiny, shinyBS, devtools, openxlsx, gdata, DT, tidyllm, ollamar, markdown, httr, ggplot2, shinyjs, shinyWidgets, rvest, writexl, shinydashboard, esquisse, ggsci, ggpubr, rstatix, Seurat, data.table, purrr, ceLLama, tidygraph, ggraph, corrr, tidyverse, ConsensusClusterPlus, LSD, pheatmap, grid, ggplotify, akima, ggcorrplot, psych, reshape2, igraph, dplyr, ape, cluster, limma, samr, DNB, ellipse, GGally, patchwork, funnelR, ggseqlogo, ggtree, clusterProfiler, tidyr, GRA, factoextra, alr3, ggpmisc, ggExtra, Mfuzz, impute, Amelia, ropls, FactoMineR, ade4, vegan, ggpie, pwr, ggradar, ggforce, ggdist, ggghalves, ggrepel, plotRCS, ggvegan, ggridges, pROC, RRHO2, scales, ggsankey, mdatools, survminer, survival, ggtern, timecourse, Rtsne, tidyestimate, umap, ggupset, ggwordcloud). You may run the codes below to check them:

```
if(!require(pacman)) install.packages("pacman")
pacman::p_load(shiny, shinyBS, devtools, openxlsx, gdata, DT, tidyllm, ollamar, markdown,
httr, ggplot2, shinyjs, shinyWidgets, rvest, writexl, shinydashboard, esquisse, ggsci, ggpubr,
rstatix, Seurat, data.table, purrr, tidygraph, ggraph, corrr, tidyverse, ConsensusClusterPlus,
LSD, pheatmap, grid, ggplotify, akima, ggcorrplot, psych, reshape2, igraph, dplyr, ape,
cluster, limma, samr, ellipse, GGally, patchwork, funnelR, ggseqlogo, ggtree, clusterProfiler,
tidyr, factoextra, alr3, ggpmisc, ggExtra, Mfuzz, impute, Amelia, ropls, FactoMineR, ade4,
vegan, ggpie, pwr, ggforce, ggdist, ggghalves, ggrepel, ggridges, pROC, scales, mdatools,
survminer, survival, ggtern, timecourse, Rtsne, tidyestimate, umap, ggupset, ggwordcloud,
coin, survcomp)
```

Please note, if you find some packages cannot be installed directly using the above command, you can find them in the GitHub source and install them by, for example:

```
library(devtools)
devtools::install_github("CeIVoxes/ceLLama")
devtools::install_github("anspiess/propagate")
devtools::install_github("gpli/DNB")
devtools::install_github("Nisus-Liu/GRA")
devtools::install_github("ricardo-bion/ggradar")
devtools::install_github("kunhuo/plotRCS")
devtools::install_github("gavinsimpson/ggvegan")
devtools::install_github("RRHO2/RRHO2", build_opts = c("--no-resave-data", "--no-manual"))
```

```
devtools::install_github("davidsjoberg/ggsankey")
devtools::install_github("nicolash2/ggvenn")
```

d) Run this tool locally

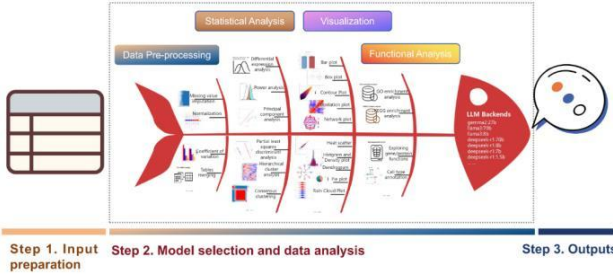
```
if(!require(WuKong)) devtools::install_github("wangshisheng/WuKong")
library(WuKong)
WuKong_app()
```

Then WuKong will be started as below, and the detailed operation about WuKong can be found in the subsequent sections.


Home
Functions
Help

Welcome to WuKong Cloud Platform!

WuKong is a cutting-edge proteomics platform integrating over 72 analysis modules for data processing, visualization, and annotation. It combines classic frameworks with customizable parameters and intuitive workflows. By incorporating local large language models (LLMs), WuKong also enables conversational interactions, simplifying complex analyses and offering real-time guidance for researchers of all expertise levels.



Step 1. Input preparation **Step 2. Model selection and data analysis** **Step 3. Outputs**

Function Modules



Conversation

Chat with local large language models that are deployed locally.

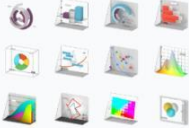
[Learn More](#)



Data Pre-processing

Normalize data, handle missing values, calculate coefficients, and more.

[Learn More](#)



Statistical Analysis

Perform PCA, clustering, regression, PLS-DA, OPLS-DA, and more.

[Learn More](#)



Functional Analysis

ID conversion, enrichment analysis, custom library enrichment, and more.

[Learn More](#)



Data Visualization

Create diverse plots with ease. Click for details.

[Learn More](#)

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2. Data Preparation

The uploaded data file formats could be .csv or .txt. Before analysis, users should prepare the proteomics expression data and sample information. The proteomics expression data required here could be readily generated based on results of several popular tools such as Spectronaut, MaxQuant. Users then can upload the data and type in right sample information into WuKong with right formats respectively and start subsequent analysis.

2.1 Expression data

Users can prepare their proteomics expression data as below from any software (e.g. Spectronaut, MaxQuant, etc.) and upload into this tool. In the situation, protein ids/names are sequentially provided in the first two columns of input file. The protein ids/names in the first column could be UniProt ids or protein names. From the second column, proteins expression intensity or signal abundance in every sample should be listed. The data structure is shown as below (rows are proteins and columns are samples):

PG Protein nGroups	DPC_0h_1	DPC_0h_2	DPC_0h_3	DPC_0h_4	DPC_3h_1	DPC_3h_2	DPC_3h_3	DPC_3h_4	DPC_6h_1	DPC_6h_2
	1	2	3	4	1	2	3	4	1	2
A0A024R8	25189.73	13440.21	19565.14	17490.47	27999.1	11120.79	8287.486	10820.16	21856.66	11681.11
A0A096LP	43690.06	40345.23	37148.71	37204.88	39428.96	39318.33	36062.08	34250.29	34548.79	30284.49
A0A0B4J2	1060724	836391.3	1103099	1063019	973396.6	821988.6	936122.9	1124465	917335	867221.2
A0A0B4J2	57807.32	60119.08	63978.09	69136.92	68128.56	31073.24	72903.39	71065.42	45719.14	48018.71
A0AV9E	14720.61	16155.04	15082.32	14969.27	14932.68	13164.33	14025.51	12534.12	14209.58	13106.73
A0AVK6	16522.56	18600.74	17865.75	18601.68	18643.31	5125.668	14807.14	16103.46	20163.18	17906.59
A0AVT1	17222.84	24763.66	30250.37	28884.49	16928.29	28179.63	22732.48	16664.99	25190.92	23202.59
A0FGH8	20893.16	17165.7	18176.4	20293.75	18935.83	12087.23	14606.6	11881.54	25152.06	17099.2
A0JLT2	37593.66	28032.06	28467.56	21959.01	34682.7	23316.45	35804.23	32016.78	44664.74	40806.98
A0MZ66	29056.75	32836.46	30119.31	30416.28	34145.86	30256.86	33450.51	30839	31779.2	37992.16
A1L0T0	35199.13	35886.66	36962.29	38643.55	29708.71	31485.84	25030.82	31670.05	36484.25	31963.77
A1L390	12021.93	12513.96	11311.04	10823.97	11329.15	9397.313	12337.43	9379.057	12087.48	10786.13
A2RRD8	46519.11	58789.84	54326.02	62714.16	57500.93	64203.45	65464.58	70372.02	79865.82	86481.73
A3KN83	26430.21	25346.23	26637.75	25969.68	26256.35	25992.66	25837	25177.89	24772.02	24262.8
A5YKK6	27706.62	27926.76	26817.8	27684.27	24960.62	26118.65	25467.98	24301.89	31291.05	30275.13
A6NQC3	35944.17	20451.28	27322.03	25434.8	41856.32	7629.363	2581.511	10482.42	26577.31	25864.84
A6NDG6	36172.88	32129.89	29492.28	36041.41	32832.6	28935.81	26838.32	24399.04	38261.67	38333.74
A6NDU8	48340.96	50244.54	40859.61	44103.27	37341.36	42296.37	38651.26	37392.52	47460.55	39691.67
A6NED2	2379.226	9749.278	11220.31	10147.28	8857.214	6130.552	6130.417	3188.316	11724.42	11549.22
A6NFI3	26165.06	32865.58	25063.64	30182.51	29647.58	31624.83	32113.08	31675.63	26228.95	32020.46
A6NHG4	10506.64	6973.583	7115.409	8546.165	3228.381	4923.56	2169.056	6023.871	7572.204	4797.863
A6NHL2	75163.5	66692.17	69335.76	77987.73	69796.09	56109.2	56733.93	61412.86	74370.55	63234.49
A6NHR9	46648.08	49067.9	47149.73	47286.44	49437.65	52611.63	50571.77	49839.58	41032.63	43034.47

2.2 Samples information data

Sample information here means that users should provide group number, replicate number, and group names, then type them in the corresponding module (see below). This tool will use these information to calculate corresponding results:

Group and replicate number:

2;4-4

Group names:

A;B

2.2.1 Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is "2;4-4".

2.2.2 Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

2.3 Download example datasets

After selecting a module, if users want to download the example datasets to their own computer and check the data format locally, they can download them from here, for example:

Step 1. Upload Data/Example Data

☐ A. Upload ☒ B. Load example data

[Download example expression data](#)

Step 2. Display Uploaded Data/Example Data

	A1	A2	A3	A4	B1	B2	B3	B4
AOA024RBG1	25189.73438	13440.20801	19565.13672	17490.47461	27999.10352	11120.79199	8287.486328	10820.16211
AOA06LP49	43690.0625	40345.23047	37148.71494	37204.87891	39428.95703	39318.33203	36062.07813	34250.29297
AOA08A12A2	1060724	836391.25	1103098.75	1063019	973396.5625	821988.5625	936122.9375	1124464.625
AOA08A12F2	57807.32422	60119.07813	63978.08594	69136.92188	68128.5625	31073.24219	72903.39063	71065.42188
AOAVN6	14720.61035	16155.04102	15082.32422	14969.2666	14932.67773	13164.33301	14025.50586	12534.11523
AOAVK6	16522.56445	18600.73633	17865.74805	18601.67969	18643.3125	5125.667969	14807.13867	16103.45508
AOAVT1	17222.83984	24763.6582	30250.36914	28884.49023	16928.29102	28179.63281	22732.48242	16664.99219
AOFG88	20893.17969	17165.69922	18176.4043	20293.74805	18935.82813	12087.23145	14606.60352	11881.54395
AOJLT2	37593.65625	28032.05664	28467.55859	21959.01367	34682.69531	23316.44727	35804.23438	32016.7832
AOIMZ66	29056.75195	32836.45703	30119.30664	30416.28125	29155.36328	30256.35938	33450.51172	30839.00195

Showing 1 to 10 of 4,277 entries

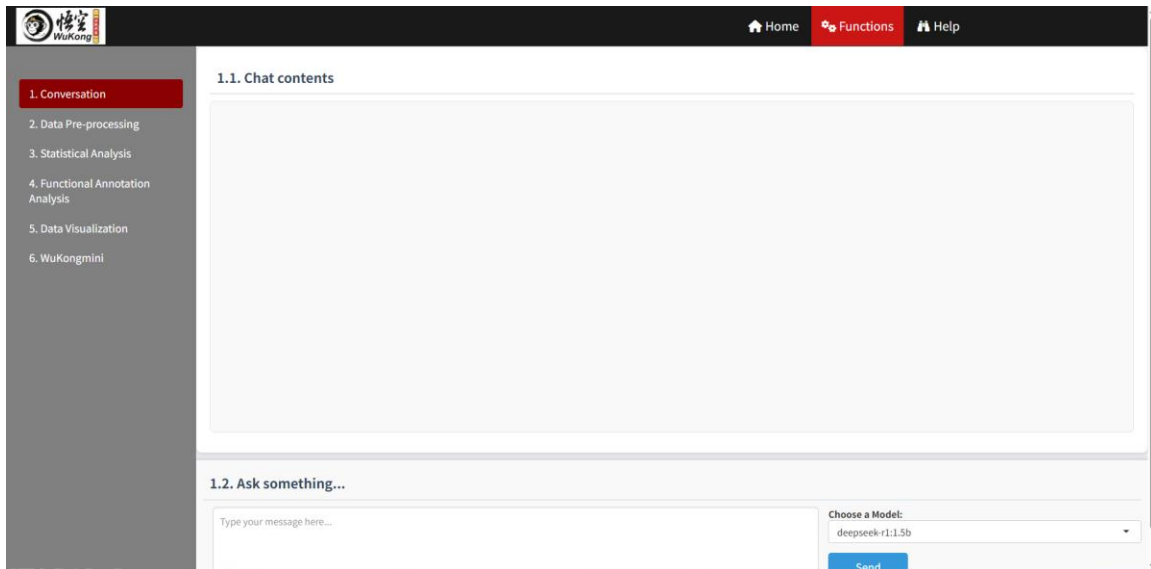
Previous 1 2 3 4 5 ... 428 Next

users can download the example data (proteomics expression data) by clicking the "Download example expression data" button. The data are saved as corresponding format and users can open them in other software, such as Excel. Then, users can check the example sample information and understand these parameters better.

3. Conversation

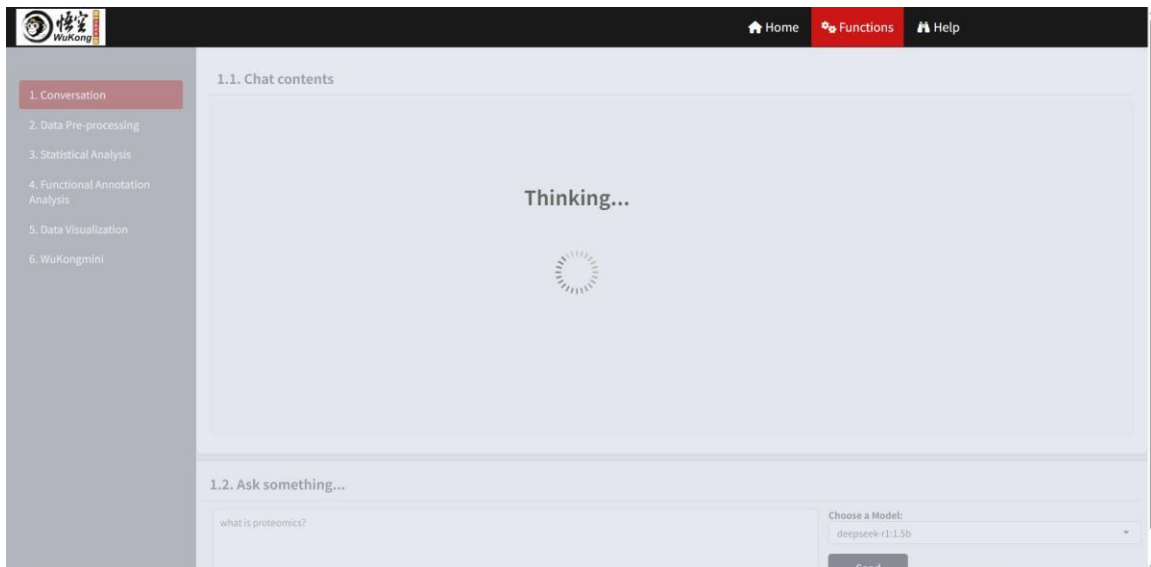
Here means users can chat with local large language models that are deployed locally. When

users click 'Functions', like below:

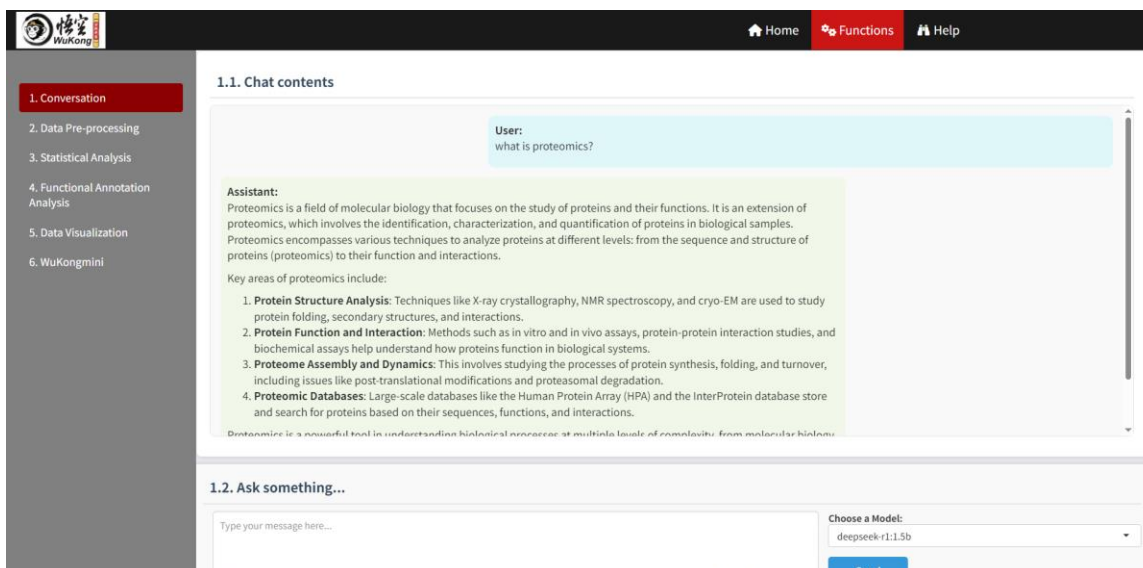


The first function, "Conversation," allows users to engage in interactive discussions. Section "1.1. Chat contents" displays the conversation history, while "1.2. Ask something..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a Model" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part).

For example, when users type one question and click "Send" button, this tool will invoke the corresponding LLM in the background, shown as below:



After a while, the conversation will be displayed as below:



How to install LLMs locally?

Utilizing LLMs on local systems is becoming increasingly popular due to enhanced privacy, control, and reliability. Herein, we primarily introduce how to install LLMs locally using Ollama, a free, open-source tool that enables users to run LLMs on their personal machines.

Step 1: Click <https://github.com/ollama/ollama>, then users can choose which version based on their own system (Windows, Linux or macOS) and download.

Ollama

Get up and running with large language models.

macOS

[Download](#)

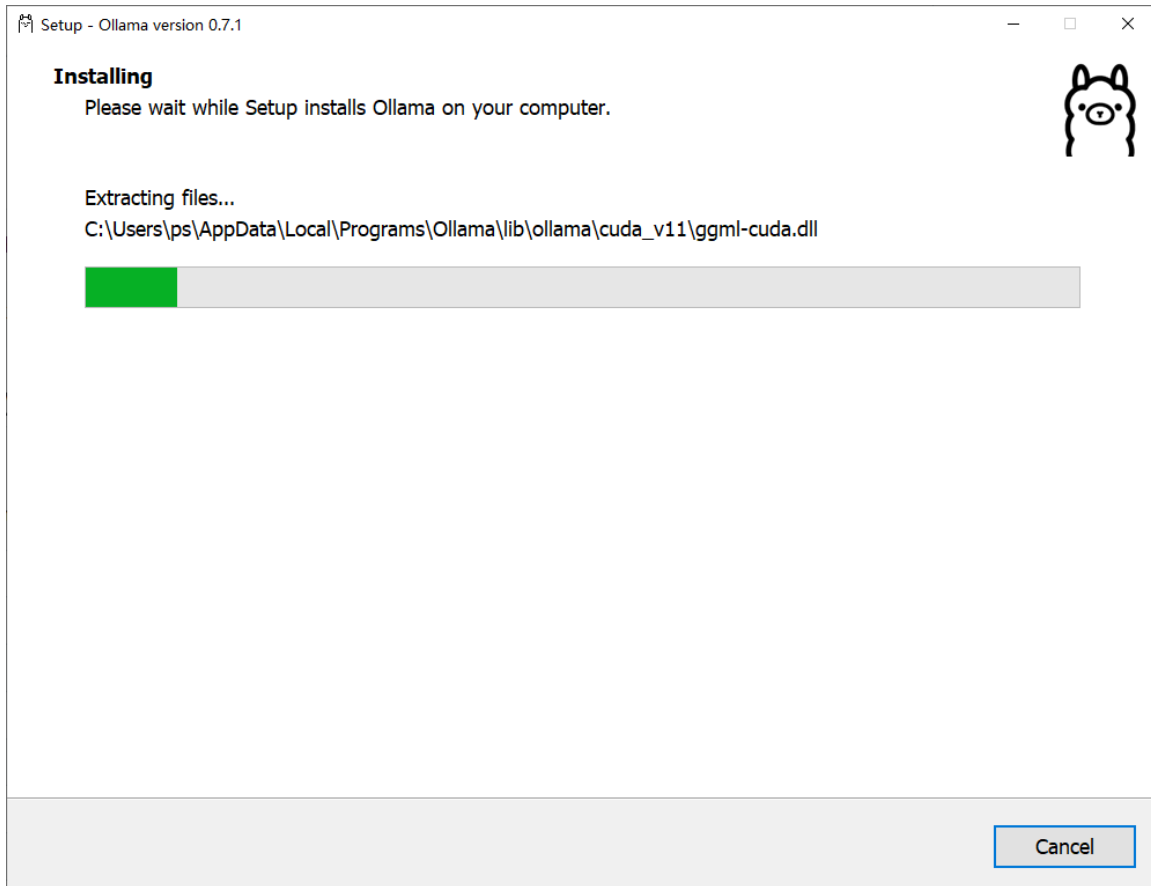
Windows

[Download](#)

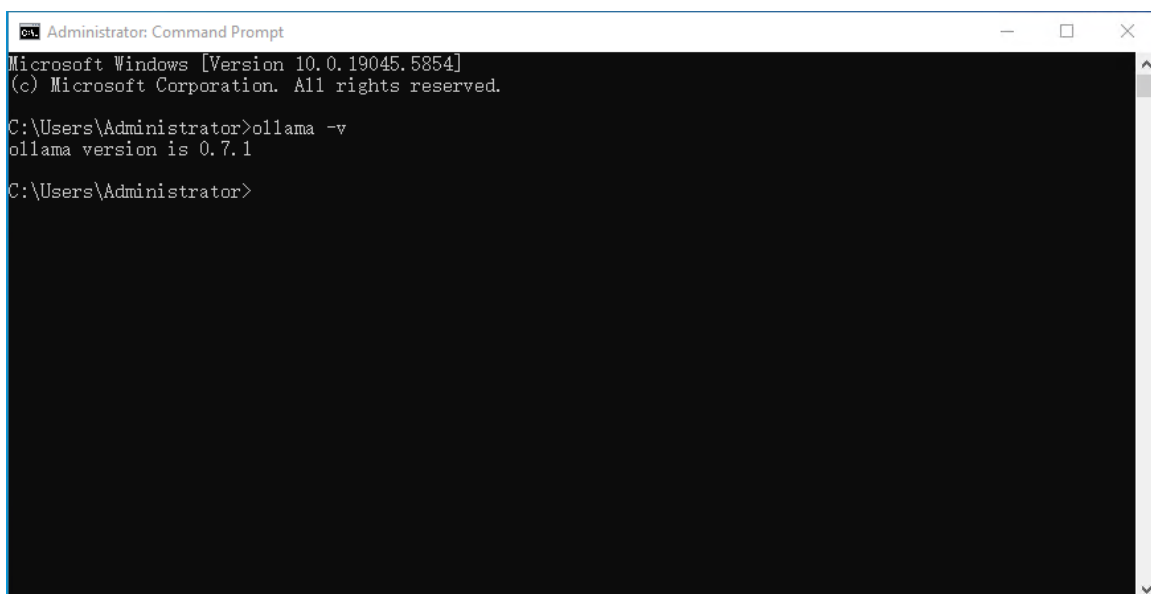
Linux

```
curl -fsSL https://ollama.com/install.sh | sh
```

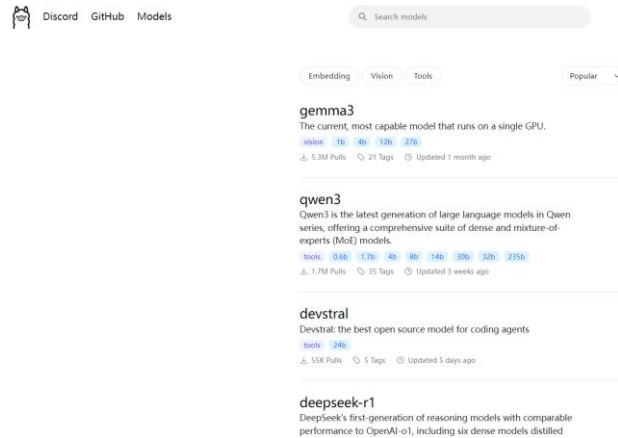
For example, after downloading the ollama for Windows, users can double click the “OllamaSetup.exe” file and the installing process like below:



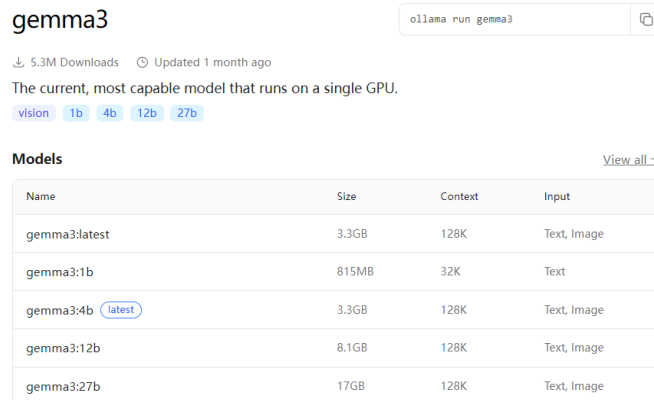
Step 2: After installation, users can open the “Command Prompt”, then type “ollama -v” to check the version of ollama.



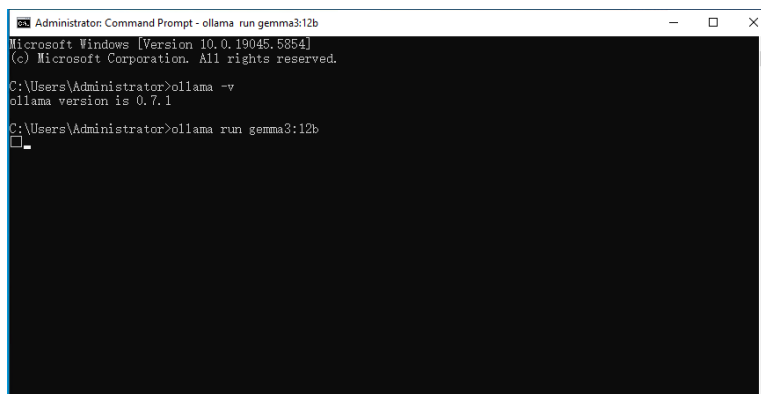
Step 3: Choose one LLM. Users can click <https://ollama.com/search>, then select any LLM they want.



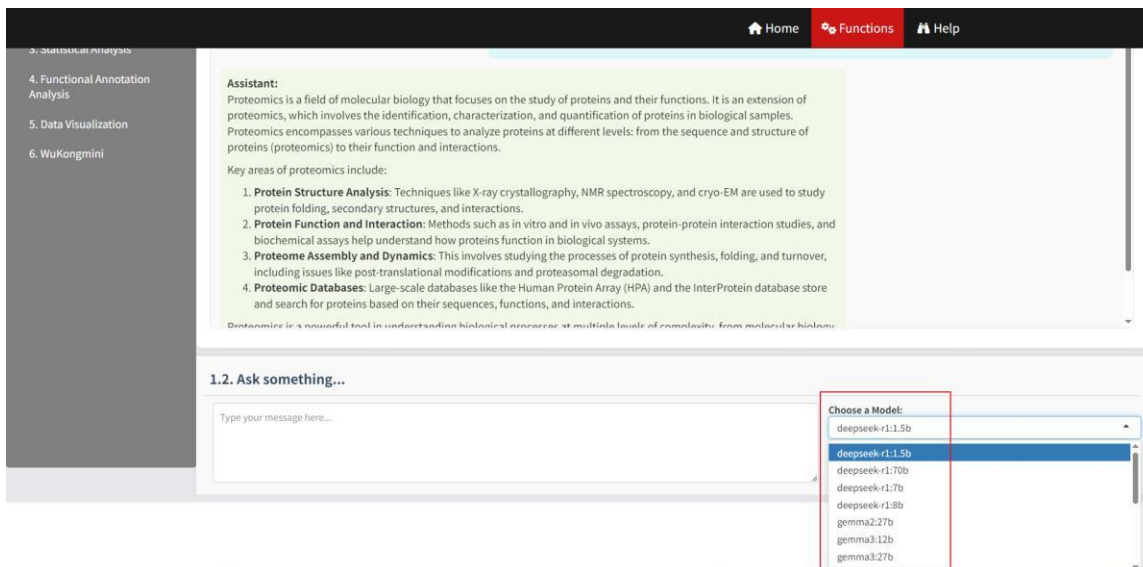
For instance, users select the "gemma3" option and then click on it. As illustrated below, there are numerous versions available with varying parameters. Consequently, users can choose any one that suits their computer configuration.



If users choose “gemma3:12b”, they can copy the command “ollama run gemma3:12b” in the “Command Prompt”. Then the model “gemma3:12b” will be installed successfully in the users’ computer.

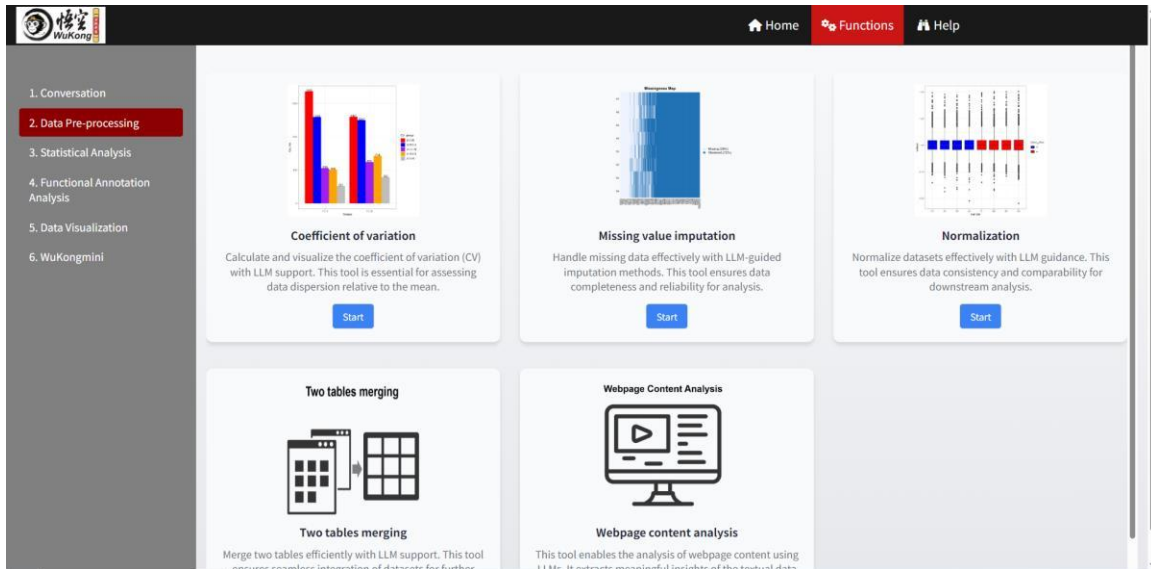


All other LLM installations are similar to the one described above. Subsequently, users will find the installed LLMs displayed in WuKong, where they can select one:



4. Data Pre-processing

Users can pre-process their data in this step, including coefficient of variation, missing value imputation, normalization, two tables merging and webpage content analysis. When users click “2. Data Pre-processing”, it is shown as below. If users want to use one module to analyze their data, just click corresponding “Start” button.



4.1. Coefficient of variation (CV)

CV is a statistical measure that represents the ratio of the standard deviation to the mean. A higher CV indicates greater variability relative to the mean, while a lower CV suggests more consistency. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the application interface for calculating the Coefficient of Variation (CV) with LLM Assistance. The interface is divided into three main sections: a sidebar on the left, a central panel for Step 1, and a right-hand panel for Step 2.

Step 1. Upload Data/Example Data

- ☒ A. Upload ☐ B. Load example data
- Select File Format:
 - ☒ .xlsx ☐ .xls ☐ .csv/txt
- Please import your data file:
 - Browse... No file selected
- ☒ Is the first row names?
- ☒ Is the first column names?
- Which Sheet to read?
 - 1

Step 2. Display Uploaded Data/Example Data

Showing 1 to 1 of 1 entries

Description
1

No data loaded. Please upload your dataset or use the example data.

Previous 1 Next

“I. Data Input” means users can upload their data or check example data from here. In the “Step 1. Upload Data/Example Data” part, users can upload their data, as below:

Step 1. Upload Data/Example Data

☒ A. Upload ☐ B. Load example data

Select File Format:

☒ .xlsx ☐ .xls ☐ .csv/txt

Please import your data file:

Browse... No file selected

☒ Is the first row names?

☒ Is the first column names?

Which Sheet to read?

1

Users can choose “A. Upload” to upload their own data here. They should select the right format (*File format: .xlsx, .xls, .csv, .txt*) based on their data and then click “Browse” button to import the data;

Is the first row names? This means whether the first row is column names. If true, you should

choose this parameter.

Is the first column names? This means whether the first column is row names. If true, you should choose this parameter.

If users do not upload any data, the “Step 2. Display Uploaded Data/Example Data” part will show “No data loaded. Please upload your dataset or use the example data.” to warn users.

If users want to download the example datasets to their own computer and check the data format locally, they can download them from here:

	A1	A2	A3	A4	B1	B2	B3	B4
ADA024RB01	25189.73438	13440.20801	19565.13672	17490.47461	27999.10202	11120.79199	8287.486328	10820.16211
ADA096LP49	43696.0625	40345.23047	37148.71484	37204.87861	39428.95703	39318.33203	36062.07813	34256.29297
ADA084J2A2	1080724	836391.25	1103096.75	1063019	973396.5625	821988.5625	936122.9375	1124464.625
ADA084J2F2	57807.32422	60115.07813	63978.08594	69136.92188	68128.5625	31073.24219	72903.39063	73065.42188
ADA096	14720.61335	16155.04182	15082.33422	14969.2666	14832.67773	13164.33303	14025.50586	12534.11523
ADA0K6	16522.56445	18600.73633	17865.74805	18601.67969	18643.3125	5125.667969	14807.13867	16103.45508
ADA0T1	17222.83984	24763.8582	30230.38914	28884.49023	16928.29102	28179.63281	22732.48242	16664.99219
ADQGR8	20893.17969	17165.69922	16176.4043	20293.74805	18935.82813	12067.23145	14006.60352	11881.54395
ADJLT2	37593.65625	28032.05664	28407.55859	21959.01367	34682.69531	23316.44727	35804.33438	32016.7832
ADH266	29056.75195	32836.45703	30119.30664	30416.28125	29336.36328	30256.35938	33450.51172	30839.60195

First, select “B. Load example data” and the example data will be shown on the right panel interactively.

Second, users can download the example data by clicking the corresponding button. The example data is saved as .csv format and users can open them in other software, such as Excel.

"II. Manual Analysis" refers to the process where users can manually analyze their data by adjusting parameters in the "Step 3. Adjust Parameters" section. Based on the adjustments made, the corresponding results will be displayed in the "Step 4. Display and Download Results" section, reflecting the parameters set in Step 3.

The parameters in the "Step 3. Adjust Parameters" part:

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

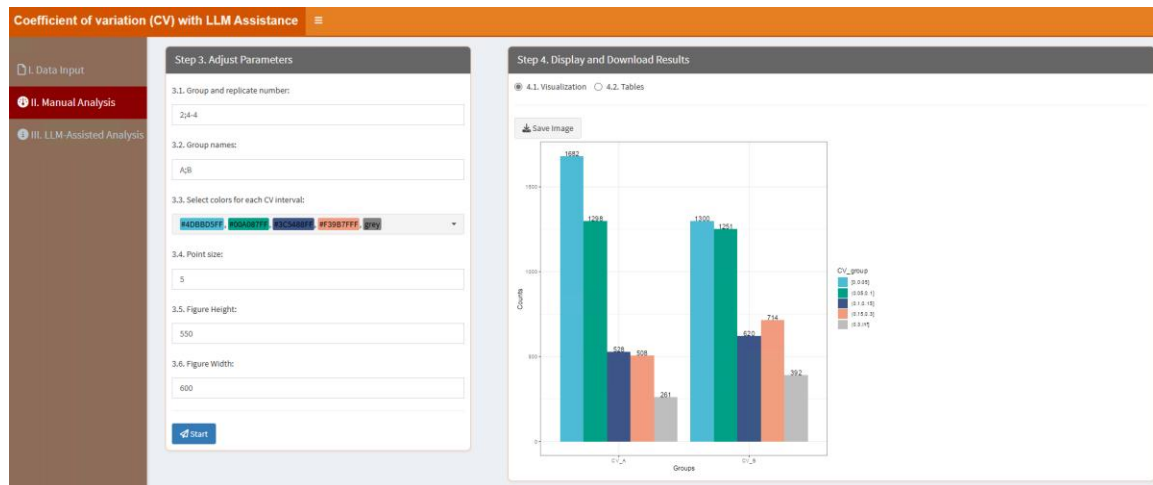
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



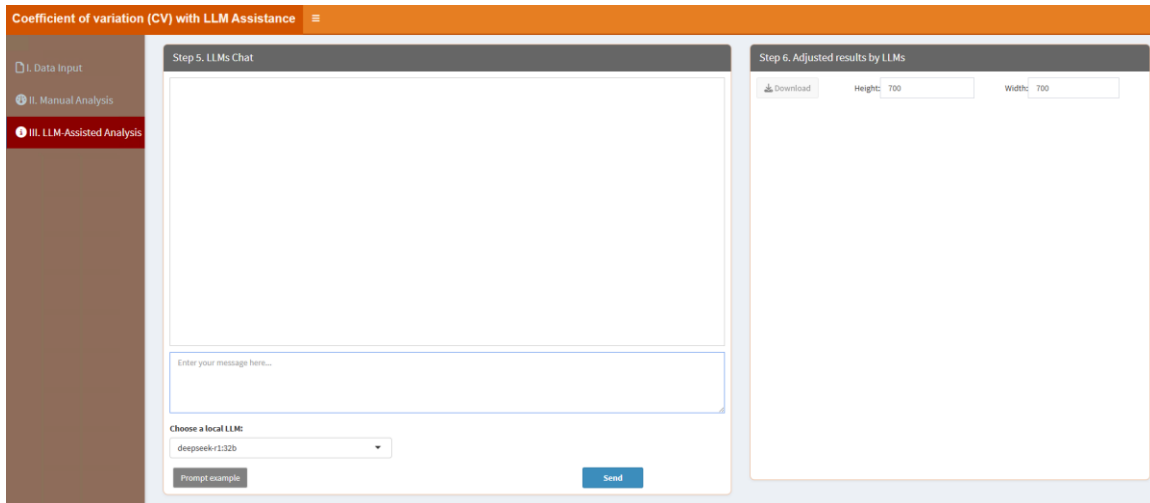
In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the CV of each group:

The screenshot shows the 'Coefficient of variation (CV) with LLM Assistance' interface. On the left, 'Step 3. Adjust Parameters' is the same as in the previous screenshot. On the right, 'Step 4. Display and Download Results' shows '4.2. Tables' selected. A 'Download' button is above the table. The table has columns for groups A1, A2, A3, A4, B1, B2, B3, B4, CV_A, and CV_B. The table contains 10 rows of data. A search bar is at the top right of the table. A 'Showing 1 to 10 of 4,277 entries' message is at the bottom left, and a pagination bar with 'Previous', '1', '2', '3', '4', '5', '...', '428', and 'Next' is at the bottom right.

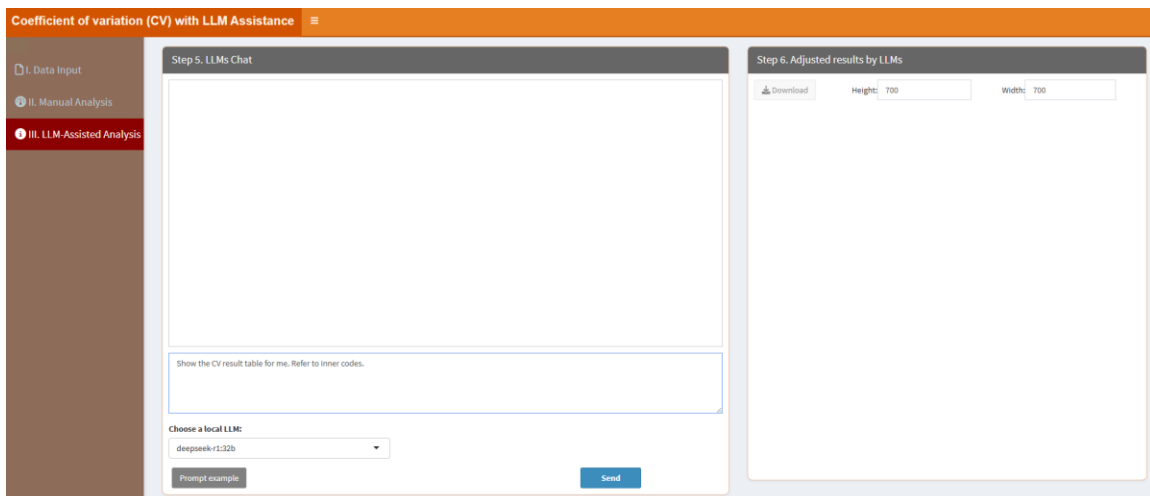
	A1	A2	A3	A4	B1	B2	B3	B4	CV_A	CV_B
ADAQ4RBG1	25189.73438	13440.20801	19565.13672	17490.47461	27999.10352	11120.79199	8287.486328	10620.16211	0.25855	0.62178
ADAQ6LP49	43690.0625	40345.23047	37148.71484	37204.87891	39428.95703	39318.33203	36062.07813	34250.29297	0.07896	0.06883
ADAQB4JDA2	1060724	836391.25	1103098.75	1063019	973396.5625	823988.5625	936122.9375	1124464.825	0.1193	0.12954
ADAQB4J2F2	57607.32422	60119.07813	63978.08594	69136.92188	66128.5625	31073.24219	72903.39063	71065.42188	0.07895	0.32751
ADAV96	14720.61035	16155.04102	15082.32422	14969.2666	14932.67773	13584.33301	14025.50586	12534.11533	0.04161	0.07637
ADAVK6	16522.56445	18600.73633	17865.74805	18601.67969	18643.3125	5123.667969	14807.13867	16103.45506	0.05476	0.43269
ADAVT1	17222.83984	24783.6582	30255.36914	28884.49023	16928.29102	28179.63281	22732.48242	16664.99219	0.23165	0.25905
ADPGR8	20893.17969	17165.69922	18176.4043	20293.74805	18935.82813	12087.23145	14606.60352	11881.54395	0.09168	0.22824
ADJX72	37593.65625	28032.05664	28467.55859	21959.01367	34682.89531	23316.44727	35804.23438	32016.7832	0.22217	0.17973
ADMZ66	29056.73195	32836.45703	30119.30664	30416.28125	29155.36328	30236.39938	33450.51172	30639.00195	0.05237	0.05893

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the

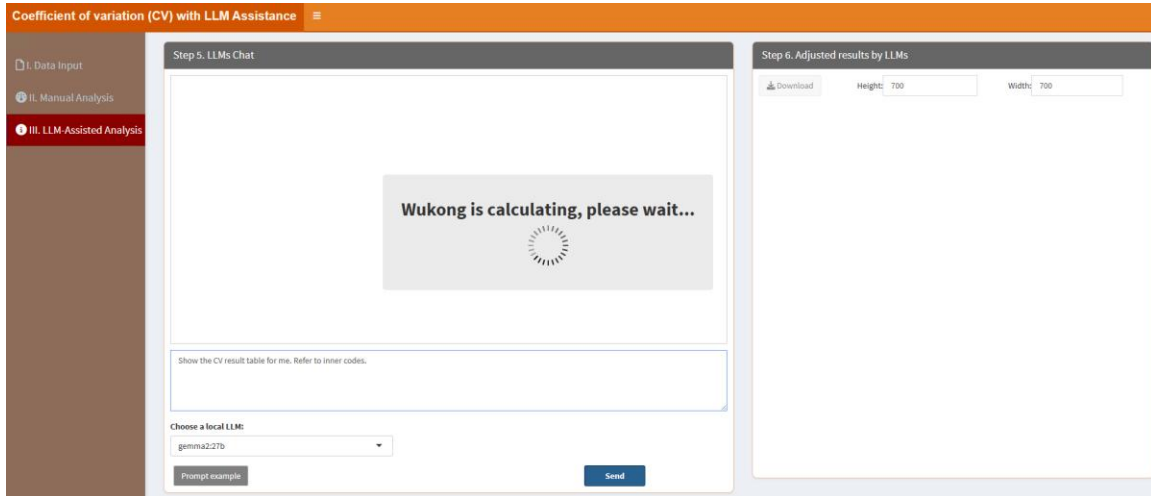
"Step 5. LLMs Chat".



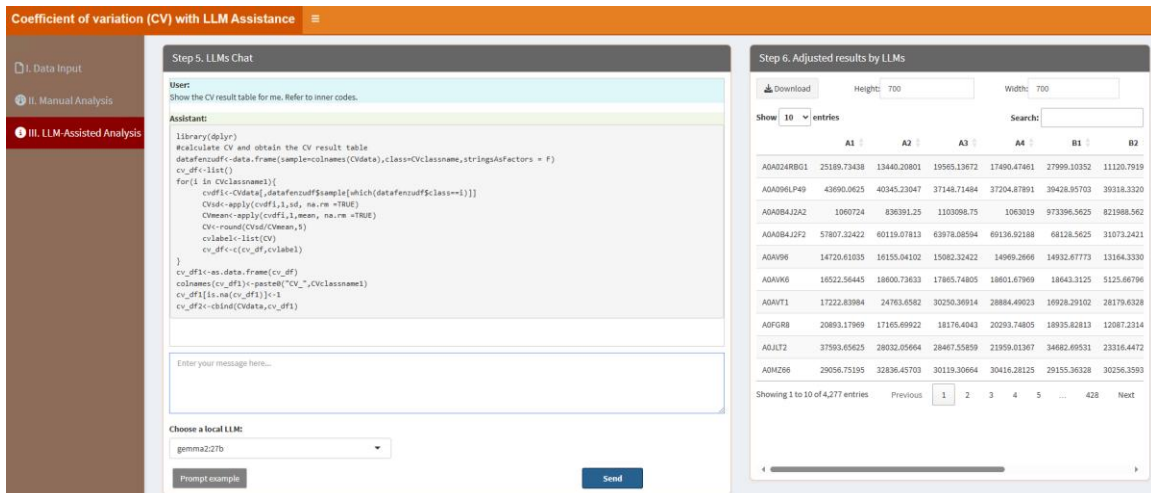
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Show the CV result table for me. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the CV table. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Show the CV result table for me. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the CV result table.



After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



4.2. Missing value imputation

Missing value imputation is the process of replacing missing or incomplete data in a dataset with substituted values to ensure the dataset remains usable for analysis. Common methods for imputation include simple approaches like replacing missing values with the mean, median, or some other advanced methods such as regression imputation, k-nearest neighbors (KNN), and machine learning-based methods. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Missing value imputation with LLM' application interface. On the left is a sidebar with three main sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The 'I. Data Input' section is currently active. The main panel is divided into two steps. 'Step 1. Upload Data/Example Data' contains options to 'A. Upload' or 'B. Load example data'. Under 'A. Upload', there is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom of Step 1 is a 'Which Sheet to read?' field with the value '1'. 'Step 2. Display Uploaded Data/Example Data' shows a 'Show: 10 entries' dropdown, a search bar, and a table with one entry: '1 No data loaded. Please upload your dataset or use the example data.' The table has pagination controls showing 'Showing 1 to 1 of 1 entries' and 'Previous 1 Next'.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Missing value imputation with LLM' application interface. The sidebar is the same as in the previous screenshot. The main panel is now on 'Step 3. Adjust Parameters'. It contains three input fields: '3.1. NA ratio:' with a value of '0.5', '3.2. Figure Height:' with a value of '600', and '3.3. Figure Width:' with a value of '900'. At the bottom of Step 3 is a blue 'Start' button. 'Step 4. Display and Download Results' is currently empty, with tabs for '4.1. Visualization' and '4.2. Tables'. A 'Save Image' button is visible in the '4.1. Visualization' tab.

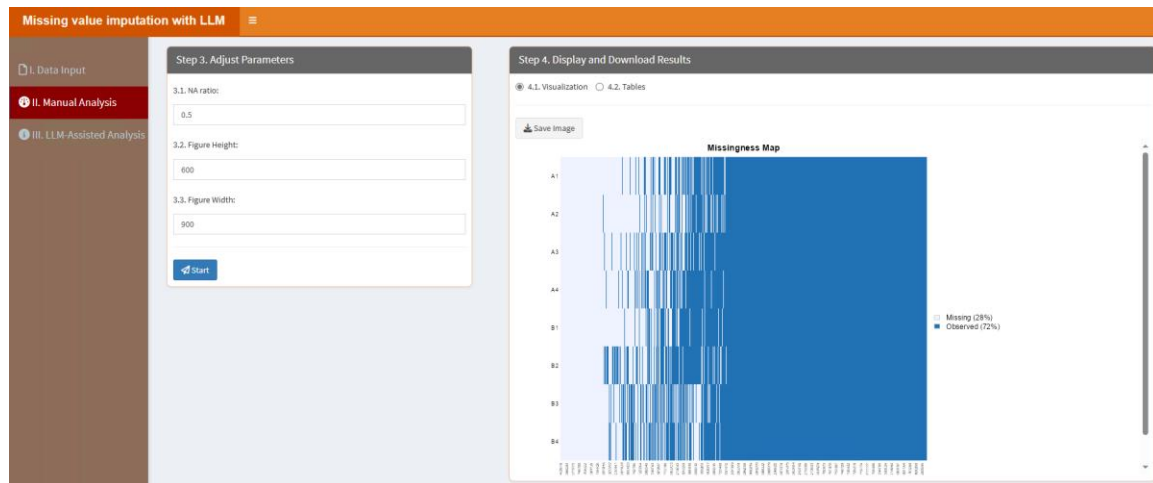
3.1. NA ratio: the threshold of NA ratio. Those peptides/proteins with NA ratio above this threshold will be removed.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the imputed results with the KNN method by default:

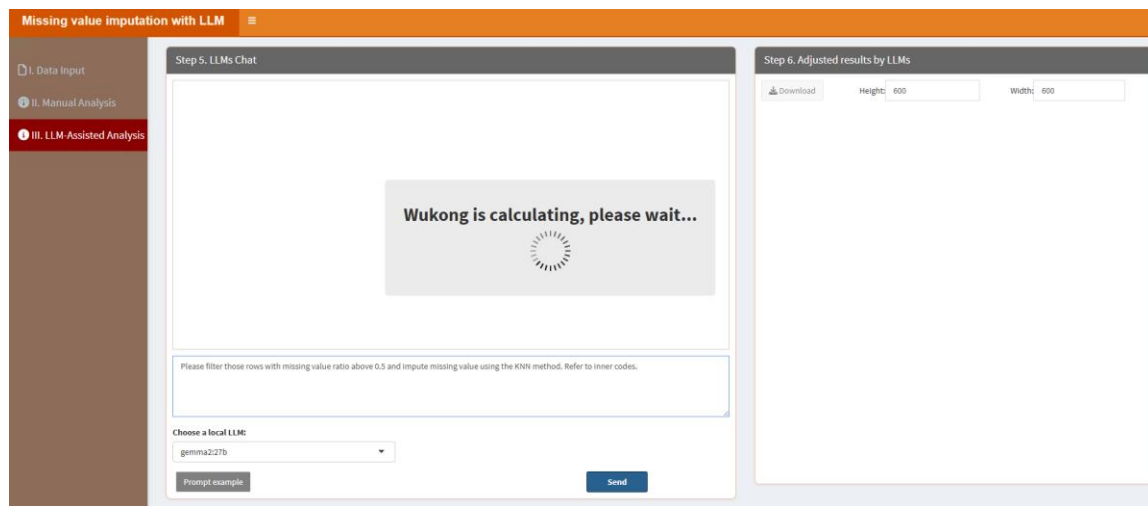
	A1	A2	A3	A4	B1	B2	B3	B4
AQA024QZP7	137800000	124650000	844030000	170240000	161460000	180970000	194200000	1703400000
AQA024R1R8	106590000	529060000	246140000	54157000	215250000	72858000	451760000	303700000
AQA024R216	61533000	43798000	45403000	85198000	70864000	74868000	39014333.33333334	36596250
AQA024R412	701770000	752100000	417480000	89648000	62925000	91468000	434328716.6666667	490541300
AQA024R4E5	2751800000	2825600000	2394500000	1573800000	1579600000	1414400000	3144000000	3262900000
AQA024RAM0	7149500000	10455000000	8402700000	8592100000	9234400000	8143000000	5975900000	6474600000
KTEN05	132210000	141820000	83057000	129620000	143190000	125790000	149720000	98778000
AQA024RAS2	1747700000	1387500000	1696100000	1250000000	1403400000	1580100000	2127000000	1807800000
AQA024RAC6	92182000	74230000	84870000	107150000	88054000	63953000	35721000	38266000
AQA024RCR8	263400000	280760000	256120000	343780000	348910000	267360000	401670000	483030000

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the

"Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please filter those rows with missing value ratio above 0.5 and impute missing value using the KNN method. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please filter those rows with missing value ratio above 0.5 and impute missing value using the KNN method. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.

Missing value imputation with LLM

I. Data Input

II. Manual Analysis

III. LLM Assisted Analysis

Step 5. LLMs Chat

User:

Please filter those rows with missing value ratio above 0.5 and impute missing value using the KNN method. Refer to inner codes.

Assistant:

```
library(impute)
nrowsum<-apply(misvaldata,1,function(x){sum(is.na(x))})/nrow(misvaldata)
datafchulic<-misvaldata[nrowsum<=0.5,]
misvaldata1<-impute.knn(as.matrix(datafchulic),k = 10, rowmax = 0.9, colmax = 0.9)
misvaldata2<-misvaldata1$data
misvaldata2
```

Enter your message here...

Choose a local LLM

gemma2.27b

Prompt example

Send

Step 6. Adjusted results by LLMs

Download

Height: 600

Width: 600

Show 10 entries

Search:

	A1	A2	A3	A4	B1	B2
AOA024QZPT	1378000000	1246500000	844030000	1702400000	1614600000	1809700000
AOA024R1R8	1065900000	529060000	2461400000	54157000	2152500000	72858000
AOA024R216	61531000	43798000	45403000	65198000	70864000	74866000
AOA024R412	701770000	752100000	417460000	89648000	62935000	91468000
AOA024R4E5	2751800000	2825600000	2384500000	1573800000	1576600000	1414400000
AOA024R4M0	71489500000	10435500000	8402700000	8592100000	9234400000	8143000000
KTEN05	132210000	141820000	83057000	129620000	143190000	125790000
AOA024RA52	1747700000	1387500000	1696100000	1250000000	1403400000	1580130000
AOA024RAC6	92182000	74230000	84870000	107150000	880540000	63953000
AOA024RCR6	263400000	280760000	256120000	343780000	348910000	267360000

Showing 1 to 10 of 4,524 entries

Previous12345...453Next

4.3. Normalization

Normalization is a crucial step in proteomics data analysis to ensure the comparability and reliability of results. In proteomics, data often comes from various experimental conditions, instruments, or runs, leading to technical variations that can obscure biological insights. Normalization aims to reduce these unwanted variations while preserving the true biological differences. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Normalization with LLM Assistance' web application. The left sidebar has three tabs: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' and 'B. Load example data'. Below these are file format selection buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Browse...' button is present, with a note 'No file selected'. There are two checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown menu is set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1 No data loaded. Please upload your dataset or use the example data.' Navigation buttons 'Previous', '1', and 'Next' are at the bottom.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Normalization with LLM Assistance' web application. The left sidebar has three tabs: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', shows a dropdown menu for '3.1. Normalization Method:' with 'Median' selected. A blue 'Start' button is at the bottom. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Tables' section with a 'Download' button.

3.1. Normalization Method: here users can select one normalization method, including: Median, mean, and total normalization involve dividing each value by the median, mean, or total of its column, respectively. Centering adjusts data by subtracting the column mean from each value, resulting in a mean of zero while maintaining the original variance. Range normalization involves subtracting the mean and dividing by the range (the difference between the maximum and minimum values) of the column. Z-score normalization standardizes data by subtracting the mean and dividing by the standard deviation, effectively scaling the data to have a mean of zero and a standard deviation of one. Quantile normalization aligns the distributions of different datasets to match a common distribution. Lastly, scaling to a range involves rescaling data to fit within a specified range, ensuring that all values fall between a predefined minimum

and maximum.

After setting parameters, click “Start” button, the results will be shown as below.

In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the table results with the selected normalization method (e.g. median normalization):

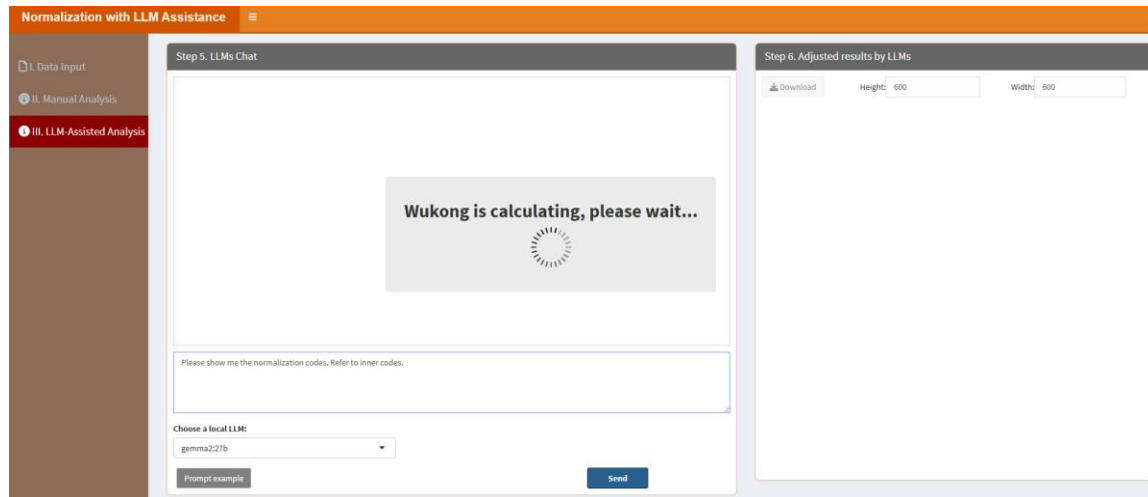
The screenshot shows a web application titled "Normalization with LLM Assistance". It has a sidebar with three sections: "I. Data Input", "II. Manual Analysis", and "III. LLM Assisted Analysis". The main content area is divided into two steps. Step 3, "Adjust Parameters", shows a dropdown menu for "3.1. Normalization Method" set to "Median" and a "Start" button. Step 4, "Display and Download Results", shows a "4.1. Tables" section with a "Download" button and a table of results. The table has columns A1 through B2 and rows of data. A search bar and pagination controls are also visible.

	A1	A2	A3	A4	B1	B2
AOA024RBG1	0.8265163110436978	0.4564046302270702	0.6714964638138178	0.5966650380563837	0.9540022563625359	0.398403871237111
AOA096LP49	1.4335422812334	1.370049479917705	1.274983712472002	1.269196576176745	1.343447083788684	1.40858454194946
AOA084J2A2	34.80408623171497	28.4023014299618	37.85953149538955	36.2635250359174	33.16615177686718	29.447850990029
AOA084J2F2	1.896752686822949	2.041532809841906	2.19579648663383	2.358517138296879	2.321317262936997	1.11320308765021
AOAV96	0.4830072591973745	0.5485953429850414	0.517641533752773	0.5106572705842173	0.5087951561655781	0.47161400358488
AOAV66	0.5421323151798849	0.6316466367431526	0.6131716231095987	0.6345723696829122	0.6352261306640531	0.18362774552046
AOAVT1	0.565109494030847	0.8409280760707926	1.038225093812308	0.9853572213797228	0.5767908896763725	1.0095391417841
A0FGR8	0.6855393368920434	0.582915428039808	0.6238336785974818	0.6922951047673092	0.6451928987372514	0.433026695909531
AOJLT2	1.233509238869348	0.9519168479955309	0.9770371247677984	0.7491034988610408	1.181729606408282	0.83531486464428
A0MZ66	0.9533994710040993	1.115065408034874	1.033726888328335	1.037612301223021	0.9933990327340818	1.08393815093141

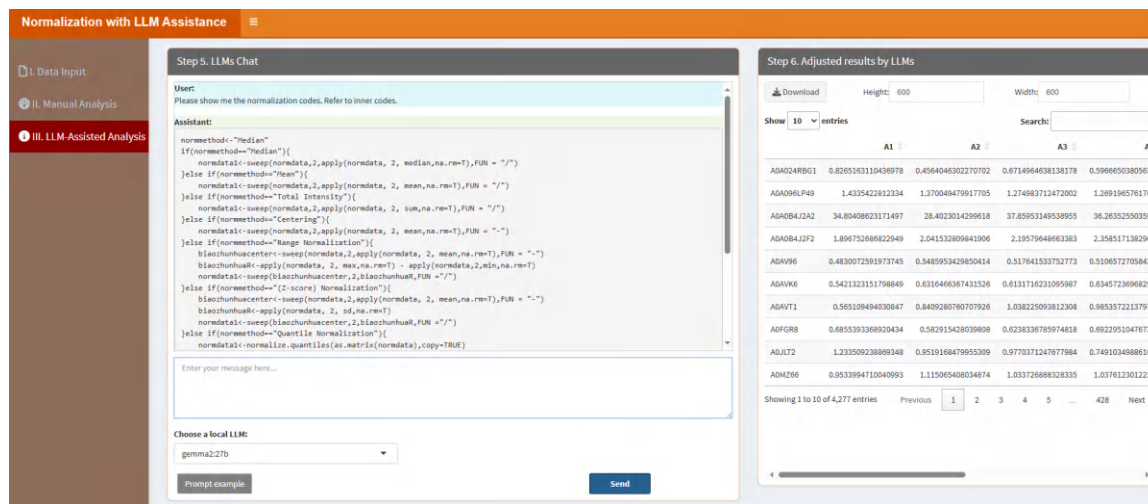
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please show me the normalization codes. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please show me the normalization codes. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



4.4. Two tables merging

Two tables merging here is the process of combining data from two separate tables into a single, unified dataset based on a common key or shared attributes. After clicking the “Start” button, it will be started in a new window like below:

Two tables merging with the assistance of LLMs

1. Upload expression data:

☒ I. Upload ☐ II. Load example data

1.1. Import table 1:

No file selected

1.2. Import table 2:

No file selected

Choose a Model:

gemma2 27b

2. Conversation:

3. Results:

3.1. Adjusted results by LLMs:

Type your message here...

Prompt example

The “1. Upload expression data:” here is similar to the “I. Data Input” part in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here. If users choose “I. Upload”, they can upload two data through “1.1. Import table 1” and “1.2. Import table 2” by clicking responding “Browse” button. Then they can click “Review table 1” and “Review table 2” to check the uploaded tables.

Next, users can choose a local LLM model from “Choose a Model” parameter and then type the prompt in the message box on the bottom, for example, the example prompt “Please merge the two uploaded tables by rownames. Refer to inner codes.” instructs LLMs to learn the internal R codes and merge the two uploaded tables by the row names. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please merge the two uploaded tables by rownames. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the merged result table.

After type the prompt, click “Send” button. It will learn like below:

Two tables merging with the assistance of LLMs

1. Upload expression data:
☐ I. Upload ☒ II. Load example data

Download example table 1

Download example table 2

Review table 1

Review table 2

Choose a Model:
gemma2.27b

2. Conversation:

Thinking.....

3. Results:

3.1. Adjusted results by LLMs

Download

Please merge the two uploaded tables by rownames. Refer to inner codes.

Prompt example

Send

The results are shown as below, then users can download the results by clicking the “Download” button in the “3. Results” section.

Two tables merging with the assistance of LLMs

1. Upload expression data:
☐ I. Upload ☒ II. Load example data

Download example table 1

Download example table 2

Review table 1

Review table 2

Choose a Model:
gemma2.27b

2. Conversation:

User:

Please merge the two uploaded tables by rownames. Refer to inner codes.

Assistant:

```
mergedata1$ID<-rownames(mergedata1)
mergedata2$ID<-rownames(mergedata2)
mergedata<-base::merge(mergedata1,mergedata2,by.x = "ID", by.y = "ID", all = FALSE,
                        all.x = FALSE, all.y = FALSE,sort =F)
mergedata
```

This code does the following:

- Adds an ID column:** It adds a column named "ID" to both `mergedata1` and `mergedata2`, populating it with the row names of each respective table.
- Merges the tables:** It uses the `merge()` function from the base R package to merge the two tables based on the "ID" column.
 - `by.x = "ID"` and `by.y = "ID"` specify that the merge should be performed using the "ID" column in both tables.
 - `all = FALSE`, `all.x = FALSE`, and `all.y = FALSE` ensure an inner join, meaning only rows with matching IDs in both tables will be included in the merged result.

3. Results:

3.1. Adjusted results by LLMs

Download

Show 10 entries

Search:

	ID	A1.x	A2.x	A3.x	A4.x	B1.x	B2.x	
1	O00232	40022.21484	38268.15234	37397.55469	37252.31641	33840.13281	33223.14063	3
2	O00233	30825.20313	30415.03516	28312.23633	28211.36281	31927.38867	26531.03711	2
3	O00255	55650.38672	31400.22266	42783.125	48400.74219	36529.41406	40784.94141	4
4	O00257	27196.81641	29829.97266	28457.19336	26979.64844	31409.35742	29636.14258	2
5	O00264	21749.39648	19034.07422	19304.38672	21512.60742	17964.30469	19966.81055	1
6	O00267	53880.75781	55765.82031	53673.48047	55927.08984	56692.33594	57982.69531	5
7	O00268	25498.04688	23963.42773	23345.96289	23373.03125	25372.68164	20313.74414	1
8	O00273	54662.63281	53106.98047	53042.30859	53599.48438	50778.10938	48472.99219	5
9	O00287	11172.78809	11407.97168	10950.60156	10236.04199	9232.460938	10103.10059	9
10	O00291	29089.50781	29933.86328	29534.35742	27817.23438	30978.83203	29862.93555	2

Showing 1 to 10 of 101 entries.

Previous

1

2

3

4

5

...

11

Next

Type your message here.

Prompt example

Send

4.5. Webpage content analysis

Webpage content analysis is a process that involves extracting the content of a webpage from a user-provided link and analyzing it using local LLMs. This module enables users to retrieve and process textual data, metadata, or other relevant information from webpages, providing insights into the content's structure, themes, or sentiment. By leveraging the capabilities of LLMs, the analysis can include summarization, keyword extraction, sentiment detection, or even deeper contextual understanding. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows a web application titled "Webpage Content Analysis with LLMs". It is divided into three main sections:

- 1. Upload Data:** Contains two radio buttons: "I. Paste Webpage Link" (which is selected) and "II. Example Webpage Link". Below them is a text input field with the placeholder "Paste the webpage link here:".
- 2. Conversation:** A section with a large, empty text area for conversation.
- 3. Interaction:** Contains a "Choose a Model:" dropdown menu with "deepseek-r132b" selected, a "Prompt example" button, a large text input field with the placeholder "Type your message here...", and a "Send" button.

“1. Upload data” here means users can type the webpage link or check the example webpage link. After typing the webpage link of interest, users can choose a local LLM from the “Choose a Model” parameter in the “3. Interaction” section and then type the prompt in the message box on the bottom. For example, the example prompt “Please extract all contents in this webpage and summary the keypoints of the contents.” instructs LLMs to learn the contents extracted from the webpage. After type the prompt, click “Send” button. It will learn like below:

Webpage Content Analysis with LLMs

1. Upload Data

☐ I. Paste Webpage Link
☒ II. Example Webpage Link

Example Webpage Link:

2. Conversation

Thinking.....



3. Interaction

Choose a Model:
deepseek-r1.32b

Prompt example

Please extract all contents in this webpage and summary the keypoints of the contents.

Send

The results are shown in the “2. Conversation” section as below:

Webpage Content Analysis with LLMs

1. Upload Data

☐ I. Paste Webpage Link
☒ II. Example Webpage Link

Example Webpage Link:

2. Conversation

User:
Please extract all contents in this webpage and summary the keypoints of the contents.
Webpage fetched successfully! The extracted contents are shown as below:
R is a programming language and software environment that has become the first choice for statistical computing and data analysis. Developed in the early 1990s by Ross Ihaka and Robert Gentleman, R was built to simplify complex data manipulation and create clear, customizable visualizations. Over time, it has gained popularity among statisticians, data scientists and researchers because of its capabilities and the vast array of packages available. As data-driven decision-making has grown, R has established itself as an important tool in various industries, including finance and healthcare, due to its ability to handle large datasets and perform in-depth statistical analysis. R is a unique language that offers a wide range of features for data analysis, making it an essential tool for professionals in various fields. Here's why R is preferred: To understand how R works, here's a basic example where we calculate the mean and standard deviation of a dataset: Output: [1] "Mean: 27.5 [1] "Standard Deviation: 15.1382517704875" R is used in a variety of fields, including: In this article, we've seen how R is an tool for data analysis, statistical computing and visualization. Its open-source nature, comprehensive feature set and active community make it an excellent choice for both beginners and seasoned professionals. A

3. Interaction

Choose a Model:
deepseek-r1.32b

Prompt example

Type your message here...

Send

The conversation history like below:

Webpage Content Analysis with LLMs

2. Conversation

User:

Please extract all contents in this webpage and summarize the keypoints of the contents.

Webpage fetched successfully! The extracted contents are shown as below:

R is a programming language and software environment that has become the first choice for statistical computing and data analysis. Developed in the early 1990s by Ross Ihaka and Robert Gentleman, R was built to simplify complex data manipulation and create clear, customizable visualizations. Over time, it has gained popularity among statisticians, data scientists and researchers because of its capabilities and the vast array of packages available. As data-driven decision-making has grown, R has established itself as an important tool in various industries, including finance and healthcare, due to its ability to handle large datasets and perform in-depth statistical analysis. R is a unique language that offers a wide range of features for data analysis, making it an essential tool for professionals in various fields. Here's why R is preferred: To understand how R works, here's a basic example where we calculate the mean and standard deviation of a dataset: `Output [1] "Mean: 27.571"` "Standard Deviation: 15.1362517704875" R is used in a variety of fields, including in this article, we've seen how R is an tool for data analysis, statistical computing and visualization. Its open-source nature, comprehensive feature set and active community make it an excellent choice for both beginners and experienced professionals. A

Assistant: Okay, so the user has provided a query where they want to extract all contents from a webpage and summarize the key points. The response given is a structured summary with sections like introduction, why R is preferred, example, applications, conclusion and references.

First, I need to understand what exactly the user is asking for. They seem to be looking for a way to process web content by extracting text and then condensing it into main ideas. This could be useful for someone who wants to quickly grasp the content without reading through the entire webpage.

Looking at the response, it's clear that the assistant has broken down the information into digestible parts. The introduction explains what R is and its creators. Then, it lists reasons why R is preferred, which includes open-source nature, extensive packages, visualization tools, community support, flexibility, and integration capabilities. An example code snippet is provided to show a practical use case.

The applications section highlights various industries where R is used, such as finance, healthcare, marketing, education, and government. The conclusion reinforces why R is valuable for both newbies and experts, mentioning its features and community support.

Now, thinking about the user's possible scenario, they might be someone who needs to analyze web content quickly, perhaps a student, researcher, or professional looking to gather information without manually reading through each webpage. They could be working on a project that requires summarizing data from multiple sources efficiently.

The user didn't specify any particular format for the summary, but the response provided is well-organized with clear headings and bullet points, making it easy to read. This suggests that the user values clarity and structure in their summaries.

I should consider if there are any deeper needs here. Maybe the user wants this process automated for multiple webpages or perhaps they need more detailed analysis beyond just summarizing key points. They might also be interested in how R compares to other tools, which isn't covered in the provided response but could be a valuable addition.

Another angle is whether the user has any specific technical requirements, like handling different types of content or extracting data from images or tables within webpages. The current response focuses on text extraction and summarization, so if the user needs more advanced features, that might require additional functionality.

In terms of potential improvements, adding examples for each application area could make the summary even more informative. Also, including a brief explanation of how R's open-source nature benefits users compared to proprietary software might add value.

Overall, the response seems comprehensive for the given query, but understanding the user's specific needs beyond the initial request would help tailor the solution better. For instance, if they're looking to integrate this into an application or use it for academic research, the approach might vary slightly.

Summary of Key Points:

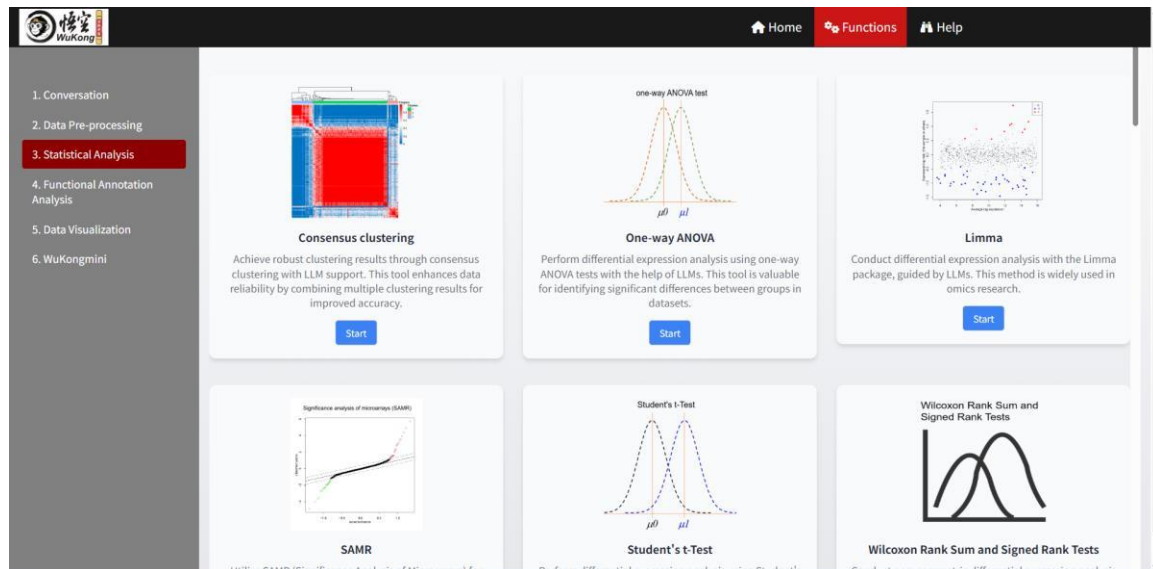
- Introduction to R**
 - R is a programming language and software environment primarily used for statistical computing and data analysis.
 - Developed in the early 1990s by Ross Ihaka and Robert Gentleman.
 - Designed to simplify complex data manipulation and create customizable visualizations.
- Popularity of R**
 - Gained popularity among statisticians, data scientists, and researchers due to its capabilities and extensive array of packages.
 - Widely used in industries like finance and healthcare for handling large datasets and performing in-depth statistical analysis.
- Why R is Preferred**
 - Open-source nature allows for flexibility and customization.
 - Extensive library of packages for various data analysis tasks.
 - Strong visualization tools for creating clear and customizable plots.
 - Active community support and continuous development.
 - Flexibility in handling both small and large datasets.
 - Integration with other tools and platforms.
- Example Code**
 - Example provided to calculate the mean and standard deviation of a dataset.
 - Output demonstrates basic functionality for statistical calculations.
- Applications of R**
 - Used in various fields, including finance, healthcare, marketing, education, and government.
 - Ideal for data-driven decision-making across industries.
- Conclusion**
 - R is a powerful tool for data analysis, statistical computing, and visualization.
 - Its open-source nature, comprehensive feature set, and active community make it suitable for both beginners and experienced professionals.

References:

- The webpage provides an overview of R's features, applications, and why it is preferred in various industries.

5. Statistical Analysis

Statistical analysis in proteomics plays a pivotal role in extracting meaningful insights from complex biological data. This section encompasses a wide array of models, including Consensus Clustering, One-way ANOVA, Limma, SAMR (Significance Analysis of Microarrays), Student's t-Test, Wilcoxon Rank Sum and Signed Rank Tests, Dynamic Network Biomarkers (DNBs), Factor Analysis, Grey Relational Analysis (GRA), Hierarchical Cluster Analysis (HCA), K-means Clustering, Lack of Fit F-test, Linear Regression, Logistic Regression, Mfuzz, Network Degree Matrix (NDM), OPLS-DA (Orthogonal Partial Least Squares Discriminant Analysis), Principal Component Analysis (PCA), Principal Coordinates Analysis (PCoA), PLS-DA (Partial Least Squares Discriminant Analysis), Power Analysis, Restricted Cubic Spline (RCS) Analysis, Redundancy Analysis (RDA), Rank Rank Hypergeometric Overlap (RRHO) Analysis, SIMCA (Soft Independent Modelling by Class Analogy), Time Course Data Analysis, t-SNE (t-Distributed Stochastic Neighbor Embedding), Tumor Purity Estimation, and UMAP (Uniform Manifold Approximation and Projection). Users can process their data by adjusting parameters in each module through the classic framework, while also leveraging local LLMs to gain deeper insights into each model and enhance their data analysis capabilities.



5.1. Consensus Clustering

Consensus clustering allows users to identify robust clusters within their data by aggregating results across multiple iterations. Users can adjust parameters like the number of clusters through the classic framework, while leveraging local LLMs to refine cluster stability and interpret the results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Consensus clustering with LLM Assistance' interface. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and 'No file selected' text. There are also two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' field has the value '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show: 10 entries' dropdown, a 'Search:' input field, and a table with one row: '1 No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' with 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Consensus clustering with LLM Assistance' interface. The sidebar is the same as in the previous screenshot. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three input fields: '3.1. Cluster number:' with the value '3', '3.2. Figure Height:' with the value '600', and '3.3. Figure Width:' with the value '600'. Below these fields is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a section for '4.1. Barplot/Visualization' with a 'Save image' button.

3.1. *Cluster number*: maximum cluster number to evaluate.

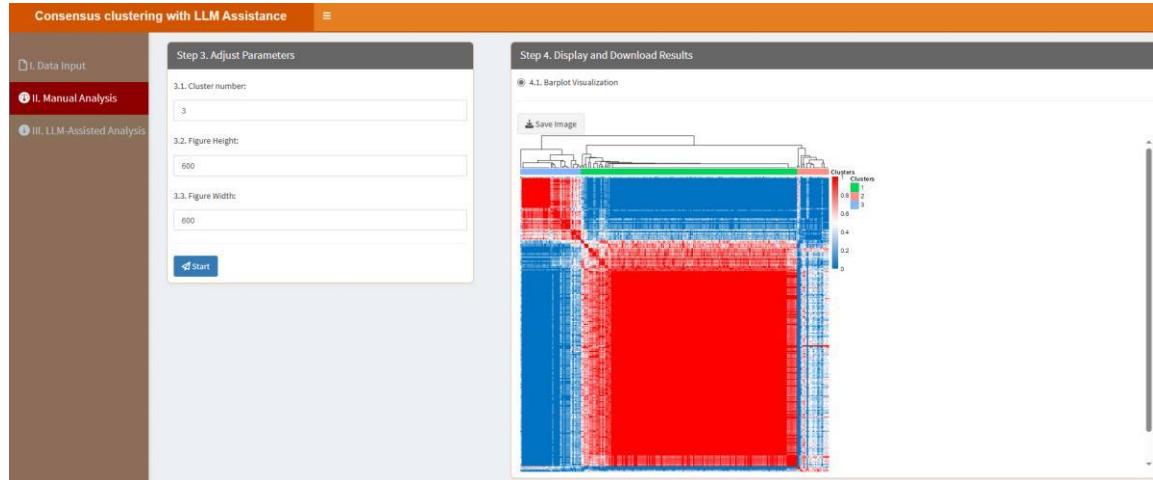
3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization,

ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

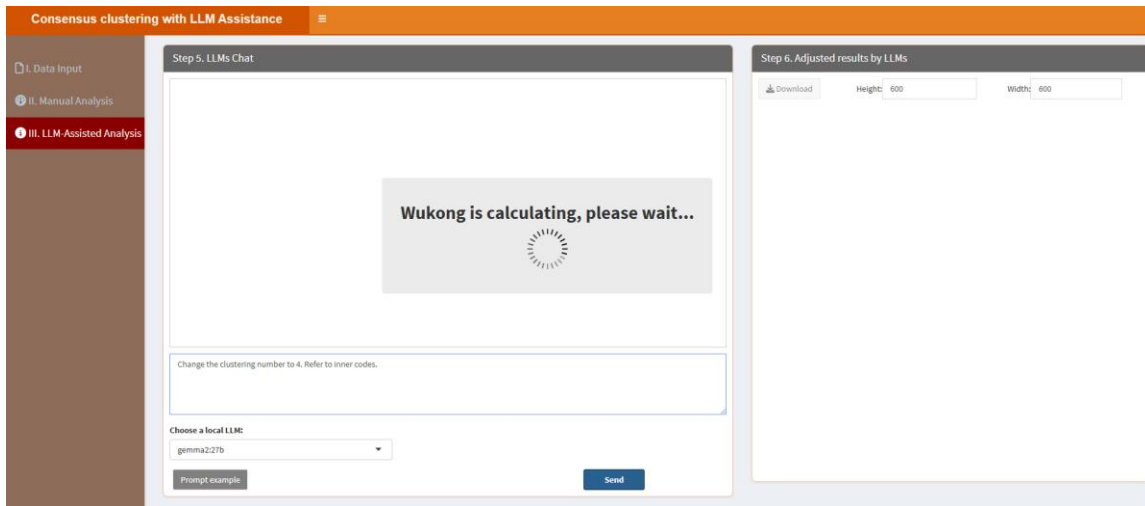
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



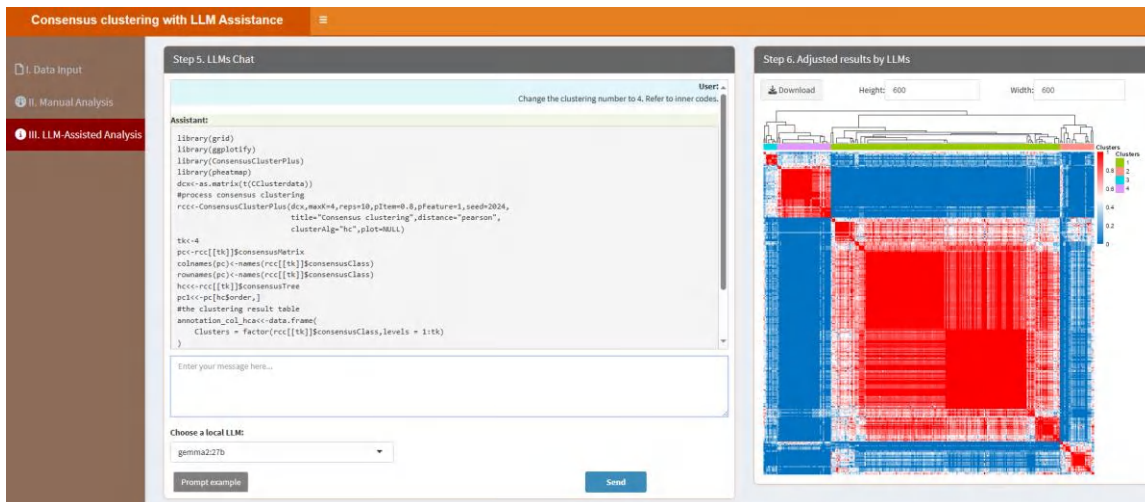
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the clustering number to 4. Refer to inner codes.” instructs LLMs to learn the internal R codes and change the clustering number to 4. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the clustering number to 4. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.2. One-way ANOVA

One-way ANOVA is a statistical method for comparing means across multiple groups. Users can modify parameters such as group definitions using the classic framework and utilize local LLMs to enhance the interpretation of significant differences. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'One-way ANOVA test with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one entry. The table has columns 'Show' (set to 10) and 'entries'. The entry is '1' with a description 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'One-way ANOVA test with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three input fields: '3.1. Group and replicate number:' with the value '3;4-4-4', '3.2. Group names:' with the value 'A;B;C', and a checked checkbox for '3.3. Log2 (base-2 logarithm) transformation used?'. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a table with one entry. The table has columns '4.1. Tables' and 'Download'. The entry is '1' with a 'Download' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. *Log2 (base-2 logarithm) transformation used?* if true, the data will be transformed to the logarithmic scale with base 2.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:

One-way ANOVA test with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 3. Adjust Parameters

3.1. Group and replicate number:

3x4-4

3.2. Group names:

A,B,C

☒ 3.3. Log2 (base-2 logarithm) transformation used?

Start

Step 4. Display and Download Results

4.1. Tables

Download

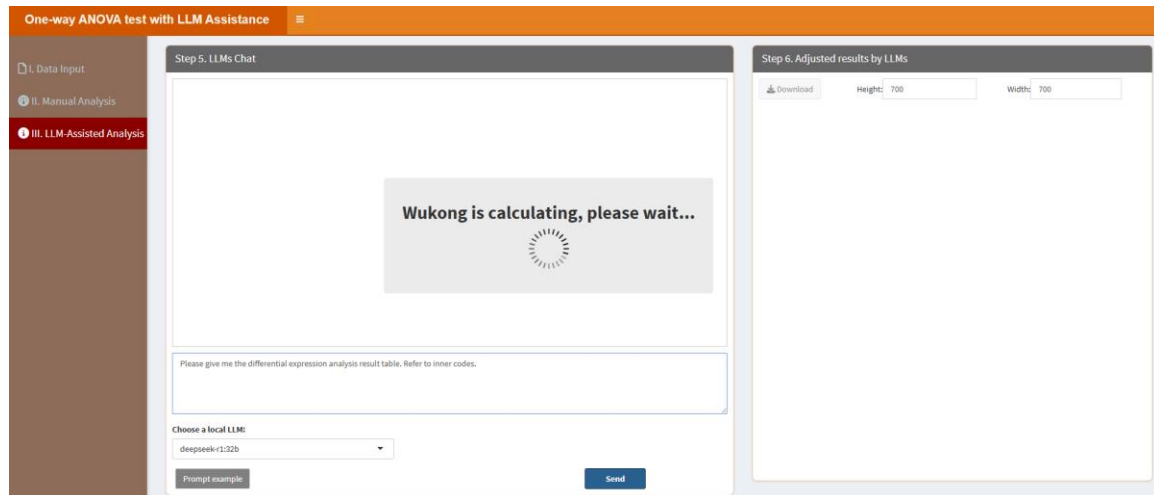
	B3	B4	C1	C2	C3	C4	pValue	p.adjust
01671886985157	13.40143440364404	14.07821189910432	14.1610406721646	14.17349125555678	14.05379598817313	0.3194883	0.337874856355094	
1.1381949161272	15.06382870819366	13.80778167064255	13.94991262623374	13.88639273452914	14.08672713447965	0	0	
83633848030308	20.10080684481088	15.89396711911008	15.80341763768959	15.89270189197624	15.08749990804194	0	0	
1.1338982932969	16.11689014189156	14.39244335017123	14.45261541023951	14.49955096560628	14.27387341441645	0.0004	0.0006030313711667255	
1.7757651852179	13.61357254099118	14.51193128615062	14.29459032174756	14.47331805684055	14.25095571748606	0.0000262	0.00004895474006116208	
8540052606822	13.97908263862132	17.2338026503974	17.43088178840561	17.53846793379661	17.64249956256775	0.000034	0.00000775562666666666	
47246761669556	14.02451302088295	14.43399121443084	14.38264500150031	14.44713940889194	13.86486274461877	0.4176738	0.4348565829113924	
83433312817034	13.53643469921329	16.10450133785368	16.35588524347304	16.30858589402025	16.46575470841427	1e-7	3.323232323232323e-7	
12784259714451	14.96854074378904	13.52475953525881	13.57102476150362	13.8450139957566	13.40846863805335	0.0000999	0.0003670990614000782	
02974096067805	14.91246845527305	15.84834277029629	15.75774645883827	16.03953005617862	15.82410142911429	2e-7	6.194062273714699e-7	

428Next

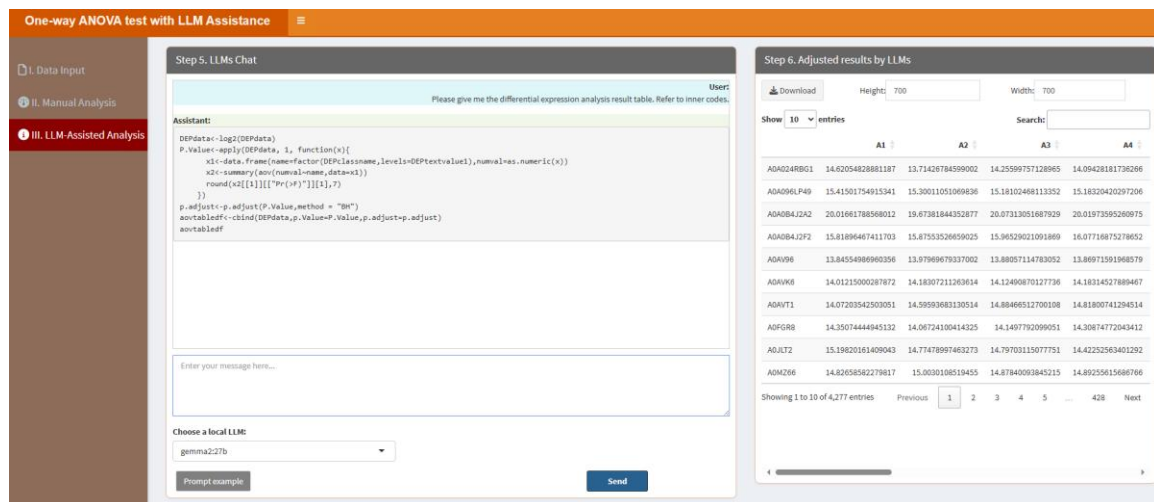
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please give me the differential expression analysis result table. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please give me the differential expression analysis result table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.3. Limma

Limma is a powerful tool for analyzing differential expression in omics datasets. Users can set parameters like contrasts and Log2 (base-2 logarithm) transformation through the classic framework, while local LLMs can help refine the analysis and provide deeper insights into the results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Limma with LLM' web interface. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' and 'Is the first column names?', and a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one entry. The table has a 'Show' dropdown set to '10 entries' and a 'Search:' input field. The table header is 'Description'. The single entry is '1 No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Limma with LLM Assistance' web interface. On the left is a sidebar with three menu items: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains four sections: '3.1. Group and replicate number:' with a text input '2;4-4'; '3.2. Group names:' with a text input 'A,B'; '3.3. Log2 (base-2 logarithm) transformation used for expression data?' with a checked checkbox; and '3.4. Log2 used for Fold Change?' with a checked checkbox. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a table with one entry '4.1. Tables'. Below the table is a 'Download' button with a download icon.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and

distinguishing the groups during data analysis and visualization.

3.3. Log2 (base-2 logarithm) transformation used for expression data? If true, the data will be transformed to the logarithmic scale with base 2.

3.4. Log2 used for Fold Change? If true, the Fold Change is calculated with log2, for example, $\text{Fold Change} = \log_2(\text{Experiment}_{\text{mean}}/\text{Control}_{\text{mean}})$, otherwise, $\text{Fold Change} = \text{Experiment}_{\text{mean}}/\text{Control}_{\text{mean}}$. $\text{Experiment}_{\text{mean}}$ is the mean value of the Experiment group, $\text{Control}_{\text{mean}}$ is the mean value of the Control group.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:

Step 3. Adjust Parameters

3.1. Group and replicate number:
2;4-4

3.2. Group names:
A,B

☒ 3.3. Log2 (base-2 logarithm) transformation used for expression data?

☒ 3.4. Log2 used for Fold Change?

Step 4. Display and Download Results

4.1. Tables

Download

Search:

B1	B2	B3	B4	Fold.Chance	p.Value	p.adjust
3093015007	13.44097191575459	13.01671880985197	13.40143449364404	-0.5132193072991509	0.2218575112975701	0.4419583492406648
3782850526	15.26291450023383	15.1383949161272	15.06382870819366	-0.0868627272434614	0.2658341002848356	0.4902214686536631
3815425873	19.64875879420833	19.83633848039398	20.10080684481088	-0.0761826312565006	0.5454479517669851	0.7272525584980989
7214603946	14.92338516003079	16.1536882932969	16.11686014169156	-0.1217667908384437	0.6682787048259151	0.8071818281108271
3527251025	13.68434680655119	13.7757651852179	13.61357254099118	-0.1589159813048422	0.0354864359972152	0.1610952572084335
7059698496	12.32352431128382	13.85400526086322	13.97508263882132	-0.5410733219333927	0.2007397854623117	0.4182096924903926
4871418762	14.78236519252264	14.47246761669556	14.02453302088295	-0.2610325629982739	0.2999662169587614	0.5221634147059919
3089625204	13.56119621629958	13.83433312617014	13.53643469921329	-0.4339293614996862	0.0250649495635852	0.131575455723306
2839919547	14.50906036120879	15.12784259714451	14.96654074378904	0.1232059319560594	0.542047594698486	0.7249335717272111
7367860924	14.88495079414307	15.02974066067805	14.91246845527305	0.01451995215998458	0.8052649105728511	0.8943438126512813

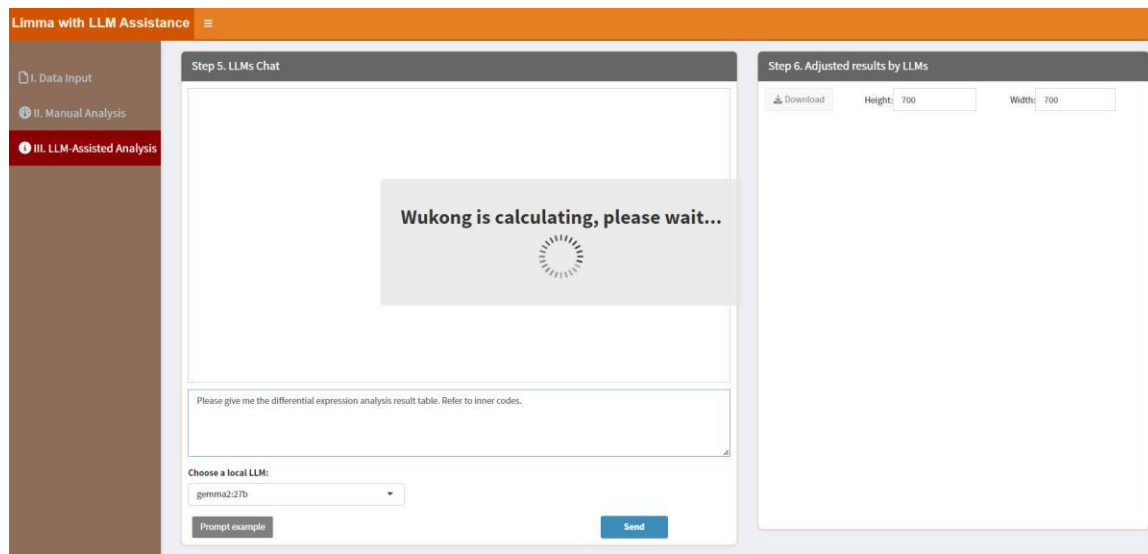
4 5 ... 428 Next

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

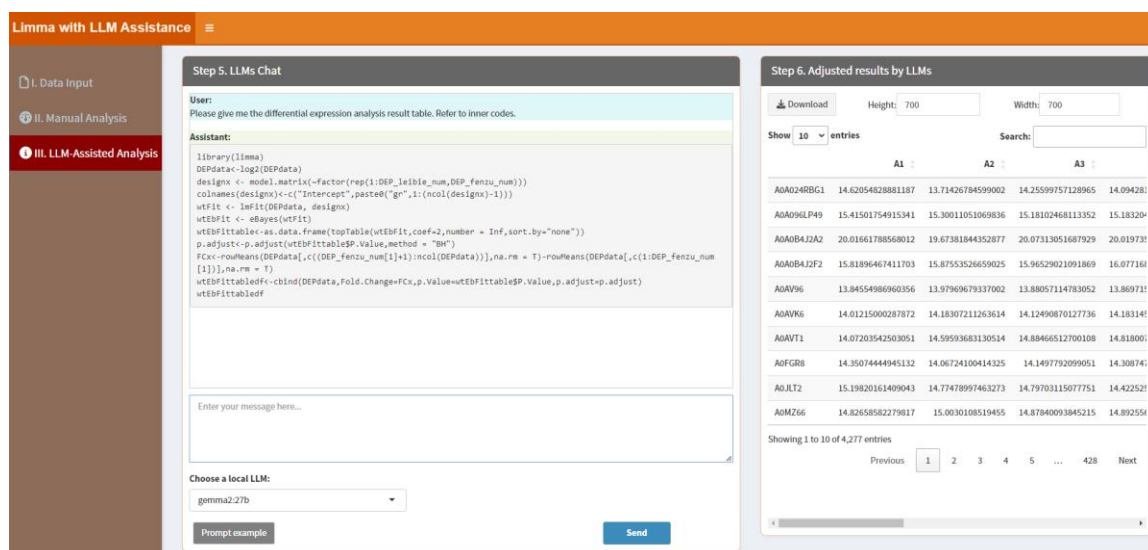
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please give me the

differential expression analysis result table. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please give me the differential expression analysis result table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.4. SAMR (Significance Analysis of Microarrays)

SAMR is a statistical technique to identify significantly expressed features in omics data. Users can adjust sample information and Log2 (base-2 logarithm) transformation using the classic framework, and local LLMs can assist in refining the analysis and interpreting the findings. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Significance analysis of microarrays (SAM) with LLM Assistance' web application. The left sidebar has three tabs: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one entry: '1 No data loaded. Please upload your dataset or use the example data.' The table has a 'Description' header. At the bottom of the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Significance analysis of microarrays (SAM) with LLM Assistance' web application. The left sidebar has three tabs: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three sections: '3.1. Group and replicate number:' with a text input field containing '2;4-4'; '3.2. Group names:' with a text input field containing 'A,B'; and '3.3. Log2 (base-2 logarithm) transformation used for expression data?' and '3.4. Log2 used for Fold Change?' both with checked checkboxes. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Tables' section with a 'Download' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and

distinguishing the groups during data analysis and visualization.

3.3. Log₂ (base-2 logarithm) transformation used for expression data? If true, the data will be transformed to the logarithmic scale with base 2.

3.4. Log₂ used for Fold Change? If true, the Fold Change is calculated with log₂, for example, $\text{Fold Change} = \log_2(\text{Experiment}_{\text{mean}}/\text{Control}_{\text{mean}})$, otherwise, $\text{Fold Change} = \text{Experiment}_{\text{mean}}/\text{Control}_{\text{mean}}$. $\text{Experiment}_{\text{mean}}$ is the mean value of the Experiment group, $\text{Control}_{\text{mean}}$ is the mean value of the Control group.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:

Significance analysis of microarrays (SAM) with LLM Assistance

Step 3. Adjust Parameters

3.1. Group and replicate number:
2,4-4

3.2. Group names:
A,B

☒ 3.3. Log₂ (base-2 logarithm) transformation used for expression data?

☒ 3.4. Log₂ used for Fold Change?

Start

Step 4. Display and Download Results

4.1. Tables

Download

Show 10 entries

Search:

	A1	A2	A3	A4	B1	B2
AD024RB01	14.6205482881187	13.71426784599002	14.25599757128965	14.09428181736266	14.773093015007	13.44097191575459
AD096LPH9	15.41501754915341	15.30011051069838	15.18102468113352	15.1832040297206	15.2696792850526	15.26291450023383
AD08BJ2A2	20.01661785588012	19.67381844352877	20.07313051687929	20.01973595260975	19.89266815425873	19.64875879420833
AD08BJ2F2	15.81896467411703	15.87533526659025	15.96529021091869	16.07716875278852	16.05597214603946	14.92138516003079
ADAV96	13.84554986960356	13.97969679337002	13.88057114763052	13.88971591968579	13.86618527251025	13.68434680655119
ADAV96	14.01215000287872	14.18307211263614	14.12490870127736	14.18314527889467	14.18637059696496	12.3252431128382
ADAVT1	14.07203542503051	14.59559683130514	14.88466512700108	14.81800741294514	14.04714071418762	14.78236519252264
ADIGR8	14.35074444945132	14.06724100414325	14.1497792099051	14.30874772043412	14.20883089625204	13.56119621629956
ADJLT2	15.19820161409043	14.77478997463273	14.79703115077751	14.42252563401292	15.08192839919547	14.50906036126879
ADH266	14.62658582279817	15.0030108519455	14.87840093845215	14.89255615686766	14.83147367860924	14.88495078414307

Showing 1 to 10 of 4,277 entries

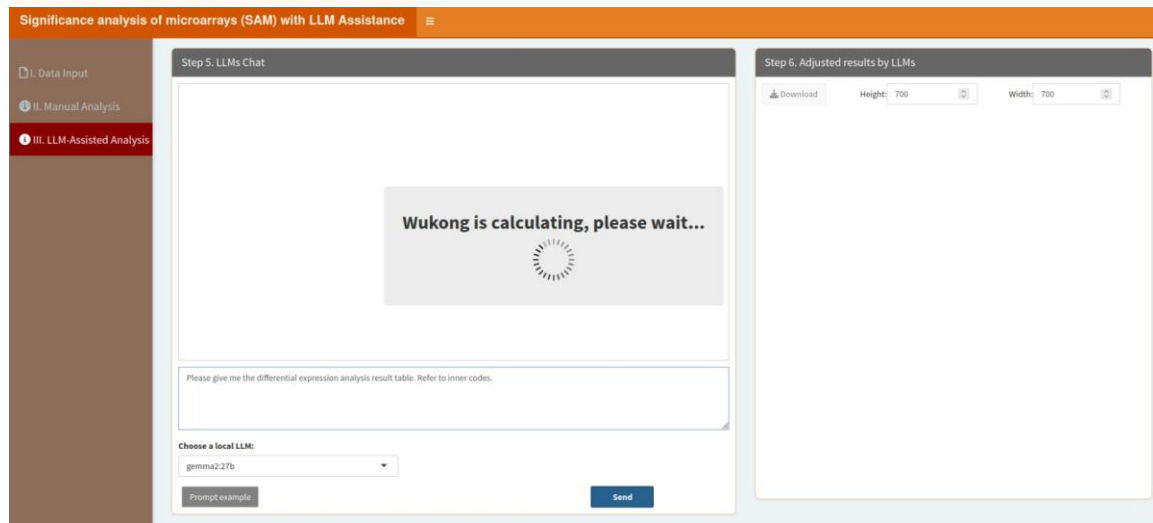
Previous 1 2 3 4 5 ... 428 Next

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

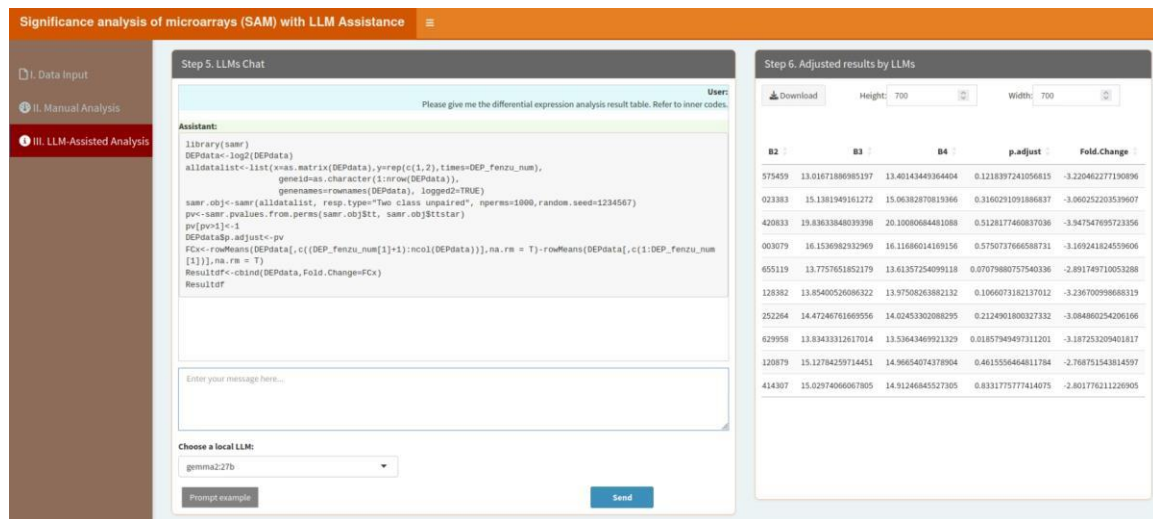
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please give me the

differential expression analysis result table. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please give me the differential expression analysis result table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.5. Student's t-Test

Student's t-Test is a fundamental statistical test for comparing two groups. Users can configure sample information and Log2 (base-2 logarithm) transformation through the classic framework, while local LLMs can aid in refining the analysis and understanding the results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Student's t-Test with LLM' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please input your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom is a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one entry: '1 No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' with 'Previous', '1', and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Student's t-Test with LLM' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three sections: '3.1. Group and replicate number:' with a text input field containing '2;4-4'; '3.2. Group names:' with a text input field containing 'A,B'; and '3.3. Log2 (base-2 logarithm) transformation used for expression data?' and '3.4. Log2 used for fold change?' both with checked checkboxes. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Tables' section with a 'Download' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. *Log2 (base-2 logarithm) transformation used for expression data?* If true, the data will be transformed to the logarithmic scale with base 2.

3.4. *Log2 used for Fold Change?* If true, the Fold Change is calculated with log2, for example, $\text{Fold Change} = \log_2(\text{Experiment}_{\text{mean}}/\text{Control}_{\text{mean}})$, otherwise, $\text{Fold Change} = \text{Experiment}_{\text{mean}}/\text{Control}_{\text{mean}}$. $\text{Experiment}_{\text{mean}}$ is the mean value of the Experiment group, $\text{Control}_{\text{mean}}$ is the mean value of the Control group.

After setting parameters, click “Start” button, the results will be shown as below:

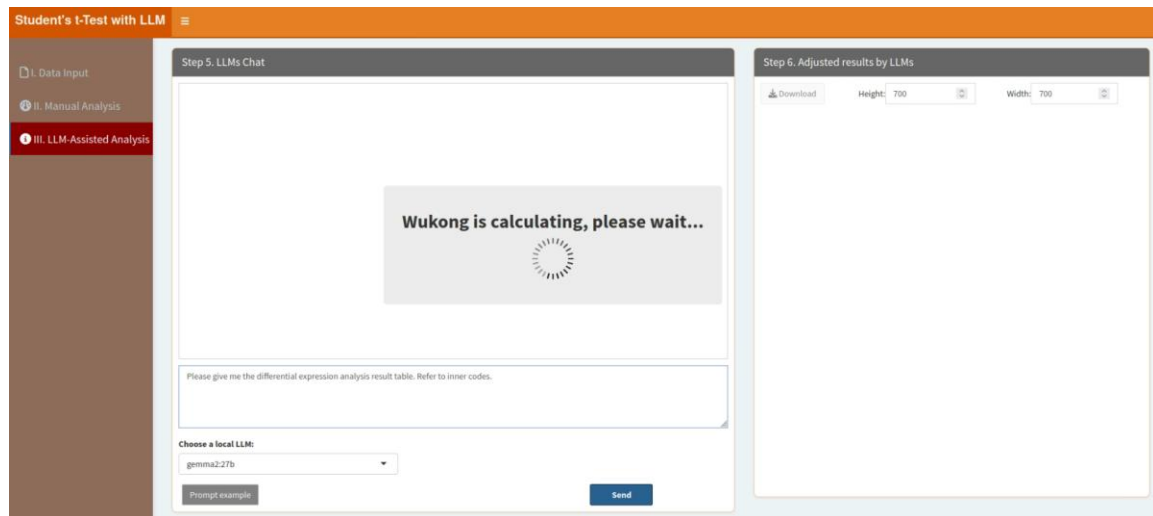
In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

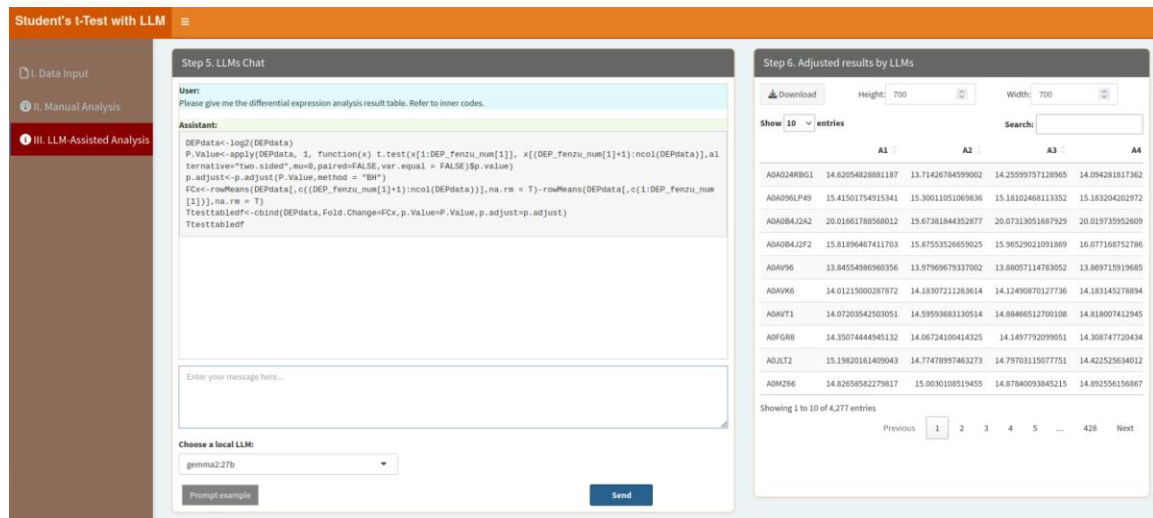
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please give me the differential expression analysis result table. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a

trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please give me the differential expression analysis result table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.6. Wilcoxon Rank Sum and Signed Rank Tests

These non-parametric tests are ideal for comparing groups without assuming normality. Users can customize test settings through the classic framework and rely on local LLMs to improve result interpretation and analysis. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' interface. On the left, a sidebar contains three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main panel has two tabs: 'A. Upload' (selected) and 'B. Load example data'. Under 'A. Upload', there is a 'Select File Format:' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are two checkboxes: 'Is the first row names?' (checked) and 'Is the first column names?' (checked). At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. On the right, 'Step 2. Display Uploaded Data/Example Data' is visible, showing a table with one entry and a 'Download' button.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' interface. On the left, the sidebar is the same as in the previous screenshot. The main panel has two tabs: 'A. Upload' and 'B. Load example data'. Under 'B. Load example data', there is a '3.1. Group and replicate number:' input field with '2;4-4' entered. Below this is a '3.2. Group names:' input field with 'A,B' entered. There are two checkboxes: '3.3. Log2 (base-2 logarithm) transformation used for expression data?' (checked) and '3.4. Log2 used for Fold Change?' (checked). At the bottom, there is a 'Start' button. On the right, 'Step 4. Display and Download Results' is visible, showing a table with one entry and a 'Download' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Log2 (base-2 logarithm) transformation used for expression data? If true, the data will be transformed to the logarithmic scale with base 2.

3.4. Log2 used for Fold Change? If true, the Fold Change is calculated with log2, for example, $\text{Fold Change} = \log_2(\text{Experiment}_{\text{mean}}/\text{Control}_{\text{mean}})$, otherwise, $\text{Fold Change} = \text{Experiment}_{\text{mean}}/\text{Control}_{\text{mean}}$. $\text{Experiment}_{\text{mean}}$ is the mean value of the Experiment group, $\text{Control}_{\text{mean}}$ is the mean value of the Control group.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:

Wilcoxon Rank Sum and Signed Rank Tests with LLM Assistance

Step 3. Adjust Parameters

3.1. Group and replicate number:
2,4-4

3.2. Group names:
A/B

3.3. Log2 (base-2 logarithm) transformation used for expression data?
☒

3.4. Log2 used for Fold Change?
☒

Start

Step 4. Display and Download Results

4.1. Tables

Download

Show 10 entries

	A1	A2	A3	A4	B1	B2	
AD024RBG1	14.62054828881187	13.71426784599002	14.2559757128965	14.09428181736266	14.773093015007	13.44097191575459	13.0167188691
AD096LP49	15.41501754915341	15.30011051009836	15.18102468113352	15.1832040297206	15.26096792850526	15.26291450023383	15.138194916
AD084J2A2	20.01661789568012	19.87381844352877	20.07313051687929	20.01973595260975	19.89266815425873	19.64875879420833	19.8363384802
AD084J2F2	15.81896467411703	15.8753526659025	15.96529021091869	16.07716875278652	16.05997214603946	14.92338516003079	16.153698293
AD0V96	13.84554986960396	13.97969679337002	13.88037114783052	13.86971391968579	13.86618527251025	13.68434686955119	13.775765182
AD0V96	14.01215000287872	14.18307211263614	14.12490670127736	14.18314527889467	14.18637059698496	12.32352431128382	13.8540052606
AD0V71	14.07203542503051	14.59593683130514	14.88466512700108	14.81800741294514	14.04714871418762	14.78236513252284	14.4724676196
AD0GR8	14.35074444945132	14.08724100414325	14.149779209051	14.30874772043412	14.20883089625204	13.56119621629958	13.8343331261
ADJLT2	15.19820161409043	14.77478997463273	14.79703115077751	14.42252563401292	15.08132839913547	14.50960636120879	15.1278425973
AD0Z66	14.82658582279817	15.0030106519455	14.87840093845215	14.89255615686766	14.83147367860924	14.88495070414307	15.0297406606

Showing 1 to 10 of 4,277 entries

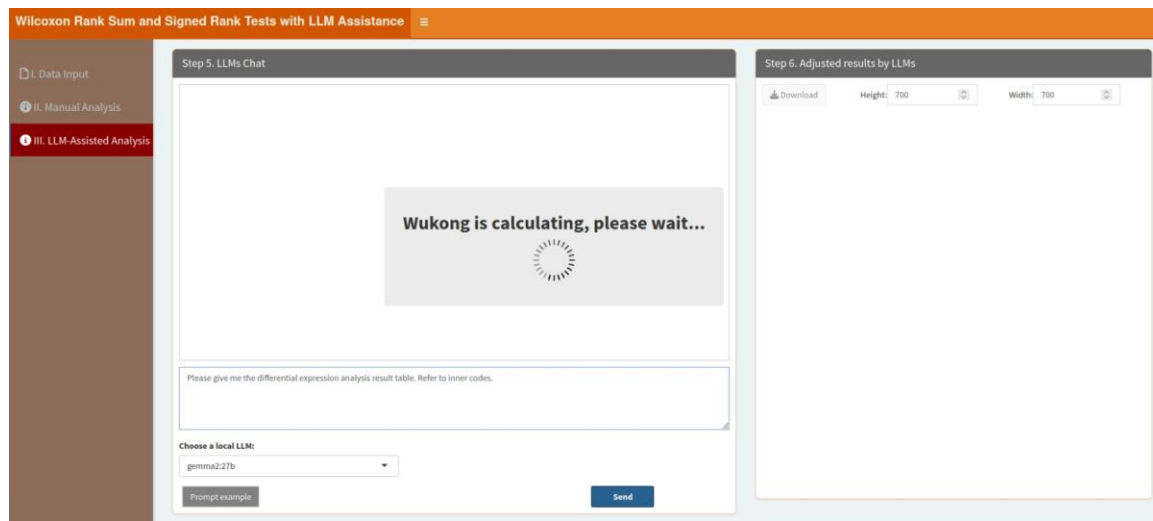
Previous 1 2 3 4 5 ... 428 Next

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

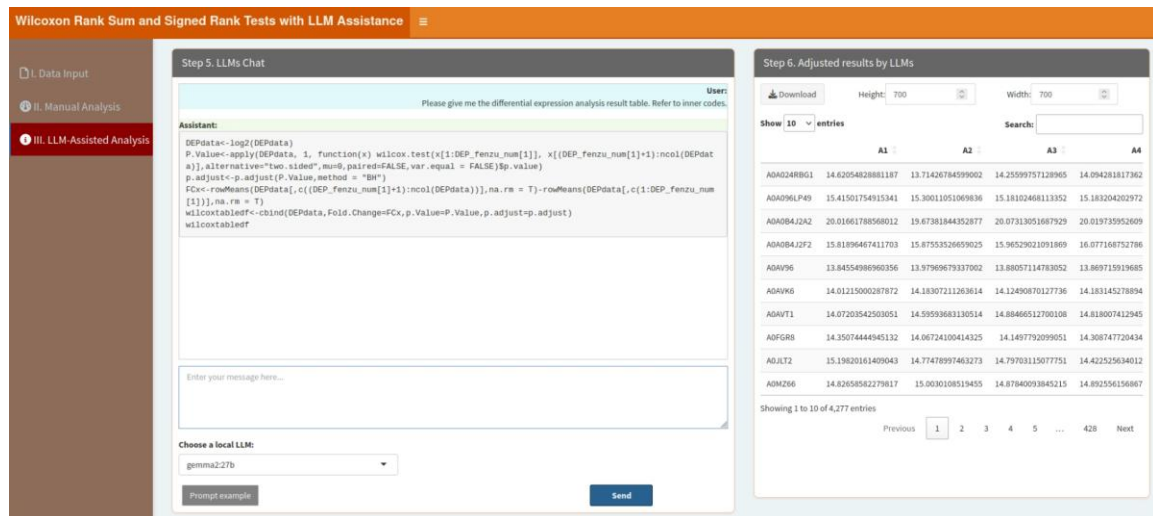
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please give me the differential expression analysis result table. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a

trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please give me the differential expression analysis result table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.7. Dynamic Network Biomarkers (DNBs)

Dynamic Network Biomarkers help identify potential biomarkers in biological systems. Users can fine-tune network parameters using the classic framework, while local LLMs provide enhanced insights into this model. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Dynamical Network Biomarkers (DNB) with LLM' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains radio buttons for 'A. Upload' (selected) and 'B. Load example data'. Below are options for file format (.xlsx, .xls, .csv/txt) and a file upload area with a 'Browse...' button. There are also checkboxes for 'Is the first row names?' and 'Is the first column names?', and a dropdown for 'Which sheet to read?'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one entry: 'No data loaded. Please upload your dataset or use the example data.' It includes a search bar, a 'Show 10 entries' dropdown, and pagination controls.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Dynamical Network Biomarkers (DNB) with LLM' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains five numbered input fields: '3.1. Time/Group and replicate number:' (with example '5;10-10-14-14-26'), '3.2. Time/Group names:' (with example 'T1;T2;T3;T4;T5'), '3.3. Select a color for the line:' (with a color picker showing '#4682B4'), '3.4. Figure Height:' (with a value of 600), and '3.5. Figure Width:' (with a value of 600). There is a 'Start' button at the bottom. The right panel, titled 'Step 4. Display and Download Results', contains radio buttons for '4.1. Visualization' (selected) and '4.2. Tables'. Below is a section for 'DNB score distribution:' with a 'Save Image' button.

3.1. Time/Group and replicate number. Type in the timepoint/group number and replicate number here. Please note, the timepoint/group number and replicate number are linked with “;”, and the replicate number of each group is linked with “-”. For example, if you have two timepoints/groups, each group has three replicates, then you should type in “2;3-3” here. Similarly, if you have 3 groups/timepoints with 5 replicates in every groups, you should type in “3;5-5-5”. The example data has 5 timepoints and (10, 10, 14, 14, 26) replicates in each group,

so here is "5;10-10-14-14-26".

3.2. *Time/Group names*: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 5 timepoints in the example data, you can type in "T1;T2;T3;T4;T5".

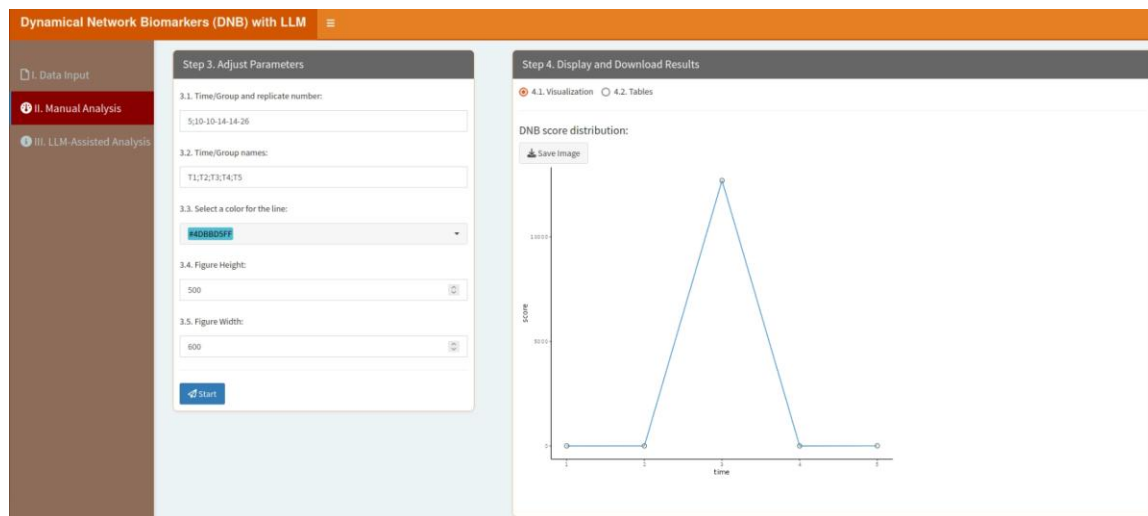
3.3. *Select a color for the line*: Select a color for the line in the figure.

3.5. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click "Start" button, the results will be shown as below:

In the "4.1. Visualization" part, users can click "Save image" button to download the figure:



In the "4.2. Tables" part, users can click "Download" button to download the table, which contains the DNB potential biomarkers:

Dynamical Network Biomarkers (DNB) with LLM

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 3. Adjust Parameters

3.1. Time/Group and replicate number:
5:10-10:14-14:26

3.2. Time/Group names:
T1;T2;T3;T4;T5

3.3. Select a color for the line:
#4DB6FF

3.4. Figure Height:
500

3.5. Figure Width:
600

Start

Step 4. Display and Download Results

4.1. Visualization 4.2. Tables

DNB potential biomarkers:

Download

Show 10 entries

Search:

	Markers
1	g.1185
2	g.442
3	g.1106
4	g.1152
5	g.207
6	g.1302
7	g.1323
8	g.326
9	g.410
10	g.768

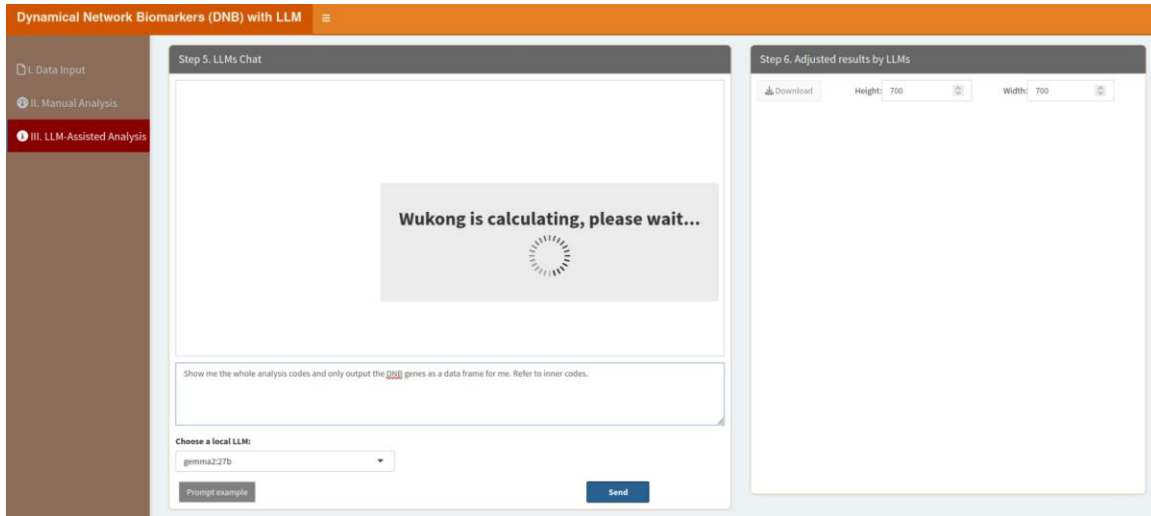
Showing 1 to 10 of 39 entries

Previous 1 2 3 4 Next

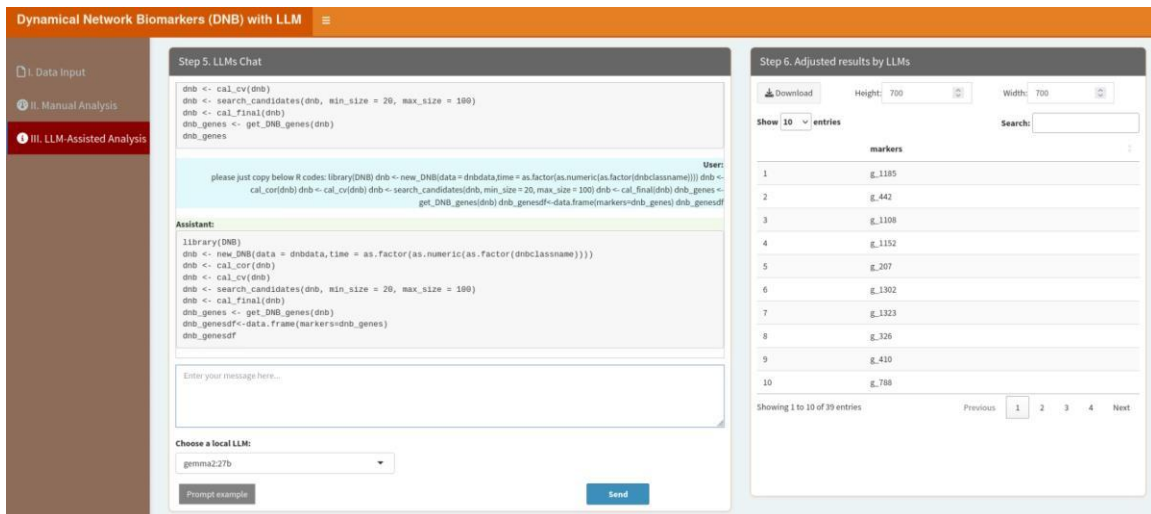
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Show me the whole analysis codes and only output the results. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Show me the whole analysis codes and only output the results. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.8. Factor Analysis

Factor analysis reduces data dimensionality by identifying latent variables. Users can adjust the number of factors and sample information via the classic framework, and local LLMs help refine the analysis and interpret the factors. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Factor Analysis with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a note 'No file selected'. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', features a 'Show 10 entries' dropdown, a search bar, and a table with one row containing the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Factor Analysis with LLM Assistance' interface, specifically Step 3: Adjust Parameters. The sidebar on the left has 'II. Manual Analysis' selected. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains several input fields: '3.1. Group and replicate number:' with the value '2;4-4', '3.2. Group names:' with the value 'A,B', '3.3. Select colors for Groups:' with a dropdown showing '1: #1f77b4' and '2: #ff7f0e', '3.4. Number of factors:' with the value '3', '3.5. Figure Height:' with the value '600', and '3.6. Figure Width:' with the value '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a '4.1. Visualization' radio button (selected) and a '4.2. Summary Results' radio button. Below these is a 'Save Image' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for groups: This parameter lets you choose specific colors for each group. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of groups, ensuring that the visual representation is both informative and aesthetically pleasing.

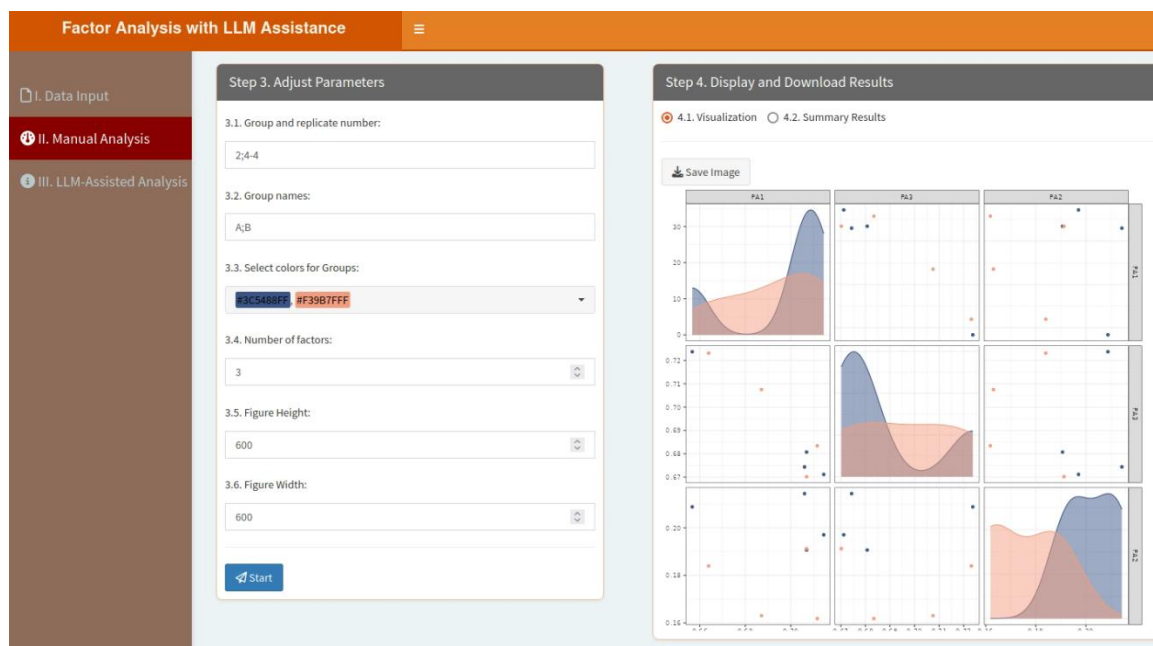
3.4. Number of factors: Number of factors to extract.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



In the “4.2. Summary Results” part, users can click “Download” button to download the table, which contains the results summarized from the factor analysis model:

Factor Analysis with LLM Assistance

Step 3. Adjust Parameters

3.1. Group and replicate number:
2x4

3.2. Group names:
A,B

3.3. Select colors for Groups:
#1C85BA #F38B1F

3.4. Number of factors:
3

3.5. Figure Height:
600

3.6. Figure Width:
600

Start

Step 4. Display and Download Results

4.1. Visualization 4.2. Summary Results

Factor Analysis using method = pa
Call: fa(r = factoranalysisdata, nfactors = input\$factormux, rotate = "varimax",
max.iter = 100, fs = "pa")
Standardized loadings (pattern matrix) based upon correlation matrix

	PA1	PA2	h2	u2	com
A1	0.71	0.67	0.21	1.00	0.00070 2.2
A2	0.66	0.72	0.21	1.00	0.00142 2.2
A3	0.71	0.67	0.20	1.00	0.00022 2.1
A4	0.71	0.68	0.19	1.00	0.00051 2.1
B1	0.71	0.67	0.19	0.99	0.03432 2.1
B2	0.66	0.72	0.18	1.00	0.00248 2.1
B3	0.69	0.71	0.16	1.00	0.00070 2.1
B4	0.71	0.68	0.16	1.00	0.00036 2.1

SS loadings
3.86 3.83 0.29
Proportion Var 0.48 0.48 0.04
Cumulative Var 0.48 0.96 1.00
Proportion Explained 0.48 0.48 0.04
Cumulative Proportion 0.48 0.96 1.00

Mean item complexity = 2.1
Test of the hypothesis that 3 factors are sufficient.
df null model = 20 with the objective function = 42.91 with Chi Square = 1952.48
df of the model are 7 and the objective function was 6.36

The root mean square of the residuals (RMSR) is 0
The df corrected root mean square of the residuals is 0

The harmonic n.obs is 50 with the empirical chi square 0 with prob < 1
The total n.obs was 50 with Likelihood Chi Square = 15.09 with prob < 0.028

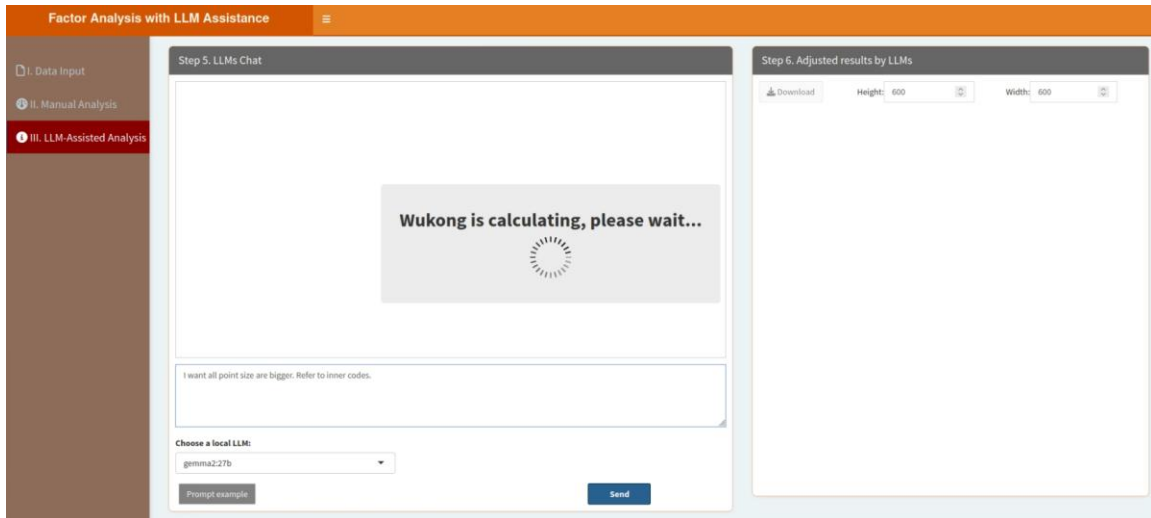
Tucker Lewis Index of factoring reliability = 0.981
RMSEA index = 0.156 and the 90 % confidence intervals are 0.049 0.266
BIC = -11.69
Fit based upon off diagonal values = 1
Measures of factor score adequacy

	PA1	PA2	PA3	PA2
Correlation of (regression) scores with factors	0.96	0.96	0.89	
Multiple R square of scores with factors	0.92	0.92	0.80	

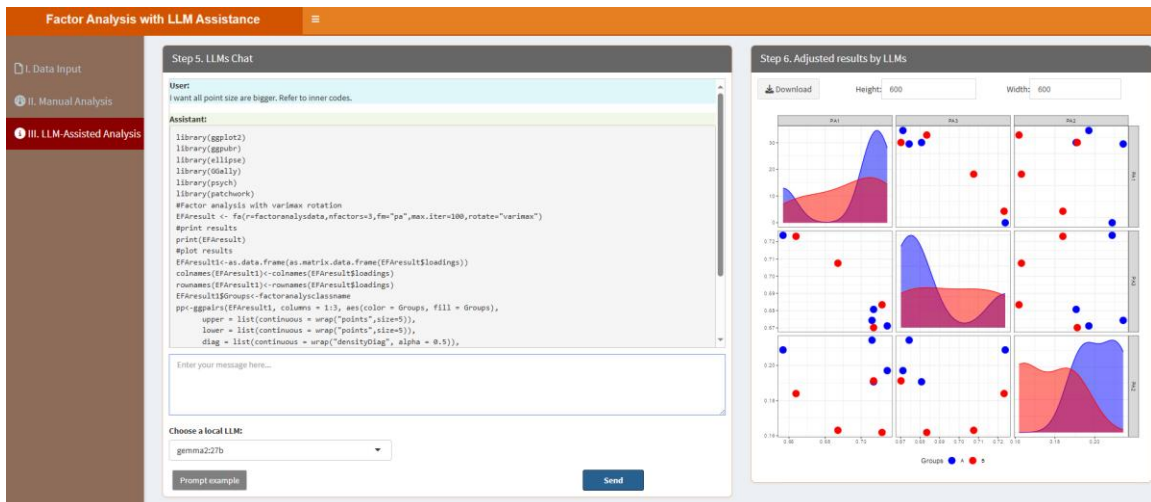
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.9. Grey Relational Analysis (GRA)

GRA evaluates relationships between variables in complex systems. Users can modify basic parameters to obtain the results directly through the classic framework, while local LLMs enhance result interpretation and analysis refinement. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Grey Relational Analysis with LLM Assistance' application. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for 'xlsx', 'xls', and 'csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are two checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' section with a text input field containing the number '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one entry: '1 No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' and 'Previous 1 Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Grey Relational Analysis with LLM Assistance' application. On the left is a sidebar with three menu items: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains a section '3.1. Select two colors for Barplot:' with a dropdown menu showing 'blue' and 'red'. Below this are two input fields: '3.2. Figure Height:' with the value '650' and '3.3. Figure Width:' with the value '600'. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section '4.1. Visualization' (selected) and '4.2. Tables'. Below this is a 'Save Image' button.

3.1. Select two colors for Barplot: This parameter lets you choose two colors for barplot, as the “4.1. Visualization” show the results using a barplot by default. The first color is for the lowest value and the second color is for the highest value.

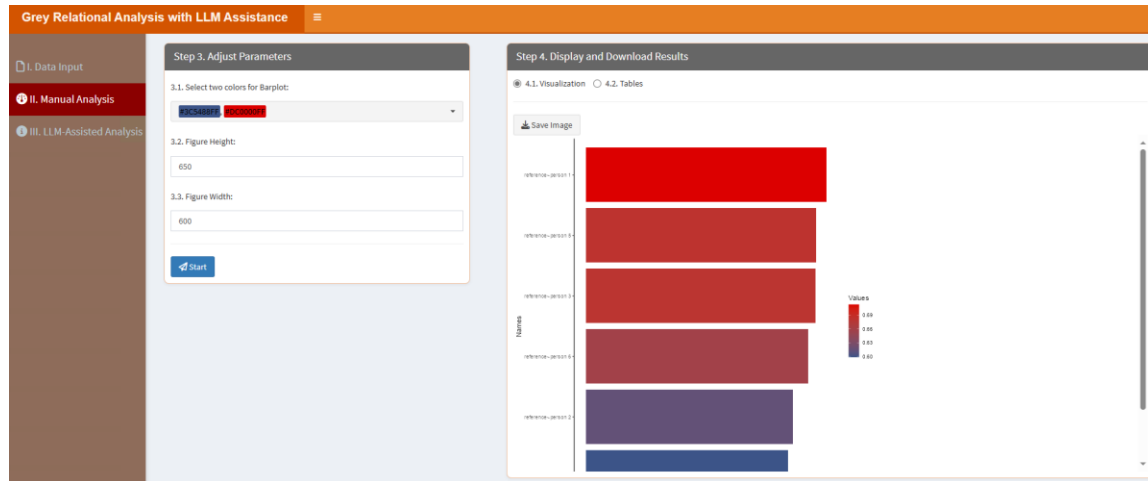
3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure

in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the results from the grey relational analysis:

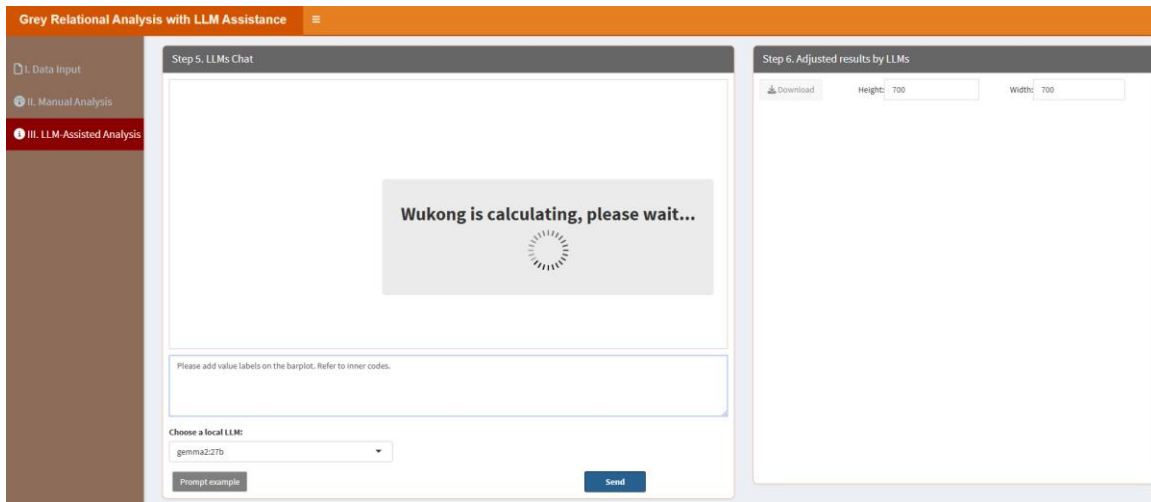
Names	Values
reference-person 4	0.5989343955386203
reference-person 2	0.61424774056353
reference-person 6	0.6584017289899643
reference-person 3	0.685202150790386
reference-person 5	0.6828682089639965
reference-person 1	0.7131313131313132

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

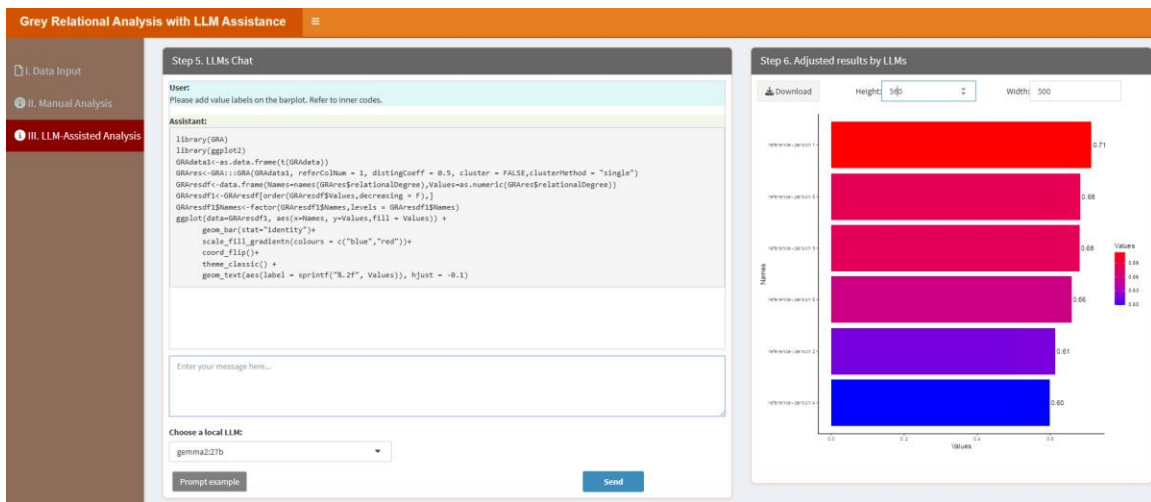
When users click the “Prompt example” button, it provides a sample prompt that users can

customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please add value labels on the barplot. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please add value labels on the barplot. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.10. Hierarchical Cluster Analysis (HCA)

HCA organizes data into a hierarchy of clusters. Users can select linkage methods and distance metrics via the classic framework, and local LLMs assist in refining cluster interpretation. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' section of the 'Hierarchical Clustering Analysis with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main panel has two tabs: 'A. Upload' (selected) and 'B. Load example data'. Under 'A. Upload', there are radio buttons for file formats: '.xlsx' (selected), '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown menu shows '1'. The right panel, 'Step 2. Display Uploaded Data/Example Data', shows a table with one entry: '1' in the 'Description' column. A message says 'No data loaded. Please upload your dataset or use the example data.' There are 'Previous', '1', and 'Next' buttons at the bottom right.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' section of the 'Hierarchical Clustering Analysis with LLM Assistance' interface. The sidebar is the same as in the previous screenshot. The main panel has two tabs: 'A. Upload' and 'B. Load example data'. The 'Step 3. Adjust Parameters' section contains several input fields: '3.1. Group and replicate number:' with the value '2;4-4'; '3.2. Group names:' with the value 'A,B'; '3.3. Select colors for groups:' with a dropdown menu showing 'EXCAV808' and 'EXCAV809'; '3.4. Select three colors for heatmap:' with a dropdown menu showing 'EXCAV808', 'white', and 'EXCAV809'; '3.6. Figure Height:' with the value '800'; and '3.7. Figure Width:' with the value '700'. At the bottom is a blue 'Start' button. The right panel, 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control"

and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for groups: This parameter lets you choose specific colors for each group. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of groups, ensuring that the visual representation is both informative and aesthetically pleasing.

3.4. Select three colors for heatmap: Colors for the heatmap plot. By default, there are three color names (#3C5488FF, white, #EE0000FF) here for the lowest (#3C5488FF), middle (white), highest (#EE0000FF) values respectively.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click "Start" button, the results will be shown as below:

In the "4.1. Visualization" part, users can click "Save image" button to download the figure:

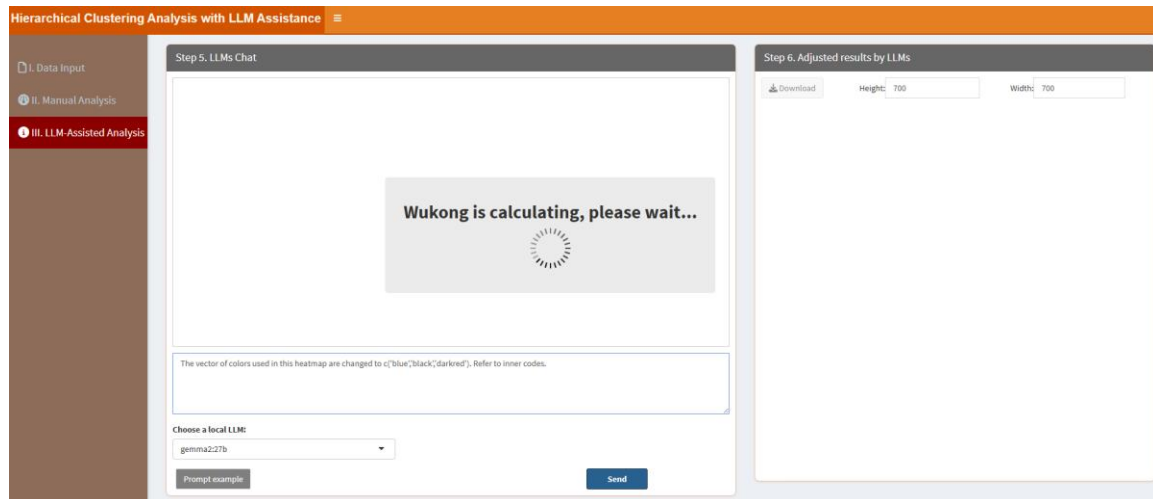


"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs"

means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "The vector of colors used in this heatmap are changed to c('blue','black','darkred'). Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "The vector of colors used in this heatmap are changed to c('blue','black','darkred'). =>", it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click "Send" button. It will learn like below:



The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.

I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis

Step 5. LLMs Chat

User:
The vector of colors used in this heatmap are changed to c("blue","black","darkred"). Refer to inner codes.

Assistant:

```

library(heatmap)
annotation_col <- data.frame(groups = factor(hc$classnames))
rownames(annotation_col) <- colnames(hc$adata)
groupcolor <- c("darkblue", "red")
names(groupcolor) <- hc$extval$cl
ann_colors <- list(groups=groupcolor)
heatmap(hc$adata, scale="row", fontsize_row = 10, fontsize_col = 10, show_rownames = F,
        show_colnames = T, cluster_rows=T, cluster_cols=F, display_numbers=F,
        annotation_col=annotation_col, border_color = NA,
        color = colorRampPalette(c("blue", "black", "darkred"))(100), annotation_colors = ann_colors,
        clustering_distance_rows = "euclidean", clustering_distance_cols = "euclidean",
        clustering_method = "complete")

```

Enter your message here...

Choose a local LLM
gemma2:27b

Prompt example
Send

Step 6. Adjusted results by LLMs

Download
Height: 700
Width: 700

68

5.11. K-means Clustering

K-means clustering partitions data into distinct groups. Users can adjust the number of clusters and initialization methods using the classic framework, while local LLMs help refine cluster analysis and provide insights. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Kmeans with LLM Assistance' application interface. On the left is a sidebar with three main sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The 'I. Data Input' section is active, showing 'Step 1: Upload Data/Example Data'. This step includes options to 'A. Upload' or 'B. Load example data', a 'Select File Format' dropdown (with .xlsx, .xls, and .csv/txt options), a 'Please import your data file:' section with a 'Browse...' button and 'No file selected' text, and checkboxes for 'Is the first row names?' and 'Is the first column names?'. Below these are 'Which Sheet to read?' and a text input field containing '1'. To the right, 'Step 2: Display Uploaded Data/Example Data' is visible, showing a 'Show' dropdown set to '10 entries', a 'Search' input field, and a table with one row: '1' and 'No data loaded. Please upload your dataset or use the example data.' The table has 'Previous', '1', and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Kmeans with LLM Assistance' application interface. The 'II. Manual Analysis' section is active, showing 'Step 3: Adjust Parameters'. This step includes five numbered parameters: '3.1. Cluster number:' with a text input field containing '2'; '3.2. Select colors for each cluster:' with a color picker showing 'Blue' and 'Red'; '3.3. Point size:' with a text input field containing '5'; '3.4. Figure Height:' with a text input field containing '600'; and '3.5. Figure Width:' with a text input field containing '600'. At the bottom of this section is a blue 'Start' button. To the right, 'Step 4: Display and Download Results' is visible, showing a '4.1. Visualization' and '4.2. Tables' section with a 'Save Image' button.

3.1. Cluster number: Maximum cluster number to evaluate.

3.2. Select colors for each cluster: This parameter lets you choose specific colors for each cluster. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of clusters, ensuring that the visual representation is both informative and aesthetically pleasing.

3.3. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which

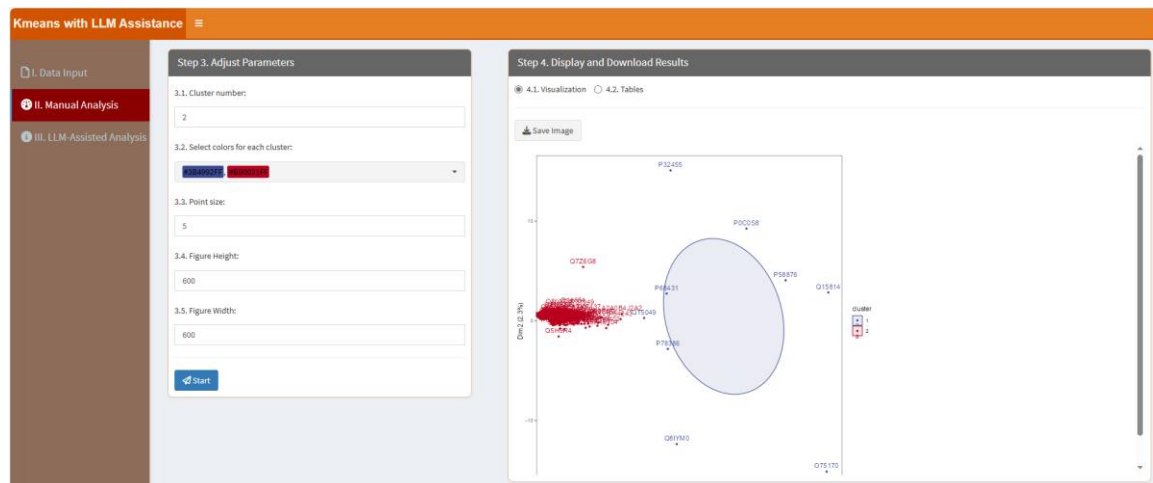
can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the results:

The screenshot shows the 'Kmeans with LLM Assistance' application interface with the '4.2. Tables' tab selected. The 'Step 3. Adjust Parameters' section remains the same. The 'Step 4. Display and Download Results' section now shows a 'Download' button and a table of results. The table has columns for 'A1', 'A2', 'A3', 'A4', 'B1', 'B2', 'B3', 'B4', and 'Clusters'. The table contains 10 rows of data. A 'Show 10 entries' dropdown is visible above the table. A search bar is also present.

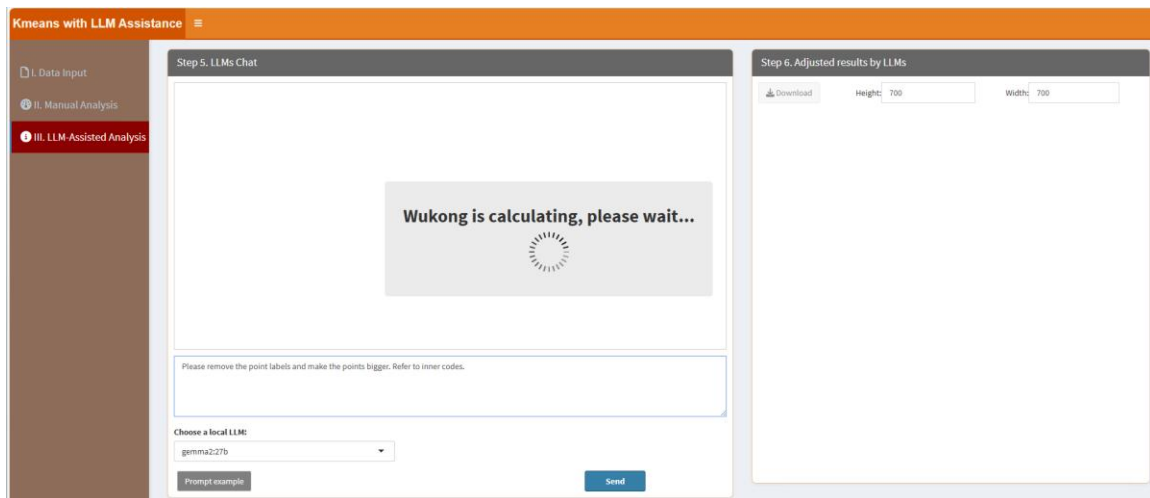
	A1	A2	A3	A4	B1	B2	B3	B4	Clusters
ADAQ24RB01	25189.73438	13440.20801	19565.13672	17490.47461	27999.10352	11120.79199	8287.486328	10820.16211	2
ADA096LP49	43690.0625	40345.23047	37148.71484	37204.87891	39428.95703	39318.33203	36062.07813	34250.29297	2
ADA084J2A2	1060724	836391.25	1103088.75	1063019	973396.5625	821988.5625	936122.8375	1124464.625	2
ADA084J2F2	57807.32422	60119.07813	63978.08594	69136.92188	68128.5625	31073.24219	72903.39063	71065.42188	2
ADA096	14720.61035	16155.04102	15082.32422	14969.2666	14932.67773	13164.33301	14025.50586	12534.11523	2
ADA0K6	16522.56445	18600.73633	17865.74805	18601.67969	18643.3125	5125.667969	14807.13867	16103.45508	2
ADA0T1	17222.83984	24763.6582	30250.36914	28884.49023	16928.29102	28179.63281	22732.48242	16664.99219	2
AD09R8	20893.17969	17165.89922	18176.4543	20293.74805	18935.82813	12087.23145	14606.60352	11881.54395	2
AD0JZ2	37593.65625	28032.05664	28467.55859	21959.01367	34682.69531	23316.44727	35804.23438	32016.7832	2
ADAZ66	29056.75195	32836.45703	30119.30664	30416.28125	29155.36328	30256.35638	33450.51172	30839.00195	2

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...”

enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "Please remove the point labels and make the points bigger. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please remove the point labels and make the points bigger. =>", it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click "Send" button. It will learn like below:



The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.

Kmeans with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 5. LLMs Chat

User:

Please remove the point labels and make the points bigger. Refer to inner codes.

Assistant:

```

library(factoextra)
library(ggplot2)
prdataread<-humans(kmeansdata,centers=clusternum)
fviz_cluster(prdataread, kmeansdata,geom = "point",show.clust.cent=F,
             ellipse = TRUE,ellipse.type = "confidence",main = "",pointsize = 8,
             ellipse.level = 0.95,ellipse.alpha = 0.1,shape=16)+
  scale_color_manual(values = colors)+
  scale_fill_manual(values = colors)+
  theme_hc()+
  theme(panel.grid.major=element_blank(),panel.grid.minor=element_blank())

```

Enter your message here...

Choose a local LLM

gemma2:27b

Prompt example

Send

Step 6. Adjusted results by LLMs

Download

Height: 700

Width: 700

5.12. Lack of Fit F-test

This test evaluates the adequacy of regression models. Users can configure model parameters through the classic framework, while local LLMs assist in interpreting the results and refining the analysis. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Lack of Fit F-test with LLM Assistance' application. The left sidebar has three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two steps. Step 1, 'Upload Data/Example Data', has two radio buttons: 'A. Upload' (selected) and 'B. Load example data'. Below these are options for file format (.xlsx, .xls, .csv/.txt) and a file upload section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which Sheet to read?' dropdown set to '1'. Step 2, 'Display Uploaded Data/Example Data', shows a table with one entry: 'No data loaded. Please upload your dataset or use the example data.' The table has a search bar, a 'Show 10 entries' dropdown, and pagination controls (Previous, 1, Next).

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

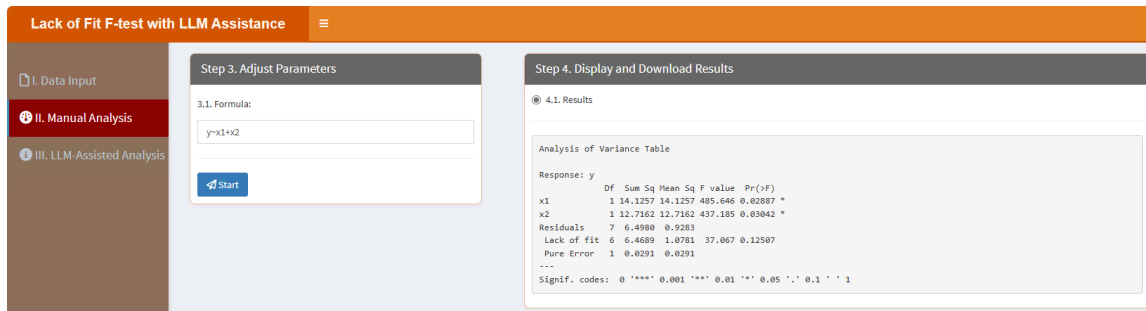
The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Lack of Fit F-test with LLM Assistance' application. The left sidebar has three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two steps. Step 3, 'Adjust Parameters', has a section '3.1. Formula:' with a text input field containing 'y~x1+x2' and a 'Start' button. Step 4, 'Display and Download Results', has a section '4.1. Results' with a large empty text area.

3.1. Formula: A symbolic description of the model to be fitted. For example, in the example data, the dependent variable is y, the independent variables are x1 and x2, so the formula should be $y \sim x_1 + x_2$.

After setting parameters, click “Start” button, the results will be shown as below:

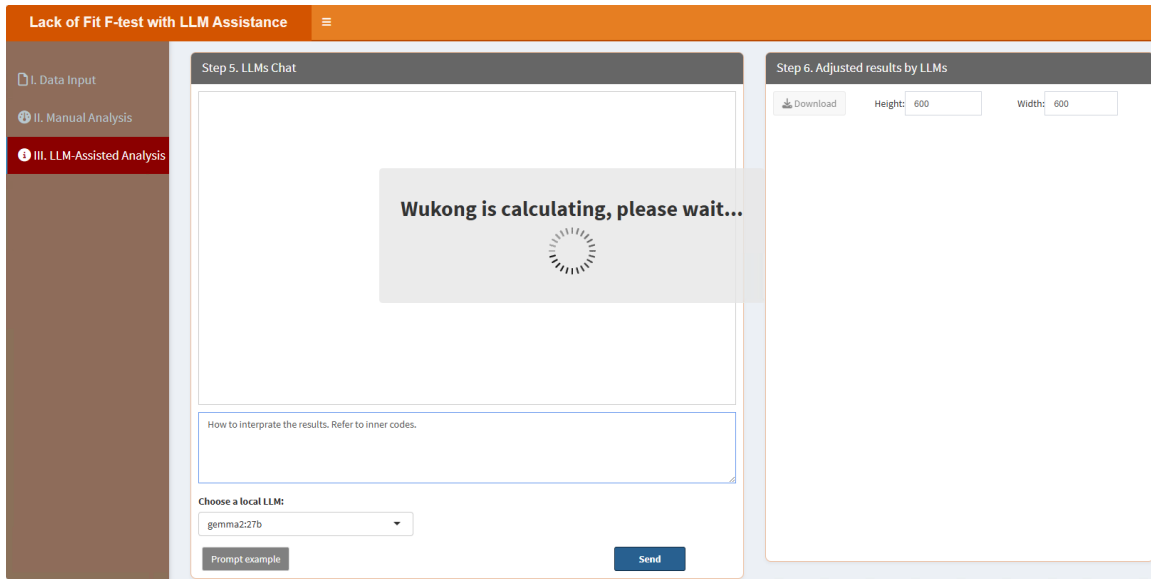
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



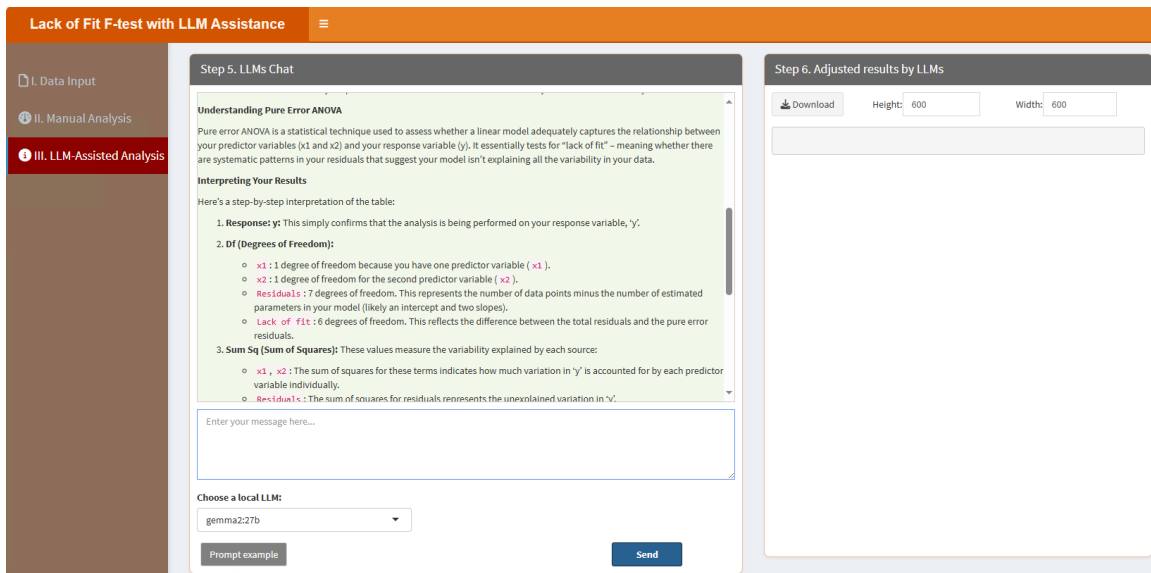
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “How to interpret the results. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “How to interpret the results. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.13. Linear Regression

Linear regression models relationships between variables. Users can set predictors and response variables via the classic framework, and local LLMs provide enhanced insights into model interpretation. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Linear Regression with LLM Assistance' application. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' status. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' field with the value '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one entry: '1 No data loaded. Please upload your dataset or use the example data.' The table has a 'Description' header. At the bottom of the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous' and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Linear Regression with LLM Assistance' application. On the left is a sidebar with three menu items: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains a '3.1. Formula:' label and a text input field with the value 'y~x'. Below the input field is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Results' label and a large empty text area for displaying results.

3.1. Formula: A symbolic description of the model to be fitted. For example, in the example data, the dependent variable is y, the independent variable is x, so the formula should be $y \sim x$.

After setting parameters, click “Start” button, the results will be shown as below:

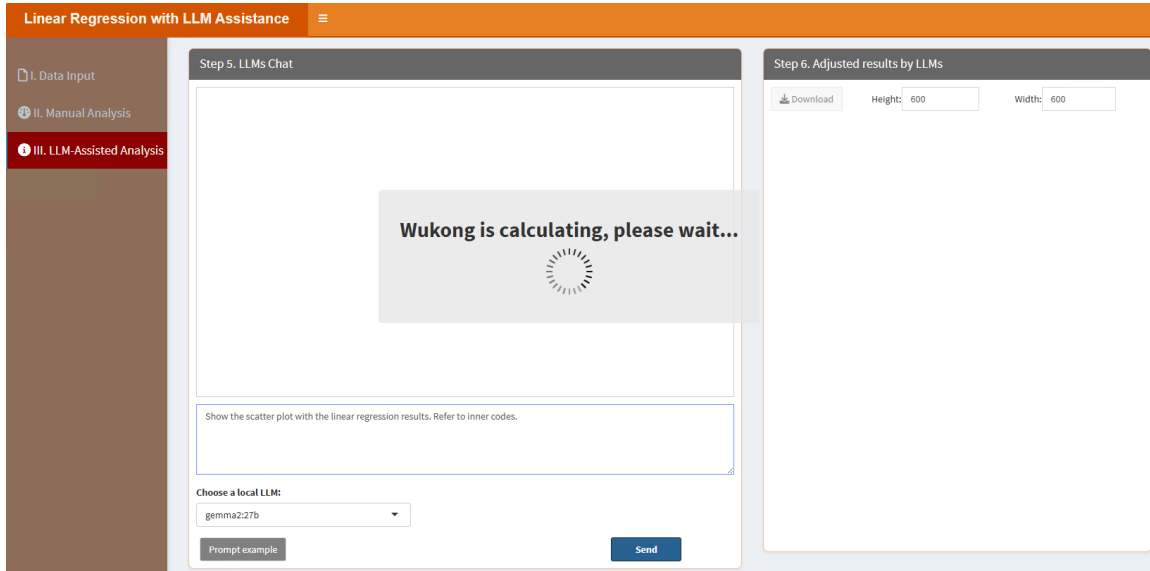
In the “4.1. Results” part, users can check the linear regression results:



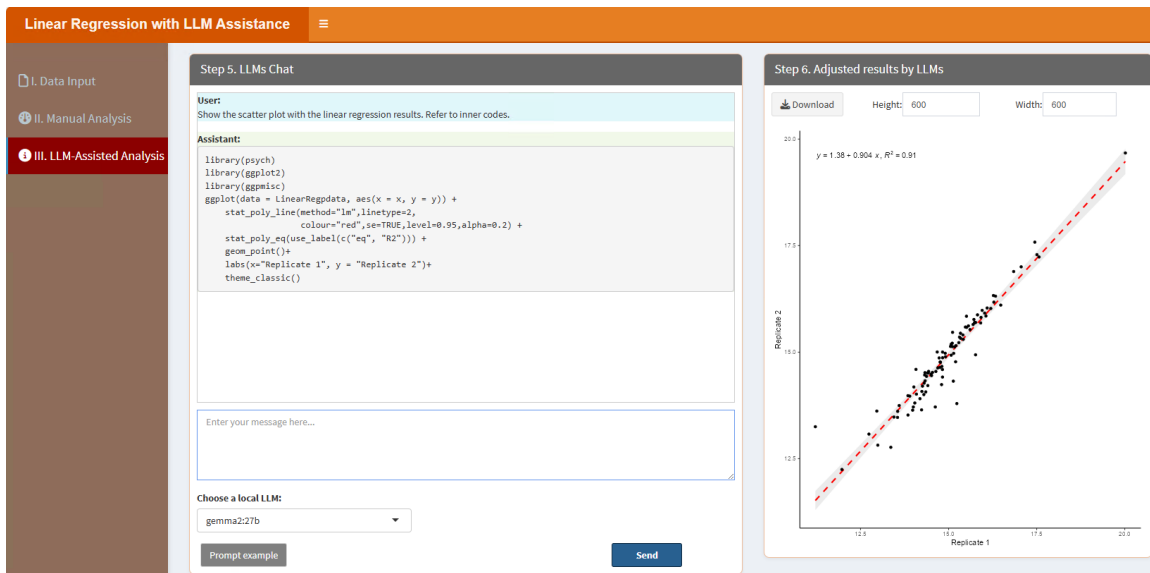
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Show the scatter plot with the linear regression results. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Show the scatter plot with the linear regression results. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.14. Logistic Regression

Logistic regression predicts binary outcomes. Users can configure model parameters through the classic framework, while local LLMs refine the analysis and improve interpretability. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Logistic Regression with LLM Assistance' application. The left sidebar has three items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two steps. Step 1, 'Upload Data/Example Data', has two radio buttons: 'A. Upload' (selected) and 'B. Load example data'. Under 'A. Upload', there is a 'Select File Format' section with radio buttons for '.xlsx' (selected), '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and 'No file selected' text. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which Sheet to read?' dropdown menu with '1' selected. Step 2, 'Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the 'Description' column with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' and 'Previous 1 Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

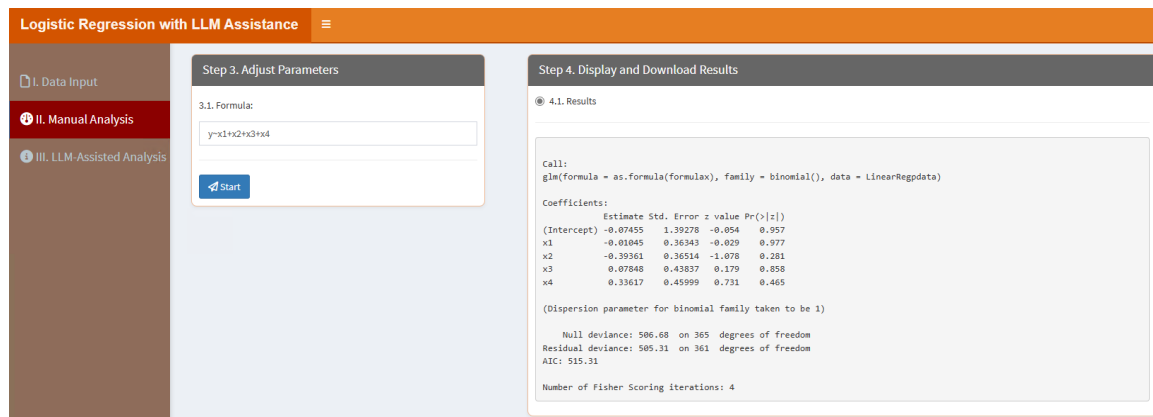
The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Logistic Regression with LLM Assistance' application. The left sidebar has three items: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two steps. Step 3, 'Adjust Parameters', has a '3.1. Formula:' label and a text input field containing the formula 'y~x1+x2+x3+x4'. Below the input field is a blue 'Start' button. Step 4, 'Display and Download Results', is currently empty.

3.1. Formula: A symbolic description of the model to be fitted. For example, in the example data, the dependent variable is y , the independent variables are x_1 , x_2 , x_3 and x_4 , so the formula should be $y \sim x_1 + x_2 + x_3 + x_4$.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Results” part, users can check the logistic regression results:



“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please calculate the odd ratios and relative 95% confidence intervals. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please calculate the odd ratios and relative 95% confidence intervals. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

Logistic Regression with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 5. LLMs Chat

Wukong is calculating, please wait...

Please calculate the odd ratios and relative 95% confidence intervals. Refer to inner codes.

Choose a local LLM: gemma2:27b

Prompt example

Send

Step 6. Adjusted results by LLMs

Download

Height: 600

Width: 600

The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.

Logistic Regression with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 5. LLMs Chat

User: Please calculate the odd ratios and relative 95% confidence intervals. Refer to inner codes.

Assistant:

```
logisticreg<- glm(as.formula(formulax),data=uploaddatax, family=binomial())
logisticres<- as.data.frame(exp(coef(logisticreg)))
logisticres<- cbind(logisticres,exp(confint(logisticreg)))
colnames(logisticres)[1]<- "Odd.Ratio"
logisticres
```

Enter your message here...

Choose a local LLM: gemma2:27b

Prompt example

Send

Step 6. Adjusted results by LLMs

Download

Height: 600

Width: 600

Show 10 entries

Search:

	Odd.Ratio	2.5 %	97.5 %
(Intercept)	0.928163404541594	0.05954640300147563	14.42226697664275
x1	0.9896055327228896	0.4758024125766887	2.061426992084425
x2	0.6746153086395513	0.3128182090974416	1.344327815724158
x3	1.081645967220545	0.4531908086698448	2.590666458110479
x4	1.399579196735318	0.5694635352555436	3.535208029010167

Showing 1 to 5 of 5 entries

Previous 1 Next

5.15. Mfuzz

Mfuzz is used for clustering time-series data. Users can adjust fuzzification parameters and cluster numbers through the classic framework, while local LLMs help refine the analysis and interpret clusters. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Mfuzz with LLM Assistance' web application. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' status. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown and a search bar. Below is a table with one entry: '1 No data loaded. Please upload your dataset or use the example data.' At the bottom of the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous', '1', and 'Next' navigation links.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Mfuzz with LLM Assistance' web application. On the left is a sidebar with three menu items: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains four numbered sections: '3.1. Cluster number:' with a text input field containing '9'; '3.2. Select colors for each cluster:' with a color picker showing '90C548F8', 'white', and '8B0000'; '3.3. Figure Height:' with a text input field containing '600'; and '3.4. Figure Width:' with a text input field containing '900'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains two radio buttons: '4.1. Visualization' (selected) and '4.2. Tables'. Below the radio buttons is a 'Save Image' button.

3.1. *Cluster number*: maximum cluster number to evaluate.

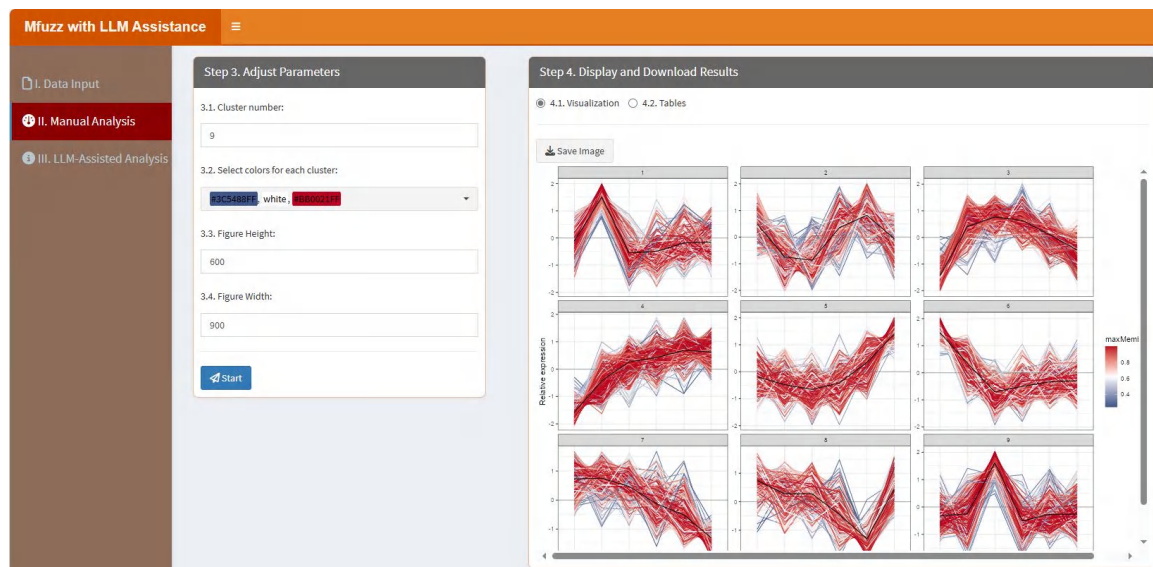
3.2. *Select colors for each cluster*: This parameter lets you choose specific colors for each cluster. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of clusters, ensuring that the visual representation is both informative and aesthetically pleasing.

3.3. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the results from the grey relational analysis:

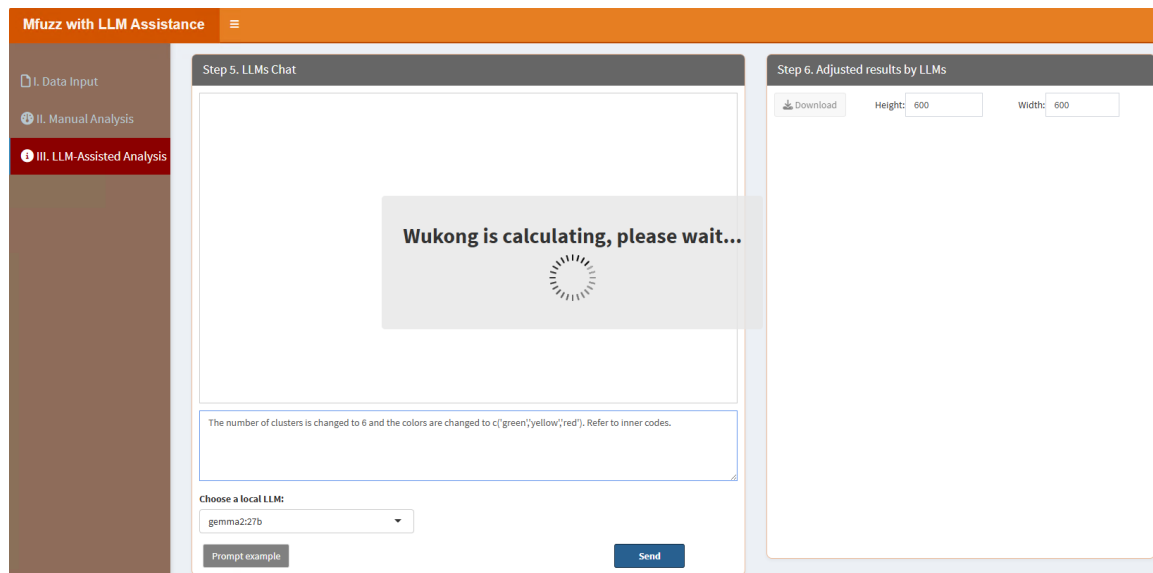
The screenshot displays the Mfuzz with LLM Assistance interface. On the left, a sidebar shows navigation options: I. Data Input, II. Manual Analysis, and III. LLM-Assisted Analysis. The main area is divided into two sections. The top section, 'Step 3. Adjust Parameters', contains input fields for '3.1. Cluster number:' (set to 9), '3.2. Select colors for each cluster:' (a dropdown menu showing 'blue', 'white', and 'red'), '3.3. Figure Height:' (set to 600), and '3.4. Figure Width:' (set to 900). A 'Start' button is at the bottom of this section. The bottom section, 'Step 4. Display and Download Results', has two tabs: '4.1. Visualization' and '4.2. Tables' (selected). Under '4.2. Tables', there is a 'Download' button and a 'Show 10 entries' dropdown. The table displays results from the grey relational analysis. The table has columns for 'IDs', 'clusterindex', 'membership1', and 'membership2'. The table shows 10 rows of data, with a search bar and pagination controls at the bottom.

IDs	clusterindex	membership1	membership2
IPI00222681.3_AW146020_169	IPI00222681.3_AW146020_169	7	0.005920382311874654
IPI00116907.8_Sh3bp5l_30	IPI00116907.8_Sh3bp5l_30	4	0.0001159672466022417
IPI00678133.3_inf2_1192	IPI00678133.3_inf2_1192	6	0.0007375259329344395
IPI00753321.2_Bod1l_2920	IPI00753321.2_Bod1l_2920	6	0.0001547593229014313
IPI00469997.4_Us20_134	IPI00469997.4_Us20_134	8	0.00353975129861875
IPI00453808.4_Ddx55_544	IPI00453808.4_Ddx55_544	8	0.1741949998151083
IPI00229859.1_Eif3b_153	IPI00229859.1_Eif3b_153	1	0.9998803602985885
IPI00453982.4_Rcor3_227	IPI00453982.4_Rcor3_227	5	0.002002500417682863
IPI00465812.2_Epb4.1l1_441	IPI00465812.2_Epb4.1l1_441	1	0.6885898822245197
IPI00308559.1_Snip1_381	IPI00308559.1_Snip1_381	2	0.0118846923724334

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “The number of clusters is changed to 6 and the colors are changed to c('green','yellow','red'). Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “The number of clusters is changed to 6 and the colors are changed to c('green','yellow','red'). =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the

“Download” button in the “Step 6. Adjust results by LLMs” section.

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 5. LLMs Chat

User

The number of clusters is changed to 6 and the colors are changed to c('green','yellow','red'). Refer to inner codes.

Assistant

```
set.seed(123)
library(hfuzz)
library(tidy)
mfuzzdata1 <- t(scale(t(mfuzzdata)))
newdataread <- new("expressionset", exprs=mfuzzdata1)
clustering <- mfuzz(newdataread, cwf, w=1.25)
clusterIndex <- clustering$cluster
membership <- clustering$membership
colnames(membership) <- paste("membership", seq_along(membership[1,]), sep = "")
exp <- mfuzzdata1
exp1 <- data.frame(exp, Identifier = rownames(mfuzzdata1), clusterIndex,
  membership, check.names=F, stringsAsFactors = F)
exp2 <- exp1 %>%
  gather(sample, expression,
    .data$Identifier,
    .data$clusterIndex,
    .contains("membership"))
exp2$sample <- factor(exp2$sample, levels = colnames(mfuzzdata1))
exp2$maxMembership <- apply(exp2%>%dplyr::select(contains("membership")), 1, max)
```

Enter your message here...

Choose a local LLM:
gemma2:27b

Prompt example

Send

Step 6. Adjusted results by LLMs

Download

Height: 600

Width: 600

5.16. Network Degree Matrix (NDM)

NDM analyzes connectivity in biological networks. Users can customize network parameters using the classic framework, and local LLMs provide deeper insights into the results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' interface. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/bit'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' status. There are two checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom is a 'Which Sheet to read?' section with a text input field containing '1'. The right panel is titled 'Step 2. Display Uploaded Data/Example Data' and shows a 'Show 10 entries' dropdown, a 'Search:' input field, and a table with one entry: '1 No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' and 'Previous 1 Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' and 'Step 4. Display and Download Results' interfaces. The sidebar on the left now has 'II. Manual Analysis' selected. The main area has two panels. The left panel, titled 'Step 3. Adjust Parameters', contains a 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section for '4.1. Tables' and a 'Download' button.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results from this module:

Network Degree Matrix (NDM) with LLM Assistance

Step 3. Adjust Parameters

Step 4. Display and Download Results

4.1. Tables

Download

Show 10 entries

Search:

	A1	A2	A3	A4	B1	B2	B3	B4
AOA024RBG1	1224	0	589	0	530	620	442	1112
AOA096LP49	314	853	318	0	533	0	840	514
AOA0B4J2A2	0	453	412	541	0	149	237	107
AOA0B4J2F2	1224	489	0	541	0	112	192	425
AOAV96	0	439	806	588	0	524	540	243
AOAVK6	0	432	0	766	530	174	582	573
AOAVT1	230	0	200	704	552	298	0	93
AOFGRR	915	0	0	756	533	911	915	466
AOJLT2	136	357	0	195	347	461	308	0
AOIMZ66	336	673	589	0	450	0	442	814

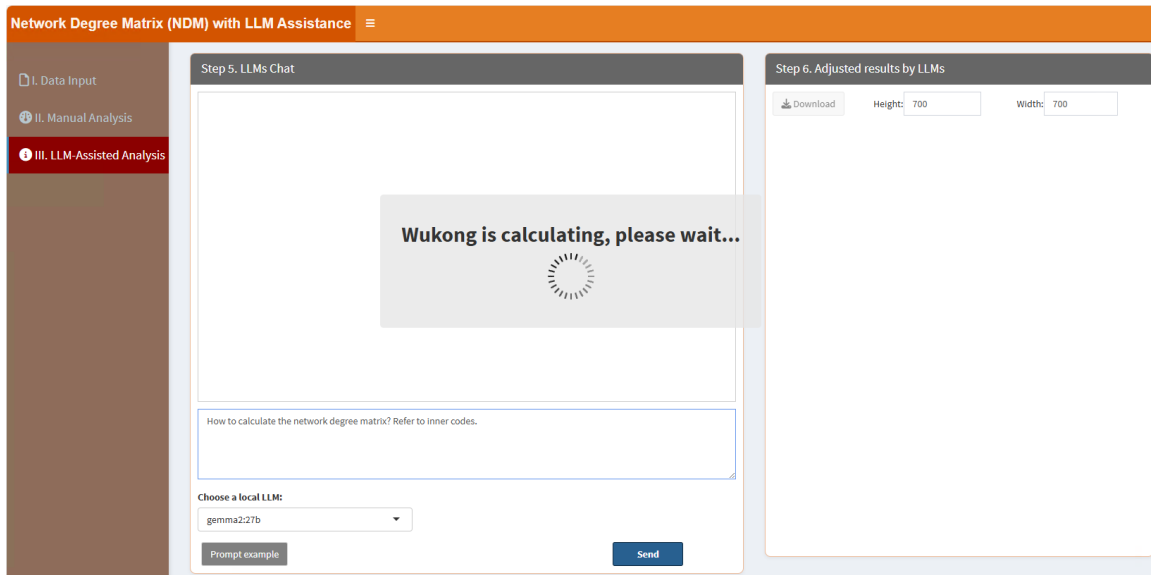
Showing 1 to 10 of 4,277 entries

Previous 1 2 3 4 5 ... 428 Next

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “How to calculate the network degree matrix? Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “How to calculate the network degree matrix? =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.

Step 5. LLMs Chat

User:
How to calculate the network degree matrix? Refer to inner codes.

Assistant:

```
#Calculate the network degree matrix
rndx <- function(x) trunc(x*sign(x)*0.5)
alpha<-0.05
boxsize<-1
upper <- lower <-matrix(0,nrow=nrow(NDMdata),ncol=ncol(NDMdata))
for(i in 1:nrow(NDMdata)){
  s1<-sort(as.numeric(NDMdata[i,]))
  s2<-order(as.numeric(NDMdata[i,]))
  nB <- ncol(NDMdata)-sum(sign(s1))
  h <- rndx(boxsize/2*sum(sign(s1)))
  k <- 1
  while(k <= ncol(NDMdata)){
    s <- 0
    while(k+s+1 <= ncol(NDMdata) & s1[k+s+1] == s1[k]){
      s = s+1
    }
    if(s >= h){
      upper[i,s2[k:k+s]]<-NDMdata[i,s2[k]]
      lower[i,s2[k:k+s]]<-NDMdata[i,s2[k]]
    }
  }
}
```

Enter your message here...

Choose a local LLM:
gemma2:27b

Prompt example Send

Step 6. Adjusted results by LLMs

Download Height: 700 Width: 700

Show 10 entries Search:

	A1	A2	A3	A4	B1	B2	B3	B4
AOA024RBG1	1224	0	589	0	530	620	442	1112
AOA096LP49	314	853	318	0	533	0	840	514
AOA0B4J2A2	0	453	412	541	0	149	237	107
AOA0B4J2F2	1224	489	0	541	0	112	192	425
AOAV96	0	439	806	588	0	524	540	243
AOAVK6	0	432	0	766	530	174	582	573
AOAVT1	230	0	200	704	552	298	0	93
A0FGR8	915	0	0	756	533	911	915	466
A0JLT2	136	357	0	195	347	461	308	0
ADMZ66	336	673	589	0	450	0	442	814

Showing 1 to 10 of 4,277 entries

Previous 1 2 3 4 5 ... 428 Next

5.17. OPLS-DA (Orthogonal Partial Least Squares Discriminant Analysis)

OPLS-DA is used for classification and biomarker discovery. Users can configure model parameters via the classic framework, while local LLMs enhance the interpretation of the results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'OPLS-DA with LLM Assistance' web application. On the left is a sidebar with three menu items: 'I. Data Input' (highlighted in red), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' and 'Is the first column names?', and a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show: 10 entries' dropdown, a search bar, and a table with one row containing the text 'No data loaded. Please upload your dataset or use the example data.' Below the table are 'Previous', '1', and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'OPLS-DA with LLM Assistance' web application. The sidebar is the same as in the previous screenshot. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains several input fields: '3.1. Group and replicate number:' with the value '2;4-4', '3.2. Group names:' with the value 'A/B', '3.3. Select colors for each CV interval:' with a dropdown menu showing 'Blue' and 'Red', '3.4. Point size:' with the value '5', '3.5. Figure Height:' with the value '550', and '3.6. Figure Width:' with the value '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains two radio buttons: '4.1. Visualization' (selected) and '4.2. Results summary'. Below these is a 'Save Image' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control"

and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

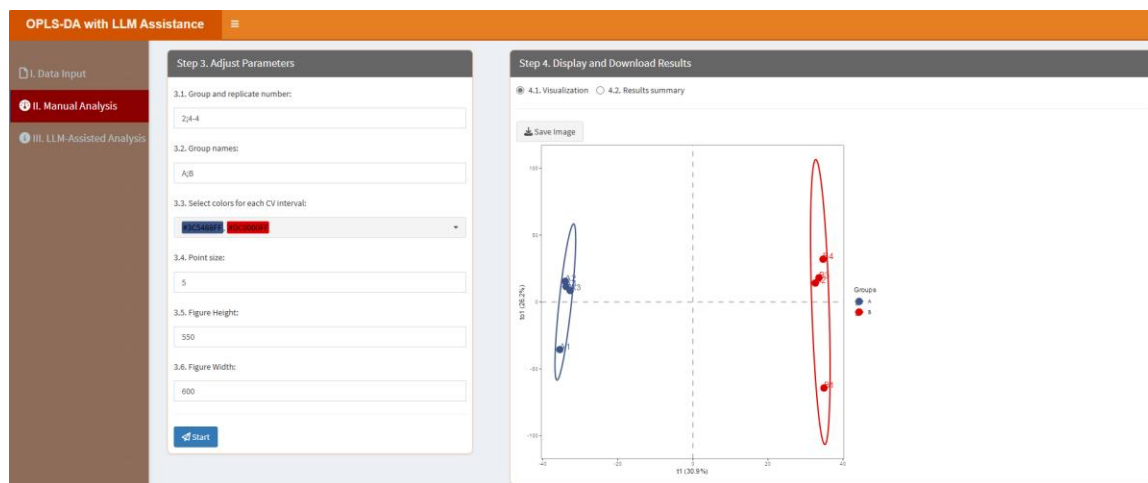
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click "Start" button, the results will be shown as below:

In the "4.1. Visualization" part, users can click "Save image" button to download the figure:



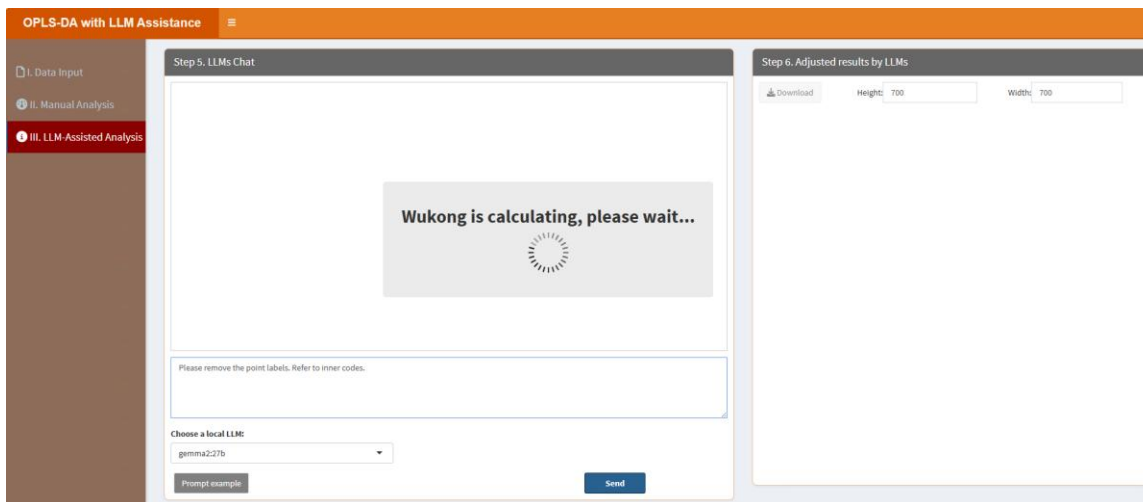
In the "4.2. Results summary" part, users can check the model summary results:

	R2X(cum)	R2Y(cum)	Q2(cum)	RMSEE	pre-ort	pR2Y	pQ2
Total	0.571	0.999	0.865	0.0377	1	1	0.85

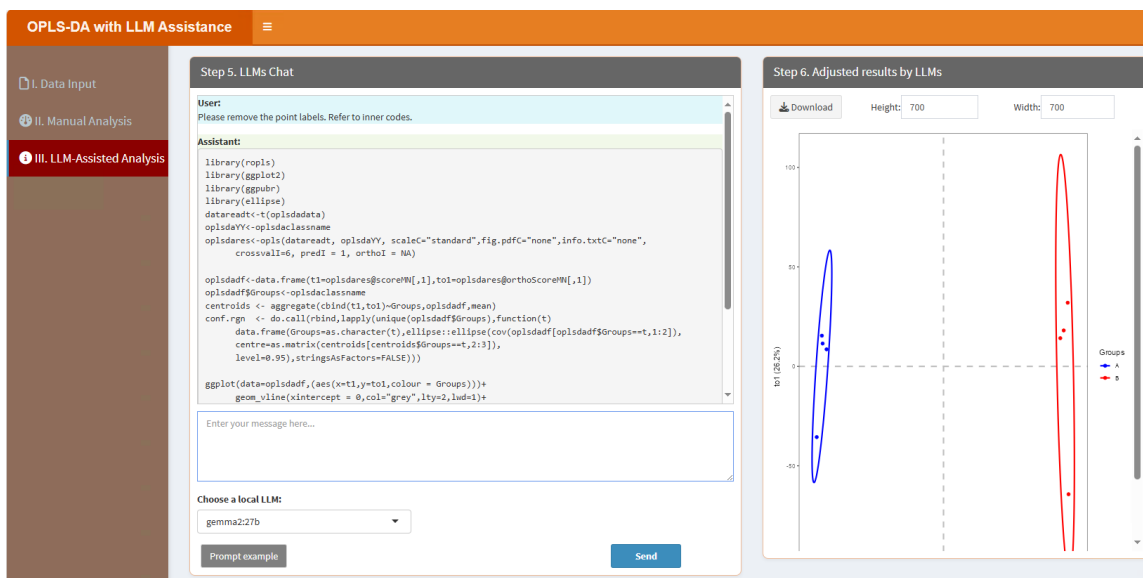
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please remove the point labels. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please remove the point labels. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.18. Principal Component Analysis (PCA)

PCA is an unsupervised dimensionality reduction technique. Users can adjust the number of components using the classic framework, and local LLMs assist in refining the analysis and interpreting the results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Principal Component Analysis (PCA) with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. Below this is a 'Select File Format:' section with radio buttons for '.xlsx', '.xls', and '.csv/bst'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' and 'Is the first column names?', both checked, and a 'Which Sheet to read?' dropdown set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show' dropdown set to '10' entries, a search bar, and a table with one entry: '1 No data loaded. Please upload your dataset or use the example data.' The table has a 'Description' header. At the bottom of the table are 'Previous', '1', and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Principal Component Analysis (PCA) with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains several input fields: '3.1. Group and replicate number:' with the value '2;4-4', '3.2. Group names:' with the value 'A;B', '3.3. Select colors for each CV interval:' with a dropdown showing '90C5488F' and '8DC6008F', '3.4. Point size:' with the value '2', '3.5. Figure Height:' with the value '600', and '3.6. Figure Width:' with the value '600'. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains two radio buttons: '4.1. Visualization' (selected) and '4.2. Results summary'. Below the radio buttons is a 'Save Image' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each,

the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

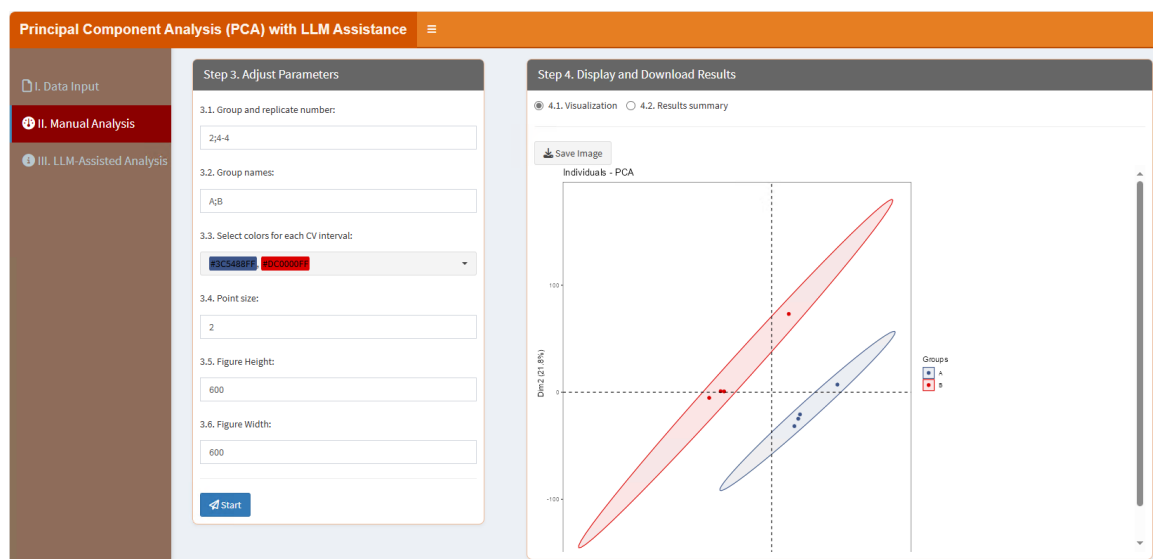
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



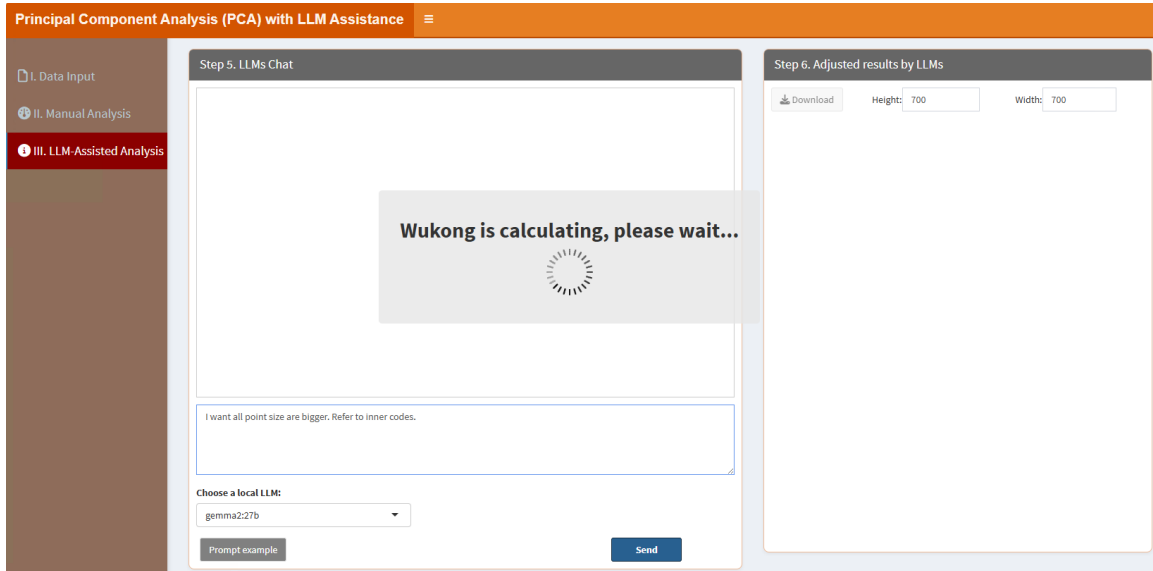
In the “4.2. Results summary” part, users can check the model summary results:

	eigenvalue	percentage of variance	cumulative percentage of variance
comp 1	1511.8662	35.348755	35.34875
comp 2	933.3082	21.821561	57.17032
comp 3	502.9924	11.768403	68.93872
comp 4	381.3154	8.915487	77.84621
comp 5	338.5086	7.914627	85.76083
comp 6	309.4509	7.235232	92.99606
comp 7	299.5583	7.003935	100.00000

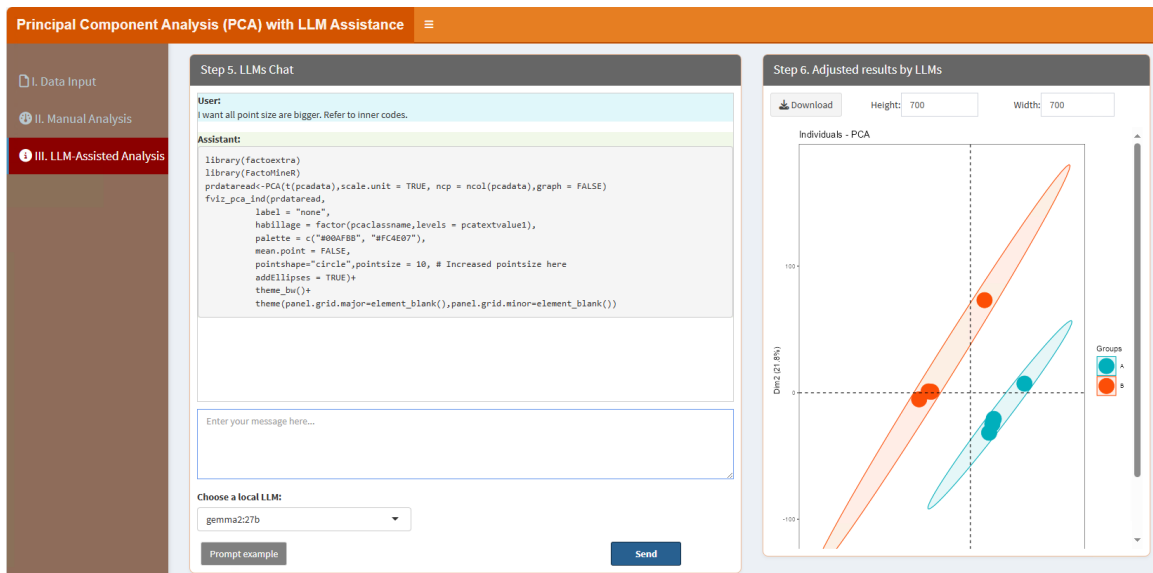
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.19. Principal Coordinates Analysis (PCoA)

PCoA visualizes distances between samples. Users can set distance metrics via the classic framework, while local LLMs help refine the analysis and provide insights. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'PCoA with LLM Assistance' web application. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains radio buttons for 'A. Upload' (selected) and 'B. Load example data'. Below are options for file format (.xlsx, .xls, .csv/bxt), a file upload button, and checkboxes for 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown is set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one entry: 'No data loaded. Please upload your dataset or use the example data.' The table has a search bar and pagination controls showing '1' of 1 entries.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'PCoA with LLM Assistance' web application. The sidebar remains the same. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains several input fields: '3.1. Group and replicate number:' with the value '2;4-4', '3.2. Group names:' with the value 'A;B', '3.3. Select colors for each CV interval:' with a dropdown showing 'PCC488FF' and 'PCC0000', '3.4. Point size:' with the value '2', '3.5. Figure Height:' with the value '550', and '3.6. Figure Width:' with the value '600'. A blue 'Start' button is at the bottom. The right panel, titled 'Step 4. Display and Download Results', contains radio buttons for '4.1. Visualization' (selected) and '4.2. Results summary'. Below is a 'Save Image' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying

experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

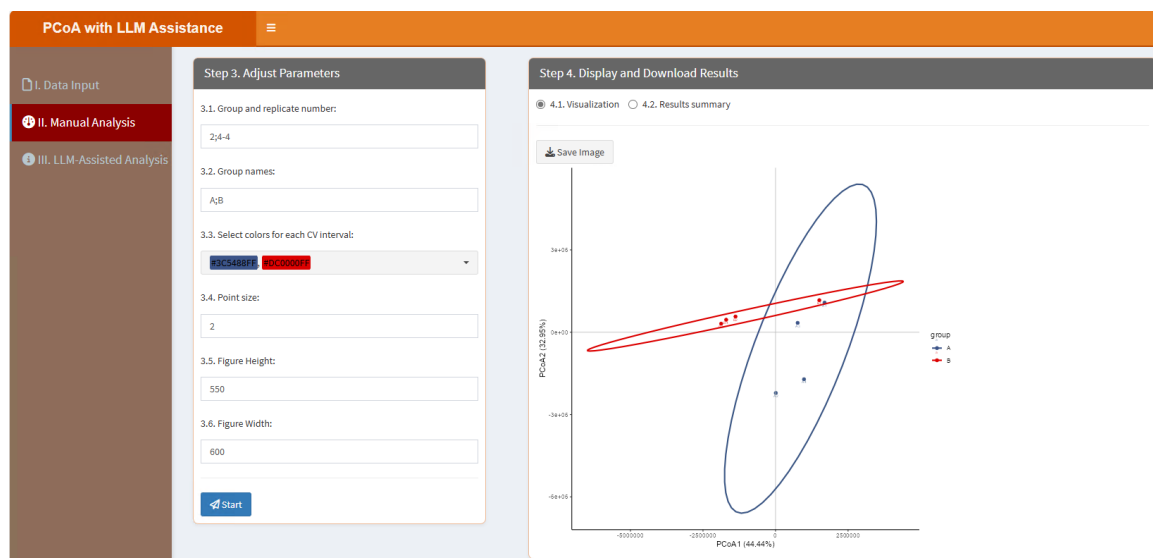
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



In the “4.2. Results summary” part, users can check the model summary results:

PCoA with LLM Assistance

Step 3. Adjust Parameters

3.1. Group and replicate number:
24-4

3.2. Group names:
A;B

3.3. Select colors for each CV interval:
pco44111 pco50001

3.4. Point size:
2

3.5. Figure Height:
550

3.6. Figure Width:
600

Start

Step 4. Display and Download Results

4.1. Visualization 4.2. Results summary

```

Class: pco dudi
Call: dudi.pco(d = df.dist, scannf = F, nf = 2)

Total inertia: 4.216e+12

Eigenvalues:
  Ax1      Ax2      Ax3      Ax4      Ax5
1.874e+12 1.389e+12 5.413e+11 2.485e+11 8.713e+10

Projected inertia (%):
  Ax1  Ax2  Ax3  Ax4  Ax5
44.441 32.947 12.840  5.894  2.867

Cumulative projected inertia (%):
  Ax1 Ax1:2 Ax1:3 Ax1:4 Ax1:5
44.44 77.39 90.23 96.12 98.19

(Only 5 dimensions (out of 7) are shown)

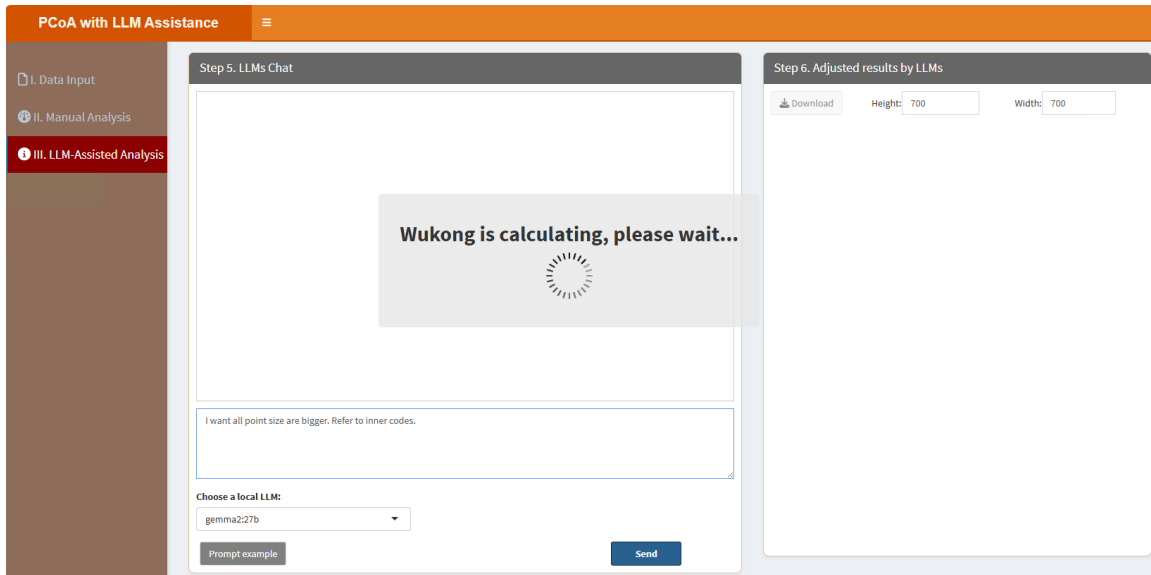
NULL

```

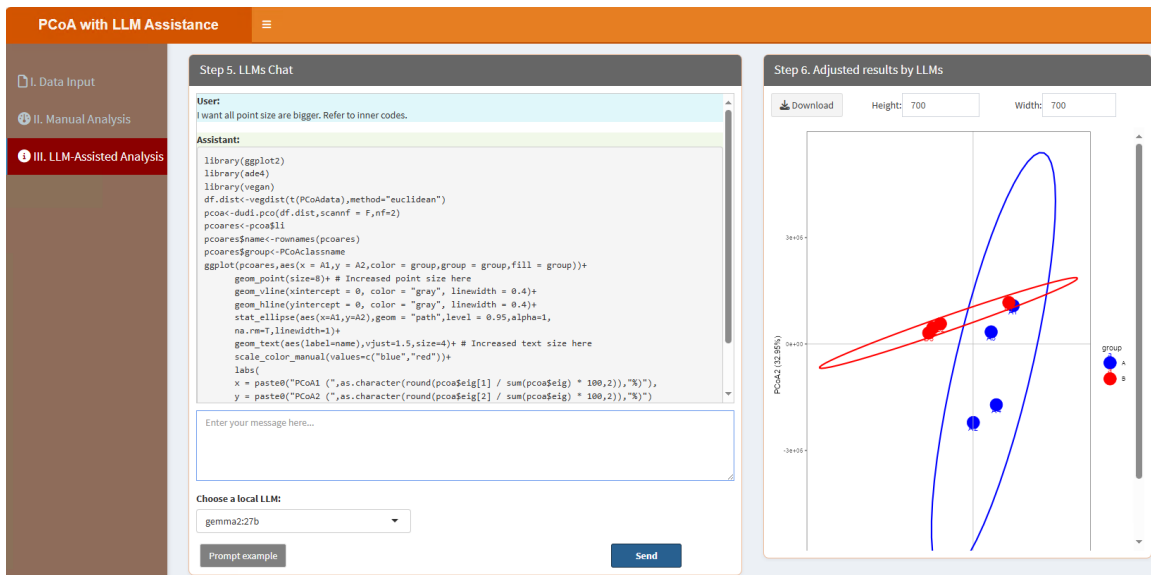
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.20. PLS-DA (Partial Least Squares Discriminant Analysis)

PLS-DA is used for classification in high-dimensional data. Users can configure model parameters through the classic framework, while local LLMs improve result interpretation. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' panel of the 'PLS-DA with LLM Assistance' application. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main panel has two tabs: 'A. Upload' (selected) and 'B. Load example data'. Under 'A. Upload', there are options for 'Select File Format' (.xlsx, .xls, .csv/txt) and a 'Please import your data file:' section with a 'Browse...' button and 'No file selected' text. Below this are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' dropdown menu set to '1'. The right panel, 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1 No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' with 'Previous', '1', and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' panel of the 'PLS-DA with LLM Assistance' application. The sidebar is the same as in the previous screenshot. The main panel has two tabs: '4.1. Visualization' (selected) and '4.2. Results summary'. The '4.1. Visualization' tab contains several input fields: '3.1. Group and replicate number:' with the value '2;4-4', '3.2. Group names:' with the value 'A;B', '3.3. Select colors for each CV interval:' with a dropdown showing 'A:CS441FF' and 'B:CC0000', '3.4. Point size:' with the value '5', '3.5. Figure Height:' with the value '550', and '3.6. Figure Width:' with the value '600'. At the bottom of this panel is a blue 'Start' button. The right panel, 'Step 4. Display and Download Results', is currently empty and shows a 'Save Image' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying

experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

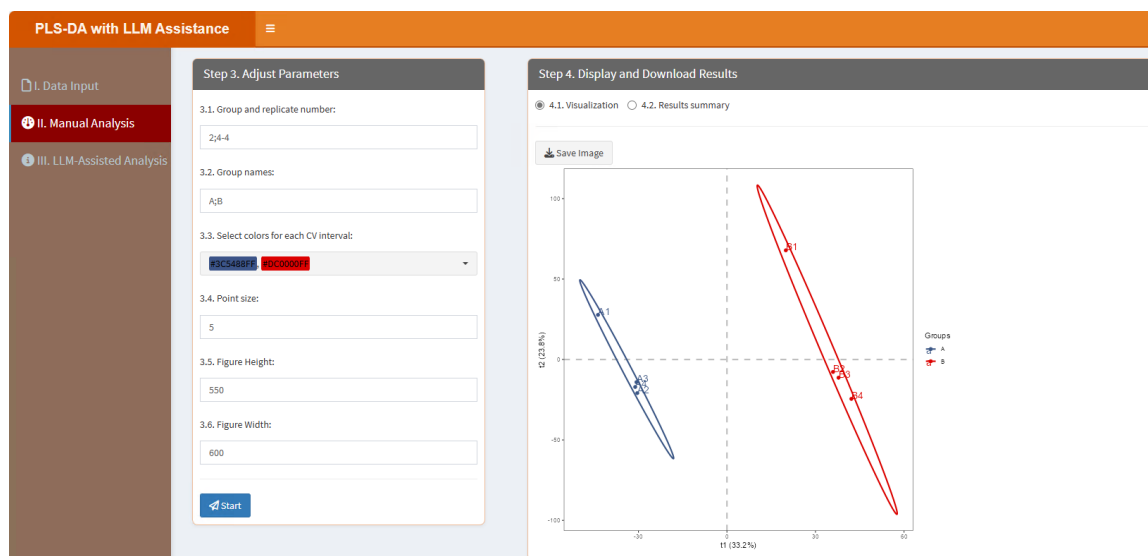
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



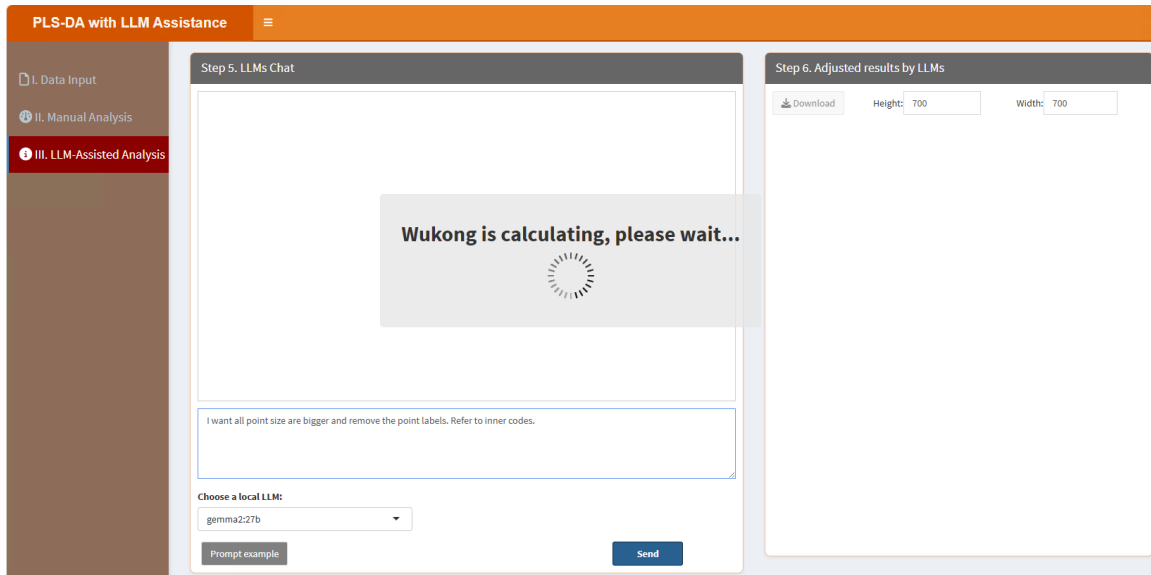
In the “4.2. Results summary” part, users can check the model summary results:

	R2X(cum)	R2Y(cum)	Q2(cum)	RMSEE	pre	ort	pR2Y	pQ2
Total	0.571	0.999	0.966	0.8177	2	0	0.05	0.05

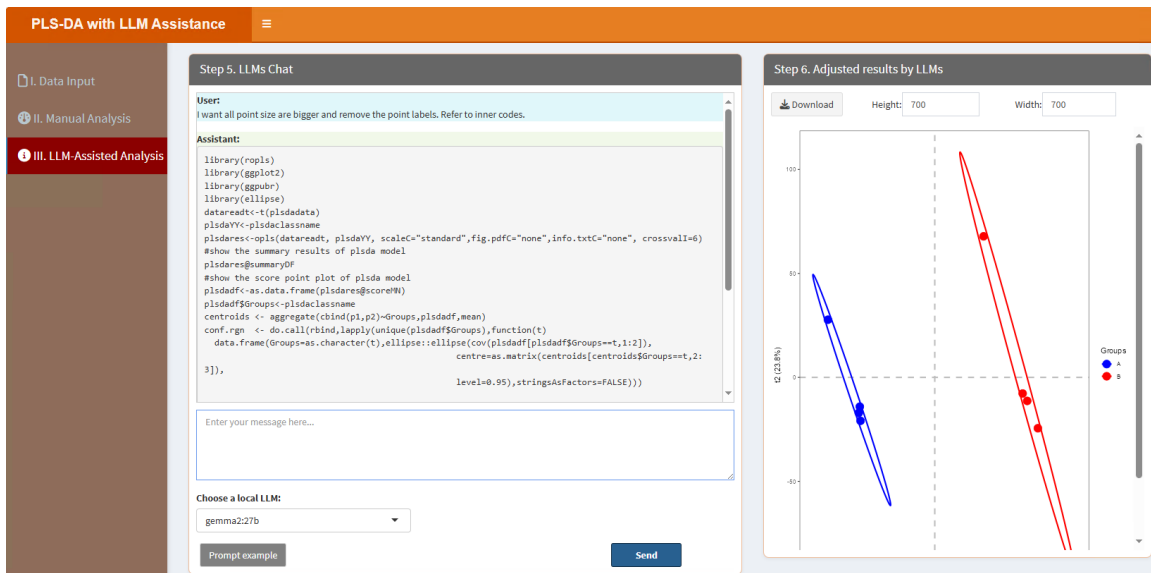
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger and remove the point labels. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger and remove the point labels. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.21. Power Analysis

Power analysis estimates sample sizes for statistical tests. Users can adjust effect sizes and significance levels via the classic framework, while local LLMs refine the analysis and provide recommendations. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Power Analysis with LLM' application interface. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/bxt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' status. There are two checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' section with a text input field containing the number '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1 No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous', '1', and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Power Analysis with LLM' application interface. On the left is a sidebar with three menu items: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three sections: '3.1. Group and replicate number:' with a text input field containing '2;4-4'; '3.2. Group names:' with a text input field containing 'A;B'; and '3.3. Log2 (base-2 logarithm) transformation used?' with a checked checkbox. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a section '4.1. Tables' with a 'Download' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. *Log2 (base-2 logarithm) transformation used?* if true, the data will be transformed to the logarithmic scale with base 2.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I need to determine the required sample size per group for a two-sided t-test with 0.85 power, an effect size of 0.5, and a significance level of 0.05. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the

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same function, so if users type “I need to determine the required sample size per group for a two-sided t-test with 0.85 power, an effect size of 0.5, and a significance level of 0.05. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.

Note=pwrres\$note,Method=pwrres\$method)); colnames(pwrresdf)<- "Values"; pwrresdf. The Step 6 panel now displays a table of results. The table has two columns: 'Values' and 'n'. The rows are: n (72.80046), d (0.5), sig.level (0.05), power (0.85), alternative (two.sided), Note (n is number in "each" group), and Method (Two-sample t test power calculation). Below the table, it says 'Showing 1 to 7 of 7 entries' with 'Previous' and 'Next' buttons."/>

	Values
n	72.80046
d	0.5
sig.level	0.05
power	0.85
alternative	two.sided
Note	n is number in "each" group
Method	Two-sample t test power calculation

5.22. Restricted Cubic Spline (RCS) Analysis

RCS analysis models non-linear relationships. Users can configure spline parameters through the classic framework, and local LLMs assist in refining the analysis. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' section of the application. It features a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area has two tabs: 'A. Upload' (selected) and 'B. Load example data'. Under 'A. Upload', there is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/bst'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' status. Further down, there are checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' section with a text input field containing the number '1'. To the right, the 'Step 2. Display Uploaded Data/Example Data' section is visible, showing a 'Show 10 entries' dropdown, a search bar, and a table with one entry: '1 No data loaded. Please upload your dataset or use the example data.' The table has a 'Description' header and a 'Showing 1 to 1 of 1 entries' footer with 'Previous' and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' section of the application. It features the same sidebar as the previous screenshot. The main content area has two tabs: 'I. Data Input' and 'II. Manual Analysis' (selected). The 'II. Manual Analysis' section contains several input fields for parameter configuration: '3.1. Outcome variable name:' with a text input containing 'status'; '3.2. Exposure variable name:' with a text input containing 'age'; '3.3. Covariate variable name:' with a text input containing 'sexrace'; '3.4. Time variable name:' with a text input containing 'time'; '3.5. Group variable name:' with a text input containing 'sex'; '3.6. Select colors:' with a color picker showing a blue and red selection; '3.7. Figure Height:' with a text input containing '600'; and '3.8. Figure Width:' with a text input containing '600'. At the bottom of this section is a blue 'Start' button. To the right, the 'Step 4. Display and Download Results' section is visible, showing a '4.1. Visualization' section with a 'Save Image' button.

- 3.1. *Outcome variable name*: The name of outcome variable in the data.
- 3.2. *Exposure variable name*: The name of exposure variable in the data.
- 3.3. *Covariate variable name*: The names of covariate variables in the data.
- 3.4. *Time variable name*: The name of time variable in the data, for Cox regressions.
- 3.5. *Group variable name*: The name of group variable in the data.
- 3.6. *Select colors*: This parameter lets you choose specific colors for each group. A color picker

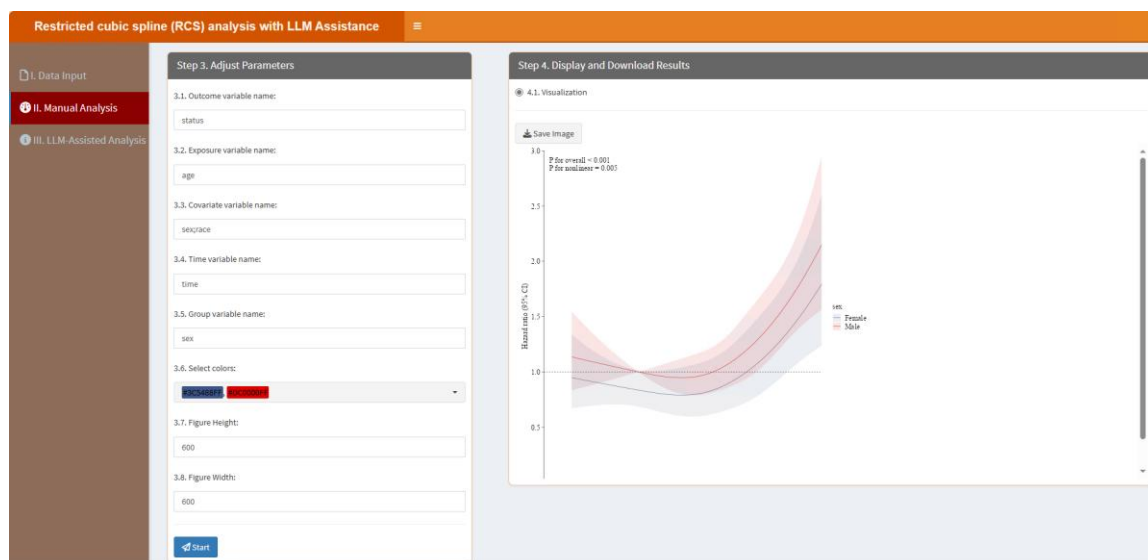
is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of groups, ensuring that the visual representation is both informative and aesthetically pleasing.

3.7. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.8. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

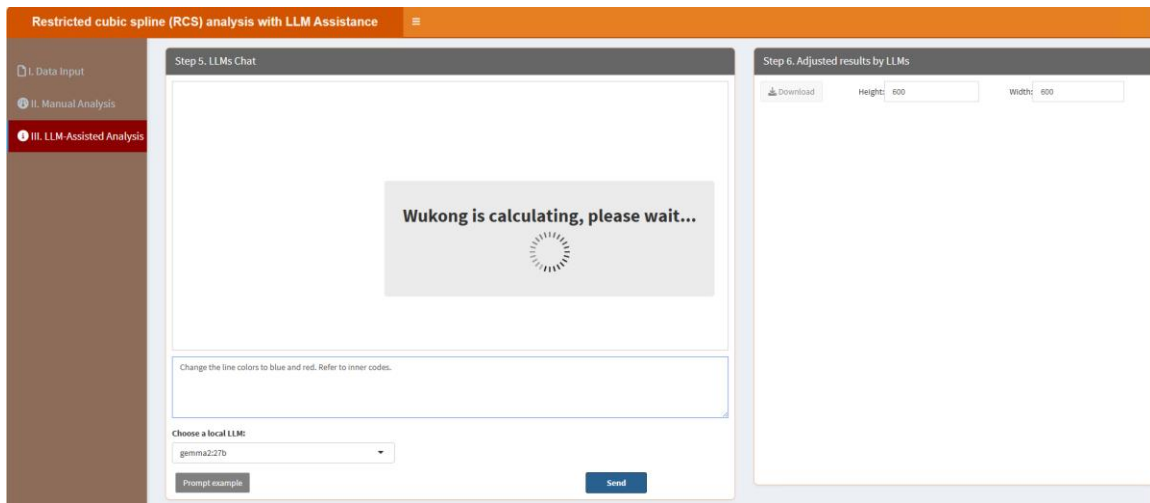


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

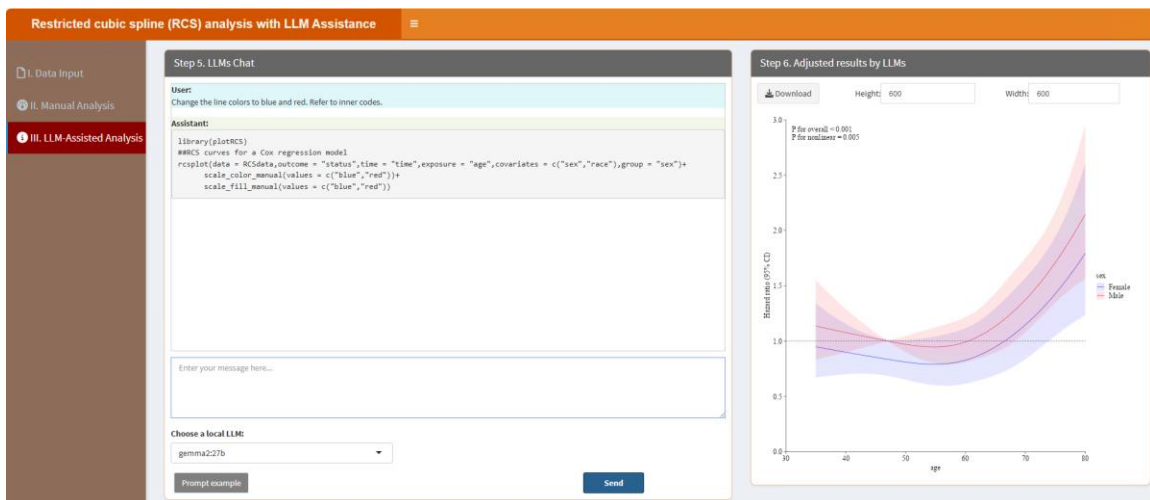
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of

LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the line colors to blue and red. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the line colors to blue and red. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.23. Redundancy Analysis (RDA)

RDA evaluates relationships between response and explanatory variables. Users can adjust parameters through the classic framework, while local LLMs enhance result interpretation. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Redundancy analysis with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one entry. The table has columns 'Show' (set to 10 entries) and 'Description'. The description text is 'No data loaded. Please upload your dataset or use the example data.' Below the table are 'Previous' and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Redundancy analysis with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains five numbered input fields: '3.1. Y variables index (Response variables index):' with '1-44' entered; '3.2. X variables index (Explanatory variables index):' with '45-58' entered; '3.3. Select a color for points:' with a color picker showing a red color; '3.4. Figure Height:' with '600' entered; and '3.5. Figure Width:' with '600' entered. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section for '4.1. Visualization' with a 'Save Image' button.

3.1. Y variables index (Response variables index): The column index for response variables. For example, in the example data, 1-44 columns are response variables, thus here should be 1-44.

3.2. X variables index (Explanatory variables index): The column index for explanatory variables. For example, in the example data, 45-58 columns are explanatory variables, thus here should be 45-58.

3.3. Select a color for points: This parameter lets you choose specific colors for each point. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select one color for the point,

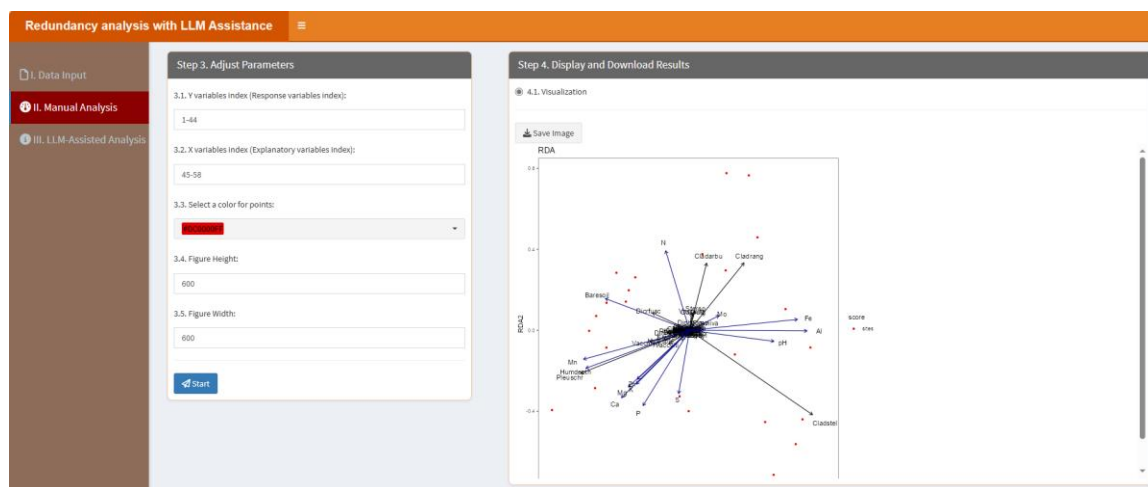
ensuring that the visual representation is both informative and aesthetically pleasing.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

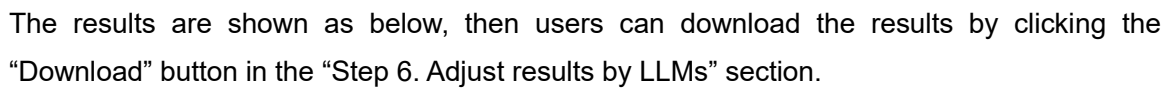
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the point size bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and

After type the prompt, click “Send” button. It will learn like below:



5.24. Rank Rank Hypergeometric Overlap (RRHO) Analysis

RRHO analysis identifies overlap between ranked datasets. Users can adjust ranking parameters via the classic framework, and local LLMs refine the analysis and provide insights. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Rank rank hypergeometric overlap (RRHO) analysis with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a message 'No file selected'. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show' dropdown set to '10' and 'entries'. It includes a 'Search:' input field and a table with one row: '1' in the 'Description' column, with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous', '1', and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Rank rank hypergeometric overlap (RRHO) analysis with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three sections: '3.1. Select colors for heatmap:' with a dropdown menu showing '#3C5488FF', 'white', and '#BB0021FF'; '3.2. Figure Height:' with a text input field containing '600'; and '3.3. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains two radio buttons: '4.1. Visualization' (selected) and '4.2. Tables'. Below these is a 'Save Image' button.

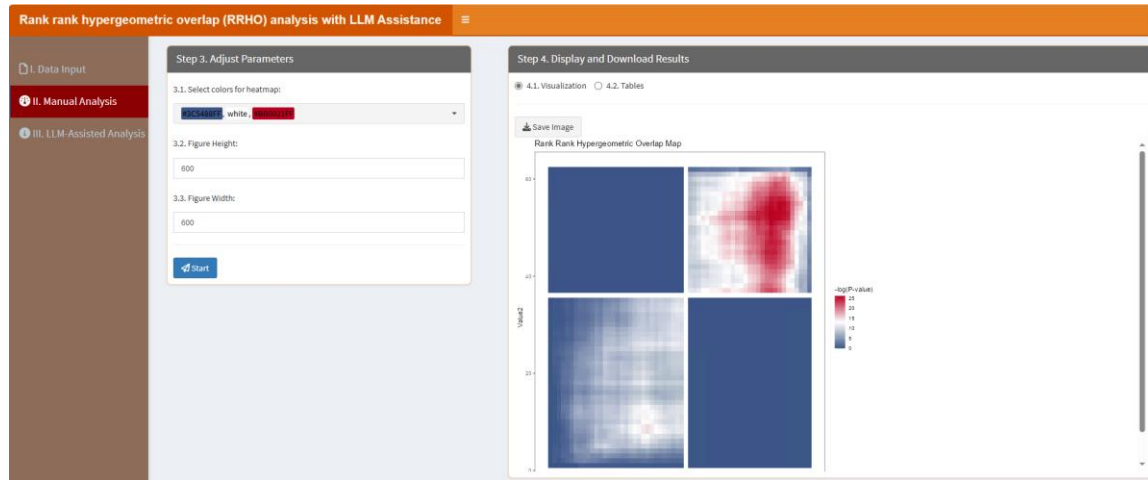
3.1. Select colors for heatmap: Colors for the heatmap plot. By default, there are three color names (#3C5488FF, white, #BB0021FF) here for the lowest (#3C5488FF), middle (white), highest (#BB0021FF) values respectively.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please provide the most significant overlap table results for me. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please provide the most significant overlap table results for me. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

5.25. SIMCA (Soft Independent Modelling by Class Analogy)

SIMCA is used for classification in multivariate data. Users can configure model parameters through the classic framework, while local LLMs improve result interpretation. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the SIMCA web interface. On the left is a sidebar with three options: "I. Data Input" (selected), "II. Manual Analysis", and "III. LLM-Assisted Analysis". The main area is divided into two panels. The left panel, titled "Step 1. Upload Data/Example Data", contains options for "A. Upload" (selected) and "B. Load example data". It includes a "Select File Format:" section with radio buttons for ".xlsx", ".xls", and ".csv/txt". Below this is a "Please import your data file:" section with a "Browse..." button and a "No file selected" message. There are also checkboxes for "Is the first row names?" and "Is the first column names?", both of which are checked. A "Which Sheet to read?" dropdown menu is set to "1". The right panel, titled "Step 2. Display Uploaded Data/Example Data", shows a table with one entry. The table has columns for "Show" (set to 10 entries) and "Description". The description for the single entry is "No data loaded. Please upload your dataset or use the example data." There are "Previous" and "Next" buttons at the bottom right of the table.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the SIMCA web interface. On the left is a sidebar with three options: "I. Data Input", "II. Manual Analysis" (selected), and "III. LLM-Assisted Analysis". The main area is divided into two panels. The left panel, titled "Step 3. Adjust Parameters", contains several input fields: "3.1. Group and replicate number:" with the value "2;4-4", "3.2. Group names:" with the value "A/B", "3.3. Select colors for each CV interval:" with a dropdown menu showing "Blue" and "Red", "3.4. Figure height:" with the value "550", and "3.5. Figure width:" with the value "600". There is a "Start" button at the bottom. The right panel, titled "Step 4. Display and Download Results", contains two radio buttons: "4.1. Visualization" (selected) and "4.2. Tables". Below these is a "Save Image" button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control"

and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

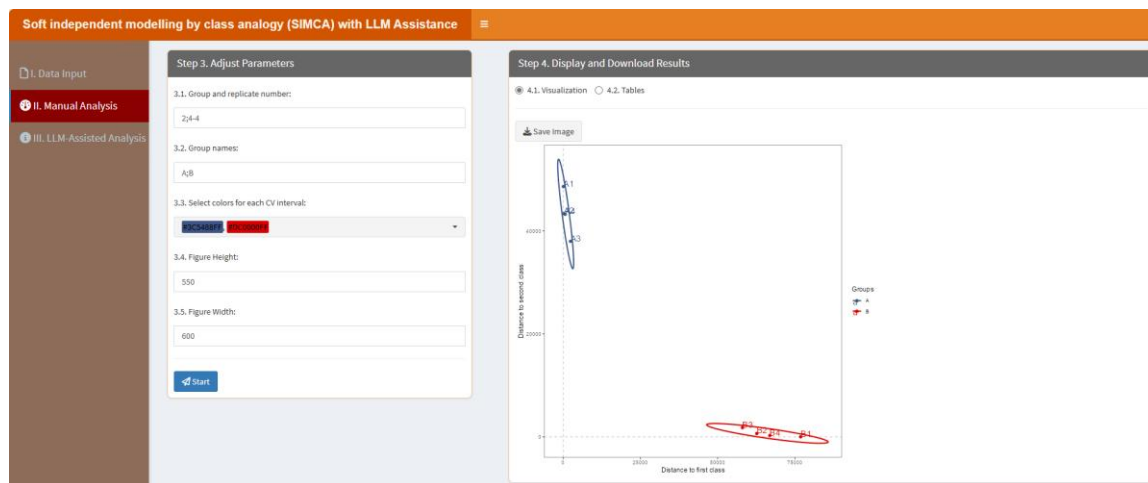
3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the CV of each group:

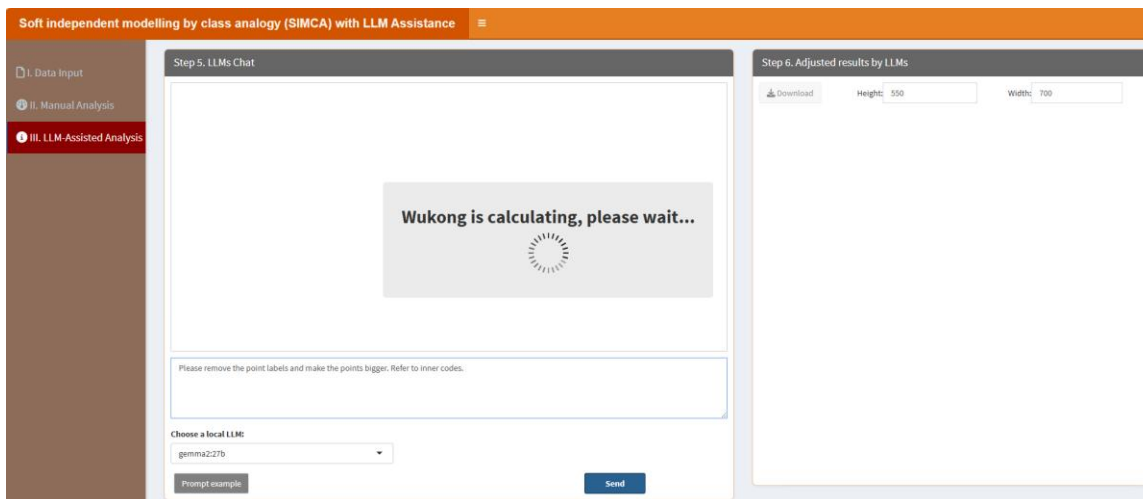
The screenshot shows the SIMCA software interface with the title "Soft independent modelling by class analogy (SIMCA) with LLM Assistance". The left sidebar has three tabs: "I. Data Input", "II. Manual Analysis", and "III. LLM-Assisted Analysis". The main area is divided into two panels. The left panel, titled "Step 3. Adjust Parameters", contains five sections: "3.1. Group and replicate number:" with a text box containing "2x4-4"; "3.2. Group names:" with a text box containing "A,B"; "3.3. Select colors for each CV interval:" with a color selection dropdown showing "Blue" and "Red"; "3.4. Figure Height:" with a text box containing "550"; and "3.5. Figure Width:" with a text box containing "600". A "Start" button is at the bottom. The right panel, titled "Step 4. Display and Download Results", has two tabs: "4.1. Visualization" and "4.2. Tables". The "4.2. Tables" tab is active, showing a table with 8 entries. The table has two columns: "Distance.to.first.class" and "Distance.to.second.class". The entries are labeled A1 through B4. A "Download" button and a "Show 10 entries" dropdown are at the top. A search bar is on the right. At the bottom, it says "Showing 1 to 8 of 8 entries" and has "Previous", "1", and "Next" buttons.

	Distance.to.first.class	Distance.to.second.class
A1	67.701658736488	48617.30272836445
A2	234.3597853350574	42356.46183482067
A3	2367.606758999479	37977.14862482518
A4	631.0718441723637	42234.23956296186
B1	76882.39778410774	0.3174900250725807
B2	62680.77384243307	646.3430440287248
B3	58025.84675941988	1782.467021453362
B4	66901.6758134878	263.4951260272435

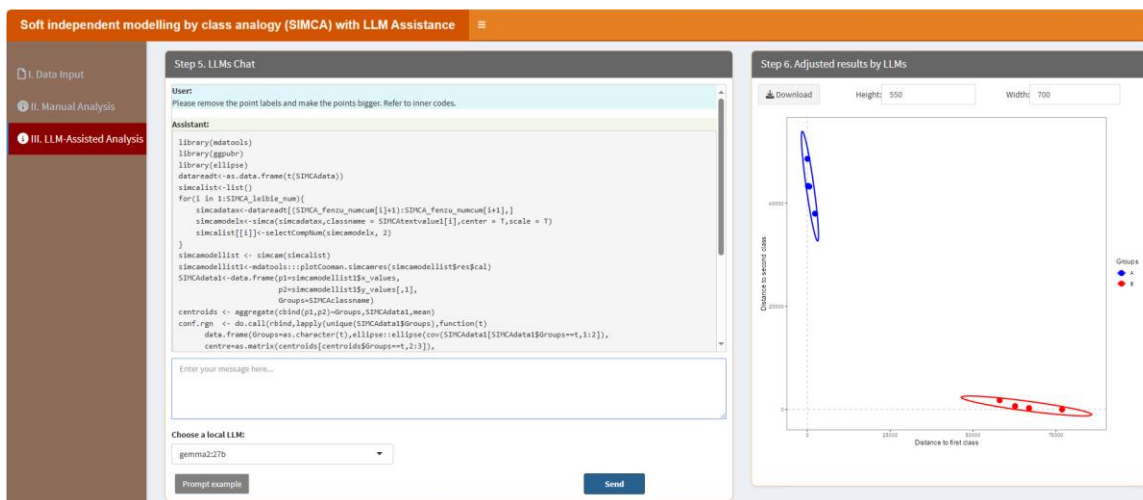
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please remove the point labels and make the points bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please remove the point labels and make the points bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.26. Time Course Data Analysis

This module analyzes time-series data. Users can adjust parameters like time intervals through the classic framework, while local LLMs assist in refining the analysis and interpreting trends. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Time course data analysis with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. It includes a 'Select File Format:' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' and 'Is the first column names?', both of which are checked. At the bottom, there is a 'Which sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one entry. The table has columns 'Show' and 'Description'. The 'Show' column has a value of '10' and a dropdown arrow. The 'Description' column has the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries'. There are 'Previous' and 'Next' buttons at the bottom right of the table.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Time course data analysis with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains several input fields: '3.1. Group and replicate number:' with the value '4;3-3-3', '3.2. Select colors:' with a dropdown menu showing 'Highlighted' and 'Not', '3.3. Point size:' with the value '3', '3.4. Figure Height:' with the value '550', and '3.5. Figure Width:' with the value '600'. At the bottom of this panel is a 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains two radio buttons: '4.1. Visualization' (selected) and '4.2. Tables'. Below these is a 'Save Image' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Select colors: This parameter lets you choose specific colors for each group (Highlighted and Not highlighted). A color picker is provided, offering a variety of predefined palettes such

as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select two colors for each group (Highlighted and Not highlighted), ensuring that the visual representation is both informative and aesthetically pleasing.

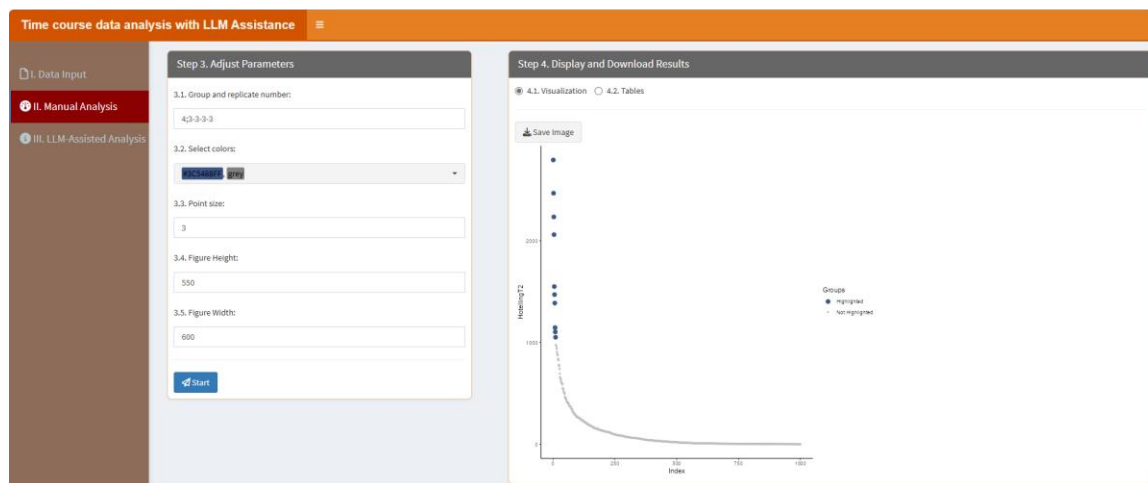
3.3. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

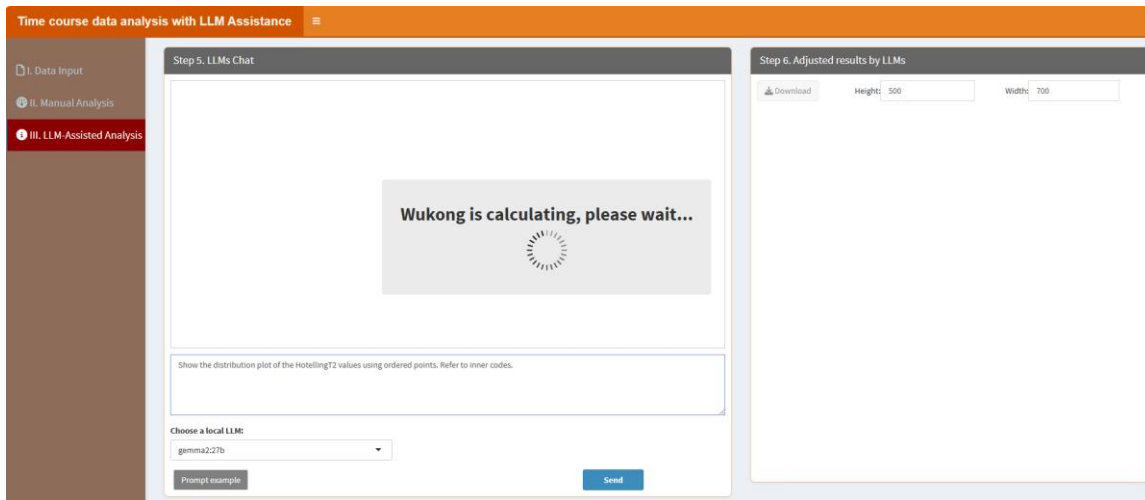


In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the CV of each group:

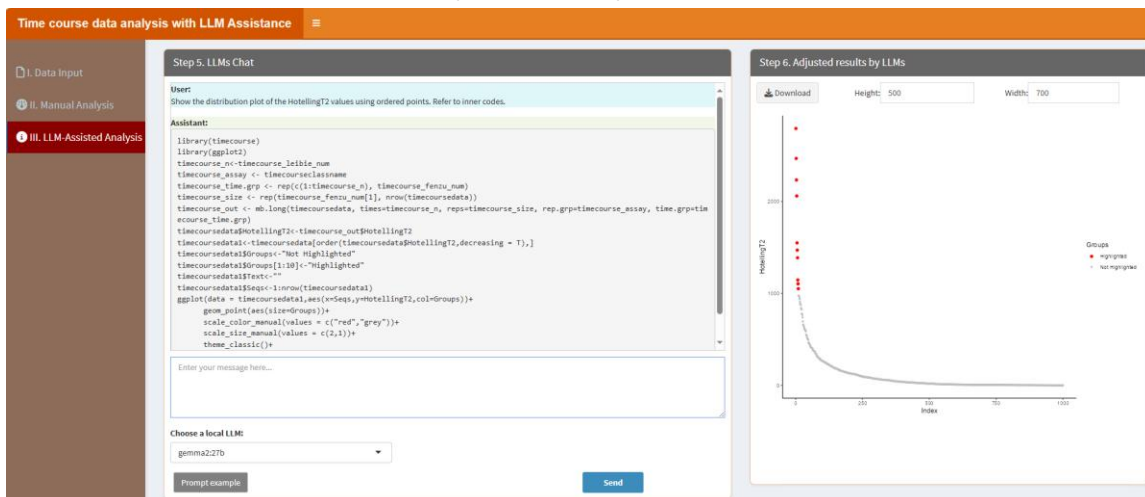
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Show the distribution plot of the HotellingT2 values using ordered points. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Show the distribution plot of the HotellingT2 values using ordered points. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.27. t-SNE (t-Distributed Stochastic Neighbor Embedding)

t-SNE visualizes high-dimensional data in 2D or 3D. Users can configure perplexity and learning rates via the classic framework, while local LLMs enhance interpretability. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 't-SNE with LLM Assistance' application interface. On the left is a sidebar with three main sections: 'I. Data Input' (highlighted in red), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' and 'Is the first column names?', both of which are checked. At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show: 10 entries' dropdown, a search bar, and a table with one row containing the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 't-SNE with LLM Assistance' application interface. On the left is a sidebar with three main sections: 'I. Data Input', 'II. Manual Analysis' (highlighted in red), and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains several input fields: '3.1. Group and replicate number:' with the value '2;4-4', '3.2. Group names:' with the value 'A;B', '3.3. Select colors for each CV interval:' with a dropdown menu showing 'Blue' and 'Red', '3.4. Point size:' with the value '5', '3.5. Figure Height:' with the value '550', and '3.6. Figure Width:' with the value '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section for '4.1. Visualization' with a 'Save Image' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control"

and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

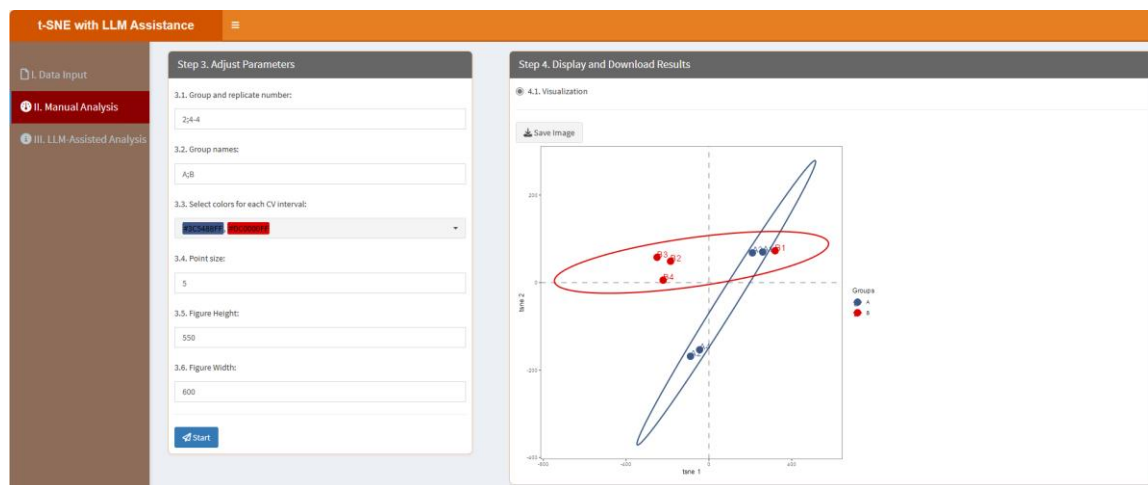
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click "Start" button, the results will be shown as below:

In the "4.1. Visualization" part, users can click "Save image" button to download the figure:

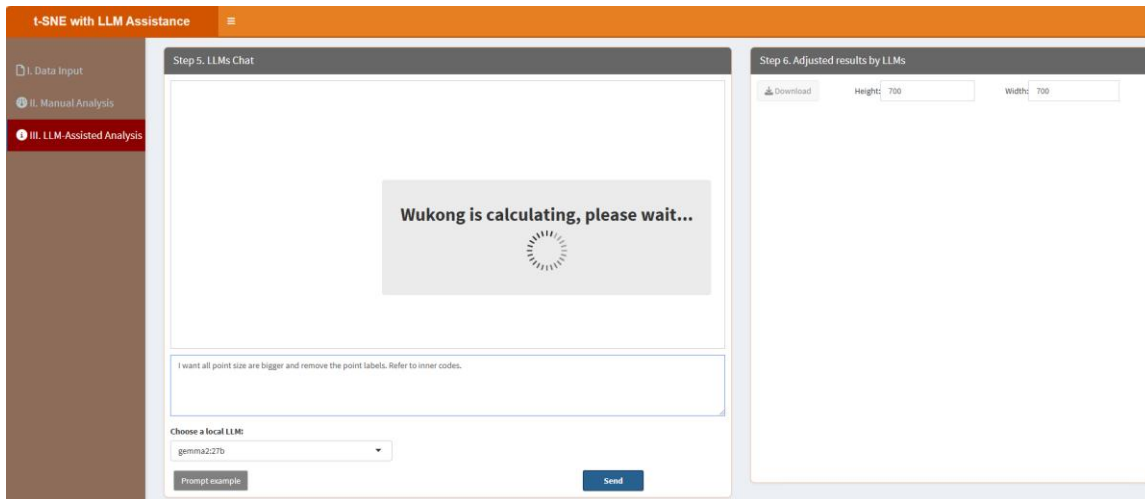


"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions

(please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger and remove the point labels. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger and remove the point labels. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.

t-SNE with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 5. LLMs Chat

User:

I want all point size are bigger and remove the point labels. Refer to inner codes.

Assistant:

```

library(tsnr)
library(ggplot2)
library(ggpubr)
library(ellipse)
set.seed(42)
dataread<-tsne(t(tsondata), perplexity=2,pca=F,pca_center=F,pca_scale=F)
tsneff<-as.data.frame(dataread$V)
tsneff$groups<-tsneclasse
colnames(tsnr)[1:2]<-c("x1","x2")
centroids <- aggregate(cbind(x1,x2)~groups,tsneff,mean)
conf.rgn <- do.call(rbind,lapply(unique(tsnr$groups),function(t)
  data.frame(groups=as.character(t),ellipse:ellipse(cov(tsnr[tsneff$groups==t,1:2]),
    centre=as.matrix(centroids[centroids$groups==t,1:2]),
    level=0.95),stringsAsFactors=FALSE)))
ggplot(data=tsneff,(aes(x=x1,y=x2,colour = groups)))+
  geom_vline(xintercept = 0,col="grey",lty=2,lwd=1)+
  geom_hline(yintercept = 0,col="grey",lty=2,lwd=1)+
  geom_point(size=5)+

```

Enter your message here...

Choose a local LLM

gemma2.2b

Prompt example

Send

Step 6. Adjusted results by LLMs

Download

Height: 700

Width: 700

Groups

• 1

• 2

5.28. Tumor Purity Estimation

Tumor purity estimation quantifies tumor content in samples. Users can adjust estimation parameters through the classic framework, and local LLMs provide refined insights into the results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Estimate tumor purity with LLM Assistance' application. The left sidebar has three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' and 'Is the first column names?', both of which are checked. At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show' dropdown set to '10' and a search bar. Below this is a table with one entry: '1' in the first column and 'No data loaded. Please upload your dataset or use the example data.' in the second column. At the bottom of the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Estimate tumor purity with LLM Assistance' application. The left sidebar has three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains four numbered sections: '3.1. Select a color for samples:' with a color picker showing a red color; '3.2. Point size:' with a text input field containing '3'; '3.3. Figure Height:' with a text input field containing '550'; and '3.4. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains two radio buttons: '4.1. Visualization' (selected) and '4.2. Tables'. Below these is a 'Save Image' button.

3.1. Select a color for samples: This parameter lets you choose a specific color for each sample point. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select one color for the sample points, ensuring that the visual representation is both informative and aesthetically pleasing.

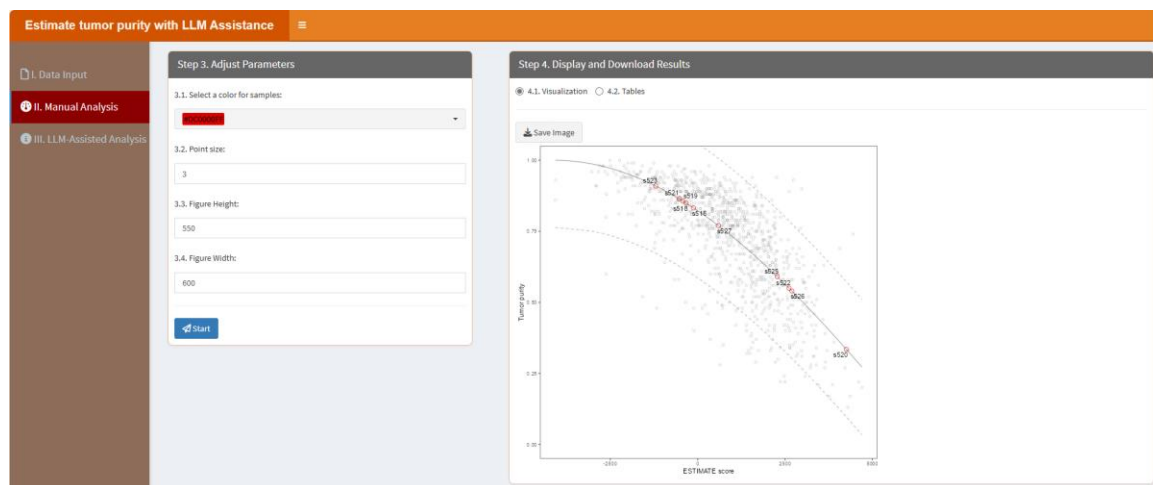
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the CV of each group:

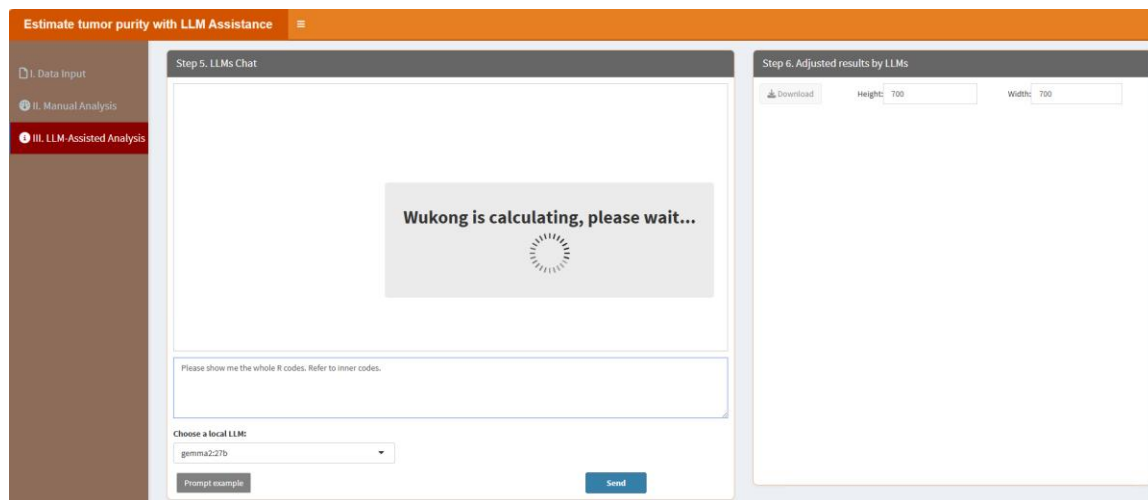
	sample	stromal	immune	estimate	purity
1	s516	-385.4984147150459	165.7506186741359	-119.74779584091	0.8323790792006818
2	s518	-429.1693135126444	99.71302086386834	-329.456292646776	0.84690420636744055
3	s519	-60.98619640768169	-366.7031429099347	-429.6893293175964	0.8567231615588794
4	s520	1927.514311332974	2326.159843492796	4293.674154825741	0.3348246206313757
5	s521	-673.8495439487133	141.7277496043046	-532.1217943424087	0.864381191015435
6	s522	1447.955173316437	1166.518535966675	2614.473709282512	0.5497247620919631
7	s523	-271.1575605983222	-928.449209409589	-1199.606770019281	0.809424238536412
8	s525	965.6180351851657	1310.2774528125	2275.895780466416	0.5905430112570948
9	s526	545.9948862349975	2149.104728704701	2695.099394939699	0.53988002472266024
10	s527	-710.4437942157576	1303.08008657397	592.6363624416399	0.7699645763184405

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions

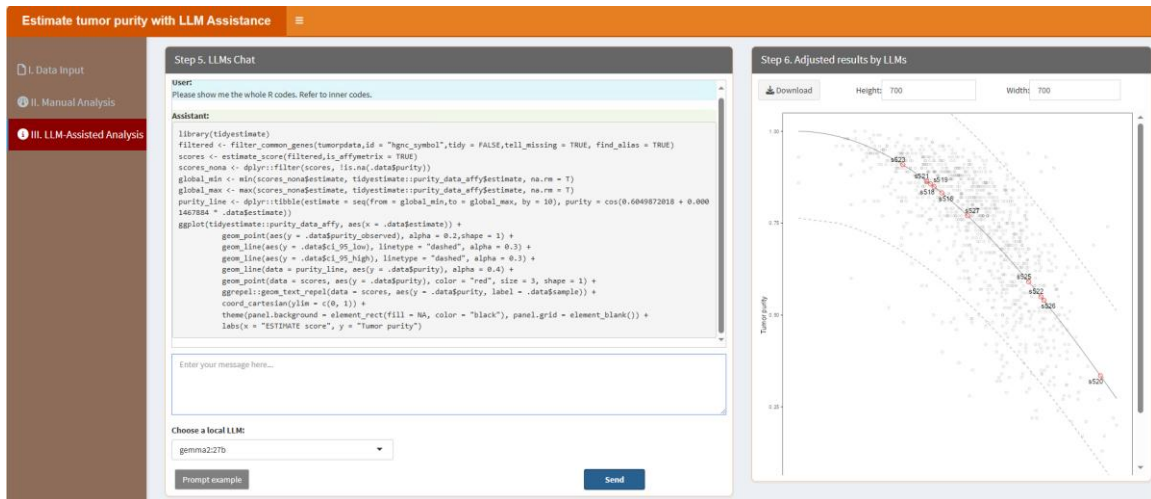
(please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please show me the whole R codes. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please show me the whole R codes. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.29. UMAP (Uniform Manifold Approximation and Projection)

UMAP is a dimensionality reduction technique for visualizing complex data. Users can configure parameters like neighbors and minimum distances via the classic framework, while local LLMs assist in refining the analysis. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'UMAP with LLM Assistance' interface. On the left is a sidebar with three tabs: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' and 'B. Load example data'. Below these are file format selection buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' and 'Is the first column names?', and a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row containing the text 'No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' and 'Previous 1 Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'UMAP with LLM Assistance' interface. The sidebar now has 'II. Manual Analysis' selected. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains six numbered input fields: '3.1. Group and replicate number:' with the value '2;4-4', '3.2. Group names:' with the value 'A;B', '3.3. Select colors for each CV interval:' with a color picker showing blue and red, '3.4. Point size:' with the value '5', '3.5. Figure Height:' with the value '550', and '3.6. Figure Width:' with the value '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a '4.1. Visualization' section with a 'Save Image' button.

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The

names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

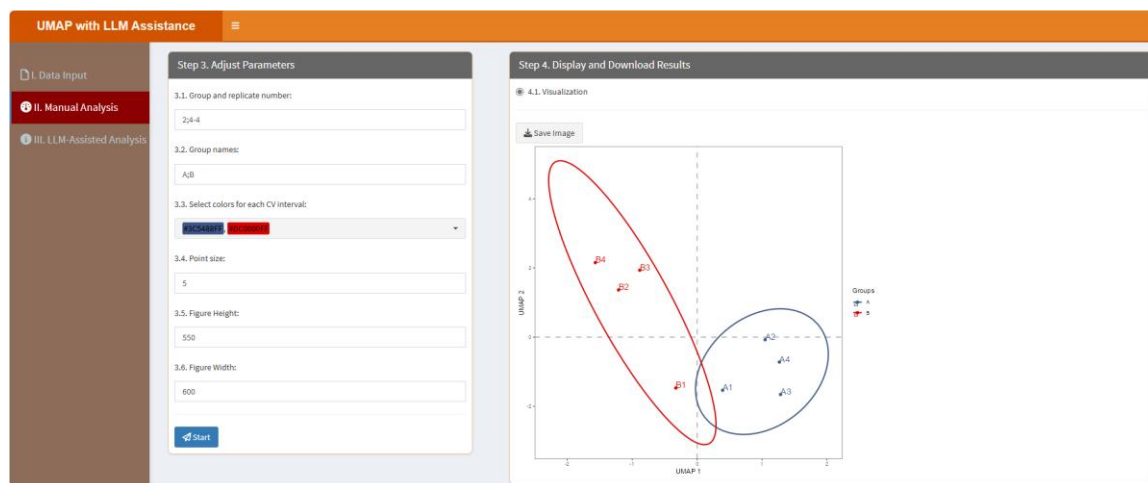
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

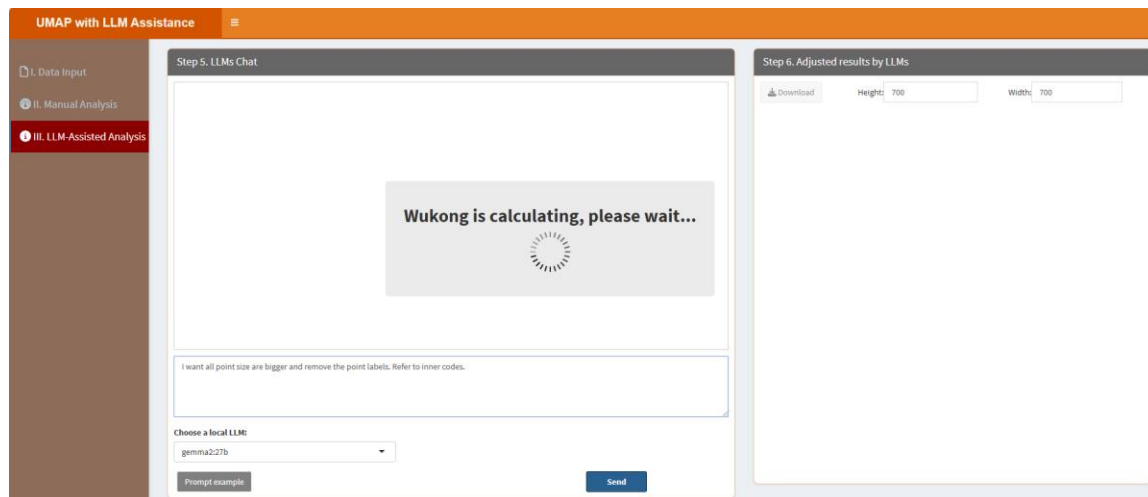


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box,

the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "I want all point size are bigger and remove the point labels. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "I want all point size are bigger and remove the point labels. =>", it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click "Send" button. It will learn like below:



The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.

UMAP with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 5. LLMs Chat

User:

I want all point size are bigger and remove the point labels. Refer to inner codes.

Assistant:

```

library(umap)
library(ggplot2)
library(ggpubr)
library(ellipse)

datareadt<-umap::umap(t(umapdata),n_neighbors=4,n_components=2,random_state = 1234)
umapdf<-as.data.frame(datareadt$layout)
umapdf$groups<- umap::lastname
colnames(umapdf)[1:2]<-c("p1","p2")
centroids <- aggregate(cbind(p1,p2)~groups,umapdf,mean)
conf.rgs <- do.call(rbind,lapply(unique(umapdf$groups),function(t)
  data.frame(groups=as.character(t),ellipse=cov(umapdf[umapdf$groups==t,1:2]),
    centre=as.matrix(centroids[centroids$groups==t,2:3]),
    level=0.95),stringsAsFactors=FALSE)))

ggplot(data=umapdf,(aes(x=p1,y=p2,colour = groups)))+
  geom_vline(xintercept = 0,col="grey",lty=2,lwd=1)+
  geom_hline(yintercept = 0,col="grey",lty=2,lwd=1)+
  geom_point(size=5)+
  geom_path(data=conf.rgs,size=1)+

```

Enter your message here...

Choose a local LLM

gemma2.27b

Prompt example

Send

Step 6. Adjusted results by LLMs

Download

Height: 700

Width: 700

Groups

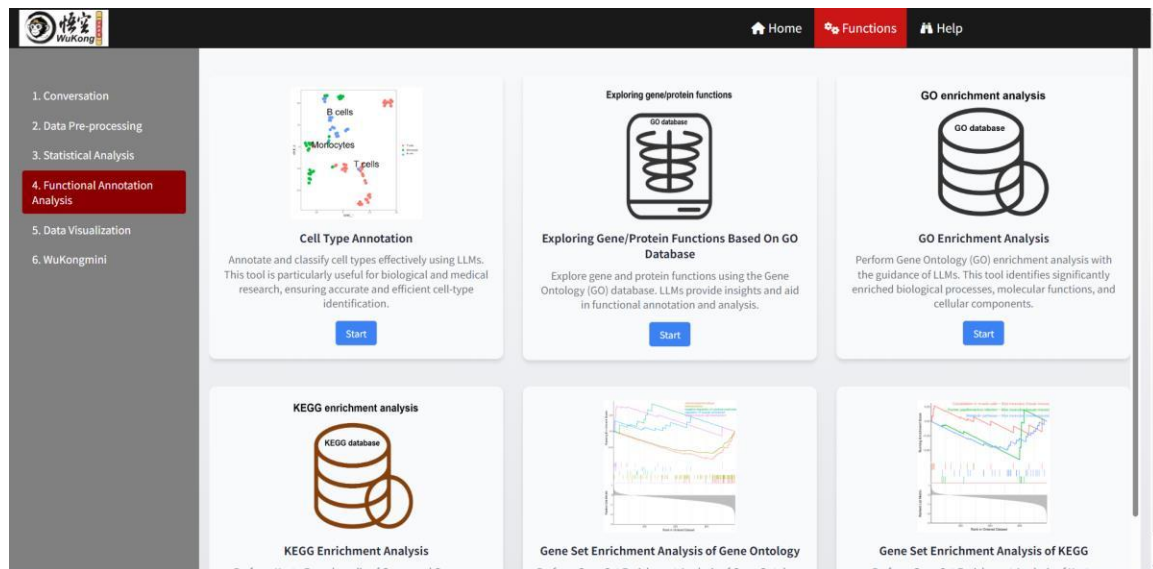
1

2

137

6. Functional Annotation Analysis

Functional Annotation Analysis in proteomics is a critical approach for understanding the biological significance of genes and proteins by exploring their functions, associated pathways, and enriched biological processes. With the integration of LLMs, researchers can not only obtain classic enrichment results but also deeply explore and interpret these findings with enhanced precision and efficiency. LLMs assist in cell type annotation, enabling accurate classification and identification of cell types, which is particularly valuable in biological and medical research. They also facilitate the exploration of gene and protein functions through the Gene Ontology (GO) database, providing insights into molecular functions, biological processes, and cellular components. Additionally, LLMs enhance GO and KEGG enrichment analyses, guiding researchers in identifying significantly enriched pathways, annotating enriched gene sets, and deriving meaningful biological insights. By contextualizing findings and suggesting potential implications, LLMs empower researchers to better understand complex biological relationships and design informed experimental validations or research directions.



6.1. Cell type annotation

Annotate and classify cell types effectively using the local LLMs. This tool is particularly useful for biological and medical research, ensuring accurate and efficient cell-type identification. Users can adjust parameters to refine their analysis and leverage LLMs to enhance the accuracy of their results. After clicking the “Start” button, it will be started in a new window like below:

Cell type annotation with the assistance of LLMs

1. Upload data:

☒ I. Upload ☐ II. Load example data

1.1 Import data:

Browse... No file selected

Review data

Choose a Model:

gemma3 27b

2. Conversation:

3. Results:

3.1 Default plot:

Download

No uploaded data and No plot here.
Please upload data or click the example data.

3.2 Adjusted results by LLMs:

Download

Type your message here...

Prompt exar

Send

6.1.1. Upload data. Herein, users can upload their data (.RDS format) from “I. Upload”. Please note the data here should be an object of class Seurat, which means users should process their single cell data in advance using Seurat package and save the results without annotation into a .RDS file. In addition, users can select “II. Load example data” and click the button “Review data” to check the example data. Next, users can select a local LLM (e.g. gemma3:27b). As below:

Cell type annotation with the a

1. Upload data:

☐ I. Upload ☒ II. Load example data

Download example data

Review data

Choose a Model:

gemma3 27b

2. Conversa

3. Results:

3.1 Default plot:

Download

Adjusted results by LLMs:

Download

Type your message here

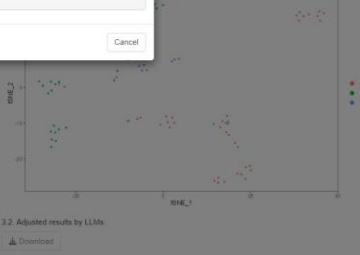
Prompt ex

Send

Uploaded data

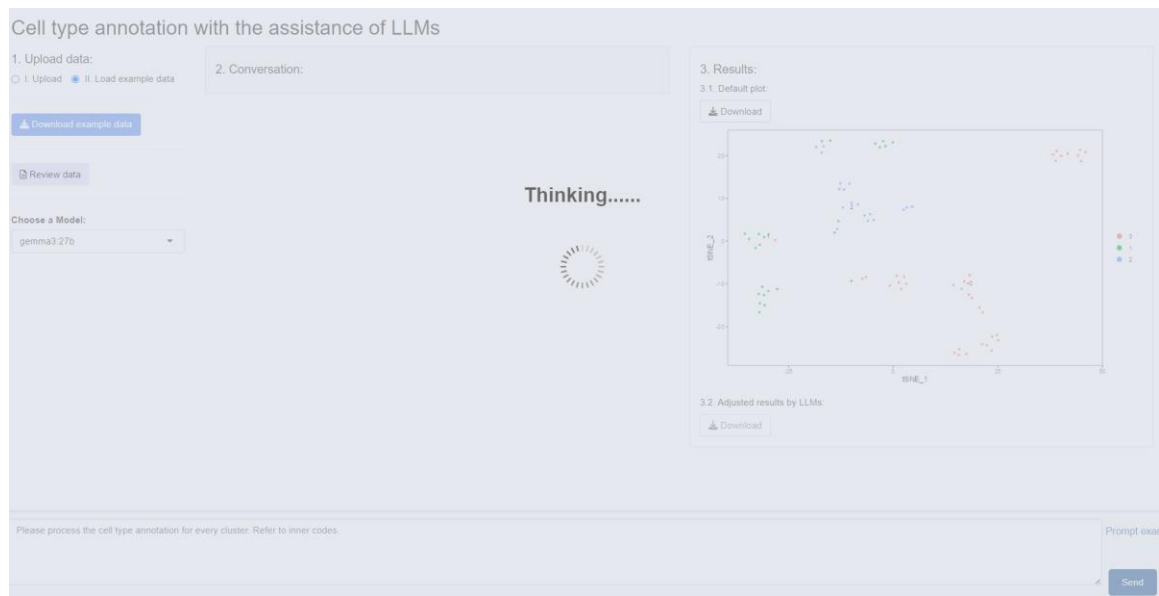
An object of class Seurat
219 features across 40 samples within 1 assay
Active assay: RNA (238 features, 28 variable features)
3 layers present: counts, data, scale.data
2 dimensional reductions calculated: pca, tsc

Cancel



6.1.2. Conversation. Here displays the conversation history, while "Type your message here..." enables users to type messages or questions in the provided text box. When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results. For instance, the example prompt "Please process the cell type annotation for every cluster. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please process the cell type annotation for every cluster. =>", it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click "Send" button. It will learn like below:



6.1.3. Results. Here shows two results: 3.1. Default plot shows the cell distribution without cell annotation. 3.2. Adjusted results by LLMs means that this panel shows the results (figures or tables) based on the conversation in the "2. Conversation". The results are shown as below.

Cell type annotation with the assistance of LLMs

1. Upload data:

☐ Upload ☒ Load example data

[Download example data](#)

[Review data](#)

Choose a Model:

gemma3.27b

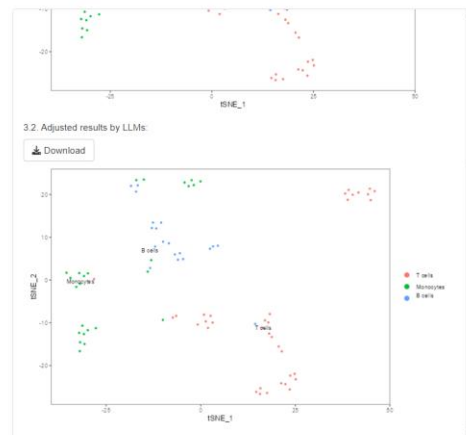
2. Conversation:

User:

Please process the cell type annotation for every cluster.
Refer to inner codes.

Assistant:

```
library(Seurat)
library(data.table)
library(purrr)
library(cellama)
# Find cluster markers
celltypedata1 <- FindAllMarkers(celltypedata, verbose =
F, min.pct = 0.5)
# split into a lists per cluster
celltypedata.list <- split(celltypedata1, celltypedata1$cl
uster)
# cell type annotation using a local large language model
celltypedata.res <- cellama(celltypedata.list, temperatur
e = 0, seed = 101, n_genes = 30, model = "gemma2:27b")
# transfer the labels
annotations <- map_chr(celltypedata.res, 1)
names(annotations) <- levels(celltypedata)
celltypedata <- RenameIdents(celltypedata, annotations)
DimPlot(celltypedata, label = 1, repel = T, label.size =
3) + theme_bw() +
  theme(panel.grid.major.element_blank(), panel.grid.mi
nor.element_blank())
```



Type your message here...

Prompt exam

[Send](#)

6.2. Exploring gene/protein functions based on GO database

Explore gene and protein functions using the GO database with the help of local LLMs. This tool allows users to adjust parameters for detailed functional annotation and analysis while leveraging LLMs to provide deeper insights into gene and protein roles. After clicking the “Start” button, it will be started in a new window like below:

Exploring gene/protein functions based on GO database with the assistance of LLMs

1. Upload Data:

- ☒ 1.1. Paste gene/protein names/UniProt IDs
- ☐ 1.2. Example gene/protein names/UniProt IDs

Pasted gene/protein names/UniProt IDs:

1.3. Choose a Model:

deepseek-r1:32b

2. Conversation

Type your message here...

Prompt example Send

In the “1. Upload Data”, users can paste gene/protein names, or UniProt IDs of interest (e.g. those proteins obtained from differential expression analysis) in the below box when selecting “1.1. Paste gene/protein names/UniProt IDs”. Or, users can select “1.2. Example gene/protein names/UniProt IDs” to check the example, as below:

Exploring gene/protein functions based on GO database with the assistance of LLMs

1. Upload Data:

- ☐ 1.1. Paste gene/protein names/UniProt IDs
- ☒ 1.2. Example gene/protein names/UniProt IDs

Pasted gene/protein names/UniProt IDs:

TP53
CTNNA1
BCL6

1.3. Choose a Model:

gemma3:27b

2. Conversation

Type your message here...

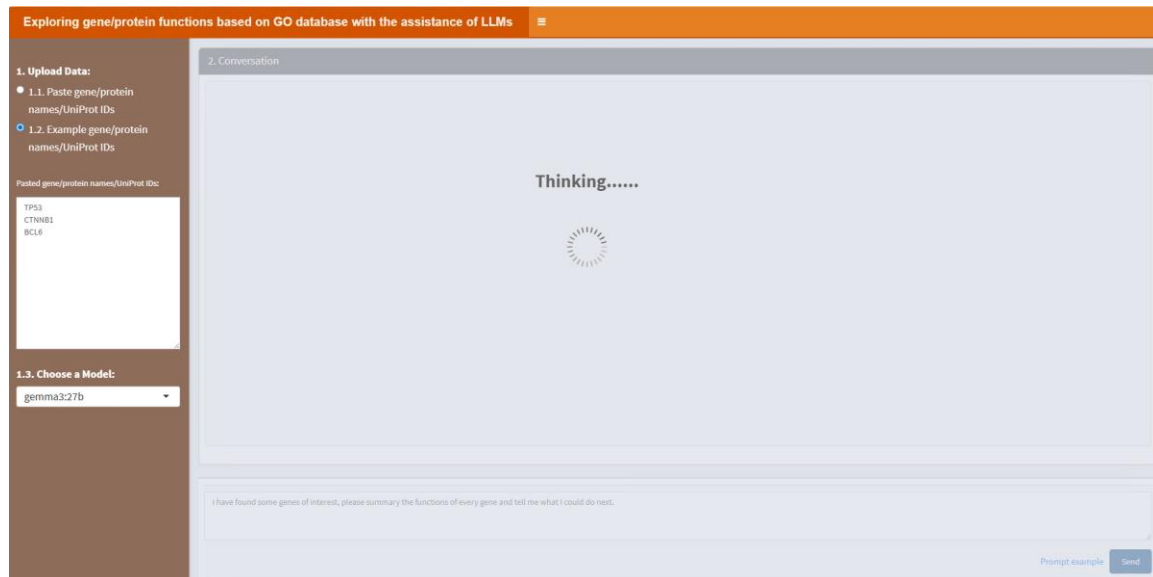
Prompt example Send

Then, users can select a pre-installed LLM from the "1.3. Choose a Model" option for their

interactions (please see below “How to install LLMs locally?” part).

In the “2. Conversation”, it displays the conversation history, while “Type your message here...” enables users to type messages or questions in the provided text box. When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. For instance, the example prompt “I have found some genes of interest, please summary the functions of every gene and tell me what I could do next.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I have found some genes of interest, please summary the functions of every gene and tell me what I could do next. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below.

Exploring gene/protein functions based on GO database with the assistance of LLMs

1. Upload Data:

1.1. Paste gene/protein names/UniProt IDs

1.2. Example gene/protein names/UniProt IDs

Pasted gene/protein names/UniProt IDs:

TP53

CTNNB1

BCL6

1.3. Choose a Model:

gemma3:27b

2. Conversation

User:

I have found some genes of interest, please summary the functions of every gene and tell me what I could do next.

TP53 CTNNB1 BCL6

Assistant:

Okay, here's a summary of the functions of TP53, CTNNB1, and BCL6, followed by suggestions for next steps in your research.

Gene Function Summaries

1. TP53 (Tumor Protein p53)

Core Function: Often called the "guardian of the genome," TP53 is a tumor suppressor gene. It plays a critical role in preventing cancer development.

How it Works:

DNA Damage Response: When DNA damage is detected, p53 is activated. It can:

Cell Cycle Arrest: Halt the cell cycle to allow time for DNA repair.

DNA Repair: Activate genes involved in DNA repair.

Apoptosis (Programmed Cell Death): If the damage is irreparable, p53 triggers apoptosis to eliminate the damaged cell.

Other Roles: p53 also participates in cellular metabolism, angiogenesis (blood vessel formation), and immune responses.

Clinical Relevance: Mutations in TP53 are found in approximately 50% of all human cancers, making it the most frequently mutated gene in cancer. Loss of p53 function allows damaged cells to proliferate unchecked, leading to tumor development.

2. CTNNB1 (β-Catenin)

Core Function: β-catenin is a crucial signaling molecule involved in cell adhesion, cell fate determination, and gene transcription. It's a key component of several signaling pathways, most notably the Wnt pathway.

How it Works:

Type your message here...

Prompt example Send

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6.3. GO enrichment analysis

Perform GO (Gene Ontology) enrichment analysis by adjusting parameters through a classic framework and refining results with local LLMs. This tool identifies significantly enriched biological processes, molecular functions, and cellular components, offering enhanced interpretability and accuracy. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'GO enrichment analysis with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. Below these are file format selection buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one entry: '1. No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' and 'Previous 1 Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'GO enrichment analysis with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains five numbered sections: '3.1. Please choose a species:' with a dropdown menu showing '9606-Homo sapiens (Human)'; '3.2. Select two colors for Barplot:' with a color selection interface showing 'Blue' and 'Red'; '3.3. Total number of GOs displayed in the Barplot:' with a text input field containing '15'; '3.4. Figure Height:' with a text input field containing '600'; and '3.5. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a '4.1. Tables' (selected) and '4.2. Visualization' section, and a 'Download' button.

3.1. Please choose a species: Select the reference species for GO term mapping and annotation.

3.2. Select two colors for Barplot: Choose two distinct colors to visually represent GO term categories in the bar plot output. These colors will be applied in a gradient or alternating format to help distinguish between different GO terms or significance levels.

3.3. Total number of GOs displayed in the Barplot: Specify the maximum number of enriched GO terms to be visualized in the bar plot. The default value here is 15, users can adjust this based on their preferences or dataset size.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table:

GO enrichment analysis with LLM Assistance

Step 3. Adjust Parameters

3.1. Please choose a species:
9006-Homo sapiens (Human)

3.2. Select two colors for Barplot:
Blue, Red

3.3. Total number of GOs displayed in the Barplot:
15

3.4. Figure Height:
600

3.5. Figure Width:
600

Start

Step 4. Display and Download Results

4.1. Tables 4.2. Visualization

Download

Show 10 entries

ID	ONTOLOGY	TERM	Count	GeneRatio	SigRatio	pvalue	p-adjust	gene
43	GO:0005634	CC nucleus	9	9/20	10044/95459	0.00008922688336629206	0.00171001261284545	OT1
44	GO:0005654	CC nucleoplasm	9	9/20	4286/95459	7.55524489561796e-8	0.000007630797344517414	OT1
48	GO:0005737	CC cytoplasm	7	7/20	8255/95459	0.001015292405188004	0.004338635731671753	OT1
31	GO:0003700	MF DNA-binding transcription factor activity	5	5/20	2768/95459	0.0002199907610479983	0.00201171795998936	AD1
51	GO:0005829	CC cytosol	5	5/20	3062/95459	0.0008990613660850661	0.003530259898729583	Q51
38	GO:0005509	MF calcium ion binding	4	4/20	2427/95459	0.001457612057503579	0.00485370488938769	B41
70	GO:0006940	BP regulation of smooth muscle contraction	4	4/20	27/95459	2.45020455445026e-11	4.940413200007705e-9	AD1
90	GO:0008270	MF zinc ion binding	4	4/20	4004/95459	0.008732245780748936	0.01547292673431023	B41
143	GO:0043565	MF sequence-specific DNA binding	4	4/20	1398/95459	0.00018401735632293572	0.00201171795998936	AD1
34	GO:0004222	MF metalloendopeptidase activity	3	3/20	423/95459	0.0000931194087193067	0.00171001261284545	B41

Showing 1 to 10 of 194 entries

In the “4.2. Visualization” part, users can click “Save Image” button to download the figure:

GO enrichment analysis with LLM Assistance

Step 3. Adjust Parameters

3.1. Please choose a species:
9006-Homo sapiens (Human)

3.2. Select two colors for Barplot:
Blue, Red

3.3. Total number of GOs displayed in the Barplot:
15

3.4. Figure Height:
600

3.5. Figure Width:
600

Start

Step 4. Display and Download Results

4.1. Tables 4.2. Visualization

Save Image

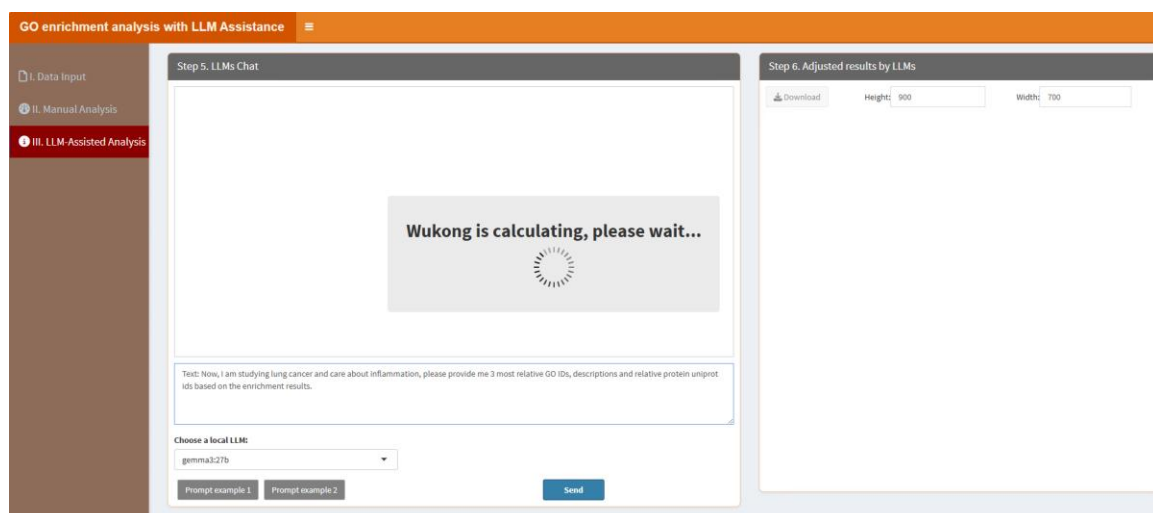
Bar chart visualization showing GO terms and their p-adjust values. The chart uses the selected colors (blue and red) to represent the terms. A legend on the right indicates the p-adjust values: 0.020 (blue), 0.010 (red), and 0.005 (dark blue).

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...”

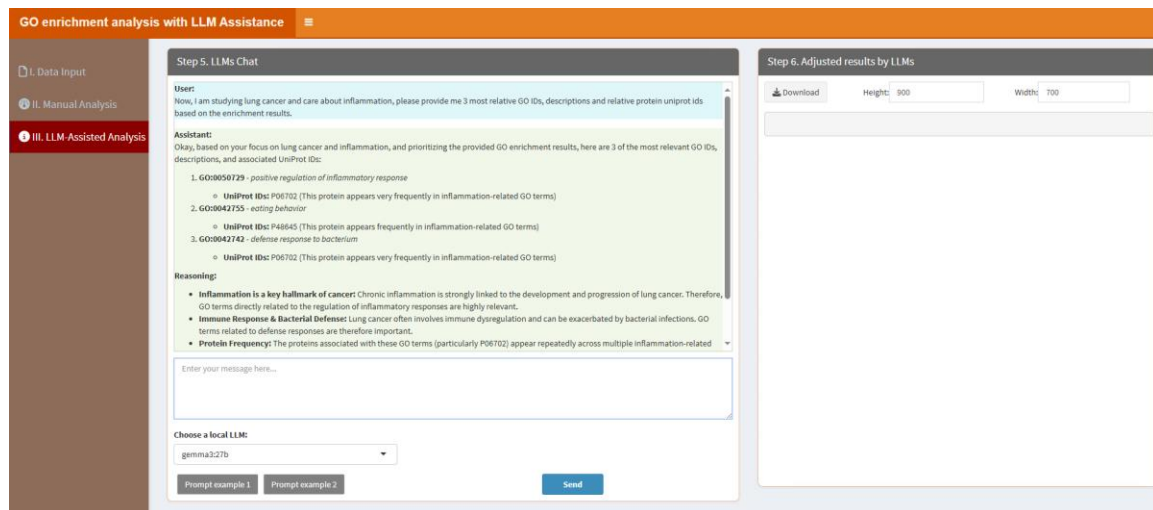
enables users to type messages or questions in the provided text box. Two types of prompts are supported: prompts beginning with "Text:" are used to request interpretation or explanation of the enrichment results (e.g. summarizing biological processes or clarifying GO term relevance), while prompts beginning with "Plot:" allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed GO terms). Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt 1 "Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results. =>", it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the first prompt, click "Send" button. It will learn like below:

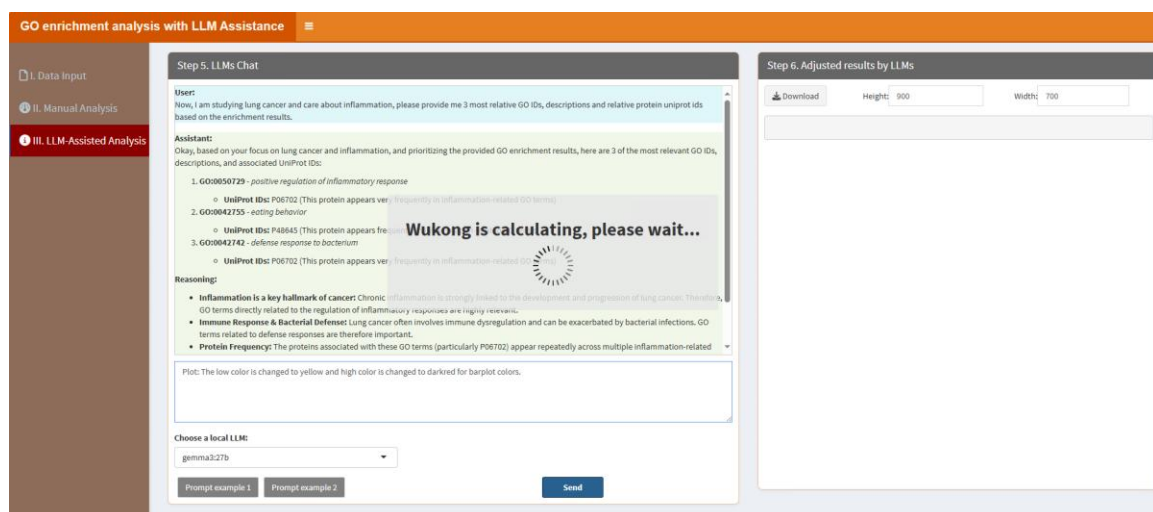


The results are shown as below:

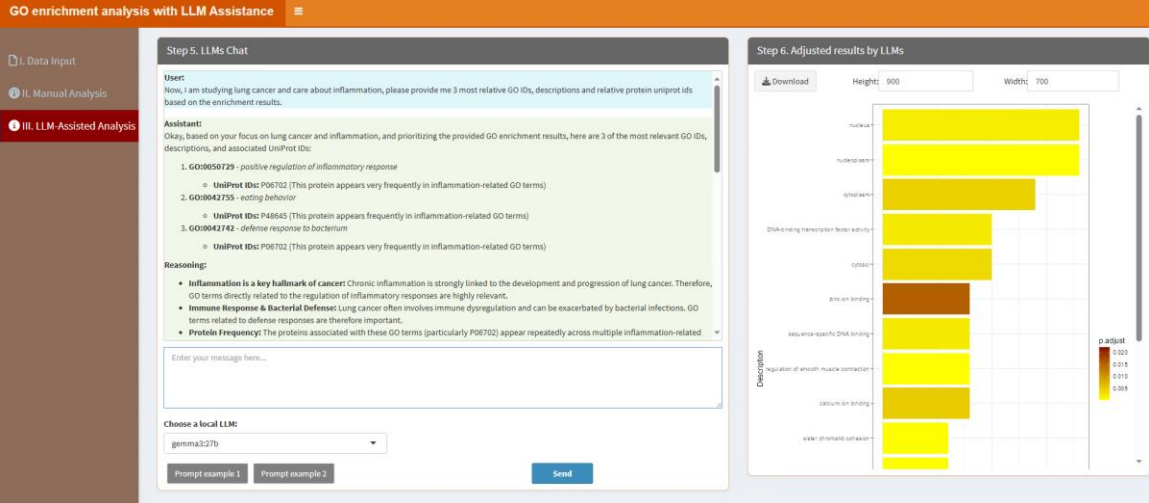


On the other hand, the example prompt 2 “Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the second prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



6.4. KEGG enrichment analysis

Perform KEGG (Kyoto Encyclopedia of Genes and Genomes) enrichment analysis by customizing parameters in a classic framework and utilizing local LLMs to refine pathway enrichment results. This tool helps identify enriched pathways and biological processes with improved precision and usability. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main panel has two tabs: 'A. Upload' (selected) and 'B. Load example data'. Under 'A. Upload', there is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and the text 'No file selected'. There are also two checkboxes: 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked). At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. On the right, 'Step 2. Display Uploaded Data/Example Data' is visible, showing a table with one entry and a 'Search:' field.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' interface. The sidebar is the same as in Step 1, but 'II. Manual Analysis' is now selected. The main panel has two tabs: '4.1. Tables' (selected) and '4.2. Visualization'. Under '4.1. Tables', there is a 'Download' button. The 'Adjust Parameters' section contains five numbered fields: '3.1. KEGG organism:' with 'hsa' entered; '3.2. Select two colors for Barplot:' with a color picker showing blue and red; '3.3. Total number of Pathways displayed in the Barplot:' with '15' entered; '3.4. Figure Height:' with '600' entered; and '3.5. Figure Width:' with '600' entered. At the bottom is a blue 'Start' button.

3.1. KEGG organism: Type the organism code (e.g. hsa for Homo sapiens, mmu for Mus musculus) corresponding to the dataset being analyzed.

3.2. Select two colors for Barplot: Choose two distinct colors to visually represent pathways in the bar plot output. These colors will be applied in a gradient or alternating format to help distinguish between different pathways or significance levels.

3.3. Total number of Pathways displayed in the Barplot: Specify the maximum number of enriched GO terms to be visualized in the bar plot. The default value here is 15, users can

adjust this based on their preferences or dataset size.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table:

KEGG enrichment analysis with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 3. Adjust Parameters

3.1. KEGG organism:

hsa

3.2. Select two colors for Barplot:

Blue

Red

3.3. Total number of GOs displayed in the Barplot:

15

3.4. Figure Height:

600

3.5. Figure Width:

600

Start

Step 4. Display and Download Results

4.1. Tables

4.2. Visualization

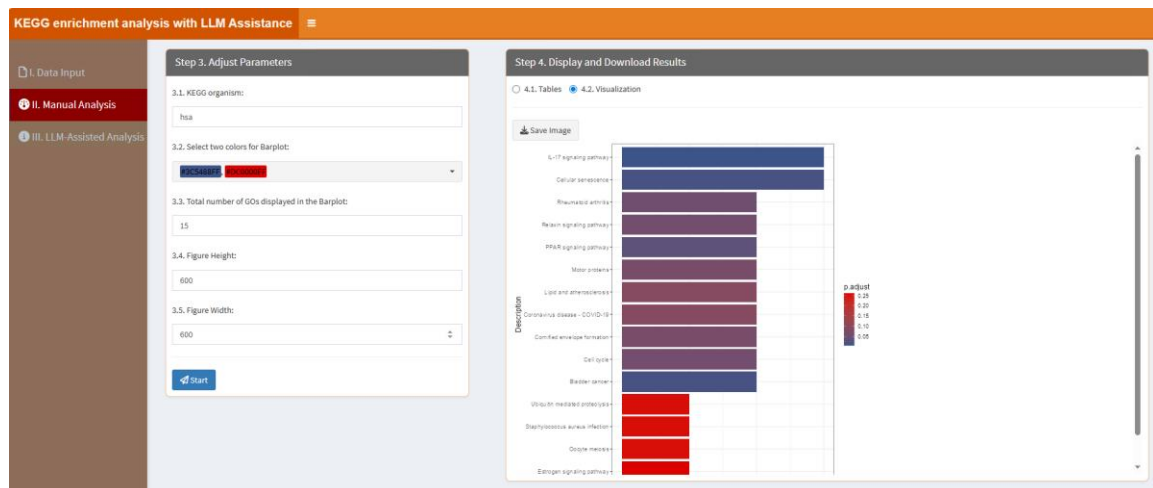
Download

Show10entries

Search:

	category	subcategory	ID	Description	GeneRatio	BgRatio	RichFactor	FoldEnrichment	ZScore
1	Organismal Systems	Immune system	hsa04657	IL-17 signaling pathway	3/12	115/10977	0.02608695652173913	23.86304347826087	8.153356226140218
2	Cellular Processes	Cell growth and death	hsa04218	Cellular senescence	3/12	245/10977	0.0122448975918367	11.20102040816326	5.341888801896871
3	Human Diseases	Cancer: specific types	hsa05219	Bladder cancer	2/12	66/10977	0.0303030303030303	27.71969696969697	7.202438730823281
4	Organismal Systems	Endocrine system	hsa03320	PPAR signaling pathway	2/12	119/10977	0.01680672268807563	15.37394957983193	5.21534488784024
5	Human Diseases	Immune disease	hsa05323	Rheumatoid arthritis	2/12	170/10977	0.01176470588235294	10.76176470588235	4.243347262237676
6	Organismal Systems	Endocrine system	hsa04926	Retinoid signaling pathway	2/12	196/10977	0.0101010101010101	9.239898989898991	3.870558191792886
7	Cellular Processes	Cell growth and death	hsa04130	Cell cycle	2/12	214/10977	0.009345794392523364	8.549065420560746	3.889281208275794
8			hsa04382	Cornified envelope formation	2/12	258/10977	0.00775193798446124	7.091085271317829	3.275184307541337
9	Cellular Processes	Cell motility	hsa04614	Motor proteins	2/12	265/10977	0.007547169811320755	6.90377384905661	3.218293004707257

In the “4.2. Visualization” part, users can click “Save image” button to download the figure:

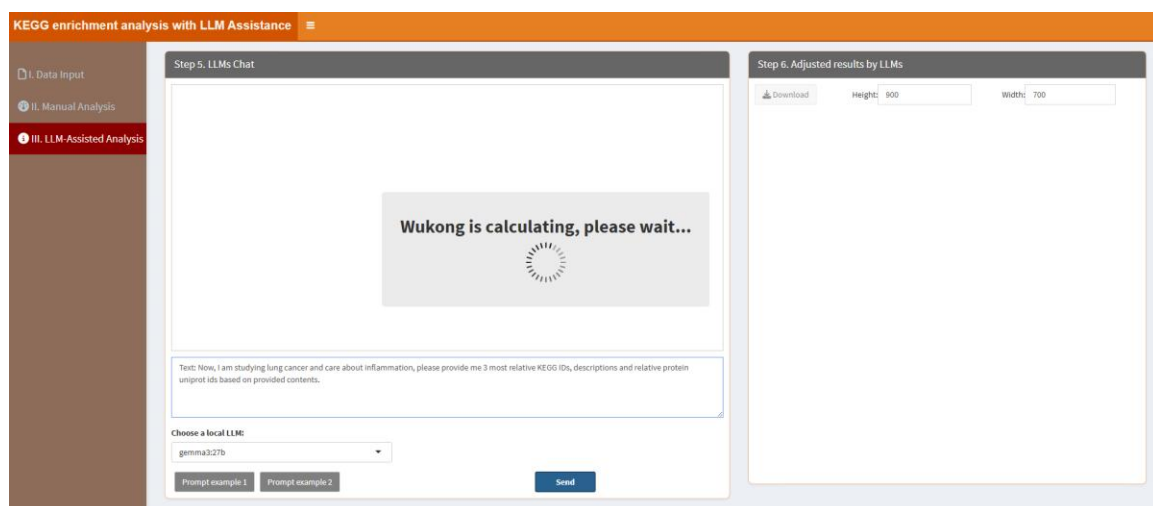


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Two types of prompts

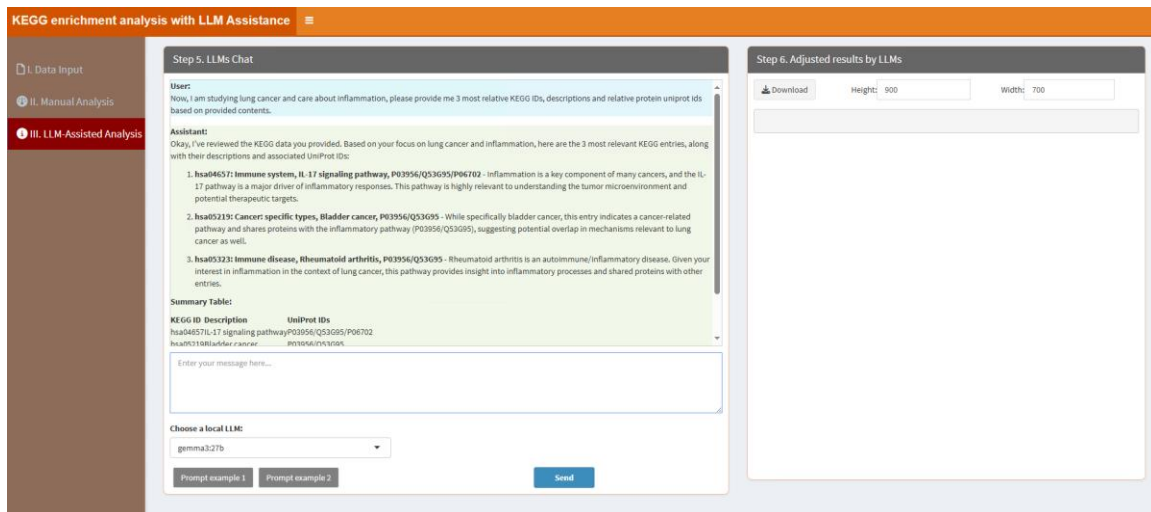
are supported: prompts beginning with "Text:" are used to request interpretation or explanation of the enrichment results (e.g. summarizing pathways relevance), while prompts beginning with "Plot:" allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed pathways). Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt 1 "Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative KEGG IDs, descriptions and relative protein uniprot ids based on provided contents." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative KEGG IDs, descriptions and relative protein uniprot ids based on provided contents. =>", it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the first prompt click "Send" button. It will learn like below:

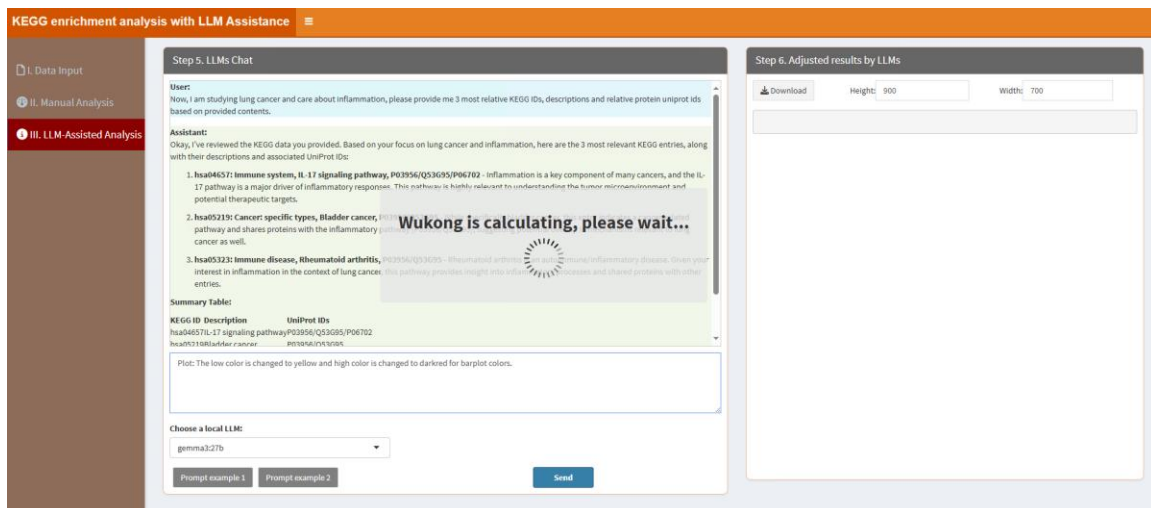


The results are shown as below:

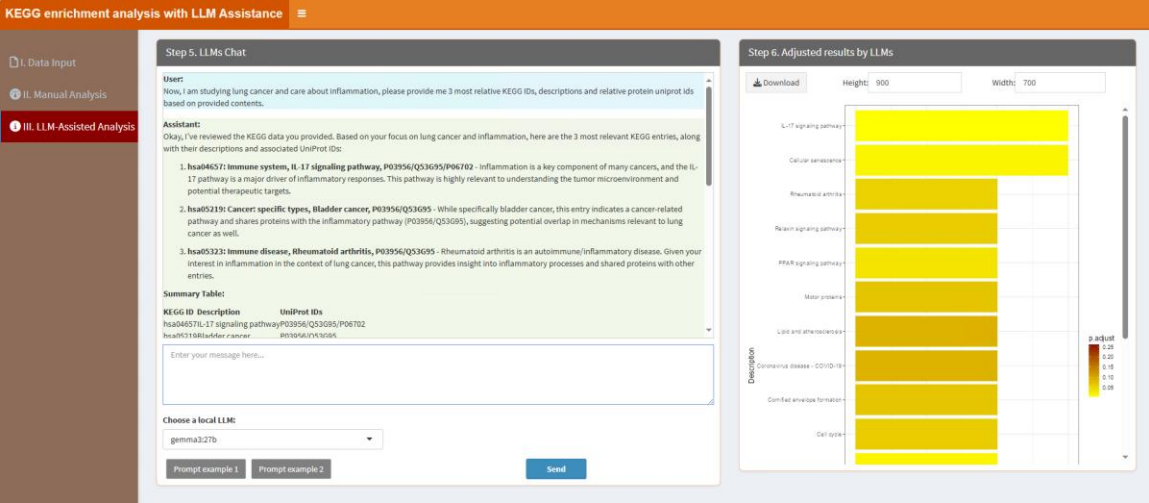


On the other hand, the example prompt 2 “Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors.=>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the second prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



6.5. Gene set enrichment analysis of GO

Perform gene set enrichment analysis based on the GO database by adjusting parameters through a classic framework and leveraging local LLMs to refine results. LLMs assist in interpreting enriched gene sets, prioritizing GO terms, and deriving meaningful biological insights. They also provide suggestions for experimental validation or further research. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Gene Set Enrichment Analysis of Gene Ontology with LLM Assistance' interface. The left sidebar has three tabs: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main panel is divided into two steps. Step 1, 'Upload Data/Example Data', has two radio buttons: 'A. Upload' (selected) and 'B. Load example data'. Below these are options for file format: '.xlsx' (selected), '.xls', and '.csv/txt'. A 'Browse...' button is present, with 'No file selected' below it. There are two checkboxes: 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked). A 'Which sheet to read?' dropdown is set to '1'. Step 2, 'Display Uploaded Data/Example Data', shows a 'Show' dropdown set to '10' entries and a search bar. Below is a table with one row: '1' in the first column and 'No data loaded. Please upload your dataset or use the example data.' in the second column. At the bottom of the table are 'Previous', '1', and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Gene Set Enrichment Analysis of Gene Ontology with LLM Assistance' interface. The left sidebar has three tabs: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main panel is divided into two steps. Step 3, 'Adjust Parameters', has five sections: '3.1. Please choose a species:' with a dropdown set to 'Mouse'; '3.2. ID type:' with a dropdown set to 'UNIPROT'; '3.3. GSEA results index for plotting:' with a text input set to '1,2,5'; '3.4. Figure Height:' with a text input set to '600' and a reset button; and '3.5. Figure Width:' with a text input set to '600' and a reset button. A 'Start' button is at the bottom. Step 4, 'Display and Download Results', has two radio buttons: '4.1. Tables' (selected) and '4.2. Visualization'. Below is a 'Download' button.

3.1. Please choose a species: Select the reference species for GO term mapping and annotation.

3.2. ID type: Select the type of gene/protein identifiers used in your dataset (e.g. UniProt ID, Entrez ID).

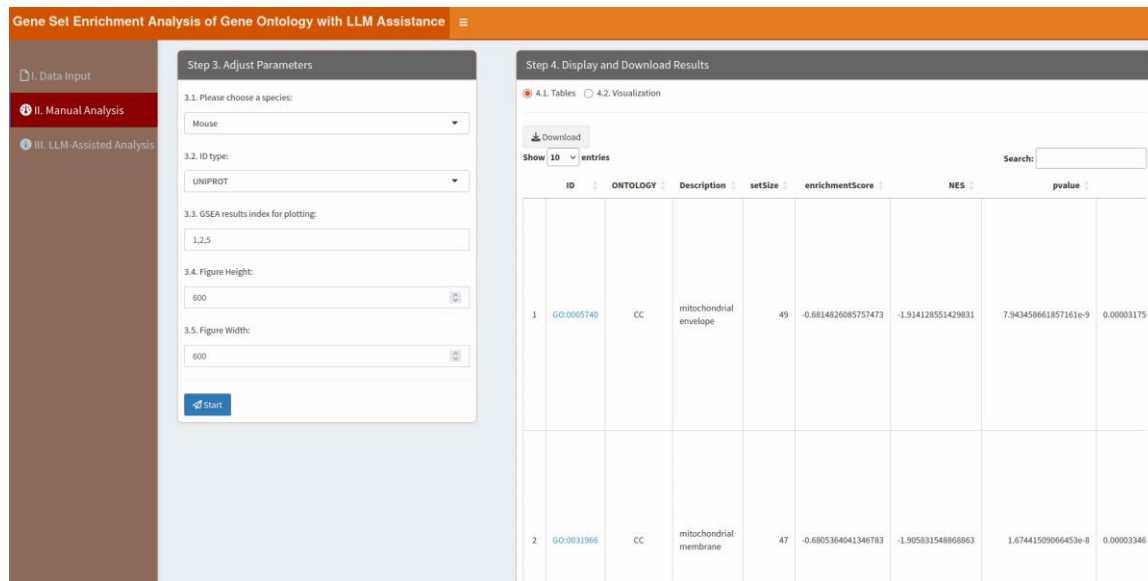
3.3. GSEA results index for plotting: Specify the index to visualize enriched GO terms. Enter '1' here to plot the first enrichment result, '1,2' to plot the first one and the second one, and so on. If you enter '1-3', it means plotting the first to the third results on one plot.

3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

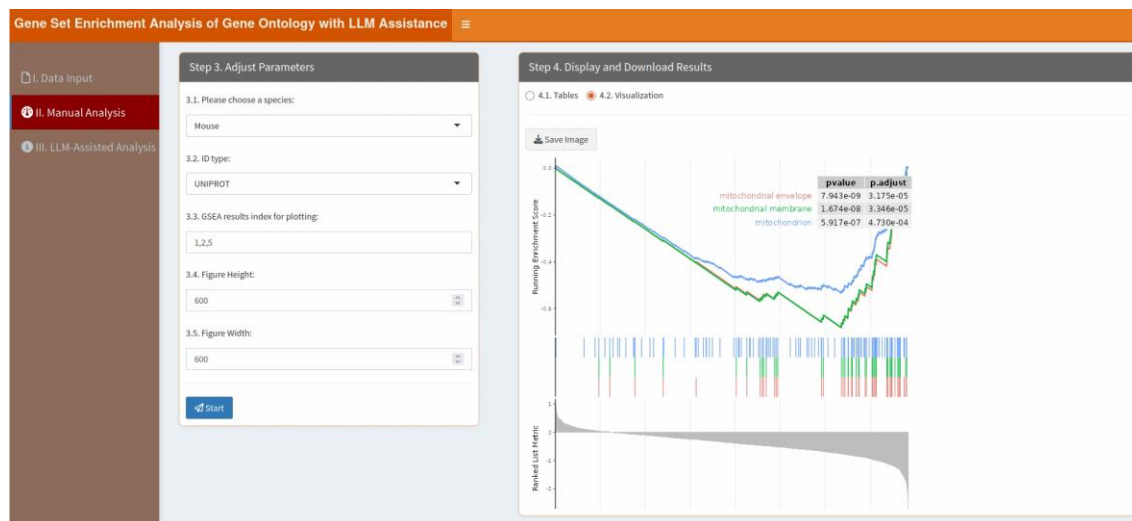
3.5. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table:



In the “4.2. Visualization” part, users can click “Save Image” button to download the figure:

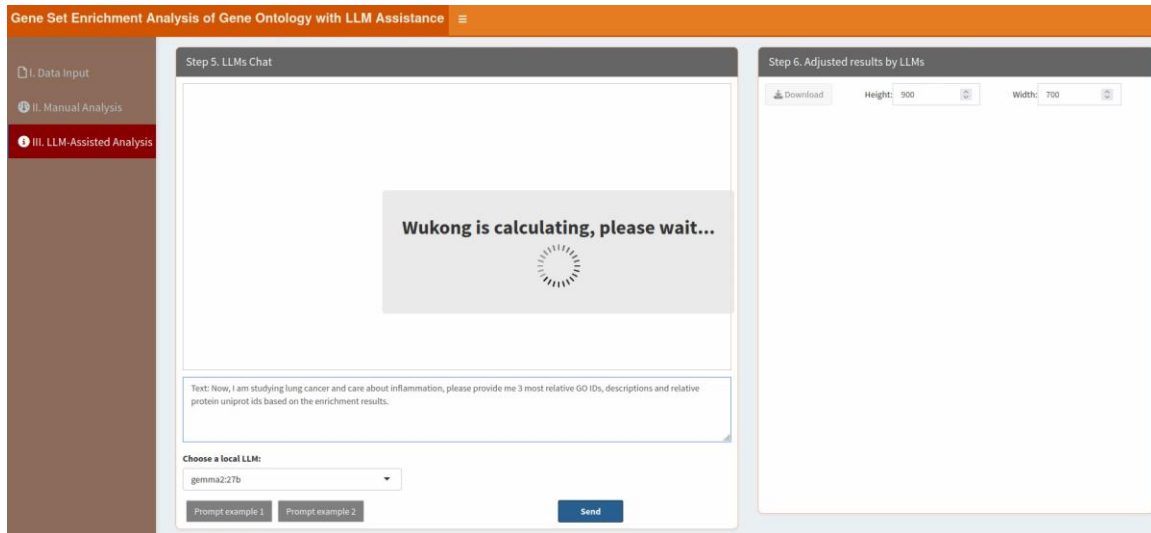


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...”

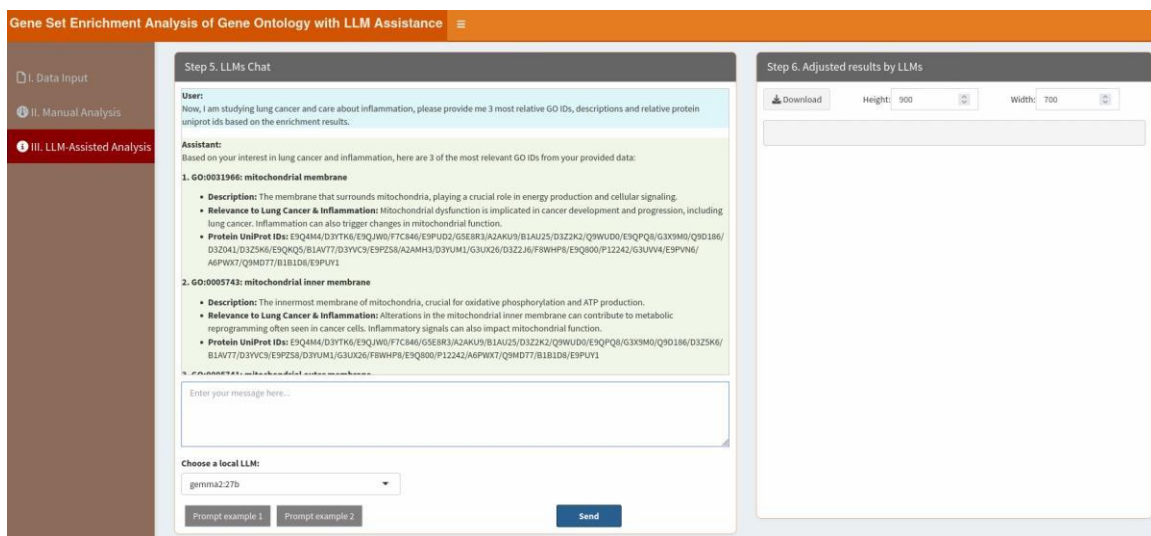
enables users to type messages or questions in the provided text box. Two types of prompts are supported: prompts beginning with "Text:" are used to request interpretation or explanation of the enrichment results (e.g. summarizing biological processes or clarifying GO term relevance), while prompts beginning with "Plot:" allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed GO terms). Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt 1 "Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results. =>", it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the first prompt, click "Send" button. It will learn like below:



The results are shown as below:



On the other hand, the example prompt 2 “Plot: Please remove the pvalue table.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Plot: Please remove the pvalue table.=>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt example 2, click “Send” button, the results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.

Gene Set Enrichment Analysis of Gene Ontology with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 5. LLMs Chat

User:

Plot: Please remove the pvalue table.

Assistant:

```
library(enrichplot)

# Assuming 'gseagresult' is your GSEA result object
gseaplot2(gseagresult, geneSetID = c(1, 5), color = "green", subplots = 1:3, ES_geom = "line", pvalue_table = FALSE)
```

Explanation:

- `library(enrichplot)` : Loads the `enrichplot` package for visualization.
- `gseaplot2(...)` : The main function to create the plot.
 - `gseagresult` : Your GSEA result object (replace with your actual variable name).
 - `geneSetID = c(1, 5)` : Specifies the gene set IDs you want to plot (adjust these numbers based on your desired sets).
 - `color = "green"` : Sets the color for the enrichment score lines.
 - `subplots = 1:3` : Controls which subplots to display (in this case, all three).
 - `ES_geom = "line"` : Uses a line plot for the enrichment scores.
 - `pvalue_table = FALSE` : Removes the p-value table from the plot.

Interact with Model:

Enter your message here...

Choose a local LLM:

gemma2-27b

Prompt example 1

Prompt example 2

Send

Step 6. Adjusted results by LLMs

Download

Height: 300

Width: 700

6.6. Gene set enrichment analysis of KEGG

Perform gene set enrichment analysis based on the KEGG database using a classic framework for parameter adjustments and local LLMs for enhanced interpretation. LLMs help identify key biological pathways, contextualize findings, and suggest potential implications or regulatory mechanisms for further exploration. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' interface. On the left, a sidebar contains three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main panel has two tabs: 'A. Upload' (selected) and 'B. Load example data'. Under 'A. Upload', there is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which Sheet to read?' dropdown menu with '1' selected. On the right, 'Step 2. Display Uploaded Data/Example Data' is visible, showing a 'Show 10 entries' dropdown, a search bar, and a table with one entry: '1 No data loaded. Please upload your dataset or use the example data.' The table has 'Previous' and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' interface. The sidebar is the same as in Step 1, but 'II. Manual Analysis' is now selected. The main panel has two tabs: '4.1. Tables' (selected) and '4.2. Visualization'. Under '4.1. Tables', there is a 'Download' button. The 'Adjust Parameters' section contains five numbered fields: '3.1. Please type a species:' with 'mmu' entered; '3.2. ID type:' with 'UNIPROT' selected in a dropdown; '3.3. GSEA results index for plotting:' with '1,2,5' entered; '3.4. Figure Height:' with '600' entered; and '3.5. Figure Width:' with '600' entered. A 'Start' button is at the bottom of the parameter section.

3.1. Please type a species: Type the organism code (e.g., hsa for Homo sapiens, mmu for Mus musculus) corresponding to the dataset being analyzed.

3.2. ID type: Select the type of gene/protein identifiers used in your dataset (e.g., UniProt ID, Entrez ID).

3.3. GSEA results index for plotting: Specify the index to visualize enriched GO terms. Enter '1' here to plot the first enrichment result, '1,2' to plot the first one and the second one, and so on. If you enter '1-3', it means plotting the first to the third results on one plot.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table:

ID	Description	setSize	enrichmentScore	NES	pvalue	p.adjust	rank	core_enr
1	Calcium signaling pathway - Mus musculus (house mouse)	4	0.8134838034578975	1.774228199643577	0.00968187564999501	0.5588387096774194	18	EPQ9K3/E
2	Alcoholic liver disease - Mus musculus (house mouse)	3	-0.8646163722137399	-1.448686490231832	0.02185048003196559	0.5588387096774194	114	EPQ9K3/E
3	Human papillomavirus infection - Mus musculus (house mouse)	8	-0.664330079609882	-1.43369232408522	0.04845054945054945	0.5588387096774194	278	EPQ9K3/E
4	Complement and coagulation cascades - Mus musculus (house mouse)	4	-0.78260895652174	-1.407670294859809	0.04096273291925466	0.5588387096774194	260	D3Y9F5/E
5	Viral life cycle - HIV-1 - Mus musculus (house mouse)	3	0.7296132879045997	1.400937540593028	0.1331028313247297	0.5588387096774194	331	EPQ9K3/E
6	Hepatitis B - Mus musculus (house mouse)	3	0.7296132879045997	1.400937540593028	0.1331028313247297	0.5588387096774194	331	Q8VCL1/C

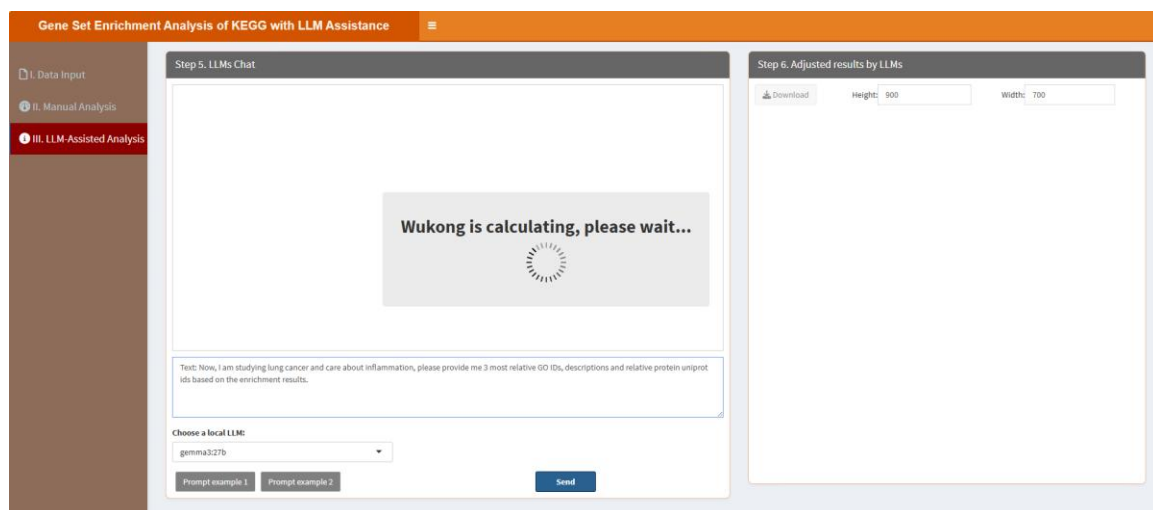
In the “4.2. Visualization” part, users can click “Save Image” button to download the figure:

“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Two types of prompts

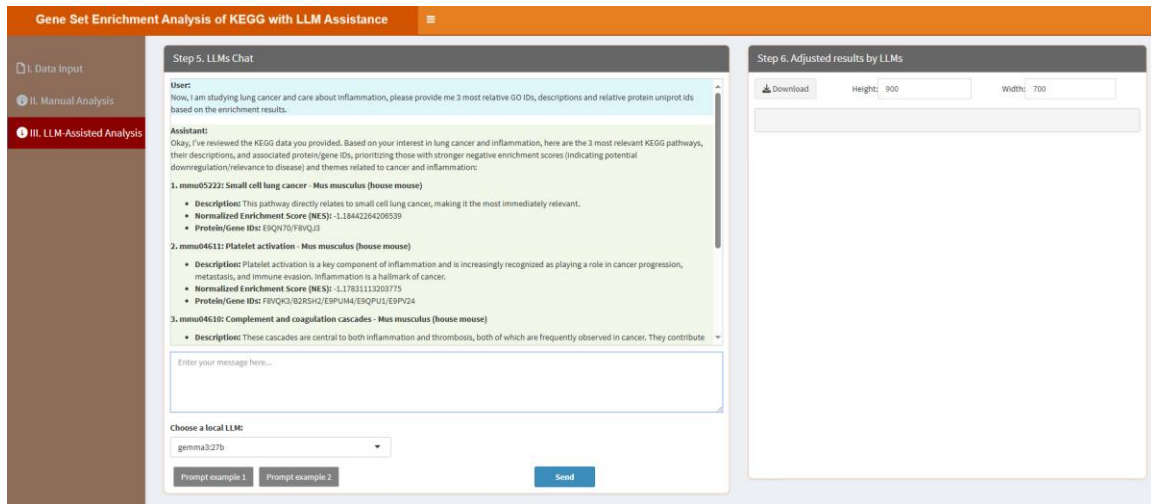
are supported: prompts beginning with “Text:” are used to request interpretation or explanation of the enrichment results (e.g. summarizing pathways relevance), while prompts beginning with “Plot:” allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed pathways). Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt 1 “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the first prompt, click “Send” button. It will learn like below:

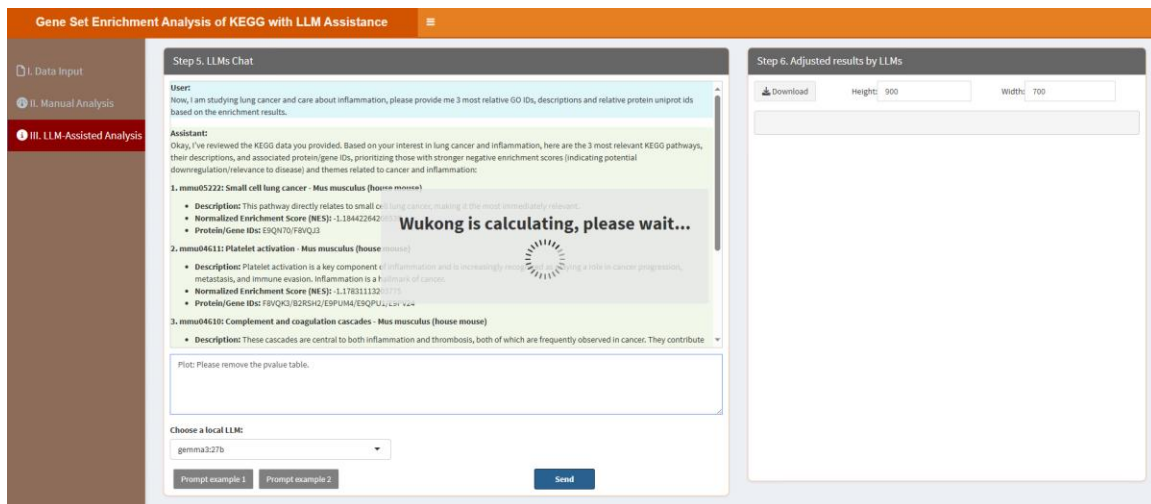


The results are shown as below:

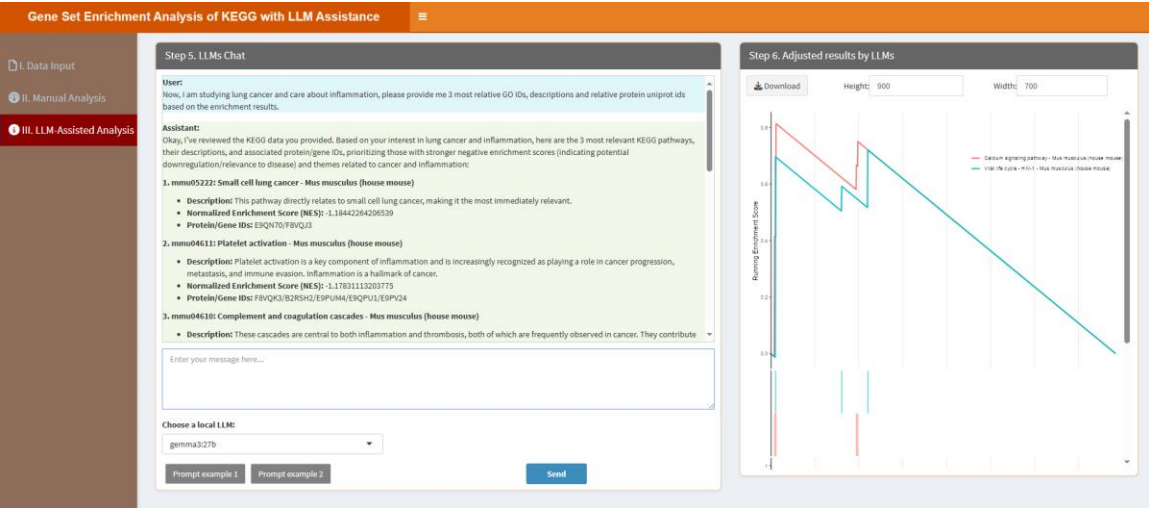


On the other hand, the example prompt 2 “Plot: Please remove the pvalue table.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Plot: Please remove the pvalue table.=>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the second prompt, click “Send” button. It will learn like below:

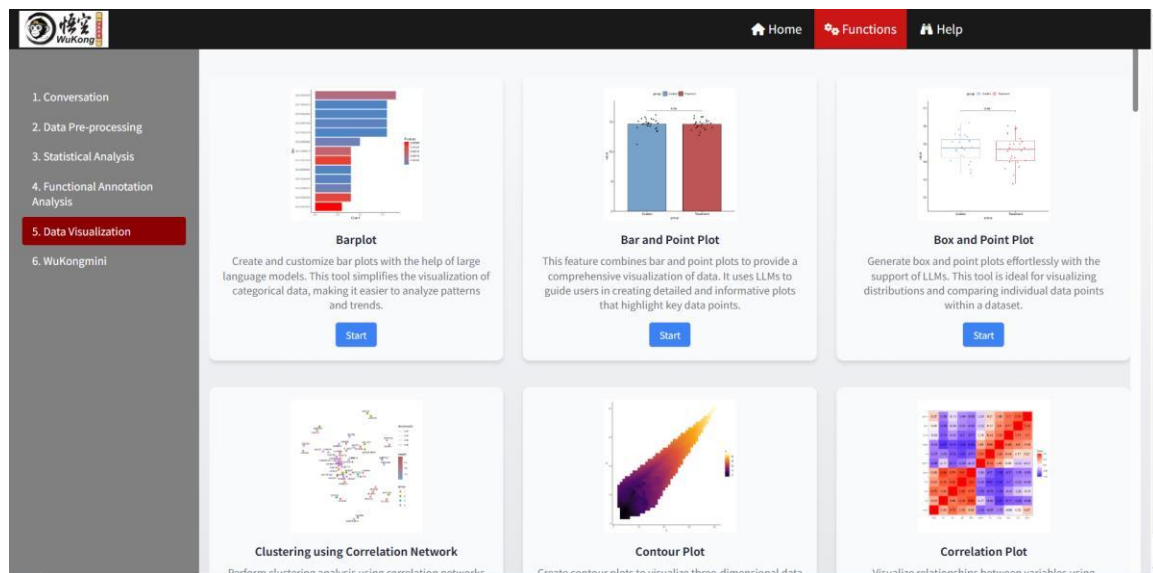


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7. Data Visualization

Data visualization in proteomics is essential for interpreting complex datasets, and a wide range of visualization modules is available to meet diverse analytical needs. Users can customize their plots by adjusting parameters through a classic framework while leveraging local large language models (LLMs) to refine visualizations based on inner R codes. These modules include bar plots, bar and point plots, and box and point plots for categorical and distributional data. Clustering can be performed using correlation networks, while contour plots and correlation plots reveal relationships in multidimensional data. Advanced modules like correlation network plots, cross error bar plots, and dendrograms (including ggtree-based dendrograms) enhance hierarchical and statistical analysis. Users can create diverging bar plots, funnel plots, protein/DNA sequence logo plots, colored scatterplots, and combined histogram and density plots for specialized insights. Additionally, lollipop charts, marginal histogram/boxplots, Nightingale rose diagrams, pair point line plots, pie plots, radar charts, rain cloud plots, rank point plots, ridge plots, and ROC plots provide tailored visualizations for different data types. Sankey charts, scatter plots with ellipses, survival plots, ternary plots, UpSet plots, Venn plots, violin plots, volcano plots, and word clouds further expand the toolkit for exploring and presenting data. By combining user-friendly parameter adjustments with LLM-driven guidance, these visualization modules empower researchers to uncover patterns, trends, and relationships within proteomics data efficiently and effectively.



7.1. Barplot

Users can create and customize bar plots to visualize categorical data by adjusting parameters through the classic framework. Additionally, local LLMs can be leveraged to refine the plot for clearer insights and better representation of patterns and trends. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Interactive Barplot Generation with LLM Assistance' interface. On the left is a sidebar with three main sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The 'I. Data Input' section is active. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Under 'A. Upload', there are radio buttons for file formats: '.xlsx' (selected), '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and 'No file selected' text. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show: 10 entries' dropdown, a search bar, and a table with one entry: '1' in the 'Description' column. Below the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous', '1', and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Interactive Barplot Generation with LLM Assistance' interface. The sidebar is the same as in the previous screenshot. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three numbered steps: '3.1. Select bar colors:' with a dropdown menu showing 'FF0000FF' and '000000FF'; '3.2. Figure height:' with a text input field containing '900'; and '3.3. Figure Width:' with a text input field containing '750'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a section for '4.1. Barplot Visualization' with a 'Save image' button.

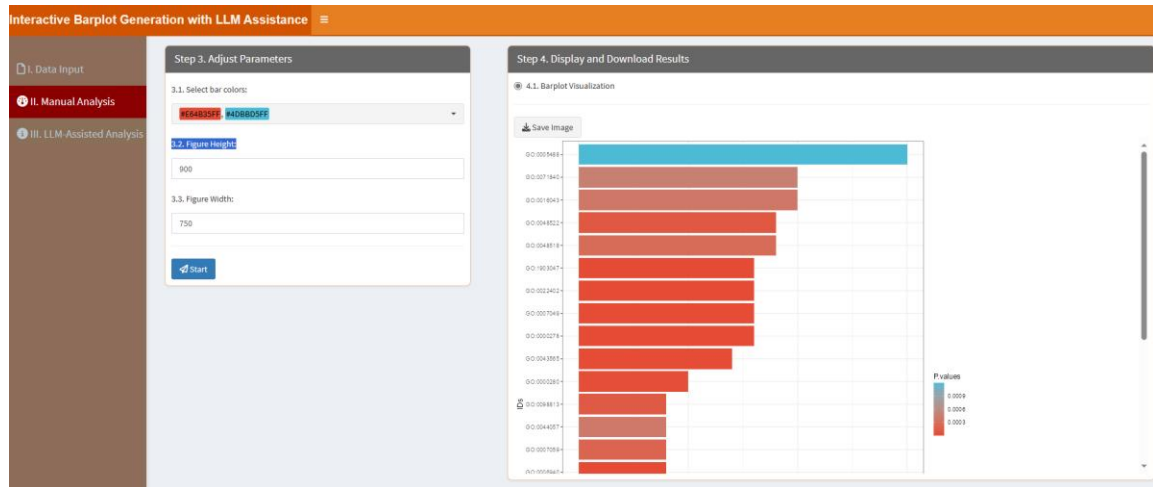
3.1. Select bar colors: Choose two distinct colors for the bar plot, one is for the lowest value and the other is for the highest value in the uploaded data.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

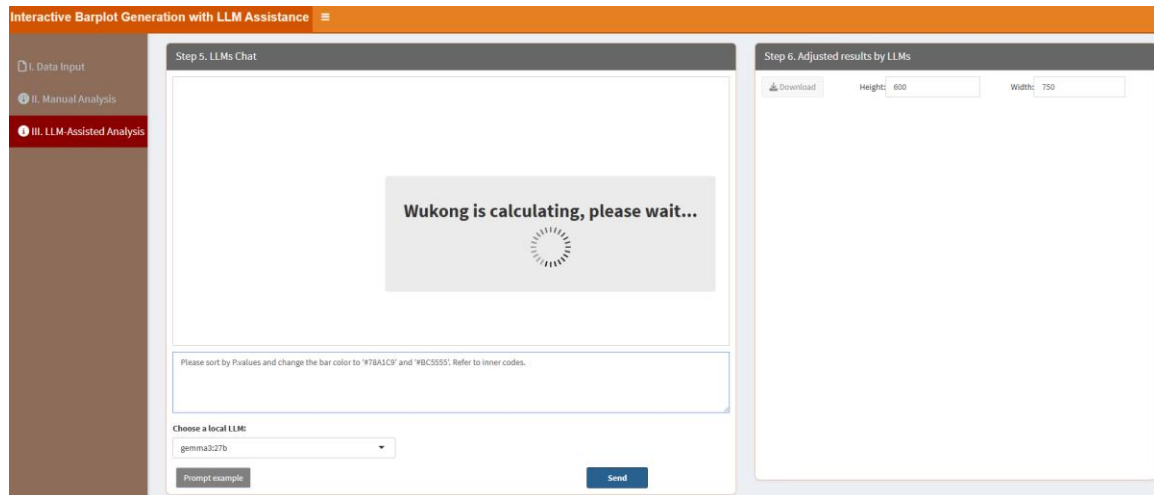
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



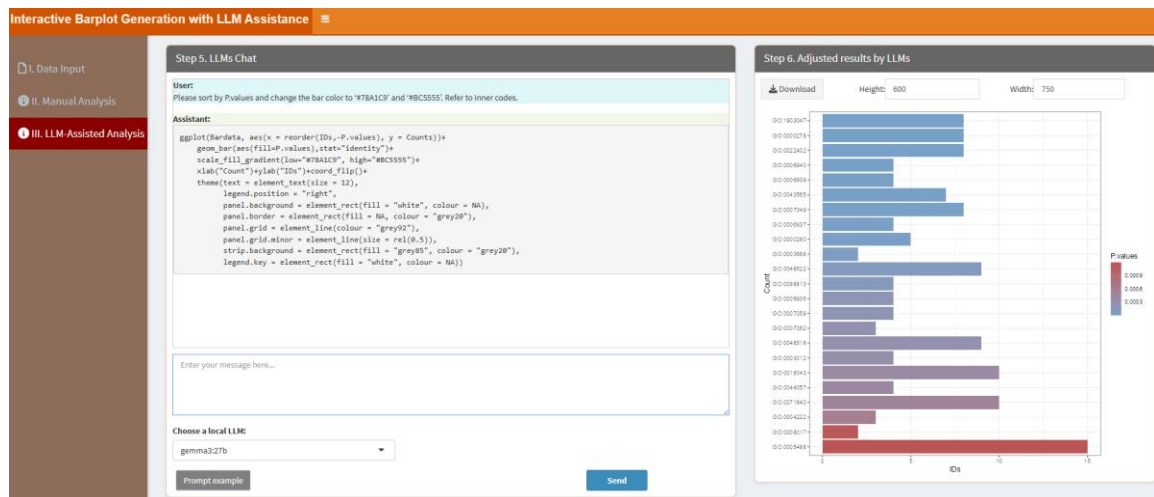
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please sort by P.values and change the bar color to ‘#78A1C9’ and ‘#BC5555’. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please sort by P.values and change the bar color to ‘#78A1C9’ and ‘#BC5555’. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.2. Bar and Point plot

This module allows users to combine bar and point plots for comprehensive data visualization. By adjusting parameters through the classic framework and utilizing local LLMs, users can refine the plot to highlight key data points with precision. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the first step of the interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is titled 'Step 1. Upload Data/Example Data'. It contains two radio buttons: 'A. Upload' (selected) and 'B. Load example data'. Below these are file format options: 'xlsx', 'xls', and 'csv/txt'. A 'Please import your data file:' section has a 'Browse...' button and 'No file selected' text. There are two checkboxes: 'Is the first row names?' (checked) and 'Is the first column names?'. A 'Which Sheet to read?' dropdown shows '1'. On the right, 'Step 2. Display Uploaded Data/Example Data' shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' and 'No data loaded. Please upload your dataset or use the example data.' Below the table is 'Showing 1 to 1 of 1 entries' and 'Previous 1 Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the third step of the interface. The sidebar now has 'II. Manual Analysis' selected. The main area is titled 'Step 3. Adjust Parameters'. It contains four sections: '3.1. Select bar colors:' with a dropdown showing 'FF4400FF' and '408080FF'; '3.2. Select a test method:' with a dropdown showing 'None'; '3.3. Figure Height:' with a text input '500'; and '3.4. Figure Width:' with a text input '500'. At the bottom is a blue 'Start' button. On the right, 'Step 4. Display and Download Results' shows a dropdown for '4.1. Bar and Point plot Visualization' and a 'Save Image' button.

3.1. Select bar colors: Choose two distinct colors for the bar plot, one is for the lowest value and the other is for the highest value in the uploaded data.

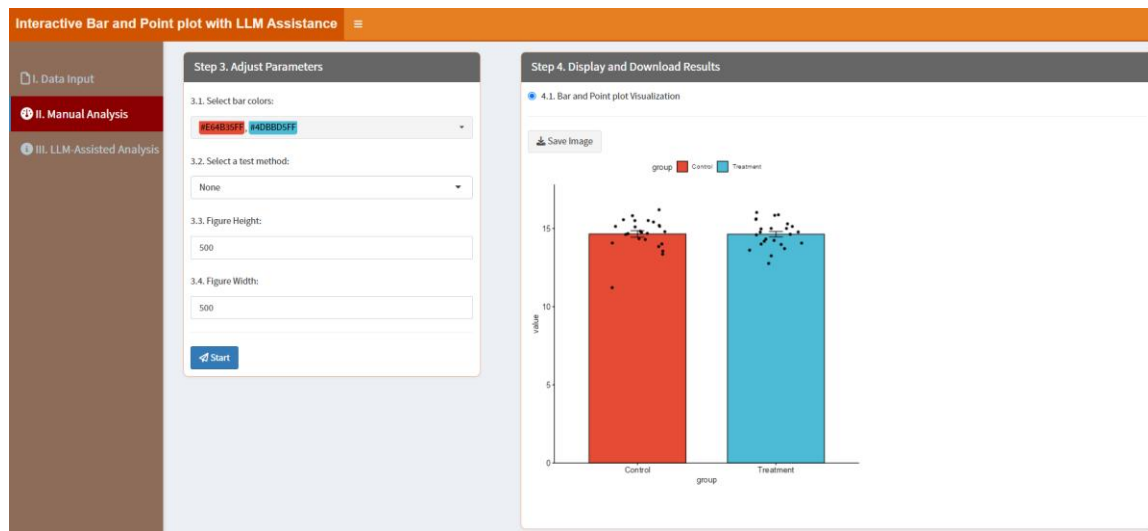
3.2. Select a test method: Specifies the statistical test to be used for comparing groups in bar and point plots. Selecting "None" means no statistical test will be performed, and the plot will not display any p-values. Choosing "t_test" applies the Student's t-test, which compares the means of two groups. The "wilcox_test" option performs the Wilcoxon rank-sum test (Mann-Whitney U test). The "anova_test" conducts a one-way ANOVA, suitable for comparing the means of three or more groups.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

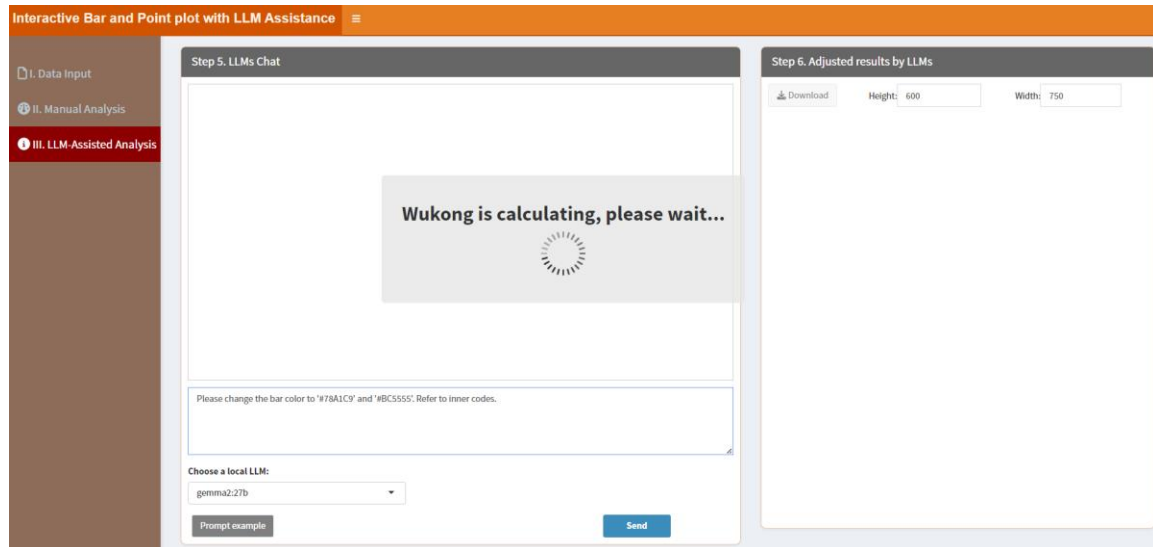


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

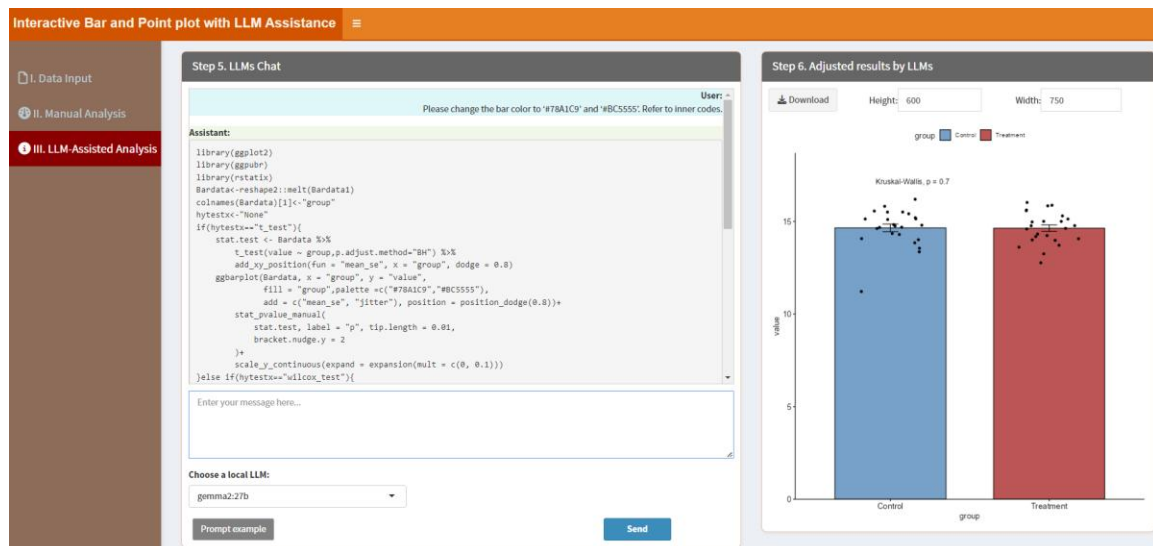
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the bar color to ‘#78A1C9’ and ‘#BC5555’. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger

for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the bar color to '#78A1C9' and '#BC5555'. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section



7.3. Box and Point plot

Box and point plots can be generated to visualize data distributions and compare individual data points. Users can adjust parameters via the classic framework and use local LLMs to enhance the plot's clarity and detail. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the first step of the interactive tool. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is titled 'Step 1. Upload Data/Example Data'. It contains two radio buttons: 'A. Upload' (selected) and 'B. Load example data'. Below these are file format options: '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are two checkboxes: 'Is the first row names?' (checked) and 'Is the first column names?'. A 'Which Sheet to read?' dropdown shows '1'. On the right, 'Step 2. Display Uploaded Data/Example Data' is partially visible, showing a 'Show: 10 entries' dropdown, a search bar, and a table with one row: '1' and 'No data loaded. Please upload your dataset or use the example data.'.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the third step of the interactive tool. The sidebar remains the same. The main area is titled 'Step 3. Adjust Parameters'. It contains four sections: '3.1. Select box colors:' with a dropdown showing 'HE44B35E' and '4AD8D2FF'; '3.2. Select a test method:' with a dropdown showing 'None'; '3.3. Figure Height:' with a text input showing '500'; and '3.4. Figure Width:' with a text input showing '500'. At the bottom is a blue 'Start' button. On the right, 'Step 4. Display and Download Results' is partially visible, showing a '4.1. Box and Point plot Visualization' section with a 'Save Image' button.

3.1. Select box colors: Choose two distinct colors for the box plot, one is for the lowest value and the other is for the highest value in the uploaded data.

3.2. Select a test method: Specifies the statistical test to be used for comparing groups in box and point plots. Selecting "None" means no statistical test will be performed, and the plot will not display any p-values. Choosing "t_test" applies the Student's t-test, which compares the means of two groups. The "wilcox_test" option performs the Wilcoxon rank-sum test (Mann-Whitney U test). The "anova_test" conducts a one-way ANOVA, suitable for comparing the

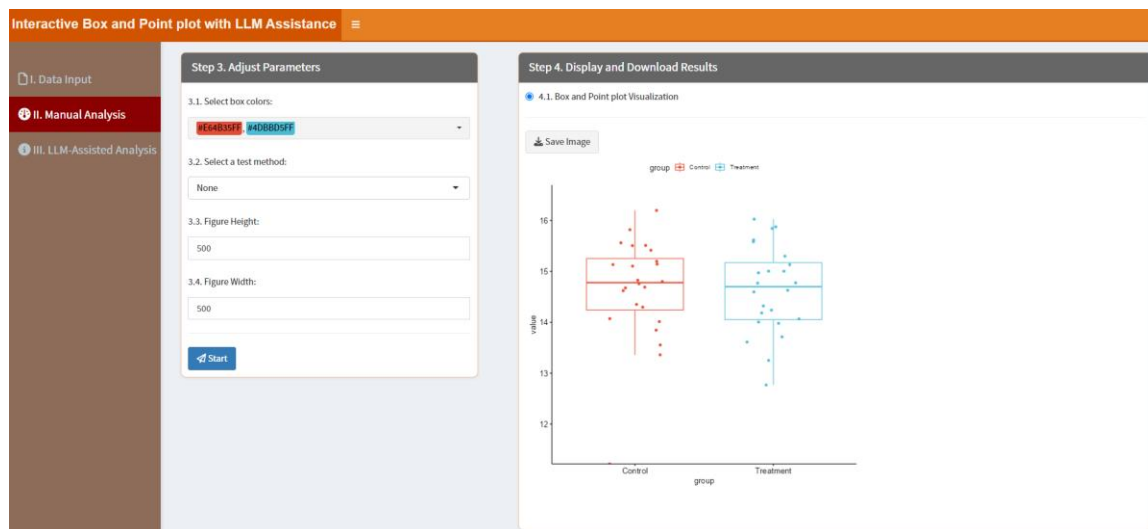
means of three or more groups.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

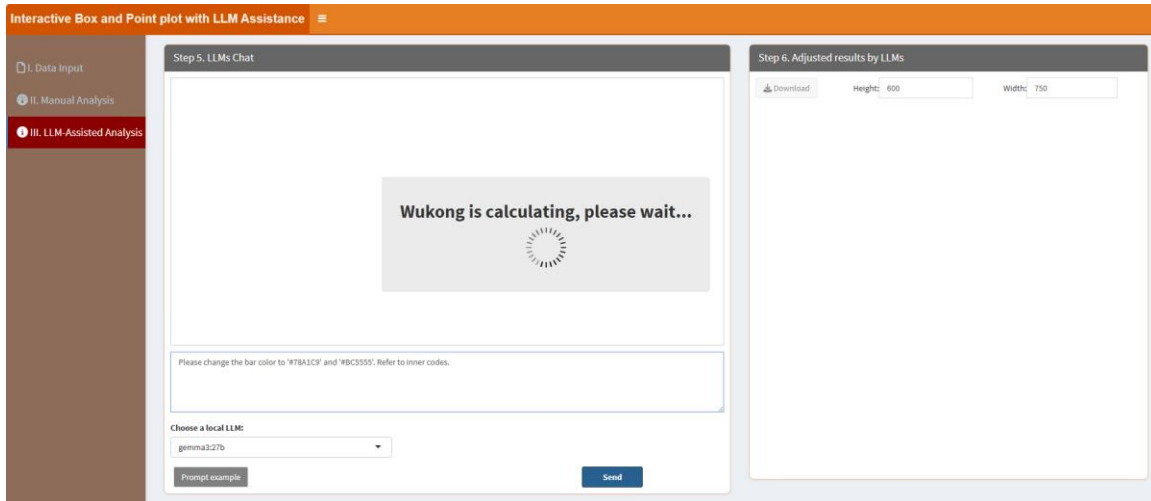


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

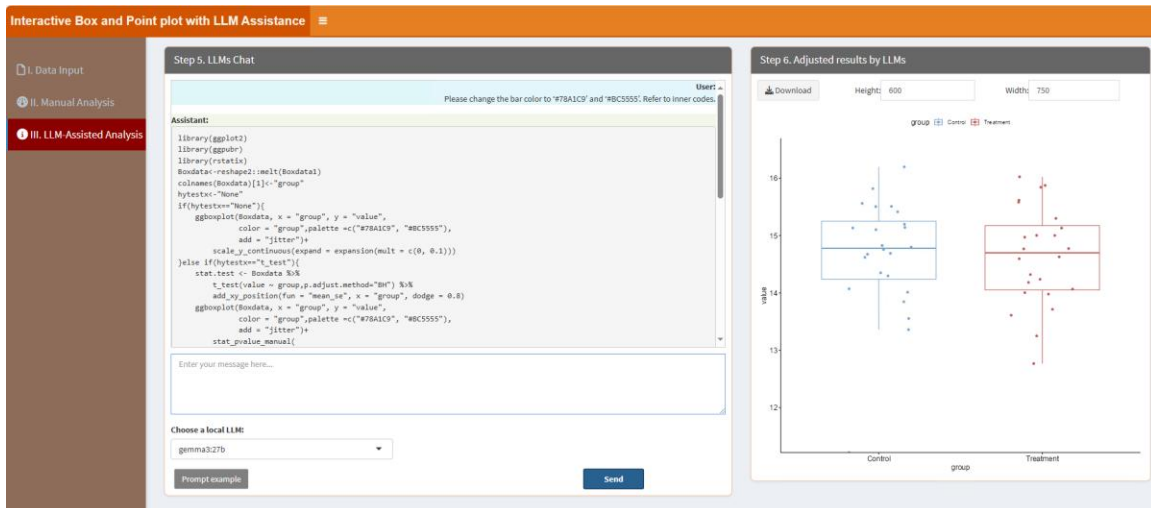
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the bar color to ‘#78A1C9’ and ‘#BC5555’. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger

for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the bar color to '#78A1C9' and '#BC5555'. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.4. Clustering using Correlation Network

This module enables clustering analysis based on correlation networks combined with K-means clustering. First, pairwise correlation coefficients are calculated between all objects in the input dataset. Correlation values below a user-defined threshold are filtered out to retain only strong associations. A correlation network is then constructed, where nodes represent individual objects and edges reflect significant correlations. K-means clustering is subsequently applied to partition the data into a specified number of groups. Cluster assignments are mapped onto the network, allowing nodes to be color-coded by cluster membership. Finally, the network is visualized with edge widths and colors representing correlation strength, and node labels and groupings highlighting cluster structures within the network. Users can adjust settings through the classic framework and rely on local LLMs to refine the visualization, uncovering relationships and groupings within complex datasets. After clicking the “Start” button, it will be started in a new window like below:

The screenshot displays a web application interface titled "Correlation Network Clustering with LLM Assistance". On the left, a sidebar contains three menu items: "I. Data Input" (highlighted in red), "II. Manual Analysis", and "III. LLM-Assisted Analysis". The main content area is divided into two panels. The left panel, titled "Step 1. Upload Data/Example Data", contains options for "A. Upload" (selected) and "B. Load example data". It includes a "Select File Format:" section with radio buttons for ".xlsx", ".xls", and ".csv/txt" (selected). Below this is a "Please import your data file:" section with a "Browse..." button and a "No file selected" message. There are also checkboxes for "Is the first row names?" (checked) and "Is the first column names?" (unchecked), and a "Which Sheet to read?" dropdown menu showing "1". The right panel, titled "Step 2. Display Uploaded Data/Example Data", features a "Show: 10 entries" dropdown, a "Search:" input field, and a table with one row containing the number "1" and the text "No data loaded. Please upload your dataset or use the example data." Below the table, it says "Showing 1 to 1 of 1 entries" and includes "Previous", "1", and "Next" navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

3.1. Cluster number: This parameter Specifies the number of clusters for the K-means method.

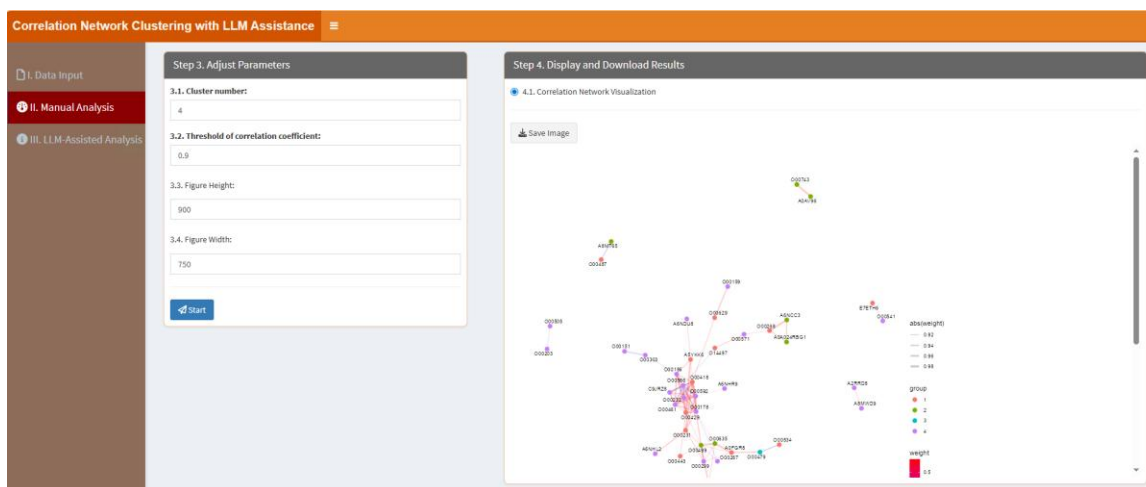
3.2. Threshold of correlation coefficient: This parameter defines the minimum absolute value of the correlation coefficient for an edge to be included in the network.

3.3. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

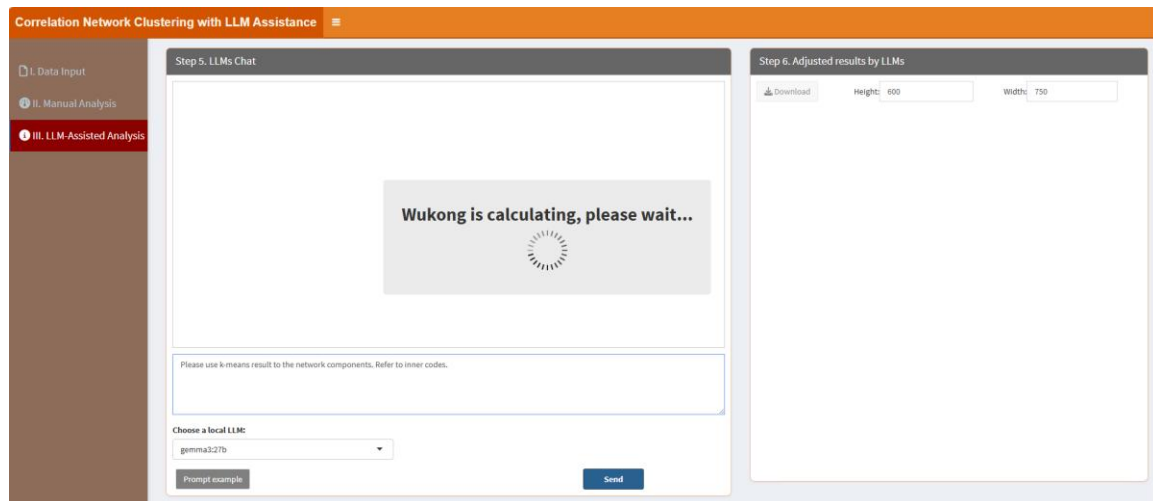


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...”

enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "Please use k-means result to the network components. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please use k-means result to the network components. =>", it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click "Send" button. It will learn like below:



The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.

Please use k-means result to the network components. Refer to inner codes.

```
library(
library(
```

Enter your message here...

gemma3:27b

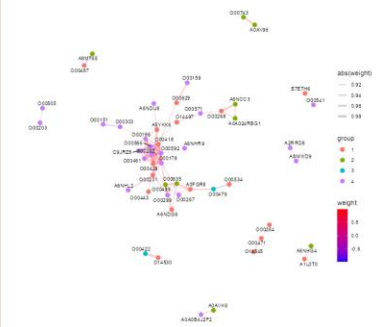
Prompt example

Send

Download

Height: 600

Width:	750
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7.5. Contour Plot

Contour plots can be used to visualize three-dimensional data in two dimensions. Users can customize the plot through the classic framework and leverage local LLMs to ensure clear and informative results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Contour Plot with LLM Assistance' interface. On the left is a sidebar with three menu items: 'I. Data Input' (highlighted in red), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. Below this is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which Sheet to read?' dropdown menu set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', features a 'Show' dropdown set to '10' and a search bar. Below is a table with one row: '1' in the first column and 'No data loaded. Please upload your dataset or use the example data.' in the second column. At the bottom of the table, it says 'Showing 1 to 1 of 1 entries'. Navigation buttons 'Previous', '1', and 'Next' are at the bottom right.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Contour Plot with LLM Assistance' interface at Step 3: Adjust Parameters. The sidebar is the same as in the previous screenshot. The main content area has two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three sections: '3.1. Select bar colors:' with a dropdown menu showing 'RED0000' and 'FA00000F'; '3.2. Figure Height:' with a text input field containing '600'; and '3.3. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section '4.1. Barplot Visualization' with a 'Save Image' button.

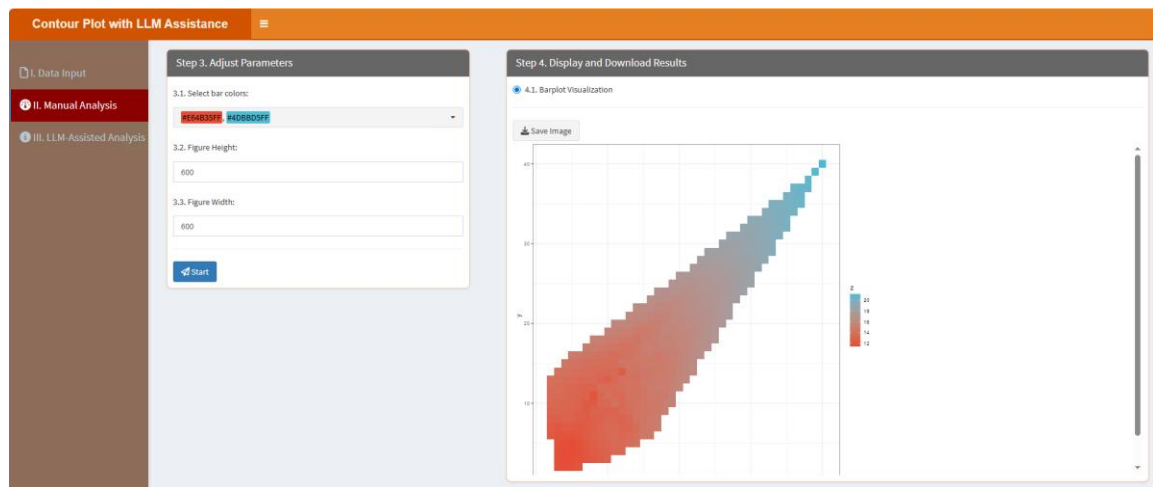
3.1. Select bar colors: Choose two distinct colors for the bar plot, one is for the lowest value and the other is for the highest value in the uploaded data.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

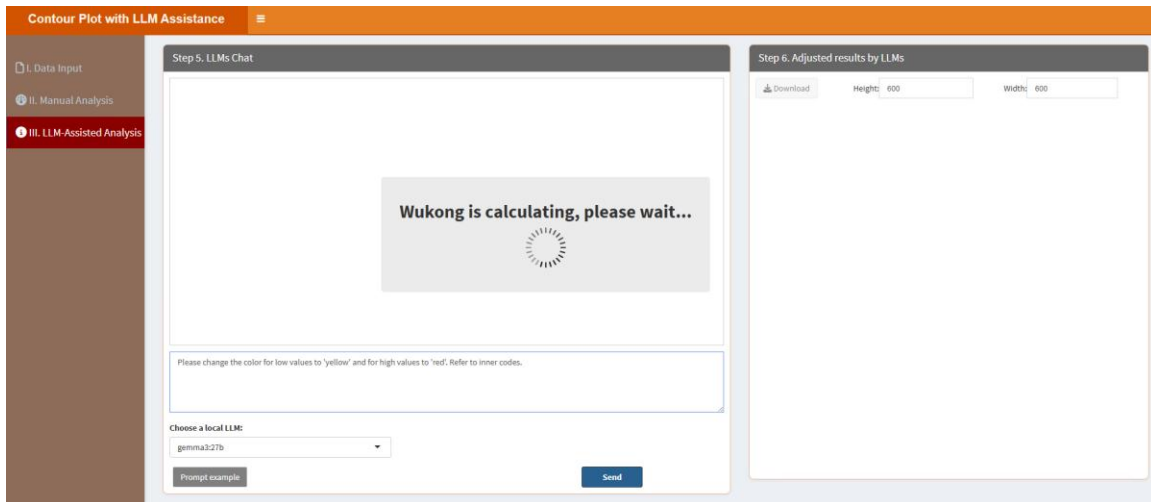
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



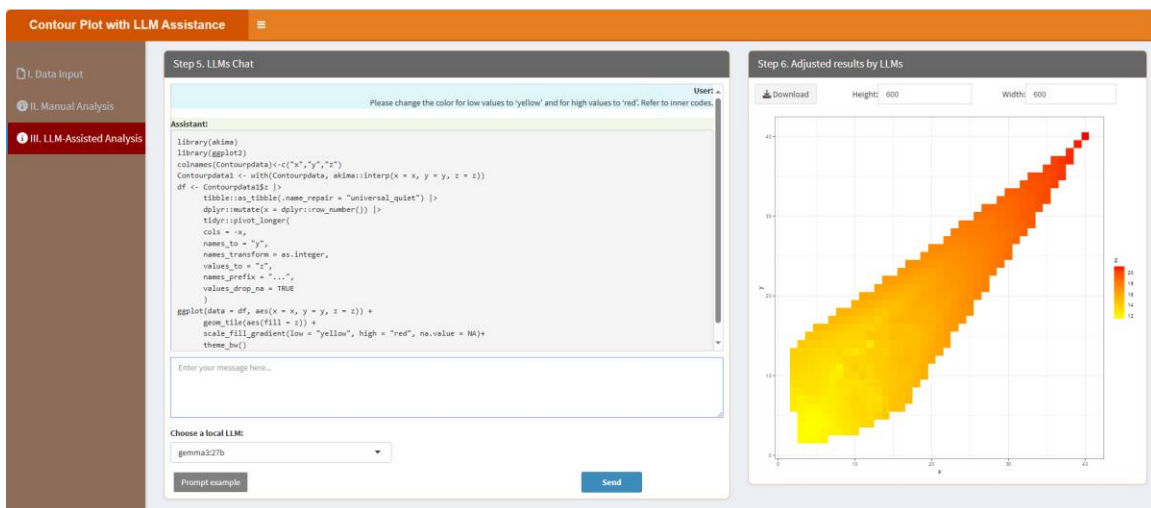
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the color for low values to 'yellow' and for high values to 'red'.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the color for low values to 'yellow' and for high values to 'red'. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.6. Correlation plot

This module helps users visualize relationships between variables using correlation plots. By adjusting parameters in the classic framework and refining with local LLMs, users can ensure accurate and insightful representations. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Correlation plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show' dropdown set to '10' and 'entries'. It includes a 'Search:' input field and a 'Description' section with a table containing one row: '1' and 'No data loaded. Please upload your dataset or use the example data.' At the bottom of the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous', '1', and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Correlation plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains several settings: '3.1. Select three colors (low, middle, high):' with a dropdown showing 'white', '#3C5488FF', and '#EE0000FF'; '3.2. Correlation method:' with a dropdown set to 'pearson'; '3.3. The visualization type:' with a dropdown set to 'square'; '3.4. The matrix type:' with a dropdown set to 'full'; '3.5. Whether correlation matrix will be ordered?' with a checked checkbox; '3.6. Limits of the scale:' with a text input '0.0:1'; '3.7. Middle value:' with a text input '0.95'; '3.8. Whether adding correlation coefficient on the plot?' with a checked checkbox; '3.9. Figure height:' with a text input '600'; and '3.10. Figure Width:' with a text input '600'. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save image' button.

3.1. Select three colors (low, middle, high): Colors for the correlation plot. By default, there are three color names (#3C5488FF, white, #EE0000FF) here for the lowest (#3C5488FF), middle (white), highest (#EE0000FF) values respectively.

3.2. Correlation method: This setting determines the statistical method used to calculate the correlation coefficients between variables. Available options typically include Pearson,

Spearman and Kendall.

3.3. The visualization type: This parameter controls how the correlations are visually represented in the plot. Users can choose from multiple styles such as circle, square.

3.4. The matrix type: This setting specifies which part of the correlation matrix is shown in the plot. Users can choose full, upper, or lower matrix views.

3.5. Whether correlation matrix will be ordered? This parameter allows users to reorder the matrix using hierarchical clustering to group variables with similar correlation patterns together. Enabling this option helps reveal clusters and hidden relationships more clearly by arranging correlated variables adjacent to each other. If disabled, the matrix retains the original input variable order, which may be preferable in some analytical contexts where order is predefined.

3.6. Limits of the scale: This option defines the minimum and maximum values of the color scale used in the correlation plot, typically ranging from -1 to 1. Customizing these limits allows users to zoom in on specific ranges of interest (e.g., focusing only on high correlations by setting limits to 0.9 and 1.0).

3.7. Middle value: This parameter sets the central reference point of the color gradient, typically defaulting to 0, which represents no correlation. Adjusting the middle value can be useful when the user wants to shift the visual emphasis in asymmetric correlation distributions.

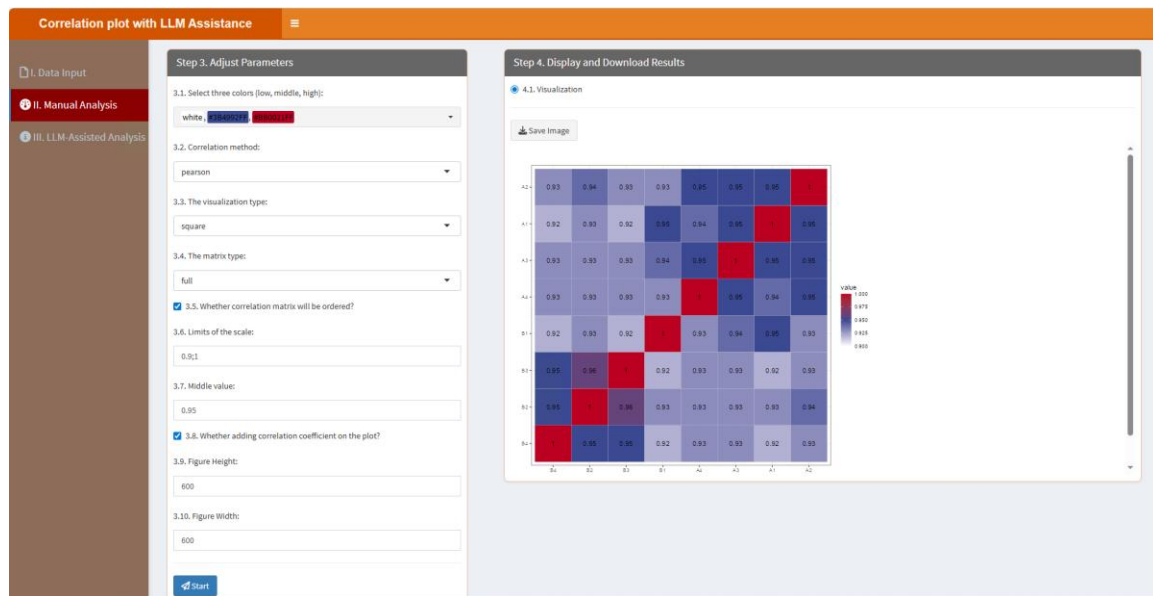
3.8. Whether adding correlation coefficient on the plot: This toggle determines whether the numerical correlation coefficients are overlaid on the visualization. When enabled, each matrix cell will display its exact correlation value, enhancing precision and analytical utility.

3.9. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.10. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

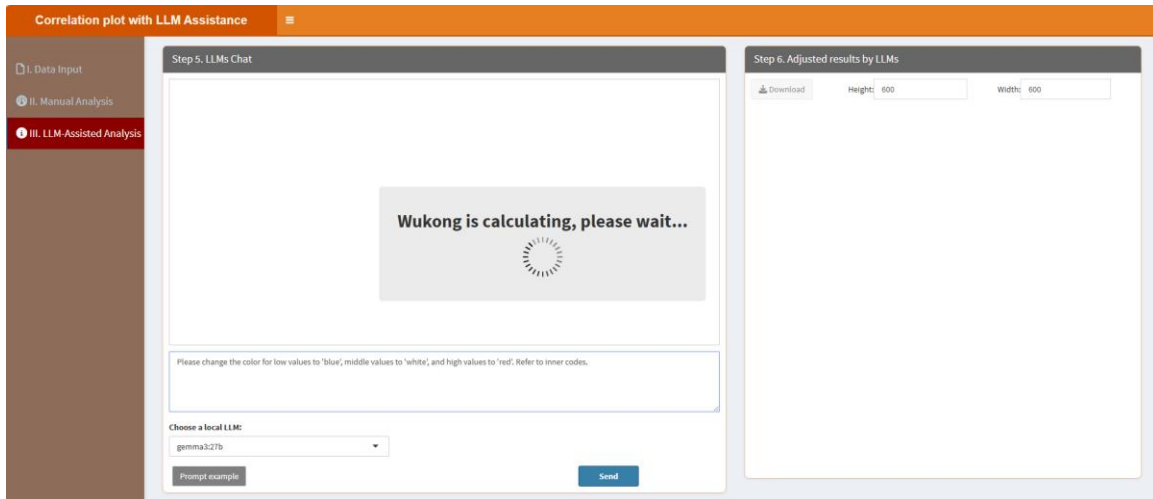
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



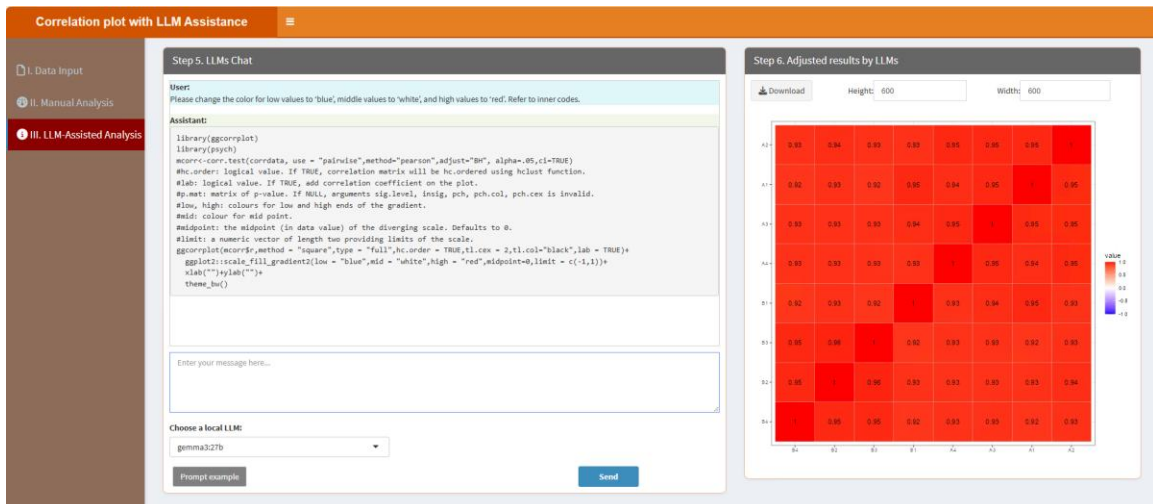
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the color for low values to 'blue', middle values to 'white', and high values to 'red'. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the color for low values to 'blue', middle values to 'white', and high values to 'red'. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.7. Correlation Network plot

Users can construct correlation network plots to visualize complex relationships between variables. The classic framework allows parameter adjustments, while local LLMs refine the visualization for better insights. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Correlation Network plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this, there's a 'Select File Format' section with radio buttons for 'xlsx', 'xls', and 'csv/txt'. A 'Please import your data file:' section has a 'Browse...' button and 'No file selected' text. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' section has a dropdown menu showing '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one entry: '1' in the 'Description' column with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous', '1', and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

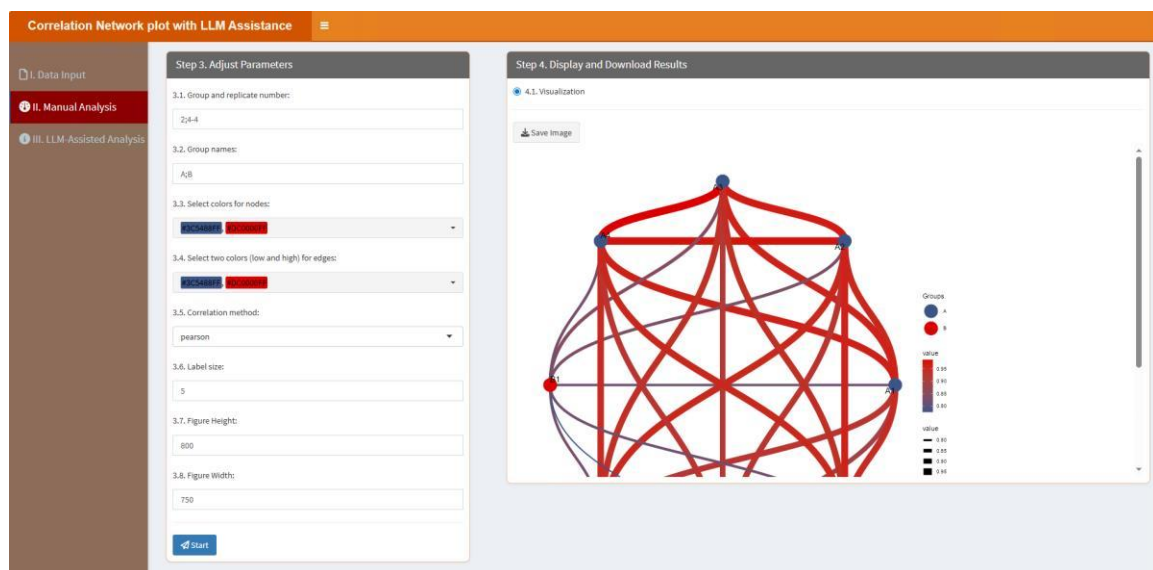
The screenshot shows the 'Correlation Network plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains several input fields: '3.1. Group and replicate number:' with the value '2;4-4', '3.2. Group names:' with the value 'A;B', '3.3. Select colors for nodes:' with a dropdown showing 'Blue' and 'Red', '3.4. Select two colors (low and high) for edges:' with a dropdown showing 'Blue' and 'Red', '3.5. Correlation method:' with a dropdown showing 'pearson', '3.6. Label size:' with the value '5', '3.7. Figure Height:' with the value '800', and '3.8. Figure Width:' with the value '750'. At the bottom is a 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save image' button.

3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

- 3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".
- 3.3. Select colors for nodes: This parameter allows users to assign specific colors to the nodes (representing variables) of each group in the network plot. The color number should be equal to the group number.
- 3.4. Select two colors (low and high) for edges: This option determines the color gradient for the edges (lines connecting nodes) based on correlation strength. The user selects two colors: one for low correlation values and one for high values.
- 3.5. Correlation method: This setting determines the statistical method used to calculate the correlation coefficients between variables. Available options typically include Pearson, Spearman and Kendall.
- 3.6. Label size: This parameter controls the font size of the node labels in the network plot.
- 3.7. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
- 3.8. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

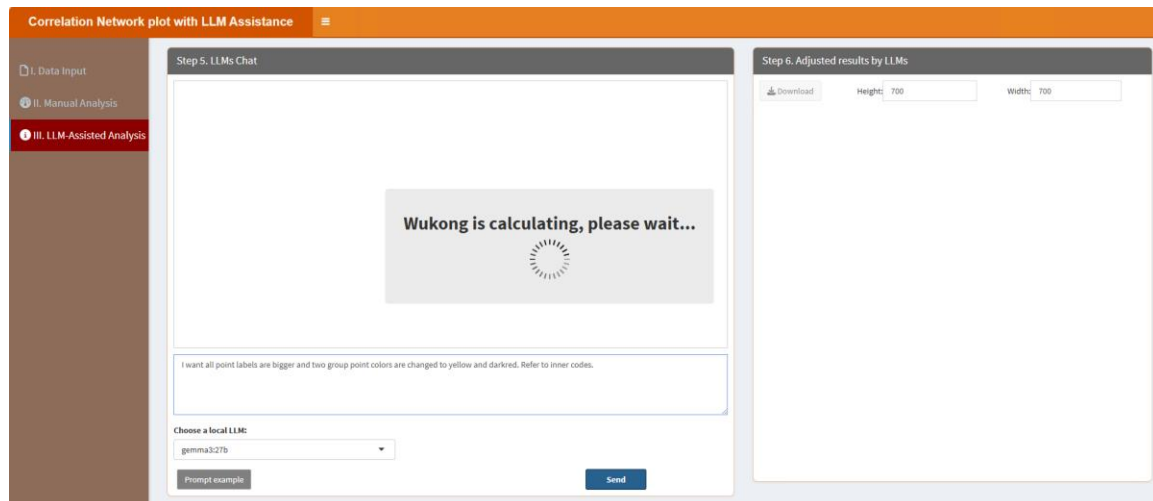


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box,

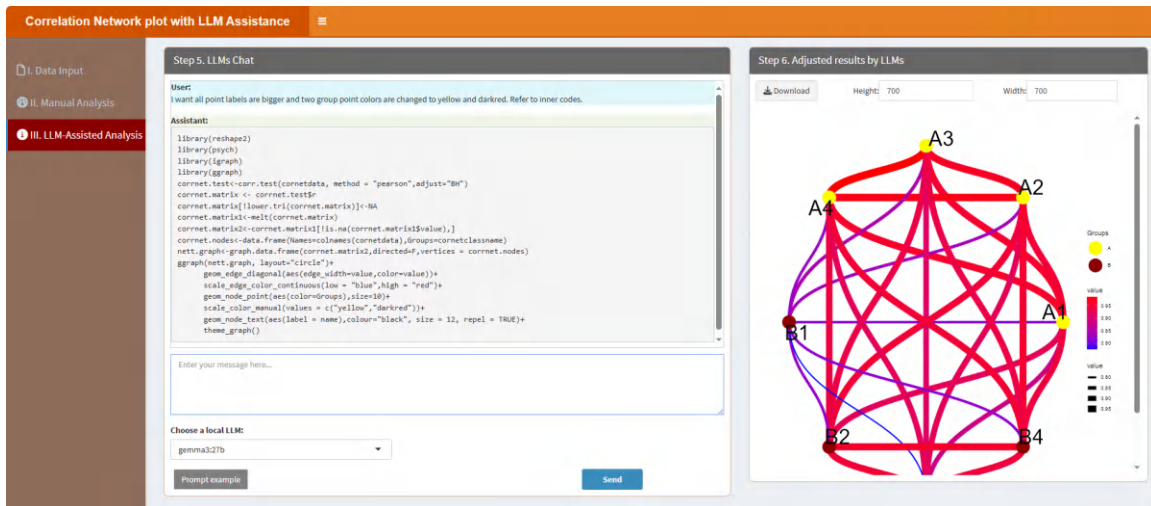
the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "I want all point labels are bigger and two group point colors are changed to yellow and darkred. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "I want all point labels are bigger and two group point colors are changed to yellow and darkred. =>", it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click "Send" button. It will learn like below:



The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



7.8. Cross Error Bar Plot

This module enables users to design cross error bar plots to represent data variability. Parameters can be adjusted through the classic framework, and local LLMs help refine the plot for precision and clarity. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Cross Error Bar plot with LLM Assistance' interface. On the left is a sidebar with three tabs: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. Below these are file format selection buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown menu shows '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', features a 'Show' dropdown set to '10' entries, a search bar, and a table with one entry: '1' with the description 'No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' and 'Previous', '1', 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Cross Error Bar plot with LLM Assistance' interface, specifically Step 3: Adjust Parameters. The sidebar on the left has three tabs: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains several input fields: '3.1. Group and replicate number:' with the value '2;4-4'; '3.2. Group names:' with the value 'A;B'; '3.3. Select colors for each point:' with a color picker showing 'BADA5500FF', 'BADA5500FF', 'BADA5500FF', and 'BADA5500FF'; '3.4. Point size:' with the value '5'; '3.5. Figure Height:' with the value '550'; and '3.6. Figure Width:' with the value '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section for '4.1. Visualization' with a 'Save image' button.

3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

3.3. Select colors for each point: One color for one point.

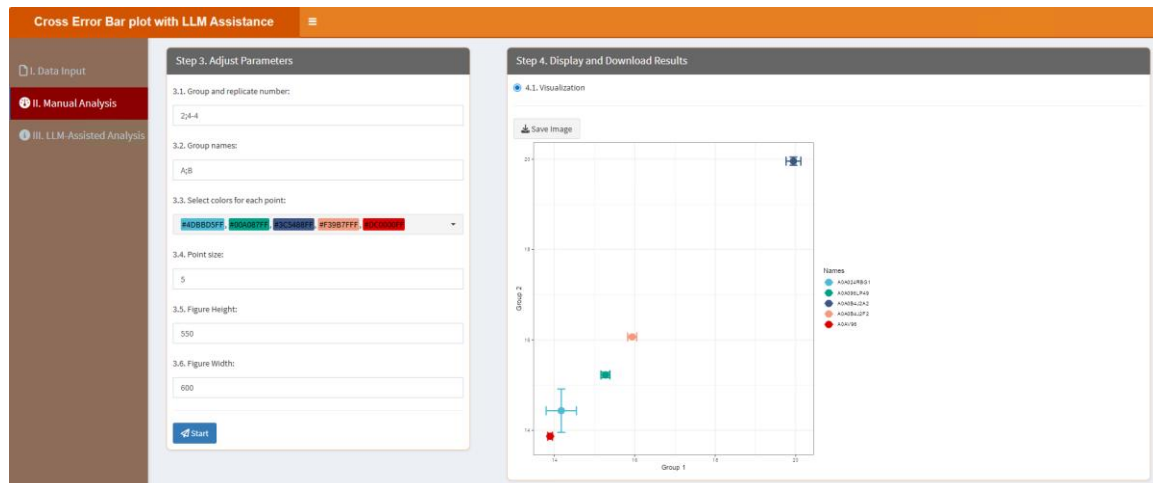
3.4. Point size: This setting controls the size of the data points shown in the plot.

3.5. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

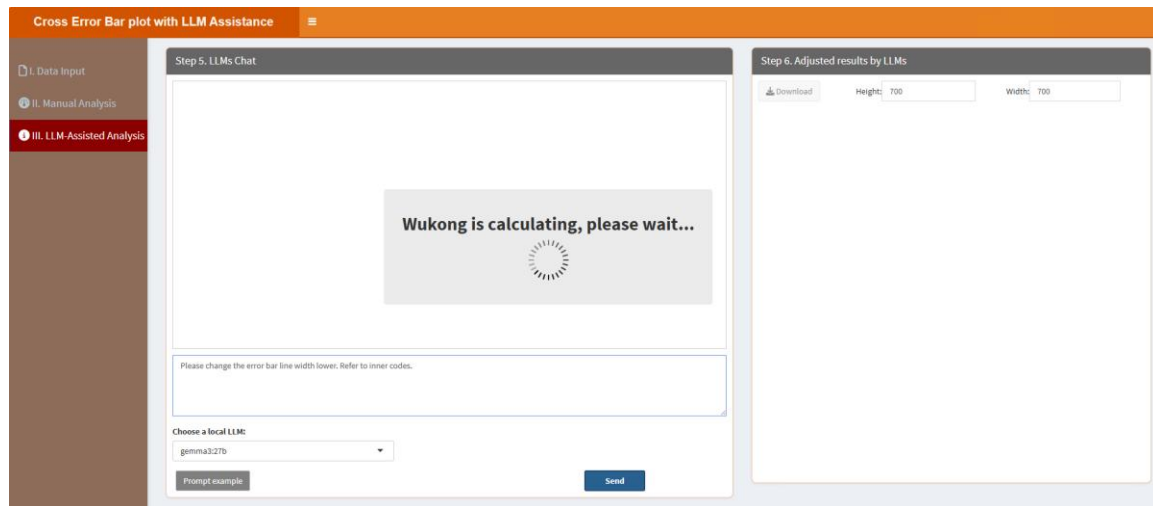


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

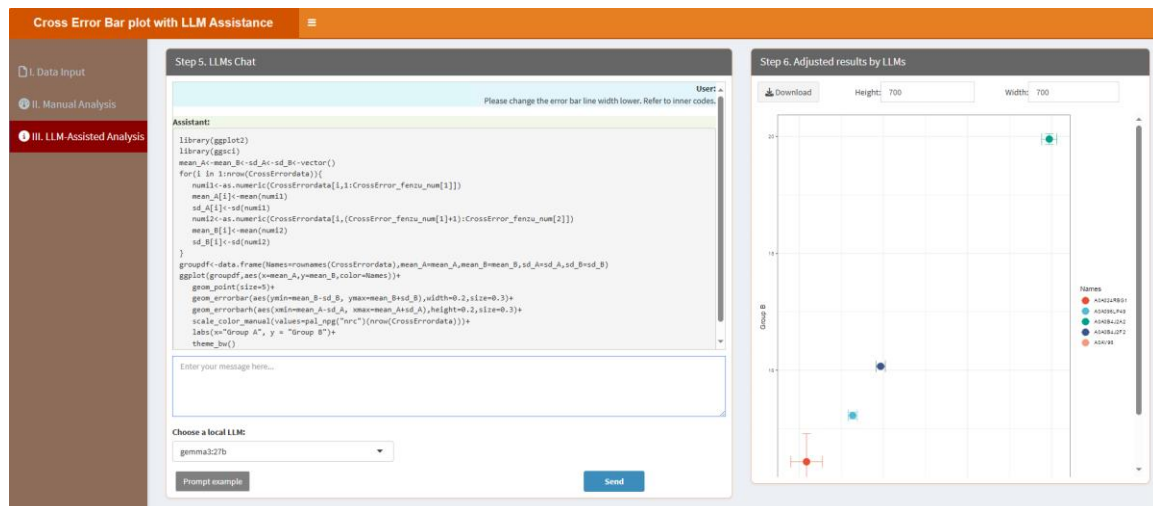
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the error bar line width lower. Refer to inner codes.” instructs LLMs to learn the internal R codes

and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the error bar line width lower. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.9. Dendrogram

Dendrograms are used for hierarchical clustering analysis. Users can customize the plot through the classic framework and enhance it with local LLMs to create detailed and interpretable visualizations. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Dendrogram with LLM' application interface. On the left is a sidebar with three main sections: 'I. Data Input' (highlighted in red), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. Below these are file format selection buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a note 'No file selected'. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown menu shows '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', features a 'Show: 10 entries' dropdown, a search bar, and a table with one row: '1' in the 'Description' column with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous' and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Dendrogram with LLM' application interface at Step 3. The sidebar now highlights 'II. Manual Analysis'. The main content area has two panels. The left panel, titled 'Step 3. Adjust Parameters', contains four numbered settings: '3.1. Cluster number:' with a text input field containing '2'; '3.2. Select colors for each cluster:' with a dropdown menu showing 'Blue' and 'Red'; '3.3. Figure Height:' with a text input field containing '600'; and '3.4. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a section for '4.1. Visualization' with a 'Save Image' button.

3.1. Cluster number: This parameter allows users to specify the number of clusters into which the data should be grouped.

3.2. Select colors for each cluster: This setting lets users assign specific colors to each identified cluster for better visual distinction.

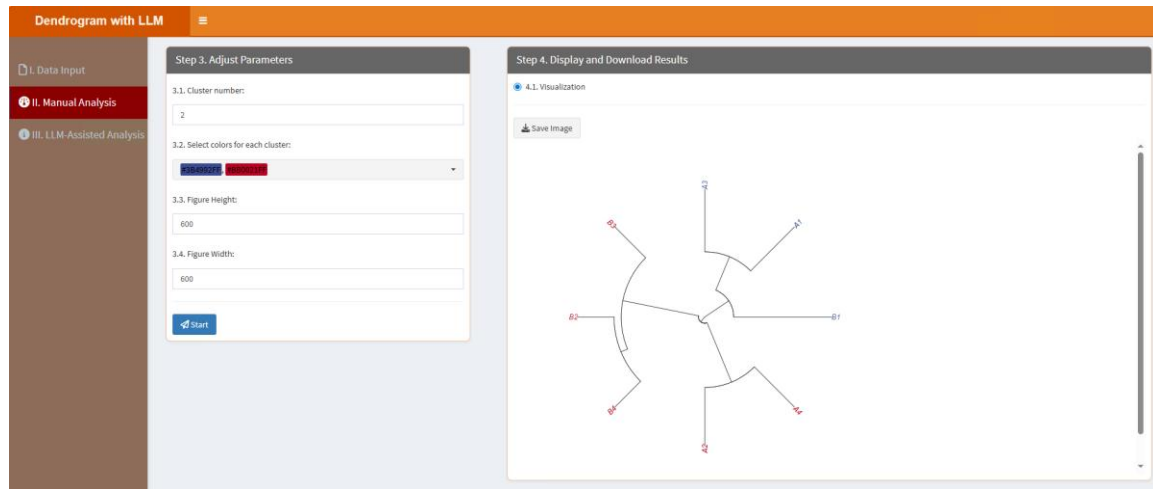
3.3. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.4. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure

in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

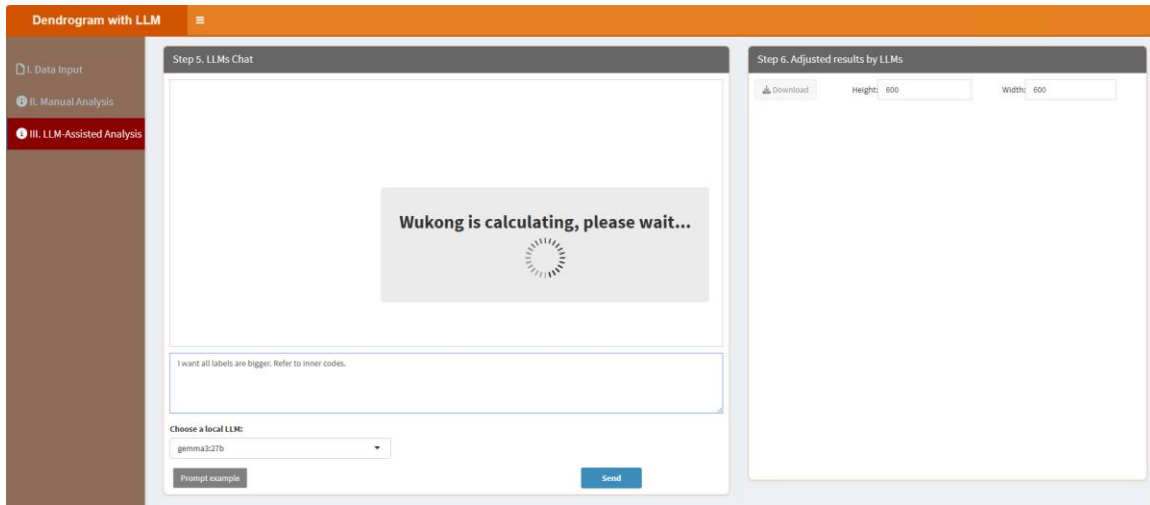
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



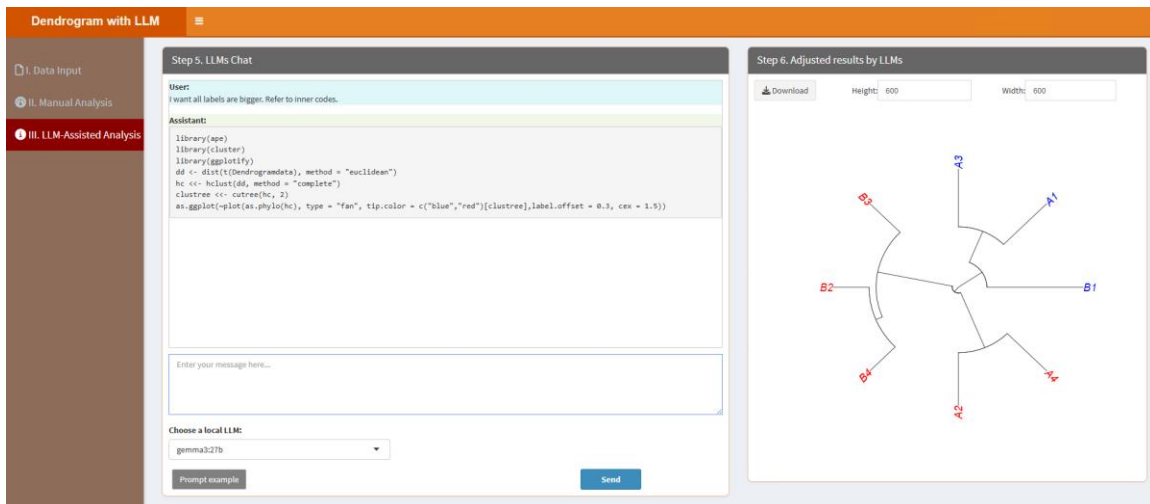
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all labels are bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all labels are bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.10. Diverging Barplot

Users can create diverging bar plots to represent data deviations. By adjusting parameters in the classic framework and refining with local LLMs, users can visualize positive and negative trends effectively. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Diverging Barplot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a message 'No file selected'. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?'. A 'Which Sheet to read?' dropdown is set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one entry: '1' with the description 'No data loaded. Please upload your dataset or use the example data.' It includes a 'Show 10 entries' dropdown, a search bar, and pagination controls showing 'Showing 1 to 1 of 1 entries'.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Diverging Barplot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three sections: '3.1. Select bar colors:' with a dropdown menu showing 'F04480FF' and 'F40800FF'; '3.2. Figure Height:' with a text input field containing '900'; and '3.3. Figure Width:' with a text input field containing '750'. There is a 'Start' button at the bottom. The right panel, titled 'Step 4. Display and Download Results', shows a section for '4.1. Barplot Visualization' with a 'Save Image' button.

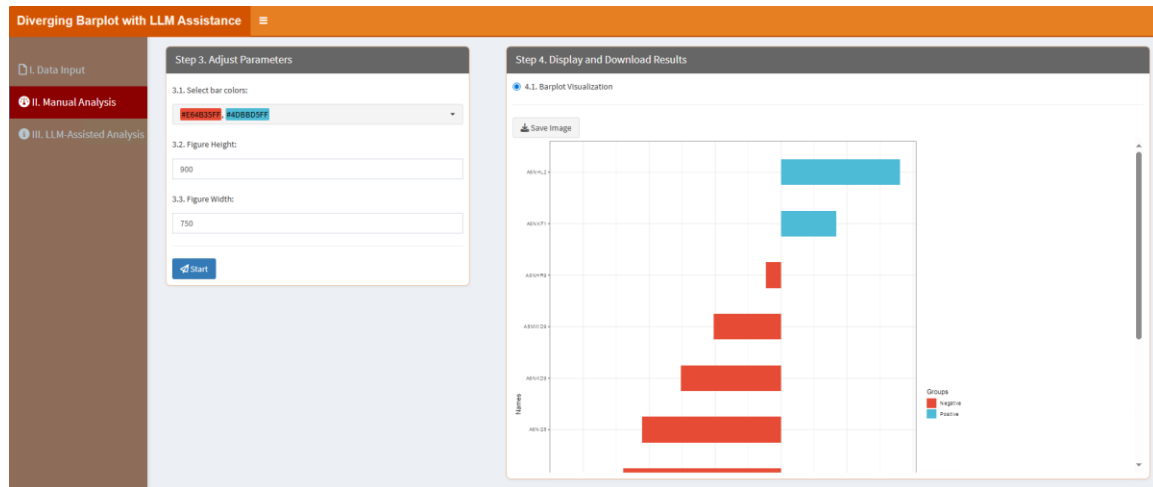
3.1. Select bar colors: Choose two distinct colors for the bar plot, one is for the negative value and the other is for the positive value in the uploaded data.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

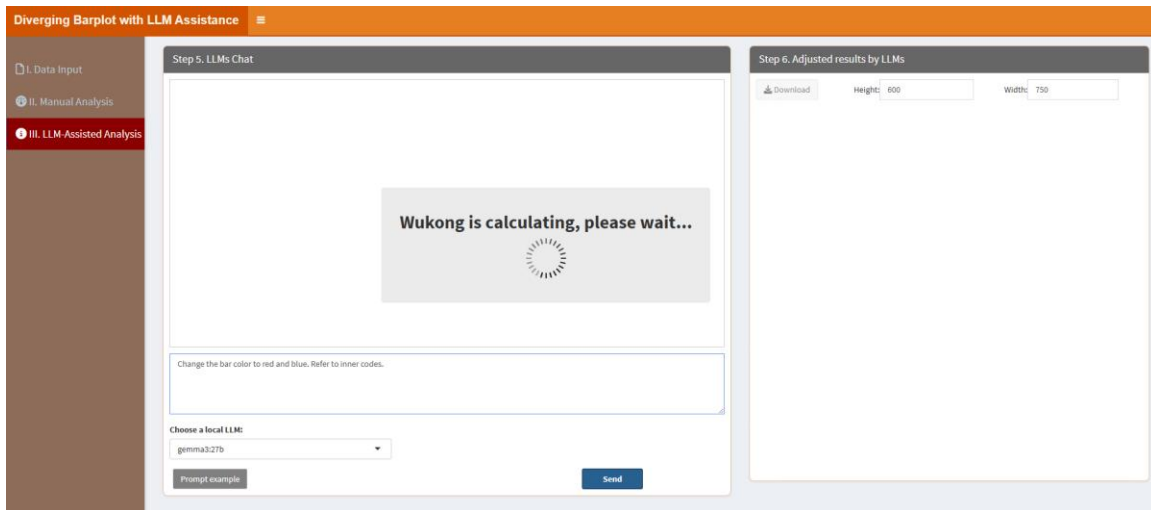
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



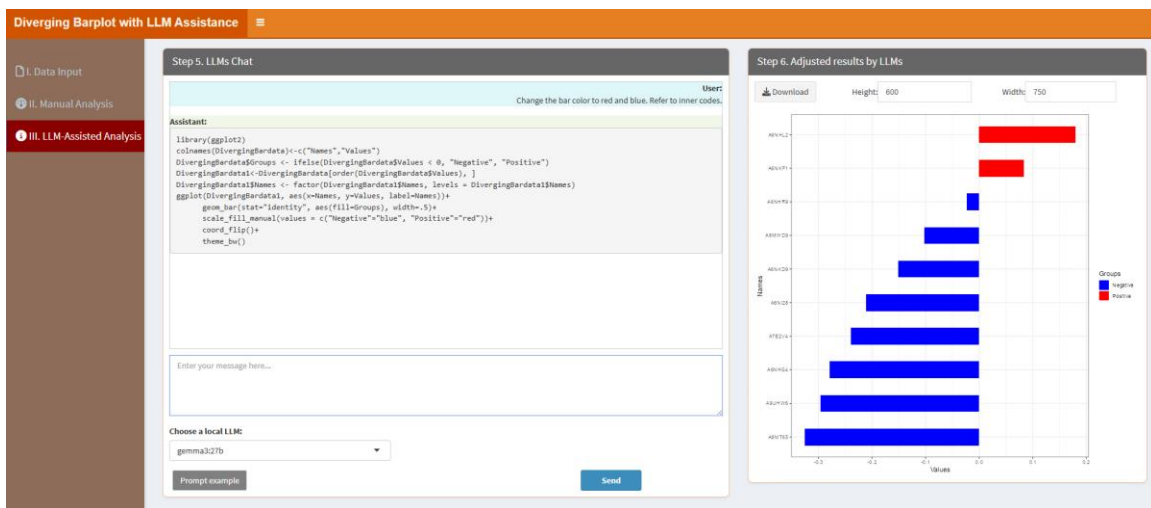
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the bar color to red and blue. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the bar color to red and blue. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.11. Funnel Plot

Funnel plots can be used to assess data variability and bias. Users can customize the plot using the classic framework and rely on local LLMs to generate clear and informative visualizations. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Funnel Plot with LLM Assistance' interface. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1: Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown is set to '1'. The right panel, titled 'Step 2: Display Uploaded Data/Example Data', shows a 'Show: 10 entries' dropdown, a search bar, and a table with one row: '1' in the first column and 'No data loaded. Please upload your dataset or use the example data.' in the second column. At the bottom of the table are 'Previous', '1', and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Funnel Plot with LLM Assistance' interface. The sidebar is the same as in the previous screenshot. The main area is divided into two panels. The left panel, titled 'Step 3: Adjust Parameters', contains three sections: '3.1. Select colors for groups:' with a color picker showing 'blue' and 'red'; '3.2. Figure Height:' with a text input field containing '600'; and '3.3. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4: Display and Download Results', contains a '4.1. Visualization' section with a 'Save Image' button.

3.1. Select colors for groups: This parameter lets you choose specific colors for each group. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of groups, ensuring that the visual representation is both informative and aesthetically pleasing.

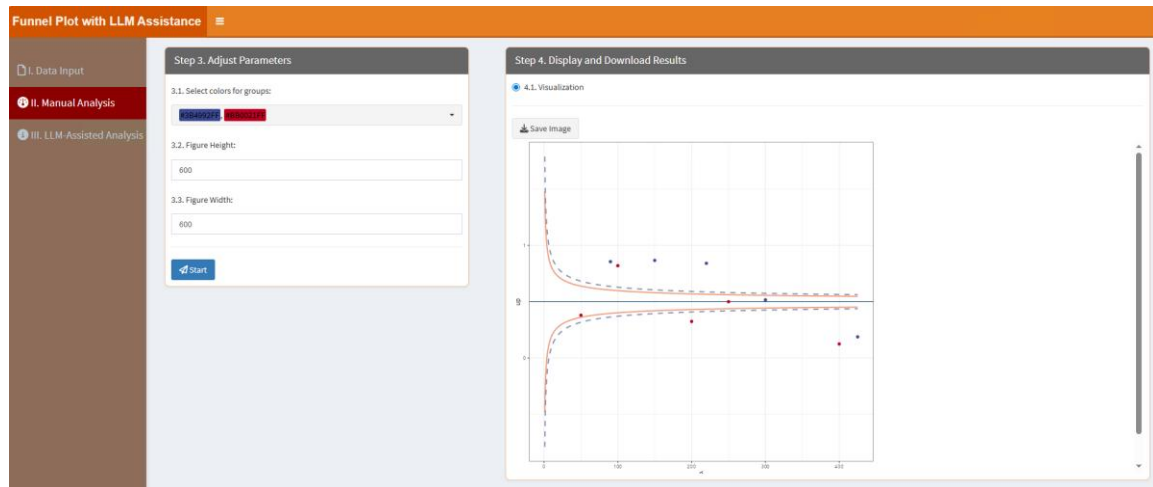
3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure

in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

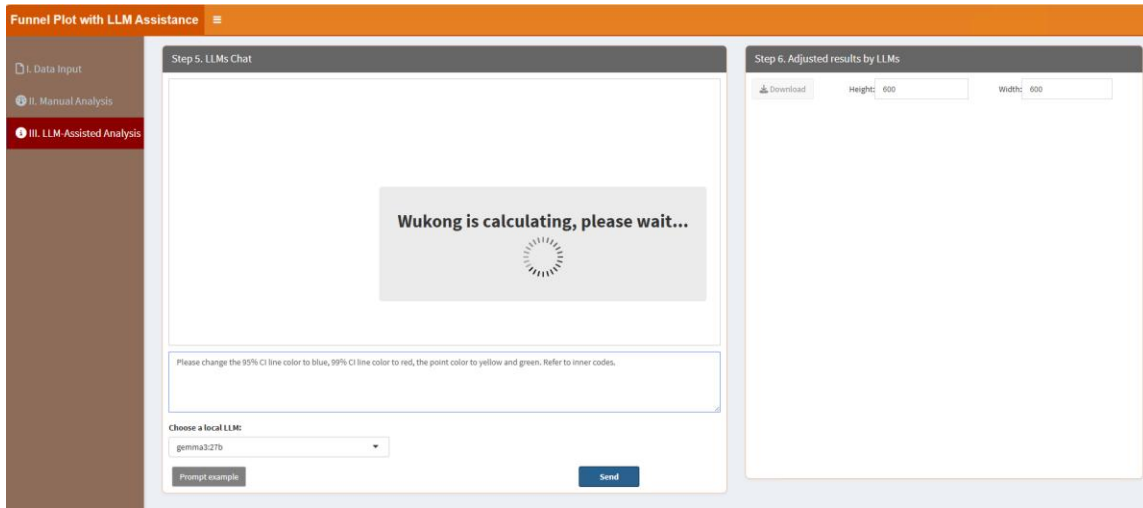
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



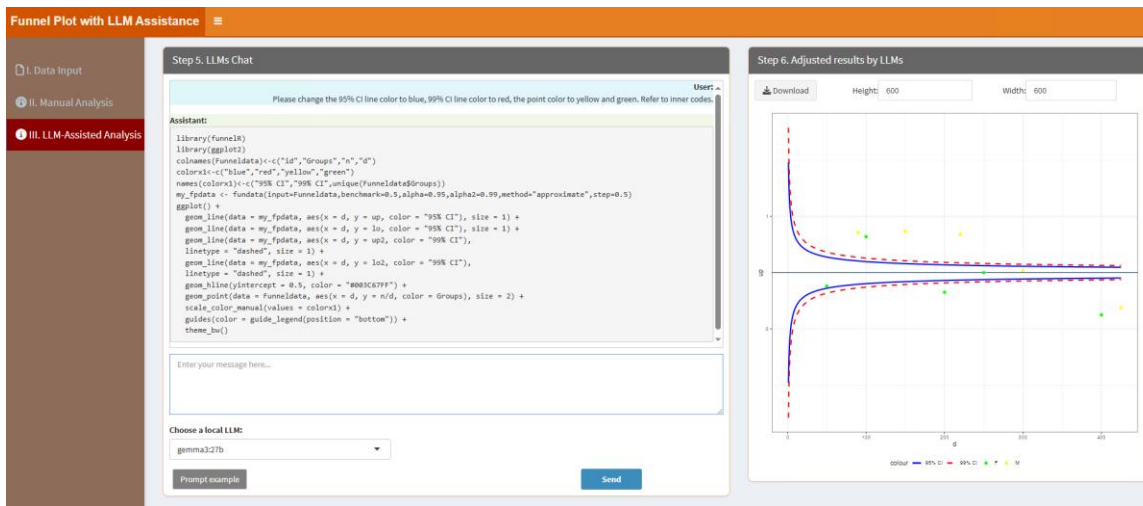
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the 95% CI line color to blue, 99% CI line color to red, the point color to yellow and green. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the 95% CI line color to blue, 99% CI line color to red, the point color to yellow and green. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.12. Protein/DNA sequence logo plot

This module allows users to generate sequence logo plots for protein or DNA sequences. Parameters can be adjusted in the classic framework, and local LLMs simplify the process to highlight conserved regions effectively. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Protein/DNA sequence logo plot with LLM Assistance' interface. The left sidebar has three tabs: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this, there's a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown is set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' and 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Protein/DNA sequence logo plot with LLM Assistance' interface. The left sidebar has three tabs: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three numbered sections: '3.1. Height method:' with a dropdown menu set to 'probability', '3.2. Figure Height:' with a text input field set to '500', and '3.3. Figure Width:' with a text input field set to '900'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section '4.1. Visualization' with a 'Save Image' button.

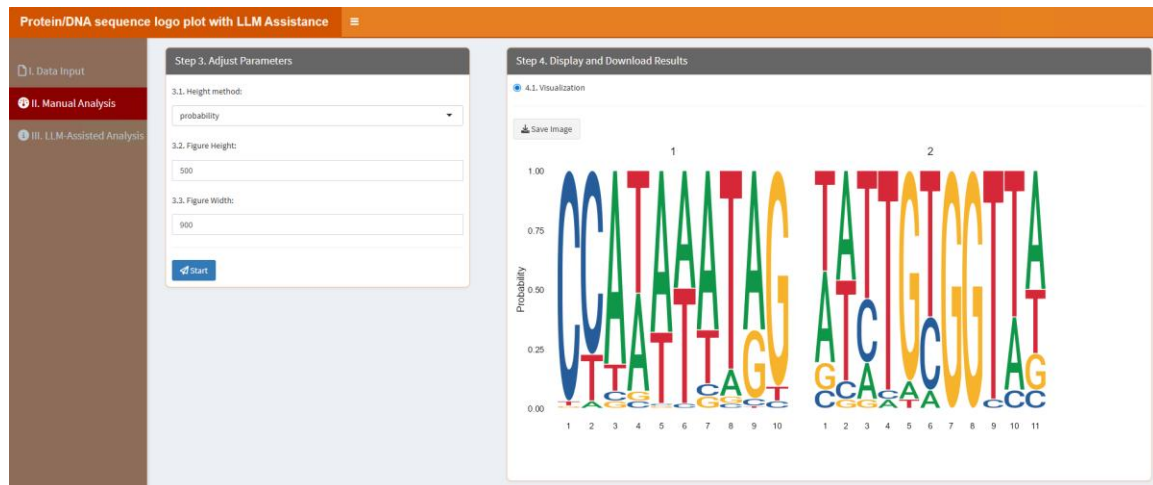
3.1. Height method: Height method, can be one of "bits" or "probability".

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

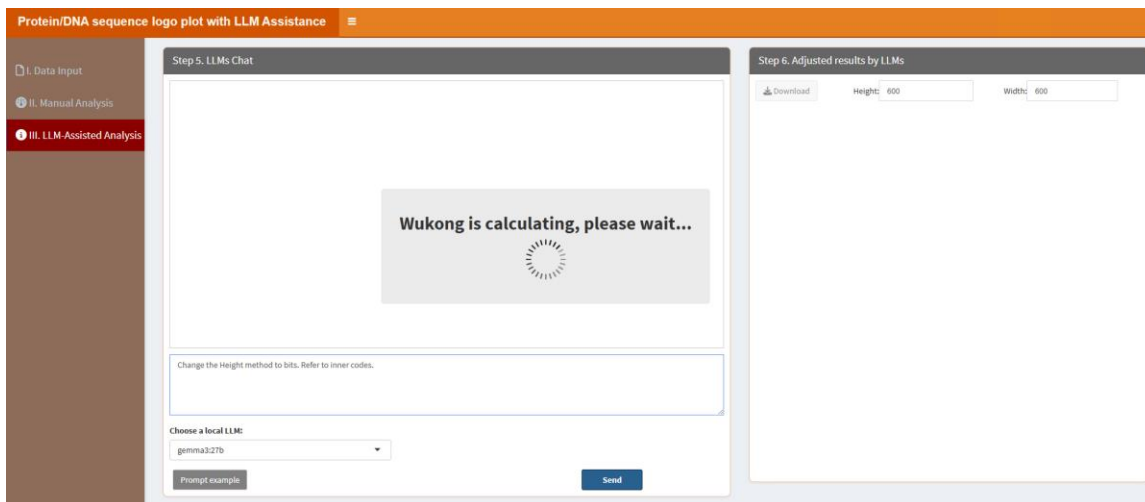
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



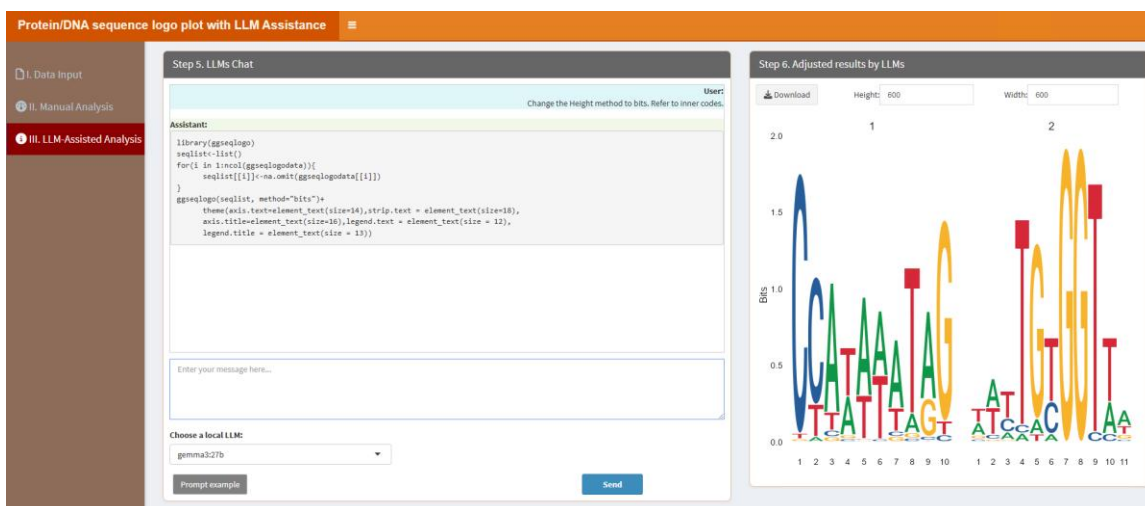
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the Height method to bits. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the Height method to bits. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.13. Dendrogram using ggtree package

Users can create advanced dendrograms using the ggtree package for hierarchical clustering visualization. The classic framework supports customization, while local LLMs assist in refining the plot for publication-ready results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Ggtree dendrogram with LLM Assistance' web application. The interface has a sidebar on the left with three main sections: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this, there's a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a note 'No file selected'. There are also checkboxes for 'Is the first row names?' and 'Is the first column names?', both of which are checked. A 'Which Sheet to read?' dropdown menu is set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row containing the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' with 'Previous' and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Ggtree dendrogram with LLM Assistance' web application. The sidebar on the left now has 'II. Manual Analysis' selected. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains four numbered input fields: '3.1. Cluster number:' with a text input containing '2'; '3.2. Select colors for each cluster:' with a color picker showing two selected colors (blue and red); '3.3. Figure Height:' with a text input containing '600'; and '3.4. Figure Width:' with a text input containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

3.1. Cluster number: Maximum cluster number to evaluate.

3.2. Select colors for each cluster: This parameter lets you choose specific colors for each cluster. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of clusters, ensuring that the visual representation is both informative and aesthetically pleasing.

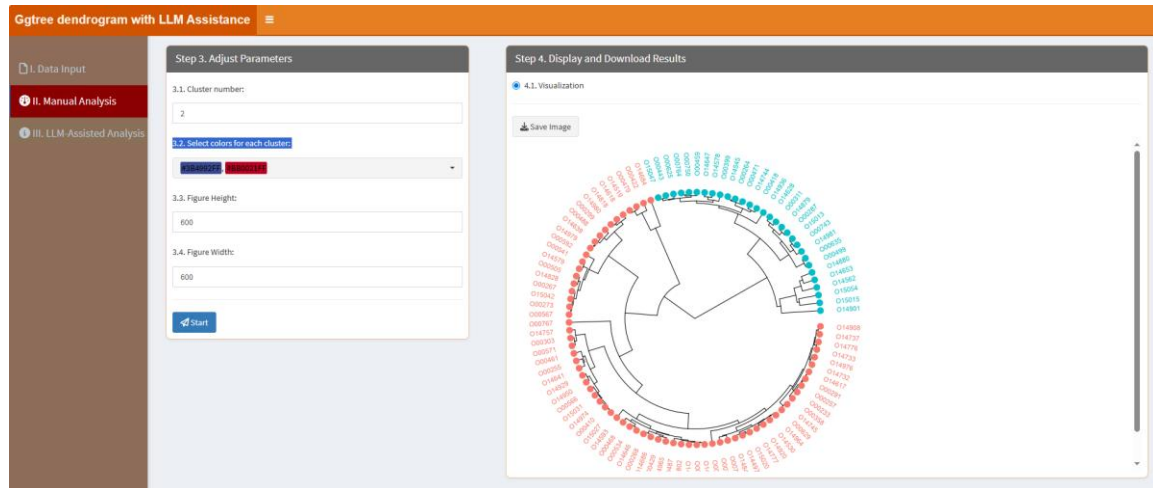
3.3. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your

presentation or analysis needs, ensuring it fits well within the intended display context.

3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

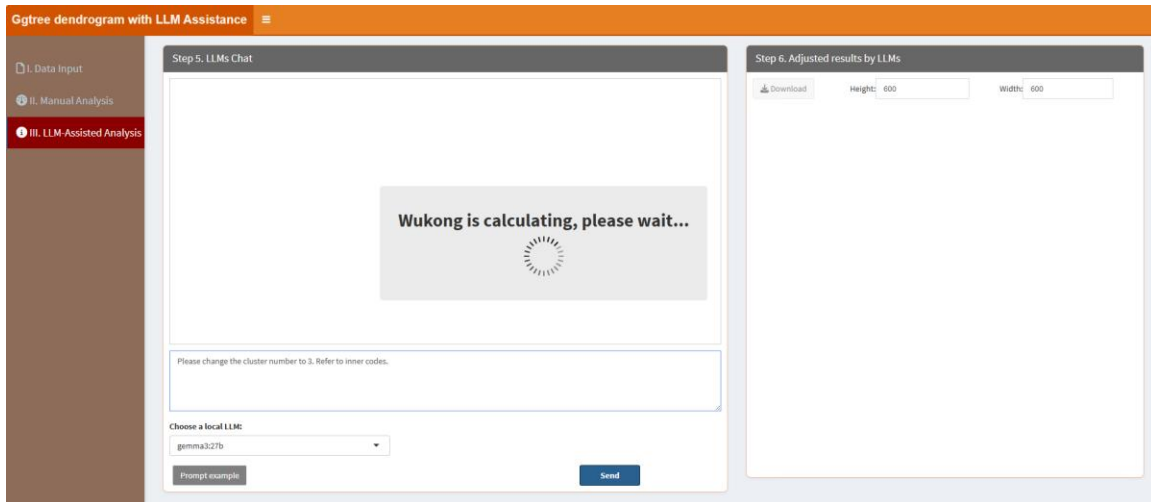
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



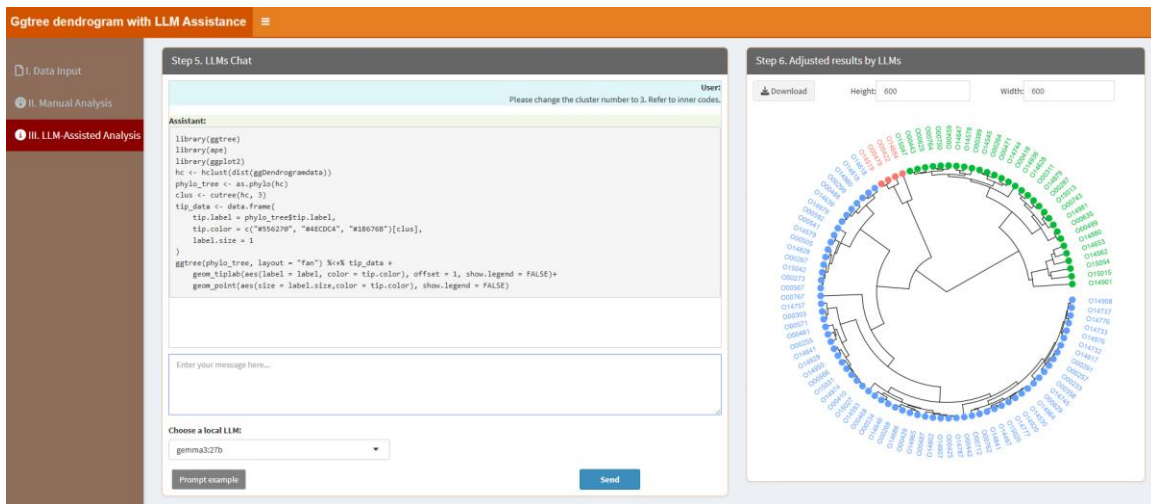
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the cluster number to 3. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the cluster number to 3. =>”, it will similarly prompt

LLMs to learn the internal R codes and produce the result table.
After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.14. Colored scatterplot

Colored scatterplots combine scatterplots with color gradients to represent data density. Users can adjust parameters in the classic framework and use local LLMs to enhance the visualization's aesthetics and informativeness. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Colored scatterplot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section has a 'Browse...' button and a 'No file selected' message. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown menu shows '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', has a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the 'Description' column with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' and 'Previous 1 Next'.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Colored scatterplot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three sections: '3.1. Select colors:' with a dropdown menu showing 'Blue' and 'Red'; '3.2. Figure Height:' with a text input field containing '600'; and '3.3. Figure Width:' with a text input field containing '600'. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section '4.1. Visualization' with a 'Save Image' button.

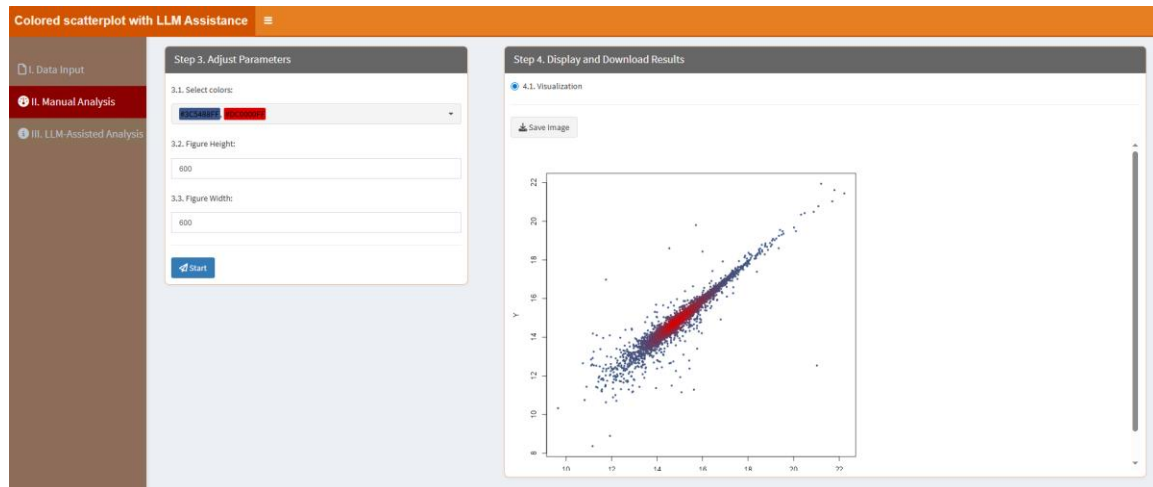
3.1. Select colors: Choose two distinct colors for the bar plot, one is for the lowest value and the other is for the highest value in the uploaded data.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

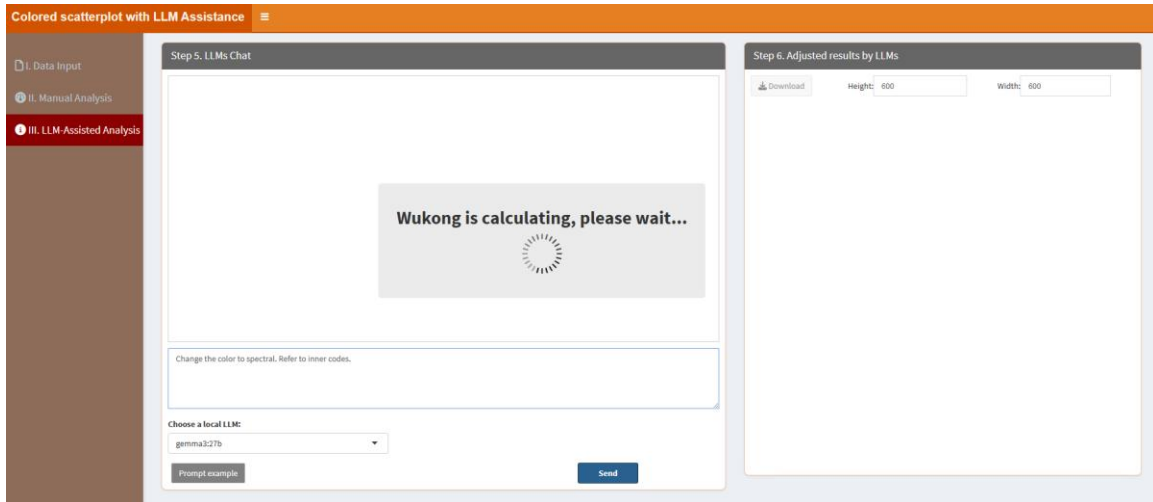
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



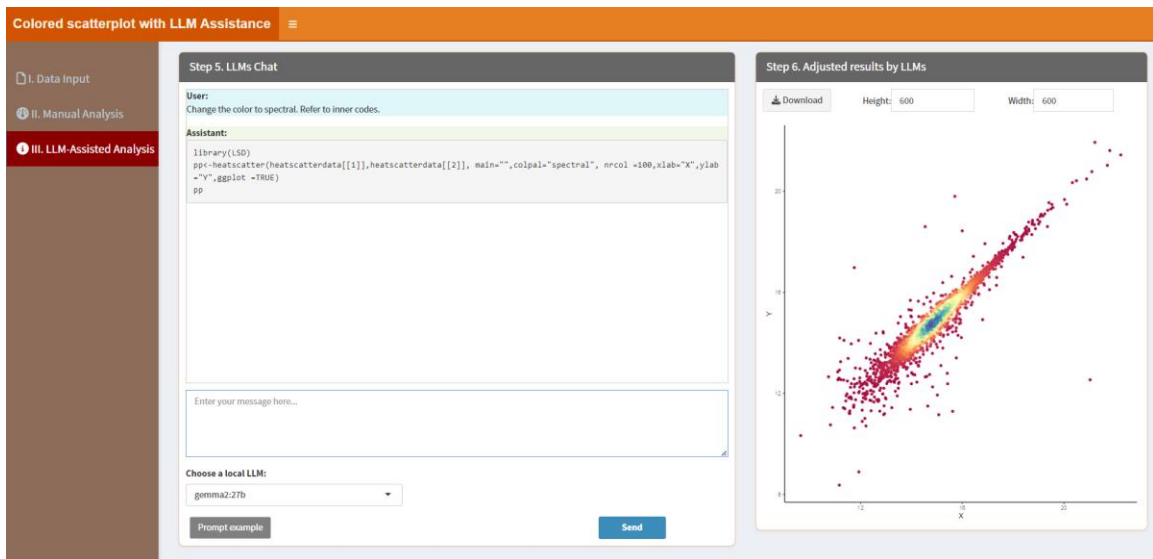
"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "Change the color to spectral. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Change the color to spectral. =>", it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click "Send" button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.15. Histogram and Density plot

This module allows users to create combined histogram and density plots to visualize data distributions. By leveraging the classic framework and local LLMs, users can generate detailed and interpretable plots. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the first step of the application. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is titled 'Step 1. Upload Data/Example Data'. It contains two radio buttons: 'A. Upload' (selected) and 'B. Load example data'. Below these are file format options: '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, a 'Which Sheet to read?' dropdown shows '1'. To the right, a preview window titled 'Step 2. Display Uploaded Data/Example Data' shows a table with one row and one column, with a message 'No data loaded. Please upload your dataset or use the example data.' and pagination controls.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the third step of the application. The sidebar now has 'II. Manual Analysis' selected. The main area is titled 'Step 3. Adjust Parameters'. It contains three numbered sections: '3.1. Select colors:' with a dropdown menu showing 'Blue' and 'Red'; '3.2. Figure Height:' with a text input field containing '600'; and '3.3. Figure Width:' with a text input field containing '600'. At the bottom of this section is a blue 'Start' button. To the right, a preview window titled 'Step 4. Display and Download Results' shows a section for '4.1. Visualization' with a 'Save Image' button.

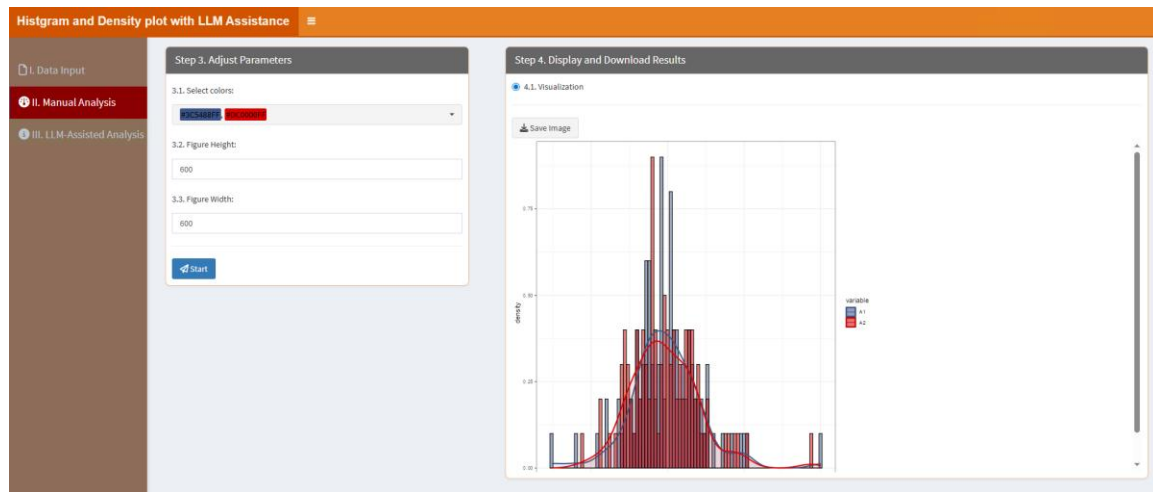
3.1. Select colors: This parameter lets you choose specific colors for each column. The color number should be equal to the column number.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

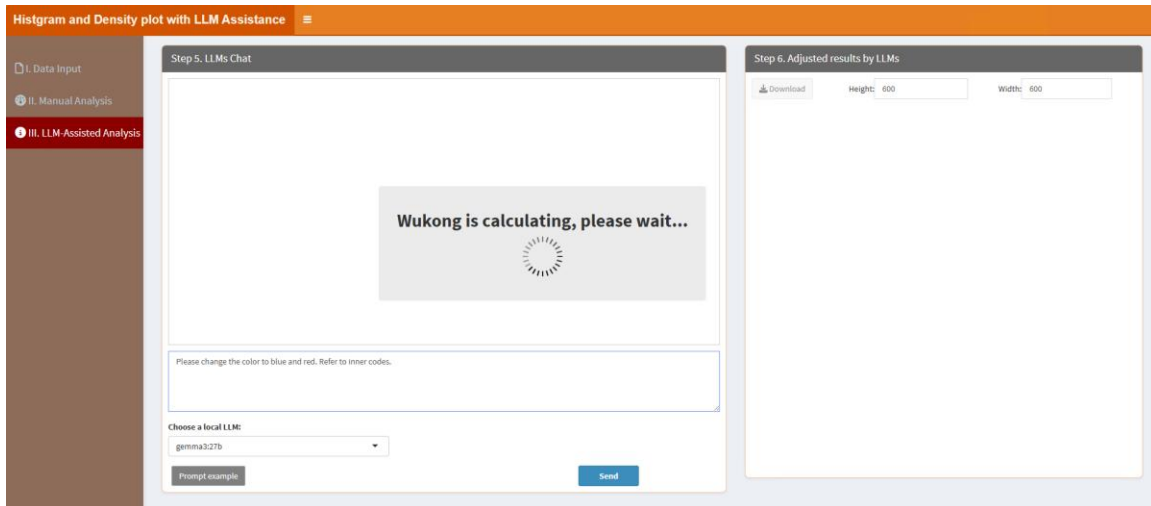
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



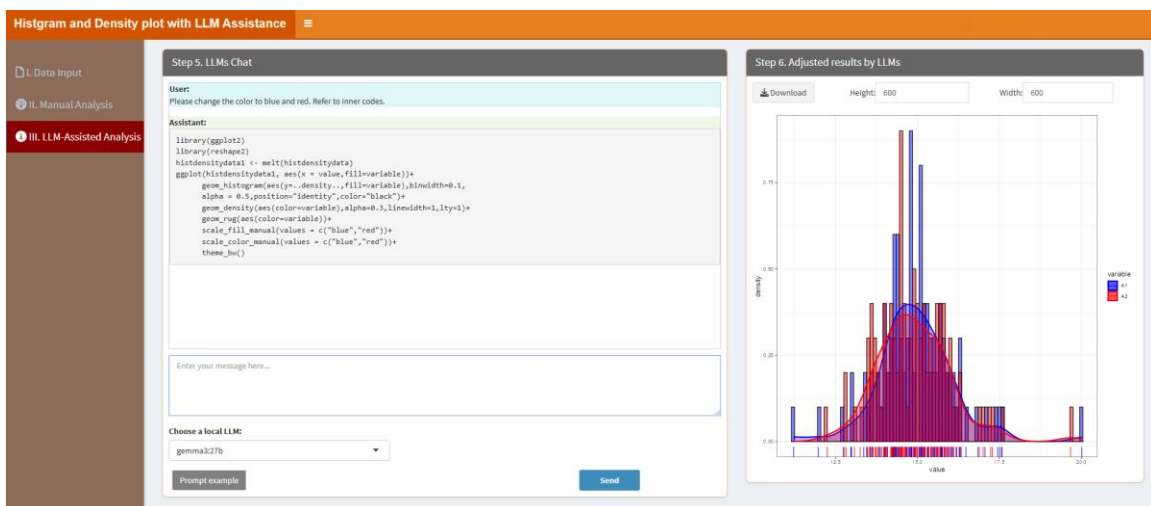
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the color to blue and red. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the color to blue and red. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.16. Lollipop Chart

Lollipop charts are ideal for comparing data points. Users can adjust settings through the classic framework and refine the visualization with local LLMs for clear and effective representation. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Lollipop Chart with LLM Assistance' application. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this, there's a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?'. A 'Which Sheet to read?' field has the value '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the 'Description' column with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous', '1', and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Lollipop Chart with LLM Assistance' application. The sidebar is the same as in the previous screenshot. The main content area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three numbered settings: '3.1. Select colors:' with a dropdown menu showing 'blue' and 'red'; '3.2. Figure Height:' with a text input field containing '600'; and '3.3. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

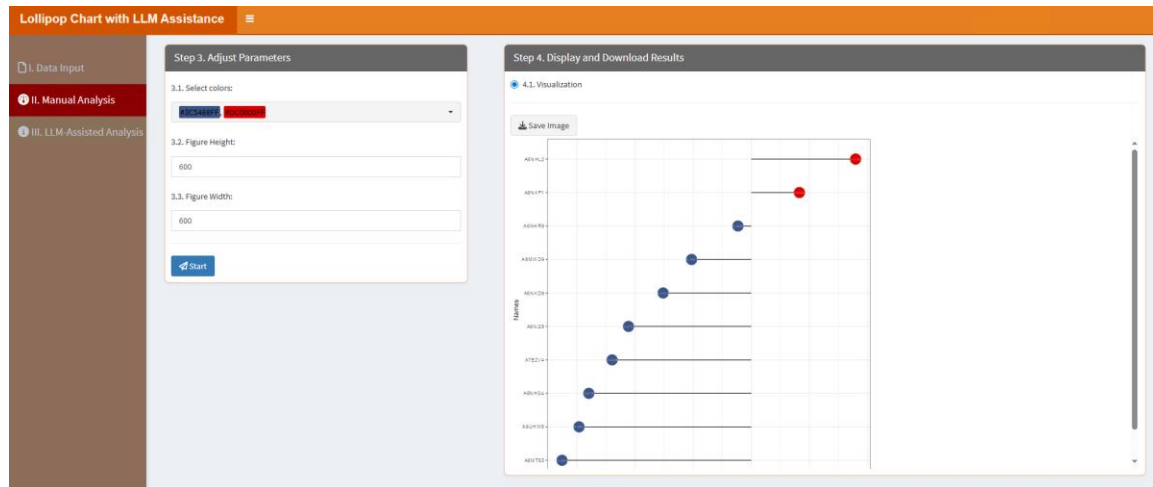
3.1. Select colors: Choose two distinct colors for the bar plot, one is for the negative value and the other is for the positive value in the uploaded data.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the color to blue and red. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the color to blue and red. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

7.17. Marginal Histogram/Boxplot

Marginal histograms or boxplots can be generated alongside scatterplots to visualize data distributions. Users can customize the plot through the classic framework and enhance it with local LLMs. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1: Upload Data/Example Data' section of the application. On the left, a sidebar contains three menu items: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this, there are radio buttons for 'Select File Format' (.xlsx, .xls, .csv, .txt) and a 'Browse...' button. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?'. A 'Which Sheet to read?' dropdown is set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one entry: '1' with the description 'No data loaded. Please upload your dataset or use the example data.' It includes a search bar, a 'Show' dropdown set to '10', and 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3: Adjust Parameters' section of the application. The sidebar is the same as in the previous screenshot. The main area has two panels. The left panel, titled 'Step 3. Adjust Parameters', contains five configuration items: '3.1. Select marginal colors:' with a dropdown showing 'blue' and 'red'; '3.2. Select point color:' with a dropdown showing 'blue'; '3.3. Select line color:' with a dropdown showing 'red'; '3.4. Figure Height:' with a text input field containing '600'; and '3.5. Figure Width:' with a text input field containing '600'. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

3.1. Select marginal colors: Choose two distinct colors for the marginal plot.

3.2. Select point color: Choose a color for the points.

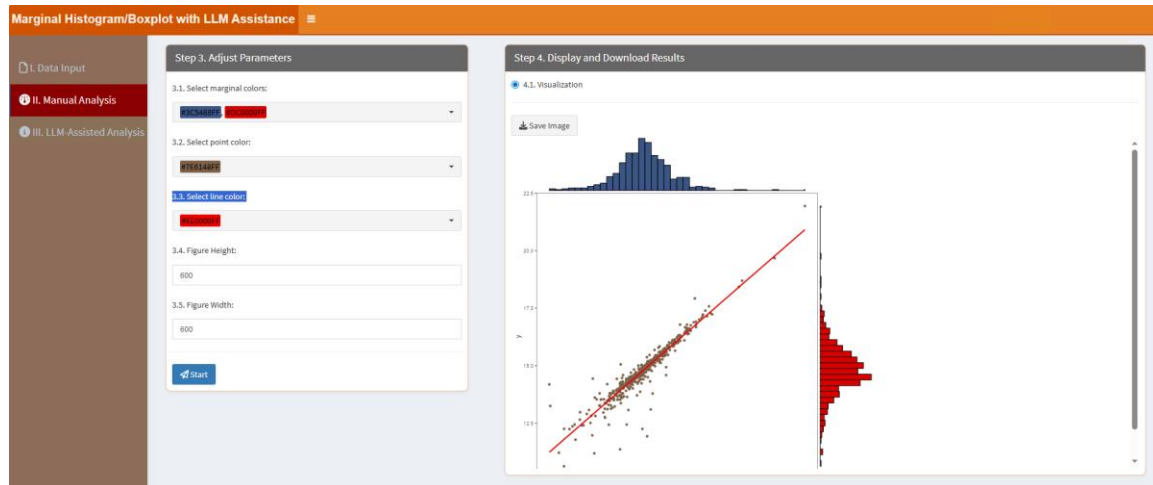
3.3. Select line color: Choose a color for the line.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

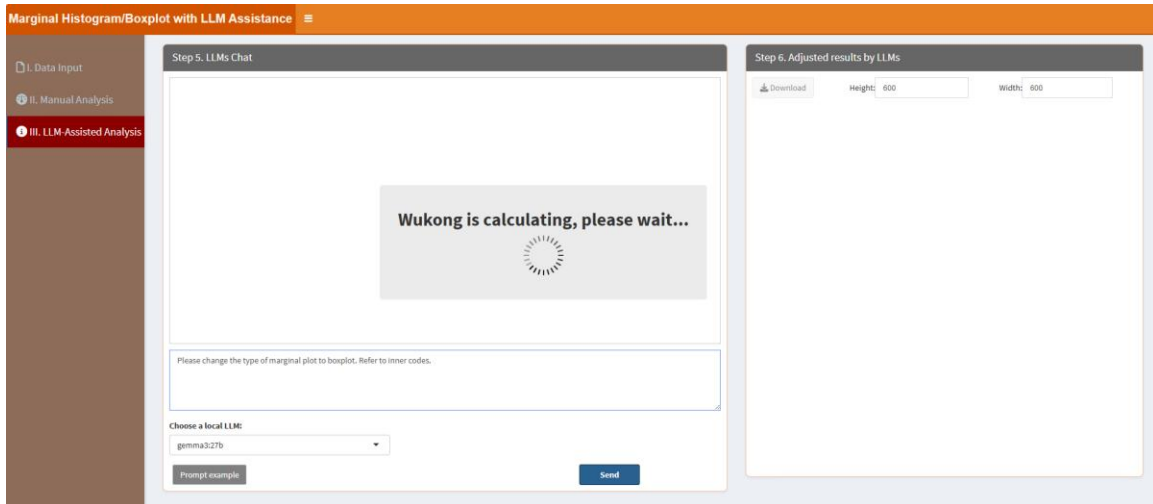
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



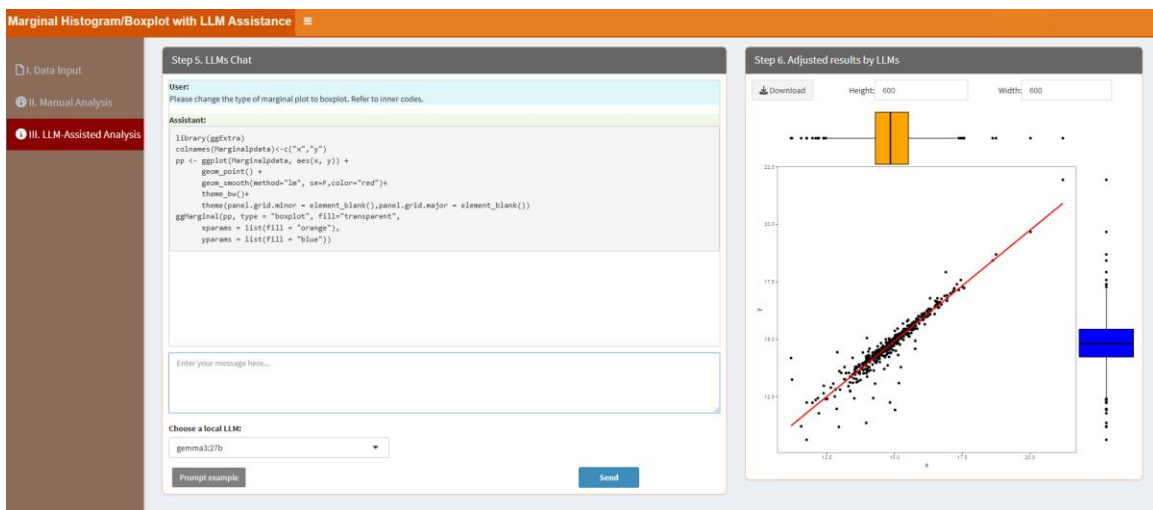
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the type of marginal plot to boxplot. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the type of marginal plot to boxplot. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.18. Nightingale Rose Diagram

Nightingale rose diagrams visualize circular data distributions. Users can adjust parameters in the classic framework and use local LLMs to ensure accurate and visually appealing results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' window of the 'Nightingale Rose plot with LLM Assistance' application. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main content area is divided into two panels. The left panel contains options to 'A. Upload' or 'B. Load example data', a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt', a 'Please import your data file:' section with a 'Browse...' button and 'No file selected' text, and checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?'. Below these is a 'Which Sheet to read?' field with the value '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show: 10 entries' dropdown, a search bar, and a table with one row: '1' in the 'Description' column with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' and 'Previous', '1', 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' window of the 'Nightingale Rose plot with LLM Assistance' application. The sidebar is the same as in the previous screenshot. The main content area has two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three numbered sections: '3.1. Select colors:' with a color picker showing three selected colors (blue, green, and red), '3.2. Figure Height:' with a text input field containing '600', and '3.3. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

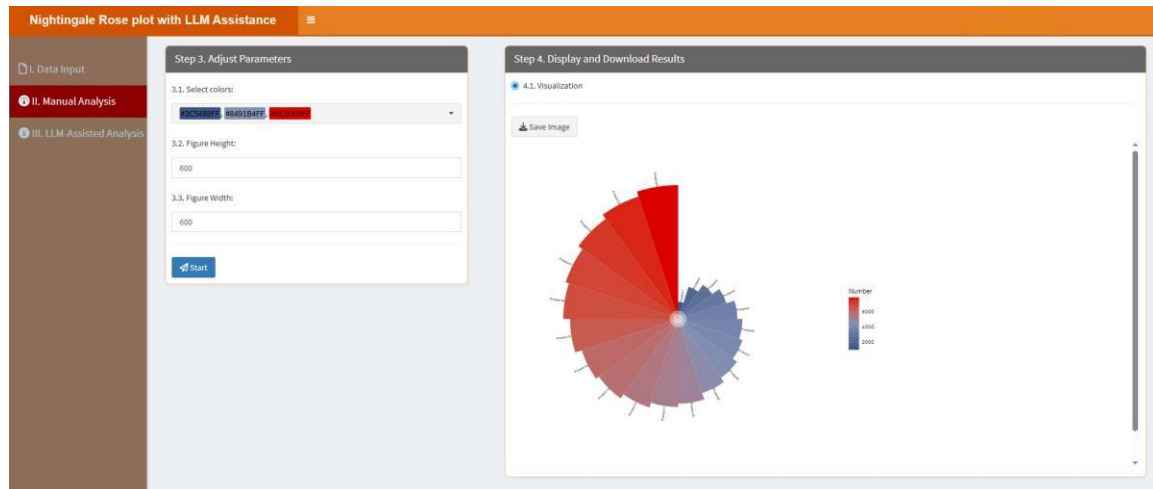
3.1. Select colors: Choose three distinct colors for the circle bar plot, one is for the lowest value, one is for the middle value and the last is for the highest value in the uploaded data.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

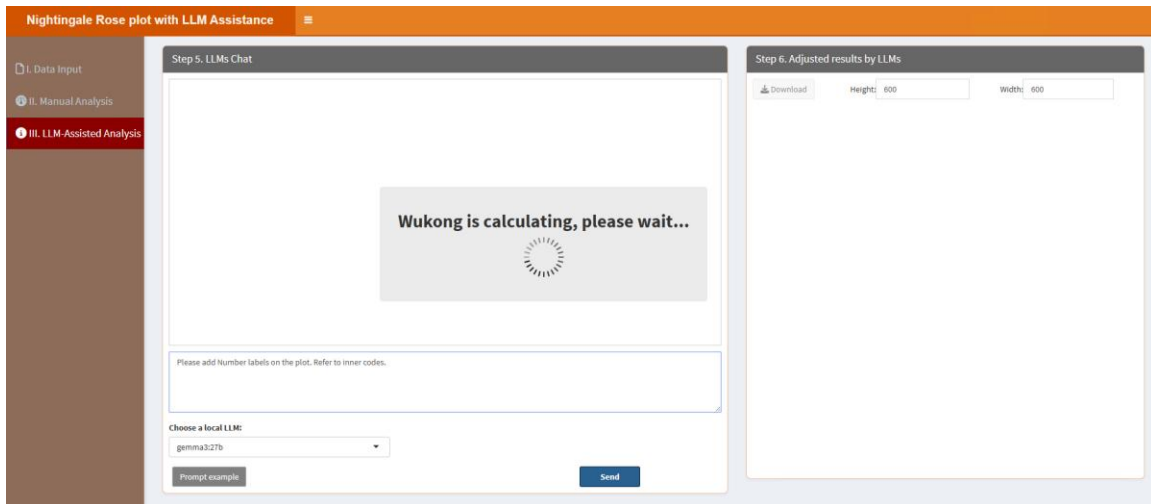
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



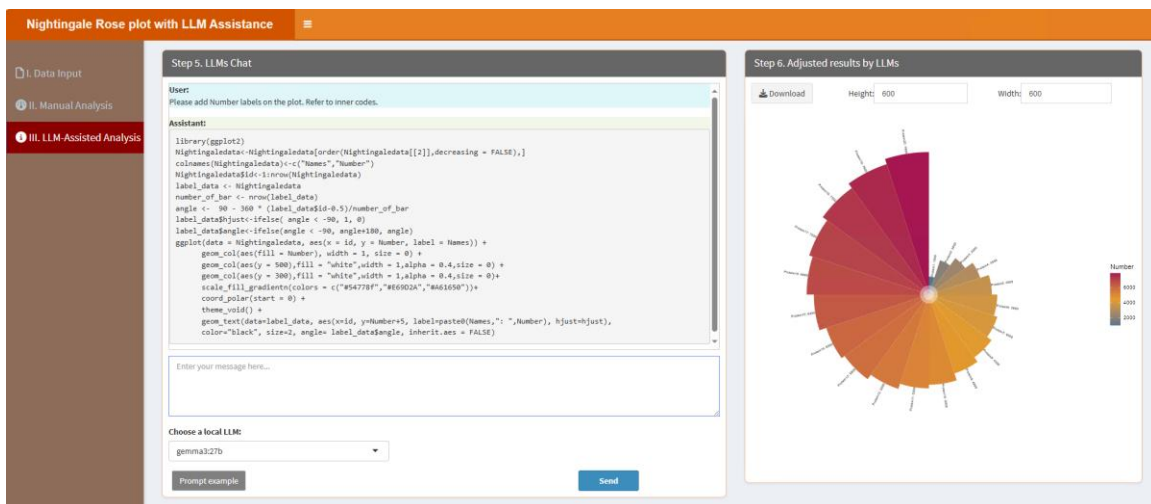
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please add Number labels on the plot. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please add Number labels on the plot. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.19. Pair Point Line plot

Pair point line plots highlight relationships between paired data points. Users can customize the plot through the classic framework and refine it with local LLMs for better clarity. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Pair Point Line plot with LLM Assistance' interface. On the left is a sidebar with three sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this, there's a 'Select File Format:' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a note 'No file selected'. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which Sheet to read?' dropdown set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one entry: '1' in the 'Description' column, with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous', '1', and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Pair Point Line plot with LLM Assistance' interface, specifically Step 3: Adjust Parameters. The sidebar on the left is the same as in the previous screenshot. The main area has two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three numbered settings: '3.1. Select two colors for point colors:' with a dropdown menu showing 'Red' and 'Blue'; '3.2. Figure Height:' with a text input field containing '600'; and '3.3. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a section for '4.1. Visualization' with a 'Save Image' button.

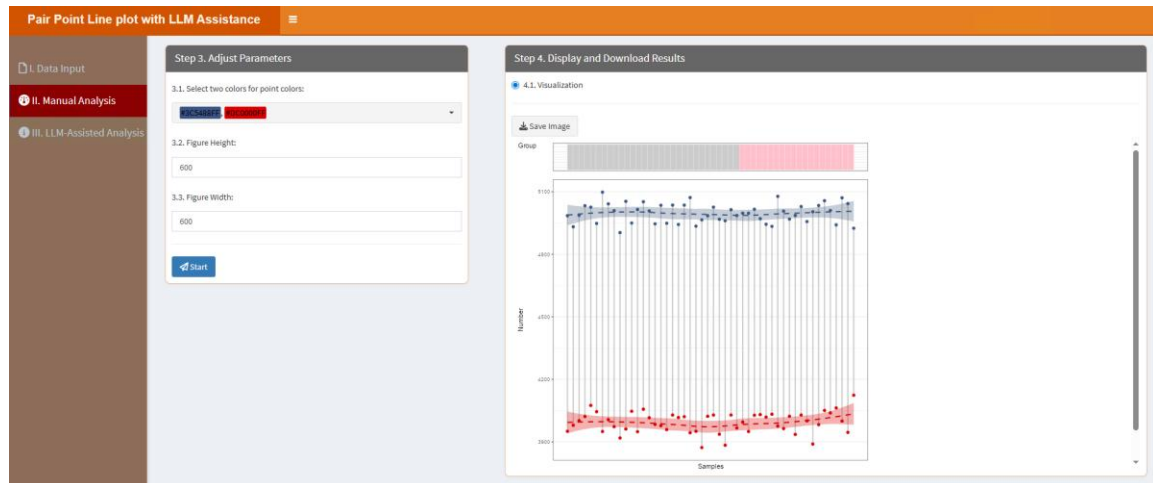
3.1. Select two colors for point colors: This parameter allows users to assign two distinct colors to visually differentiate the points representing two groups in the pair point line plot.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

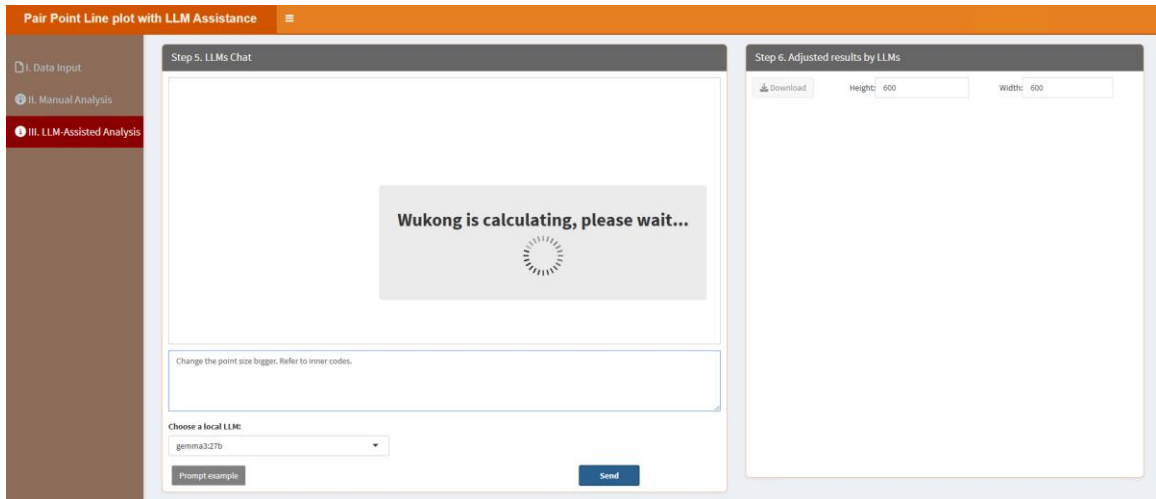
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



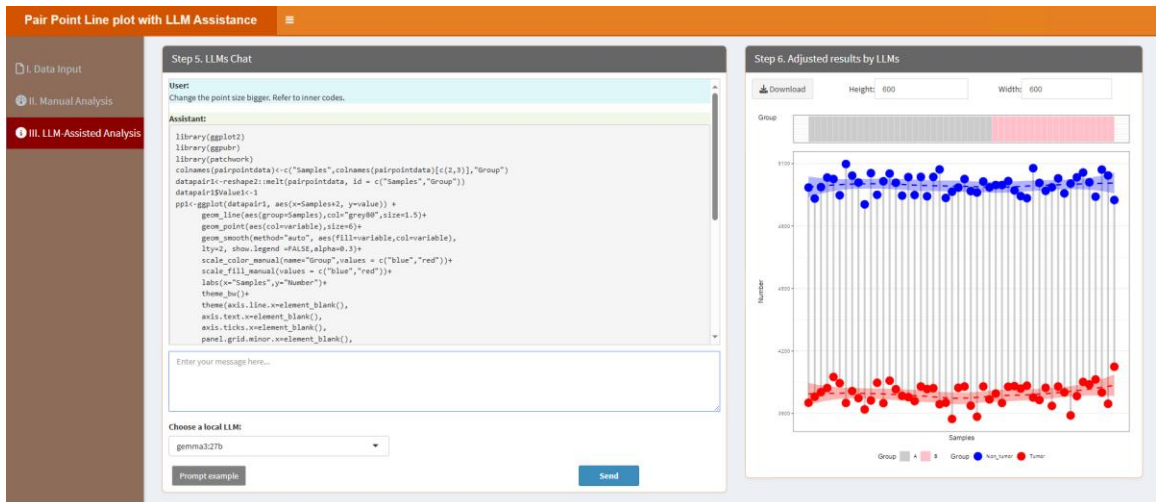
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the point size bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the point size bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.20. Pie plot

Pie charts can be used to visualize categorical data distributions. Users can adjust parameters in the classic framework and rely on local LLMs for customization and interpretation. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Pie plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. Below this is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a note 'No file selected'. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the 'Description' column. Below the table is a message: 'No data loaded. Please upload your dataset or use the example data.' and a pagination bar showing 'Showing 1 to 1 of 1 entries' with 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Pie plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three sections: '3.1. Select colors:' with a color picker showing '400B0D5E', '00A0C0', '000000', and 'FF0000'; '3.2. Figure Height:' with a text input field containing '600'; and '3.3. Figure Width:' with a text input field containing '600'. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section '4.1. Visualization' with a 'Save Image' button.

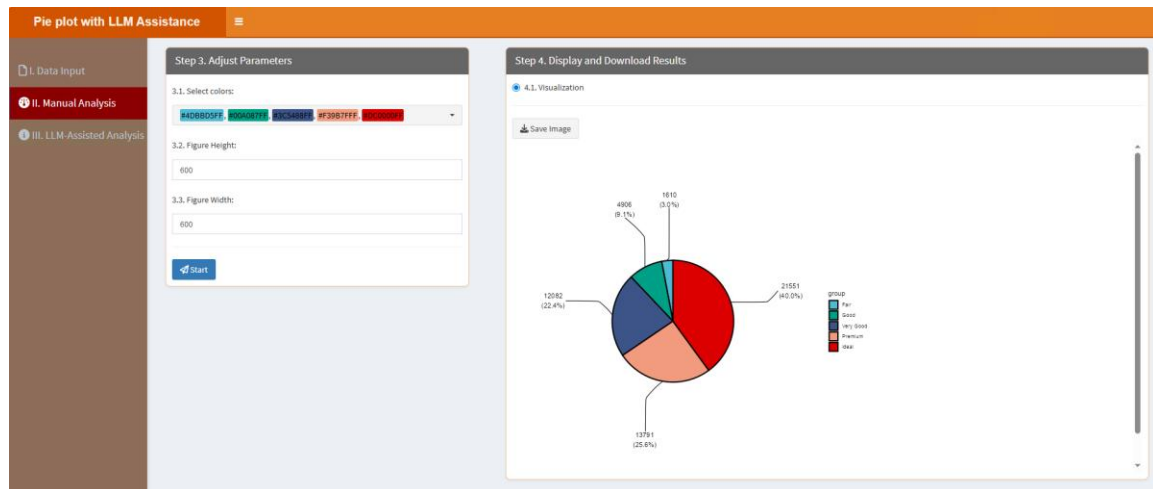
3.1. Select bar colors: This parameter allows users to choose specific colors for each category segment in the pie chart.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

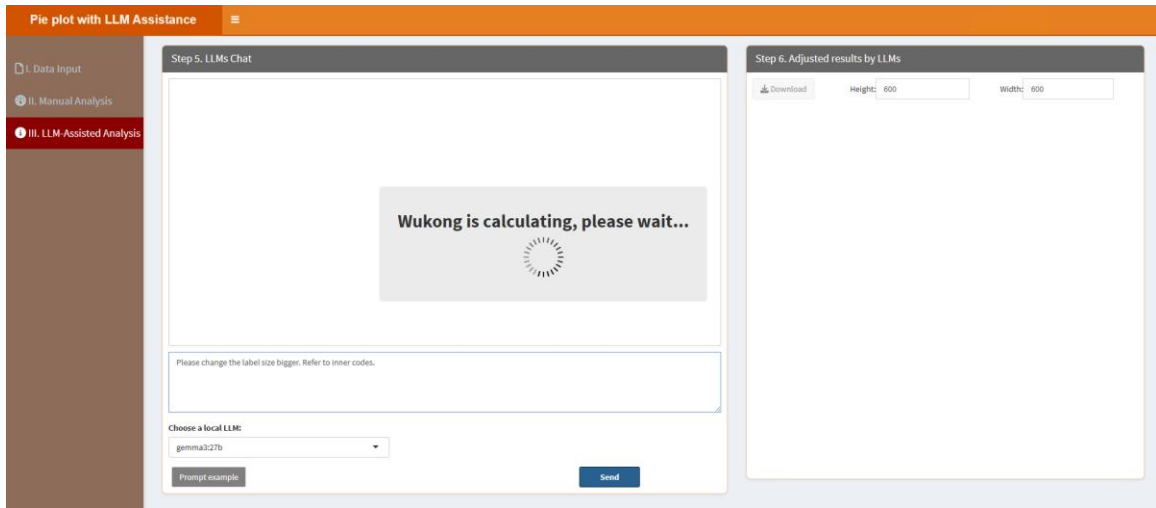
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



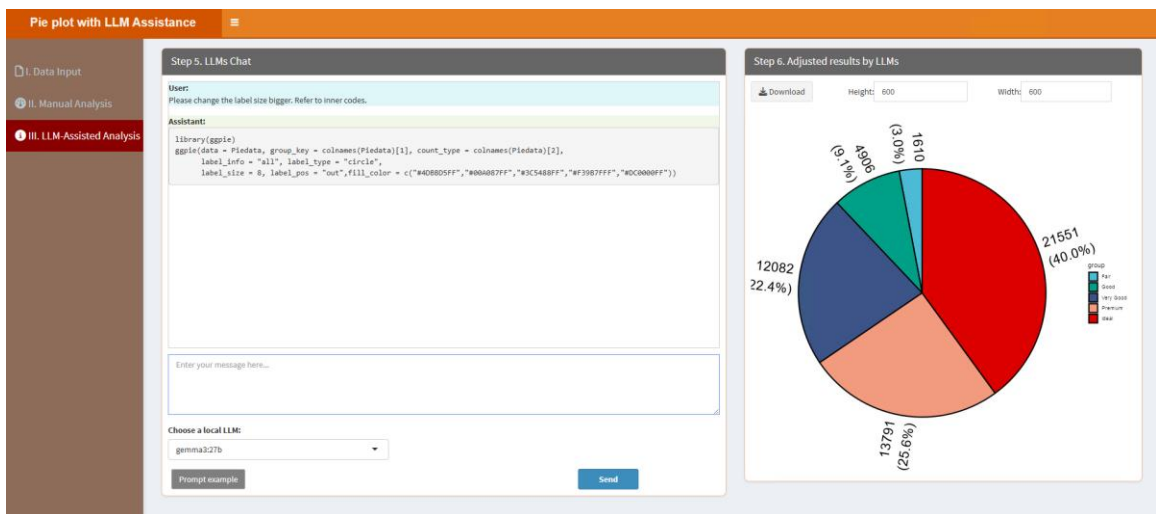
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the label size bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the label size bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.21. Radar Chart

Radar charts are useful for comparing multivariate data. Users can customize the plot using the classic framework and refine it with local LLMs to create informative and visually appealing results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1: Upload Data/Example Data' panel of the 'RadarChart with LLM Assistance' application. The left sidebar has three tabs: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main panel contains options to 'A. Upload' or 'B. Load example data'. Under 'Select File Format', there are radio buttons for '.xlsx', '.xls', and '.csv/txt', with '.xlsx' selected. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' status. Below this, there are checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?'. A 'Which Sheet to read?' dropdown menu shows '1'.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3: Adjust Parameters' panel of the 'RadarChart with LLM Assistance' application. The left sidebar has three tabs: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main panel contains three numbered sections: '3.1. Select colors:' with a dropdown menu showing '40808055', '80808080', and '80808080'; '3.2. Figure Height:' with a text input field containing '600'; and '3.3. Figure Width:' with a text input field containing '600'. A blue 'Start' button is at the bottom.

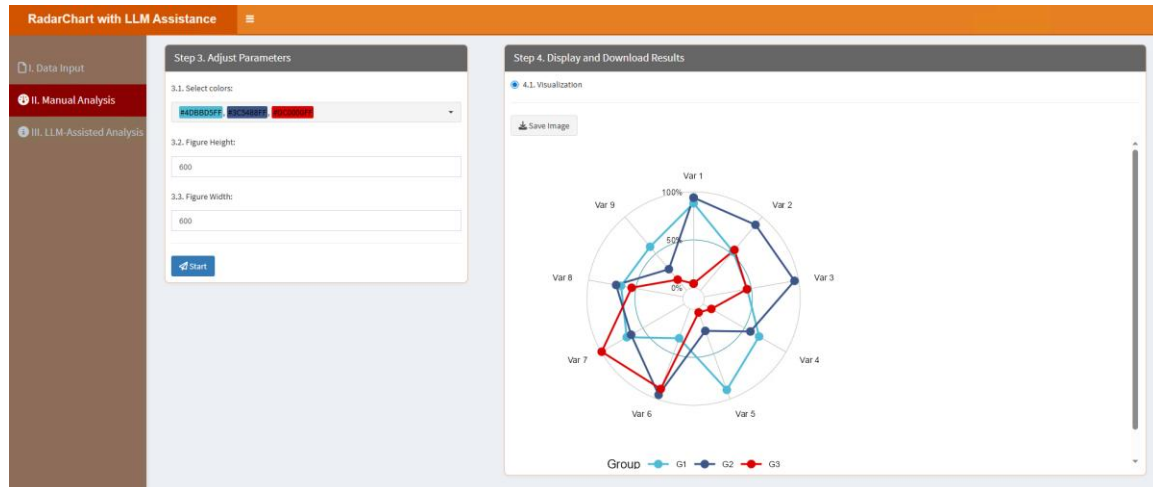
3.1. Select colors: This parameter lets you choose specific colors for each group. The color number should be equal to the group number.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

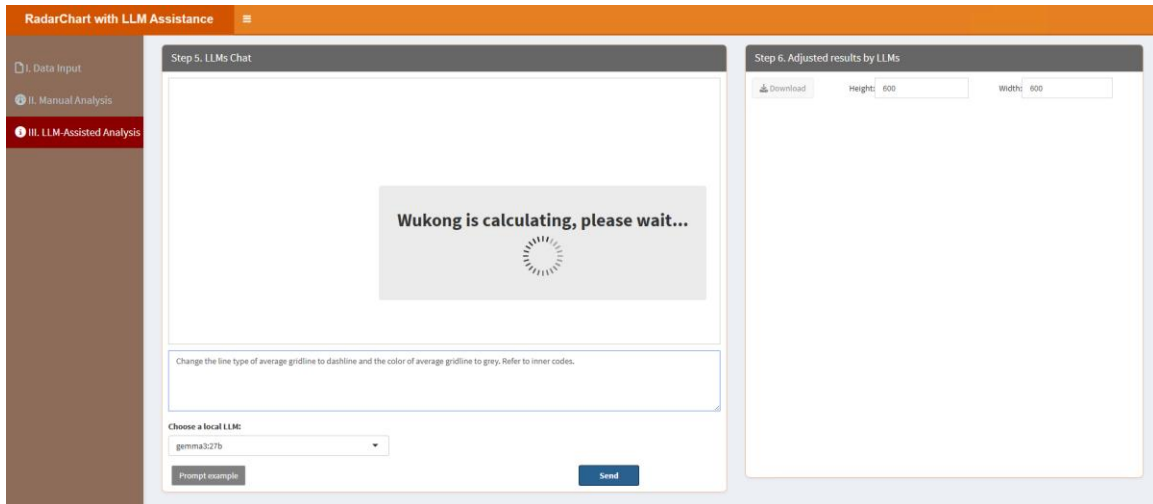
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



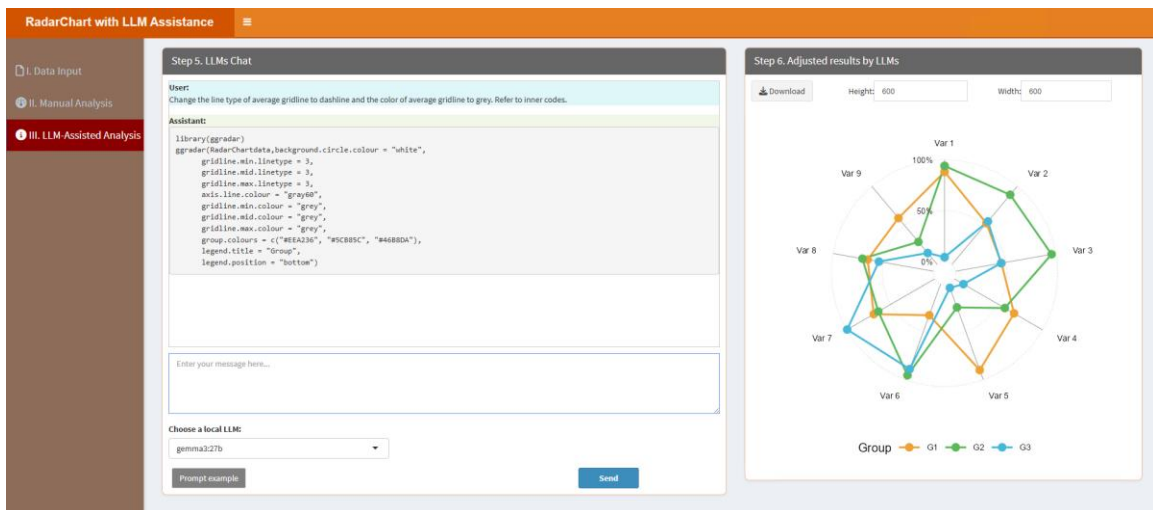
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the line type of average gridline to dashline and the color of average gridline to grey. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the line type of average gridline to dashline and the color of average gridline to grey. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.22. Rain Cloud Plot

Rain cloud plots combine density and scatter plots to visualize data distributions. Users can adjust parameters in the classic framework and use local LLMs to enhance the visualization. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main panel has two tabs: 'A. Upload' (selected) and 'B. Load example data'. Under 'A. Upload', there is a 'Select File Format:' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the 'Description' column with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous', '1', and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' interface. The sidebar is the same as in the previous screenshot. The main panel has two tabs: 'A. Upload' and 'B. Load example data'. The 'A. Upload' tab is selected. It contains five numbered input fields: '3.1. Group and replicate number:' with the value '2;4-4', '3.2. Group names:' with the value 'A;B', '3.3. Select colors for each group:' with a dropdown menu showing 'Blue' and 'Red', '3.4. Figure Height:' with the value '550', and '3.5. Figure Width:' with the value '600'. At the bottom of the form is a blue 'Start' button. The right panel, 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

3.3. Select colors for each group: This parameter lets you choose specific colors for each

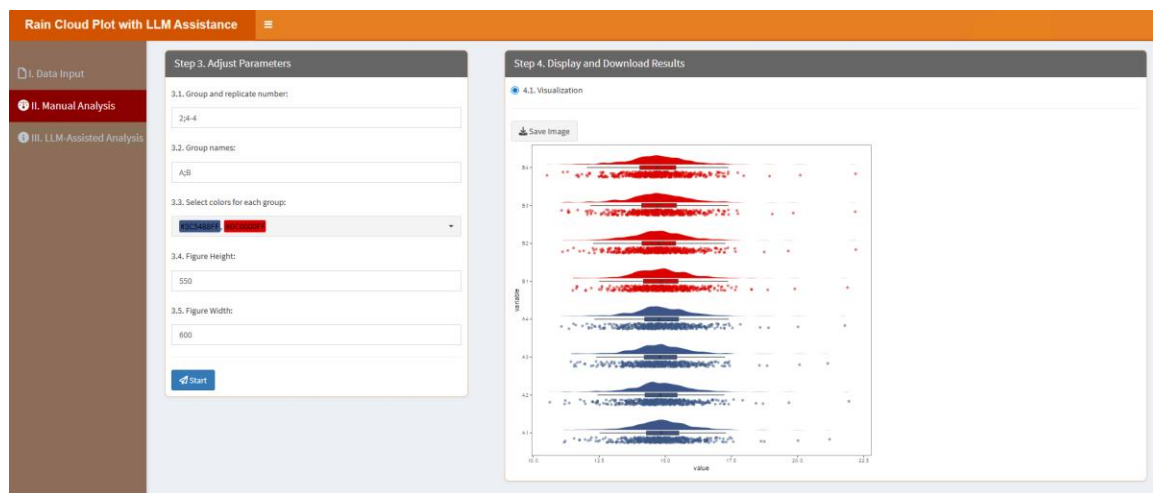
group. The color number should be equal to the group number.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

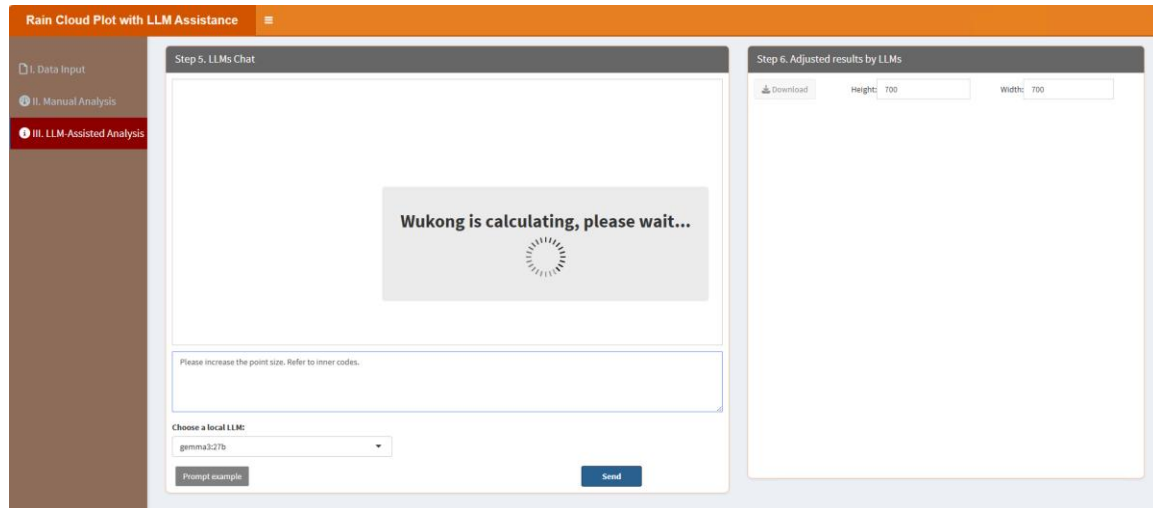


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

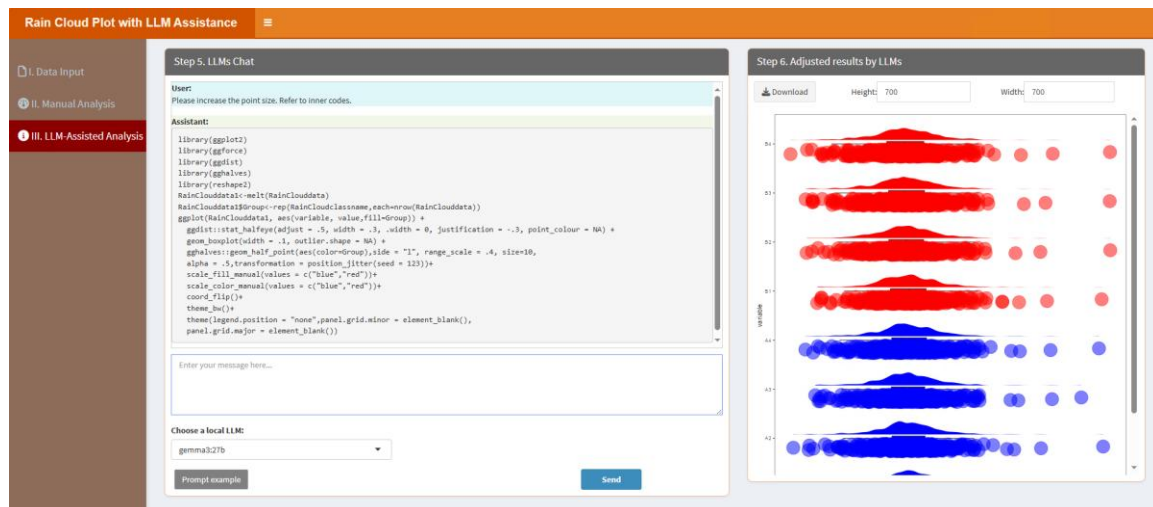
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please increase the point size. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and

utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please increase the point size. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.23. Rank Point Plot

Rank point plots highlight data rankings. Users can customize the plot through the classic framework and refine it with local LLMs for clear and effective visualization. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Rank Point Plot with LLM Assistance' interface. On the left is a sidebar with three main sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The 'I. Data Input' section is active, showing 'Step 1. Upload Data/Example Data'. This step includes options to 'A. Upload' or 'B. Load example data'. Under 'Select File Format:', there are radio buttons for '.xlsx', '.xls', and '.csv/txt', with '.xlsx' selected. A 'Please import your data file:' section contains a 'Browse...' button and a 'No file selected' message. Below this, there are checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked). A 'Which Sheet to read?' field has '1' entered. On the right, 'Step 2. Display Uploaded Data/Example Data' is visible, showing a table with one entry: '1' in the 'Description' column, with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous', '1', and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Rank Point Plot with LLM Assistance' interface, specifically 'Step 3. Adjust Parameters'. The sidebar on the left now highlights 'II. Manual Analysis'. The 'Step 3. Adjust Parameters' panel includes: '3.1. Select a color for points:' with a dropdown menu showing 'RANDOM'; '3.2. Log2 transformation?' with a checked checkbox; '3.3. Figure Height:' with a text input field containing '600'; and '3.4. Figure Width:' with a text input field containing '600'. A blue 'Start' button is at the bottom of this panel. To the right, 'Step 4. Display and Download Results' is visible, showing '4.1. Visualization' with a 'Save Image' button.

3.1. Select a color for points: Choose a color for the points.

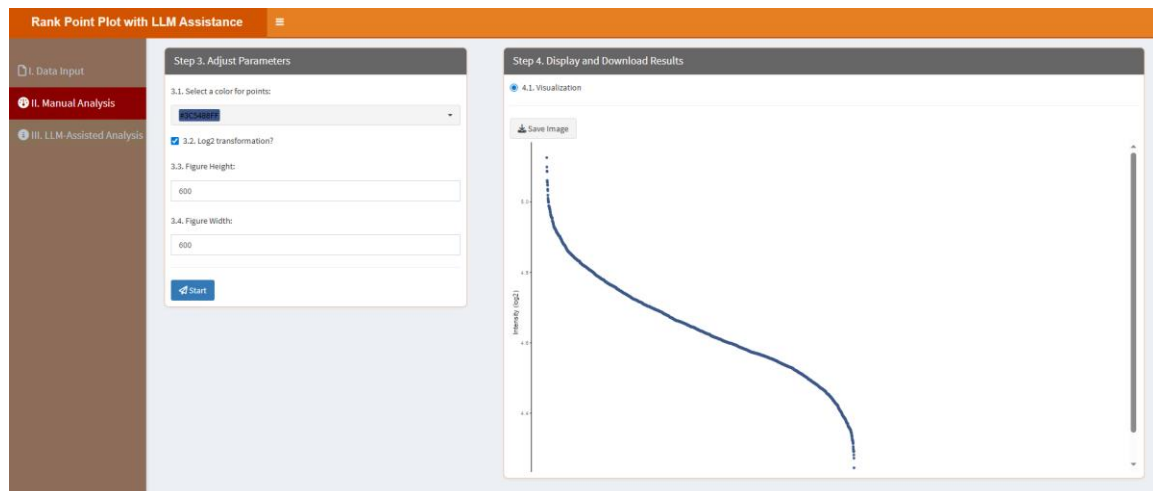
3.2. Log2 transformation? If true, all values will be transformed using the base-2 logarithm.

3.3. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

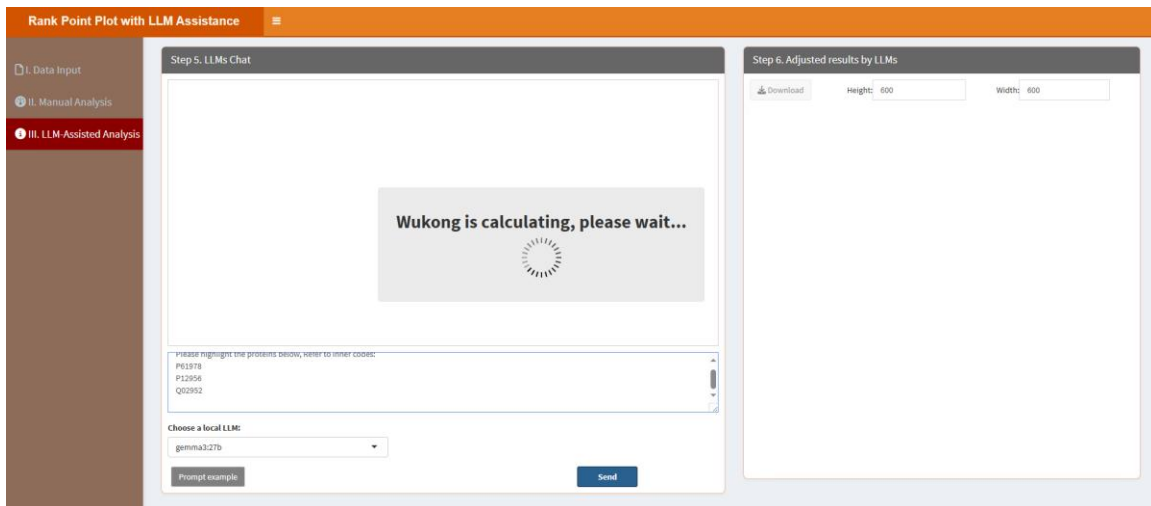
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



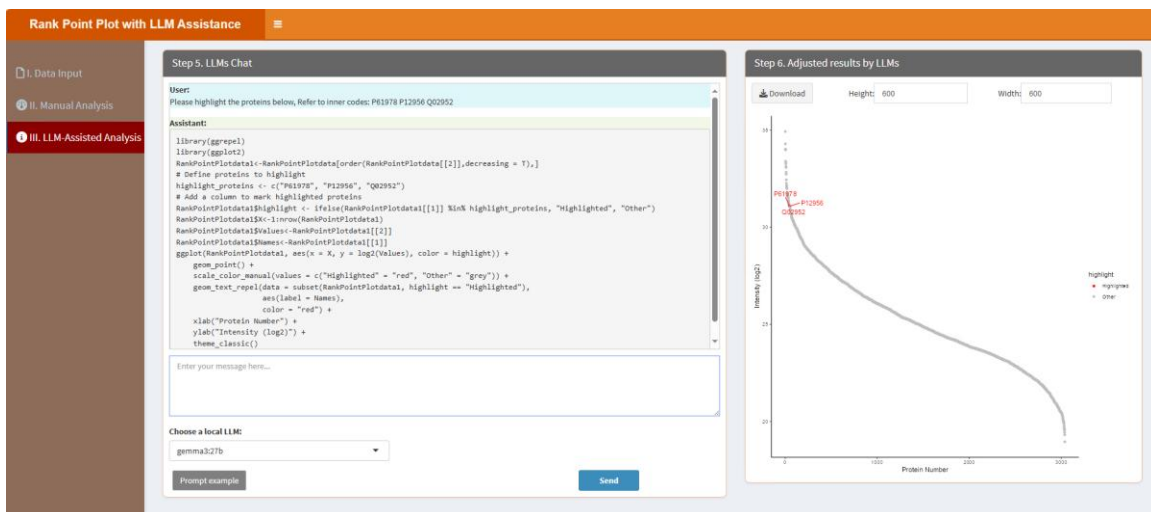
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please highlight the proteins below, Refer to inner codes: P61978, P12956, Q02952.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please highlight the proteins below, Refer to inner codes: P61978, P12956, Q02952.=>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.24. Ridge Plot

Ridge plots visualize distributions across multiple categories. Users can adjust settings using the classic framework and refine the plot with local LLMs for detailed and aesthetically pleasing results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Ridge Plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown menu is set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the 'Description' column with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Ridge Plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains five numbered input fields: '3.1. Group and replicate number:' with the value '2;4-4', '3.2. Group names:' with the value 'A;B', '3.3. Select colors for each group:' with a dropdown menu showing 'Blue' and 'Red', '3.4. Figure Height:' with the value '550', and '3.5. Figure Width:' with the value '600'. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

3.3. Select colors for each group: This parameter lets you choose specific colors for each

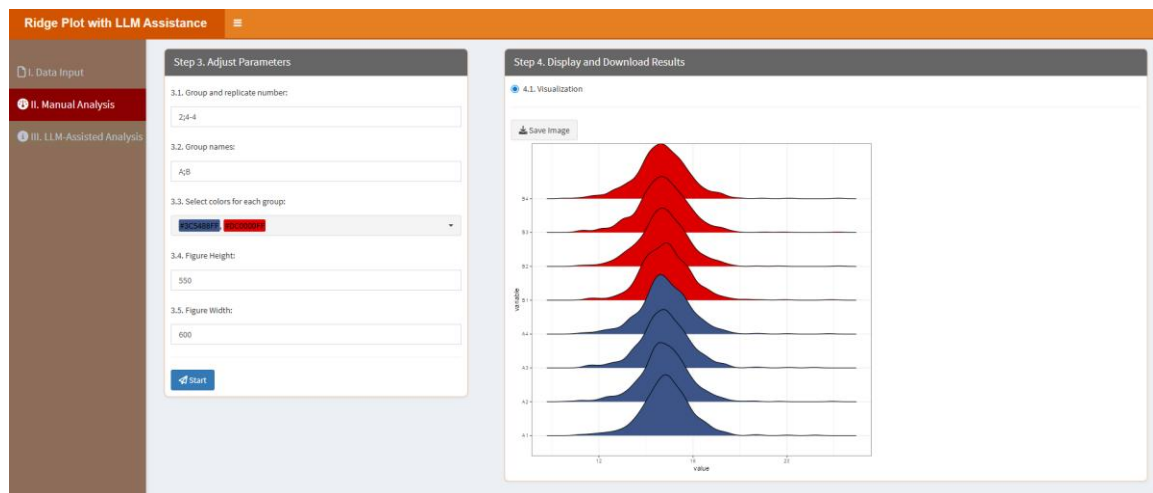
group. The color number should be equal to the group number.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

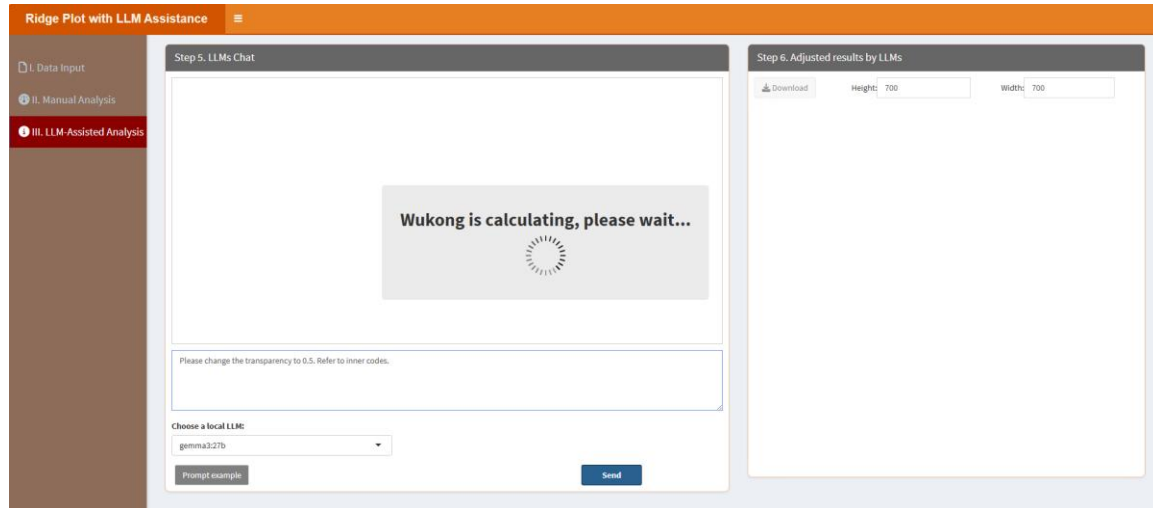


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

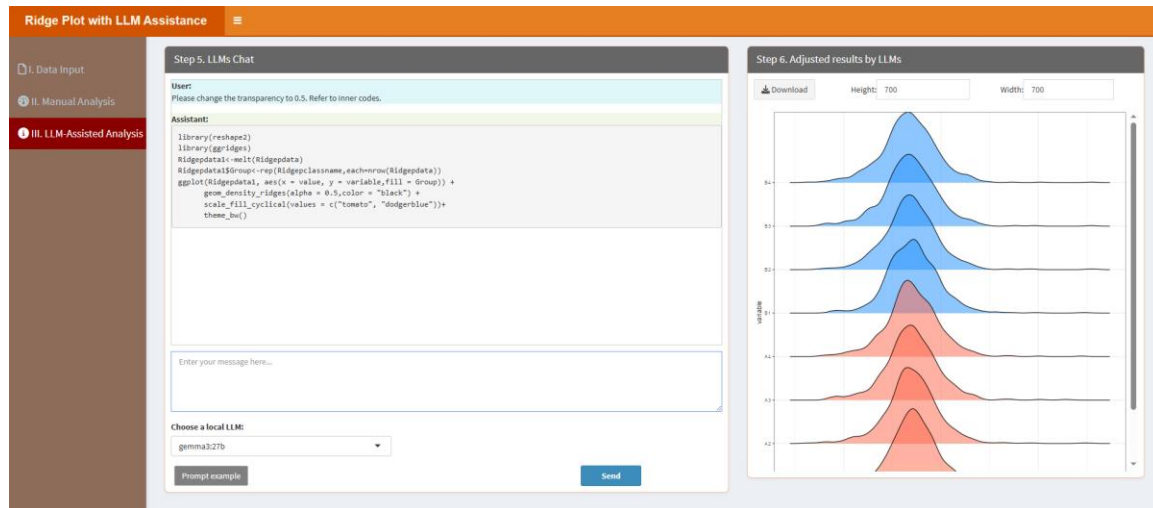
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the transparency to 0.5. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to

access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the transparency to 0.5. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.25. ROC plot

ROC plots evaluate classification model performance. Users can customize the plot through the classic framework and leverage local LLMs for accurate interpretation and visualization. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'ROC plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown menu shows '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the first column and 'No data loaded. Please upload your dataset or use the example data.' in the second column. At the bottom of the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'ROC plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three sections: '3.1. Select colors:' with a dropdown menu showing 'RANDOM', 'BLUE', and 'RED'; '3.2. Figure Height:' with a text input field containing '500'; and '3.3. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section '4.1. Visualization' with a 'Save Image' button.

3.1. Select colors: This parameter allows users to assign specific colors to each group or variable whose ROC curve will be displayed. The number of selected colors should match the number of groups or classification targets being compared.

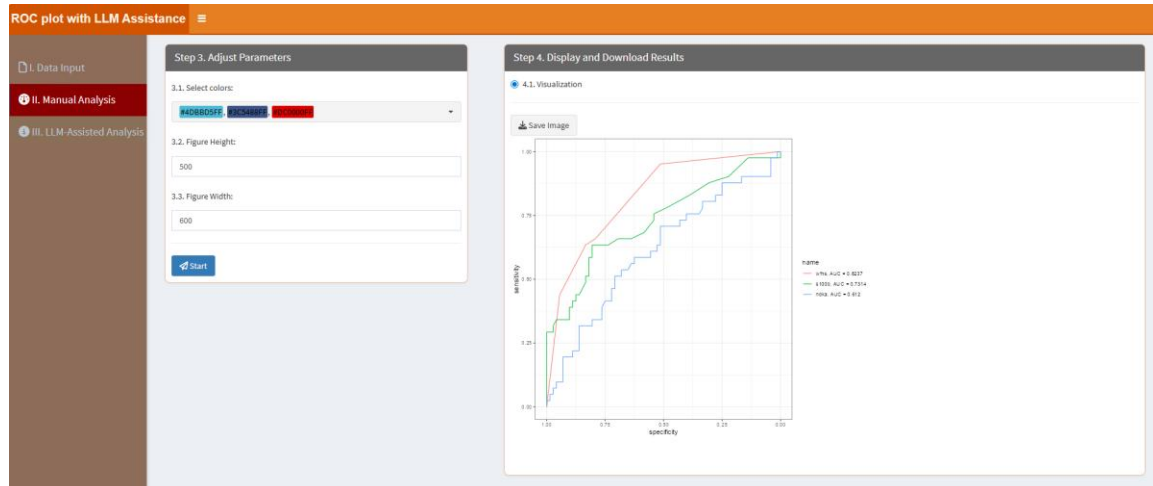
3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization,

ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

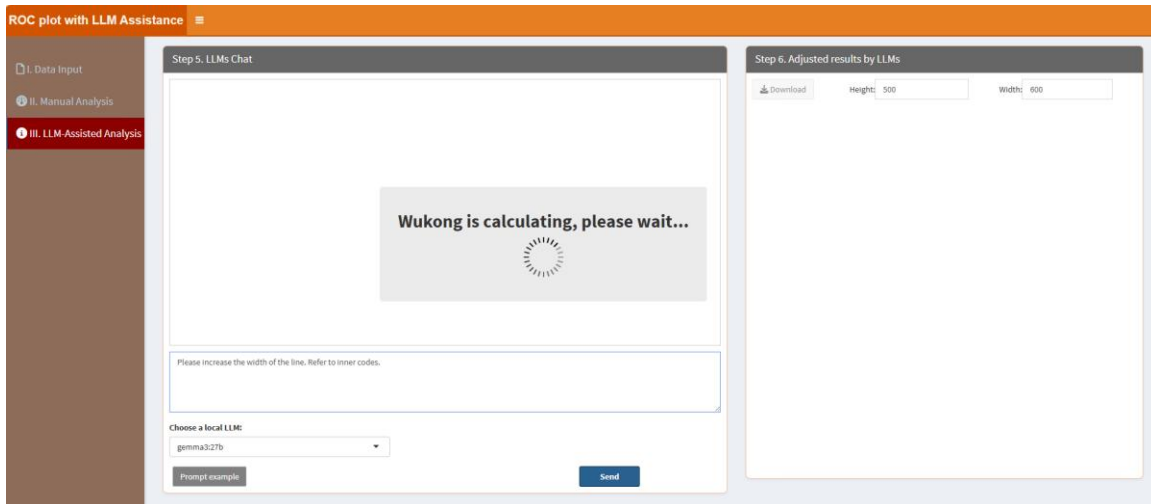
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



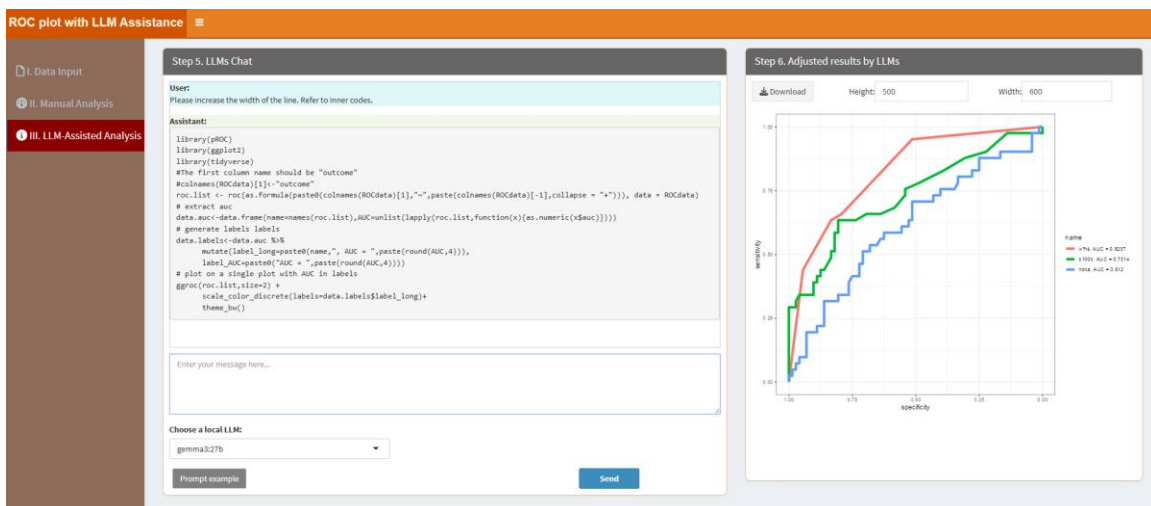
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please increase the width of the line. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please increase the width of the line. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.26. Sankey Chart

Sankey charts visualize data flows and relationships. Users can adjust parameters in the classic framework and refine the plot with local LLMs for engaging and accurate visualizations. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Sankey Chart with LLM Assistance' application. On the left is a sidebar with three main sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The 'I. Data Input' section is active. The main area is divided into two panels. The left panel, titled 'Step 1: Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. It also has a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are two checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. The right panel, titled 'Step 2: Display Uploaded Data/Example Data', shows a table with one entry. The table has columns 'Show' (set to 10) and 'entries'. The 'Description' column contains the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries'. There are 'Previous' and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Sankey Chart with LLM Assistance' application. The sidebar is the same. The main area is divided into two panels. The left panel, titled 'Step 3: Adjust Parameters', contains a '3.1. Select a palette:' section with a color palette dropdown set to 'viridis'. Below this are two input fields: '3.1. Figure Height:' with a value of 500, and '3.2. Figure Width:' with a value of 600. At the bottom is a blue 'Start' button. The right panel, titled 'Step 4: Display and Download Results', shows a section for '4.1. Visualization' with a 'Save Image' button.

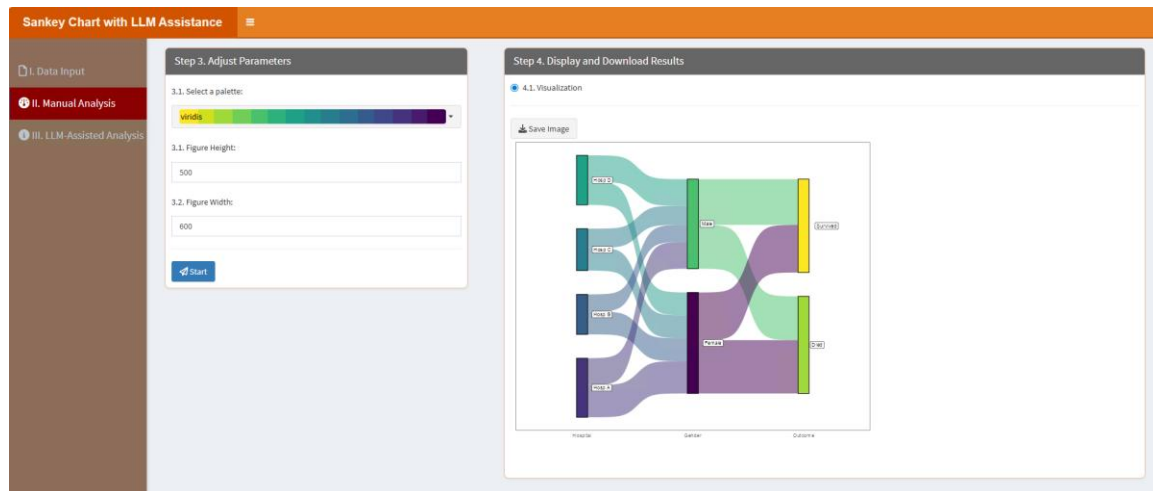
3.1. *Select a palette*: Choose a palette for the sankey chart.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

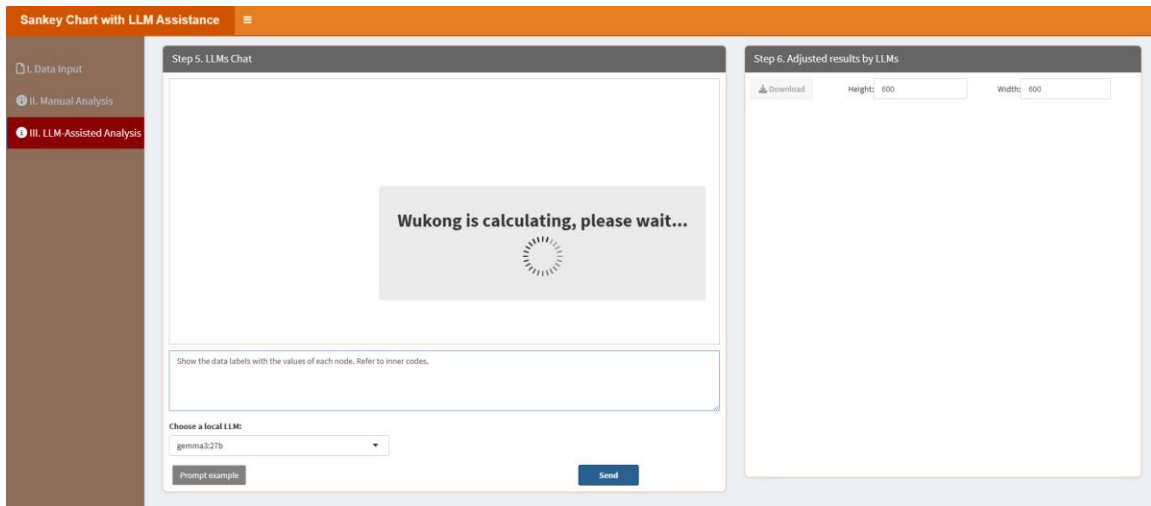
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



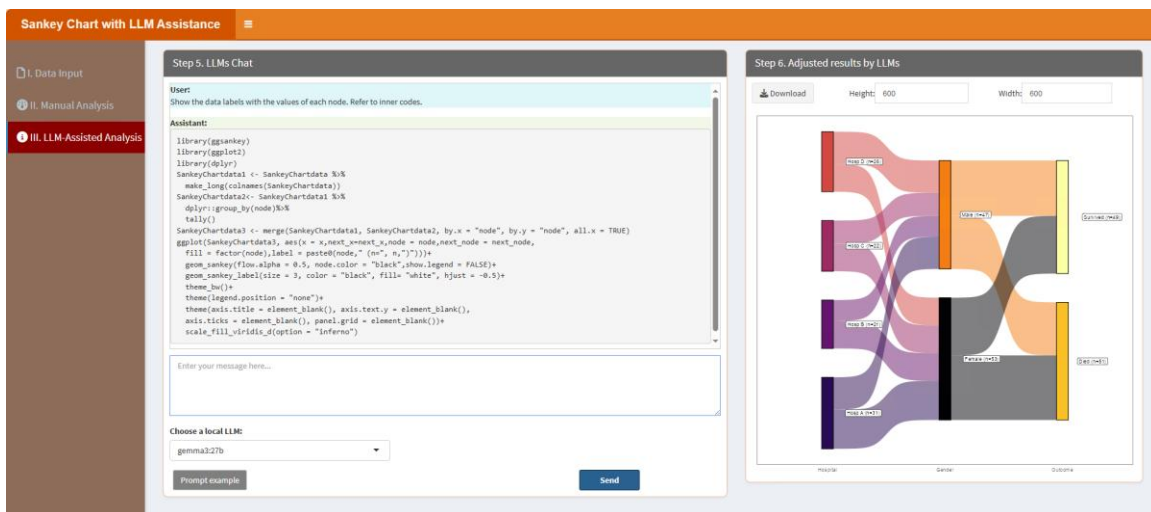
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Show the data labels with the values of each node. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Show the data labels with the values of each node. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.27. Scatter plot with ellipses

Scatter plots with ellipses highlight data groupings. Users can customize the plot through the classic framework and enhance it with local LLMs for better insights. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' section of the application. On the left, a sidebar contains three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this, there are file format selection buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Browse...' button is present, along with checkboxes for 'Is the first row names?' and 'Is the first column names?'. A 'Which Sheet to read?' dropdown is set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one row and one column. The table header is 'Description' and the row contains the text 'No data loaded. Please upload your dataset or use the example data.'.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' section of the application. The sidebar on the left now has 'II. Manual Analysis' selected. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains five numbered input fields: '3.1. Group and replicate number:' (with value '2;5-5'), '3.2. Group names:' (with value 'R;C'), '3.3. Select colors:' (with a dropdown menu showing 'R;C;B;G;Y'), '3.4. Figure Height:' (with value '600'), and '3.5. Figure Width:' (with value '600'). A 'Start' button is at the bottom. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

3.3. Select colors: This parameter lets you choose specific colors for each group. The color

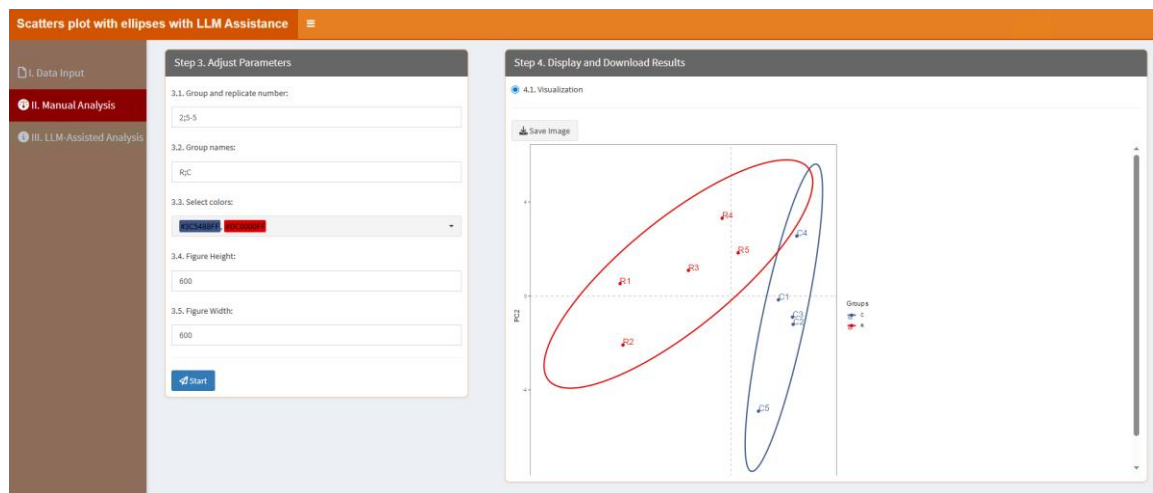
number should be equal to the group number.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

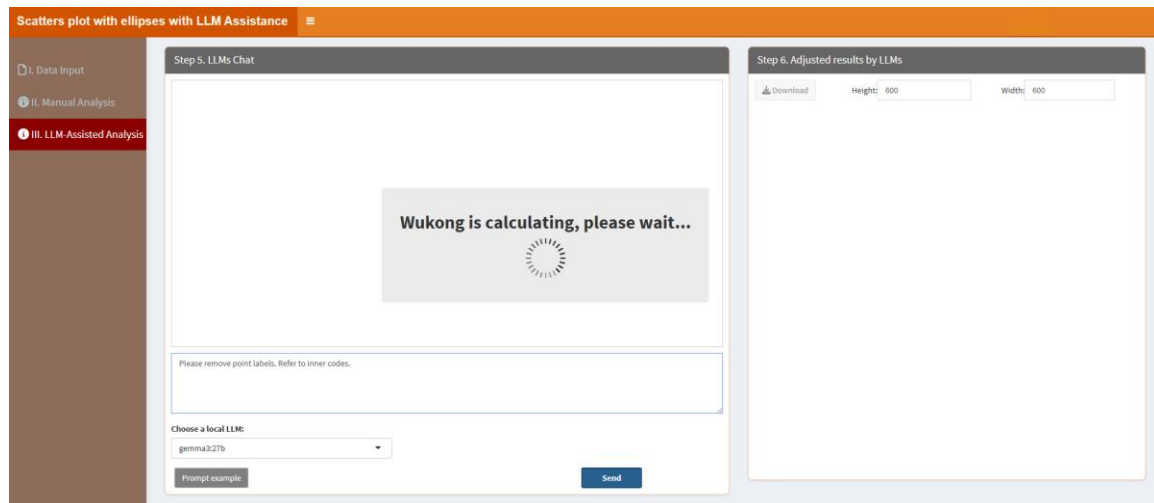


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

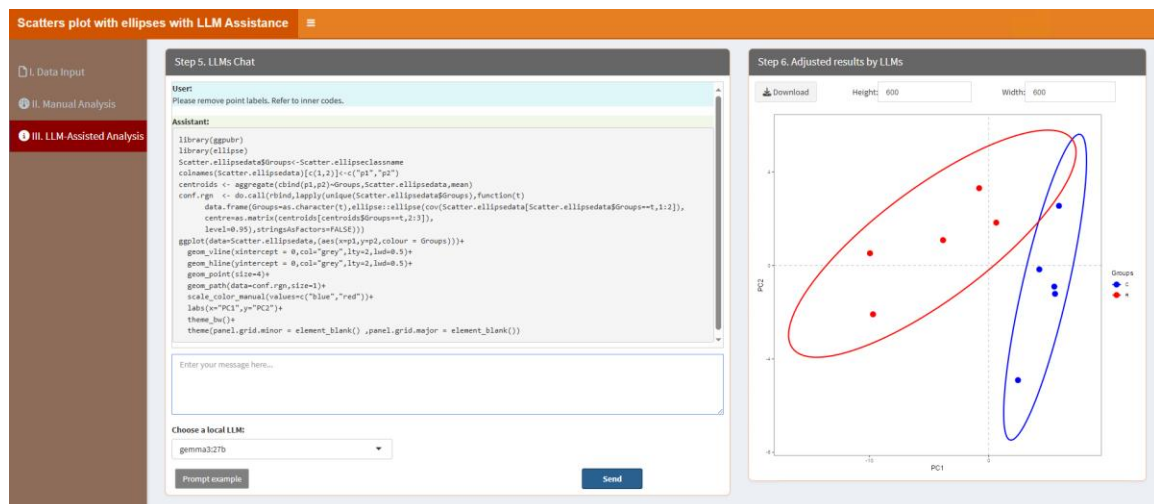
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please remove point labels. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and

utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please remove point labels. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.28. Survival Plot

Survival plots analyze time-to-event data. Users can adjust parameters in the classic framework and refine the plot with local LLMs for clear and accurate visualizations. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Survival plot with LLM Assistance' interface. On the left is a sidebar with three main sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The 'I. Data Input' section is active. The main panel is divided into two steps. 'Step 1: Upload Data/Example Data' has two radio buttons: 'A. Upload' (selected) and 'B. Load example data'. Below these are options for file format: '.xlsx', '.xls', and '.csv/txt'. A 'Browse...' button is present, with 'No file selected' below it. There are checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?'. A 'Which Sheet to read?' field has '1' entered. 'Step 2: Display Uploaded Data/Example Data' shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the first column and 'No data loaded. Please upload your dataset or use the example data.' in the second column. Navigation buttons 'Previous', '1', and 'Next' are at the bottom right of the table.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Survival plot with LLM Assistance' interface, specifically 'Step 3: Adjust Parameters'. The sidebar is the same as in the previous screenshot, but 'II. Manual Analysis' is now selected. The main panel shows four numbered parameters: '3.1. Formula:' with a text input containing 'Surv(time, status) ~ sex'; '3.2. Select colors:' with a dropdown menu showing 'Blue/Red'; '3.3. Figure Height:' with a text input containing '600'; and '3.4. Figure Width:' with a text input containing '600'. A 'Start' button is at the bottom left of the parameter section. 'Step 4: Display and Download Results' is partially visible on the right, showing a '4.1. Visualization' section with a 'Save Image' button.

3.1. Formula: This parameter allows users to define the model formula for survival analysis, typically in the format *Surv(time, status) ~ group*.

3.2. Select colors: This parameter lets you choose specific colors for each group. The color number should be equal to the group number.

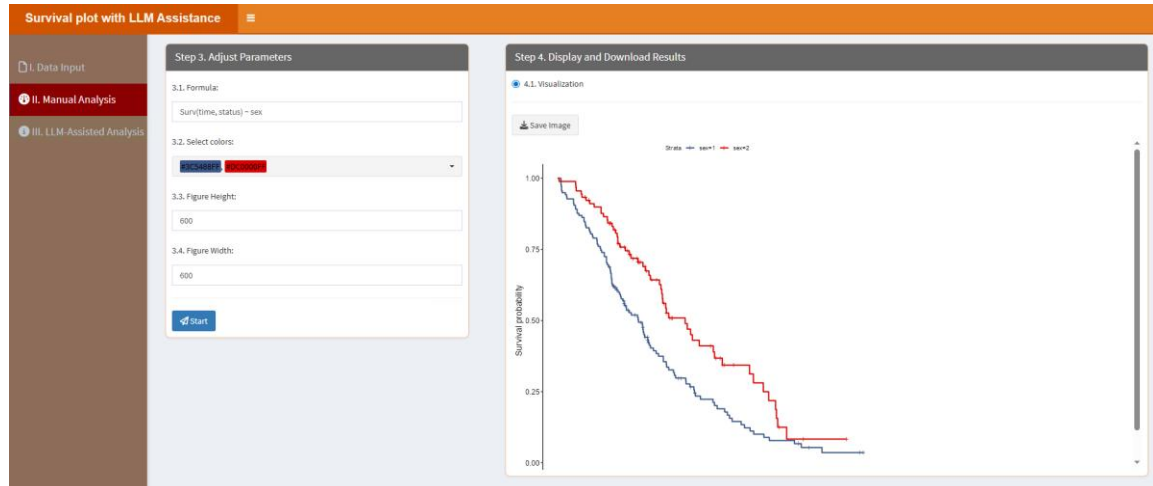
3.3. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization,

ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

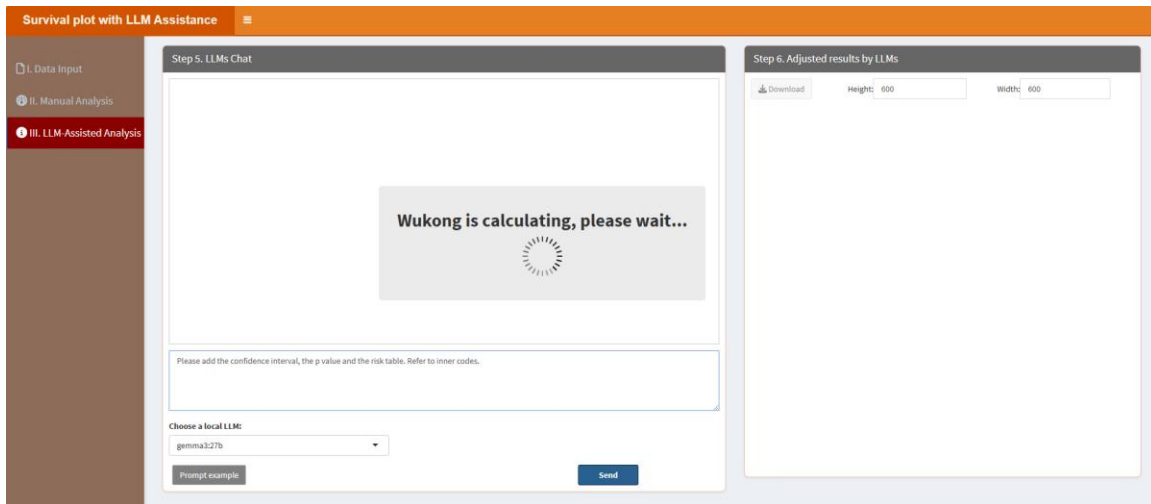
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



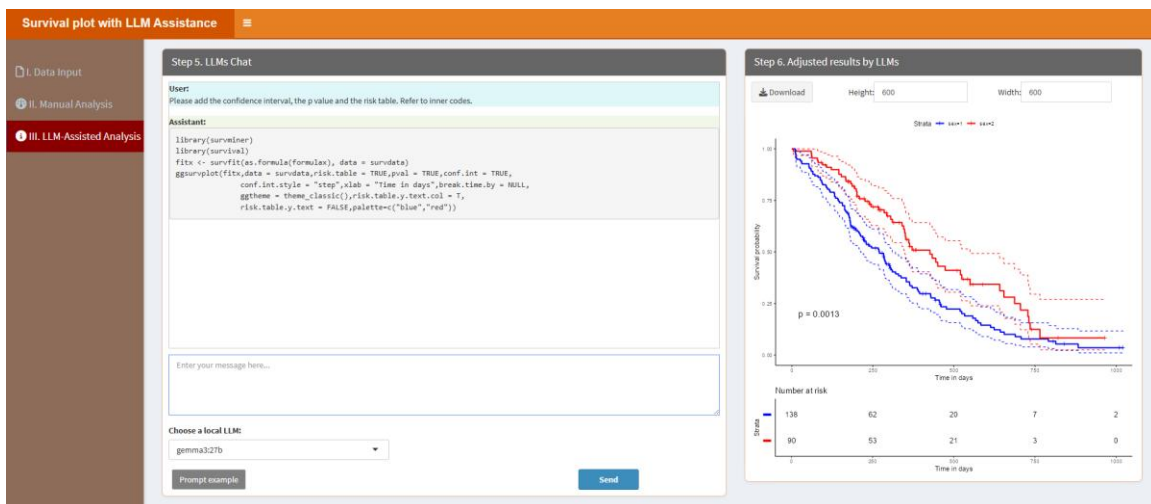
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please add the confidence interval, the p value and the risk table. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please add the confidence interval, the p value and the risk table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.29. Ternary Plot

Ternary plots visualize three-component data. Users can customize the plot through the classic framework and refine it using local LLMs for accurate and visually appealing results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Step 1. Upload Data/Example Data' panel of the 'Ternary plot with LLM Assistance' application. On the left is a sidebar with three menu items: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main panel has two tabs: 'A. Upload' (selected) and 'B. Load example data'. Under 'A. Upload', there is a 'Select File Format:' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also two checkboxes: 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked). At the bottom is a 'Which Sheet to read?' input field with the value '1'. The right panel, 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1. No data loaded. Please upload your dataset or use the example data.' Below the table is a pagination bar showing 'Showing 1 to 1 of 1 entries' with 'Previous', '1', and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Step 3. Adjust Parameters' panel of the 'Ternary plot with LLM Assistance' application. The sidebar is the same as in the previous screenshot. The main panel has two tabs: 'A. Upload' and 'B. Load example data'. Under 'A. Upload', there is a '3.1. Select colors:' section with a dropdown menu showing 'A: RED, B: BLUE, C: GREEN'. Below this is a '3.2. Figure Height:' input field with the value '600'. Below that is a '3.3. Figure Width:' input field with the value '600'. At the bottom is a blue 'Start' button. The right panel, 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

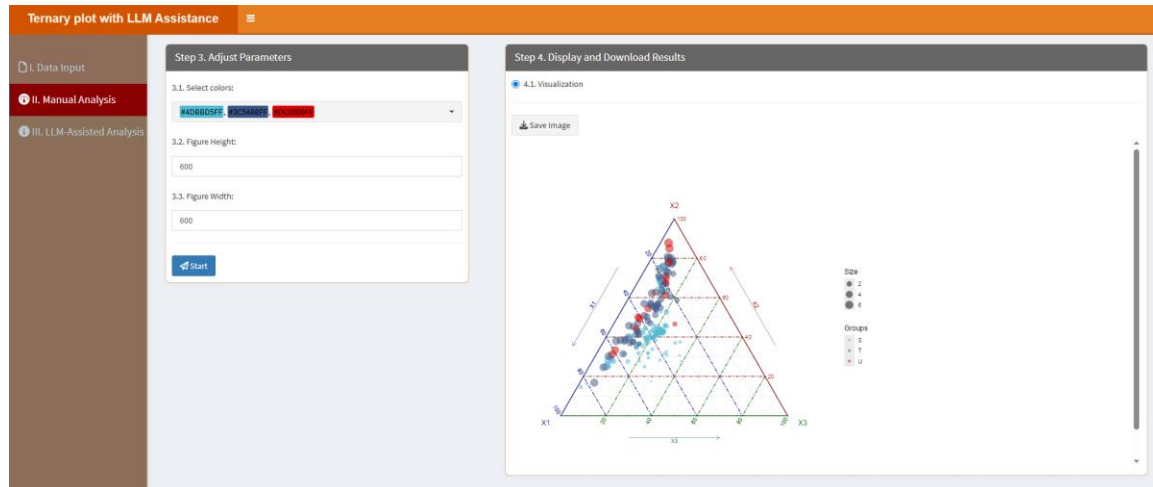
3.1. Select colors: This parameter lets you choose specific colors for each group. The color number should be equal to the group number.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

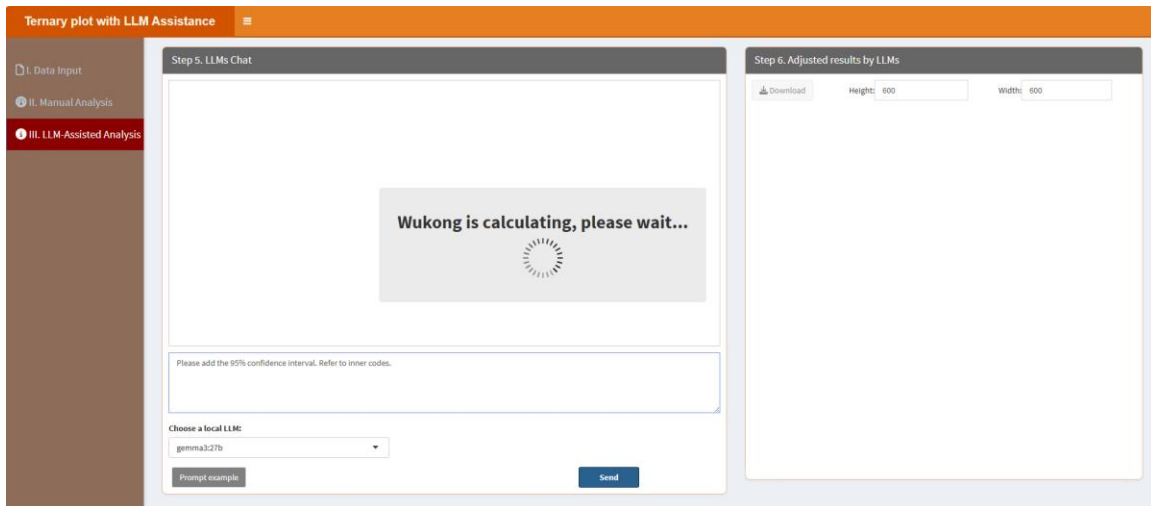
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



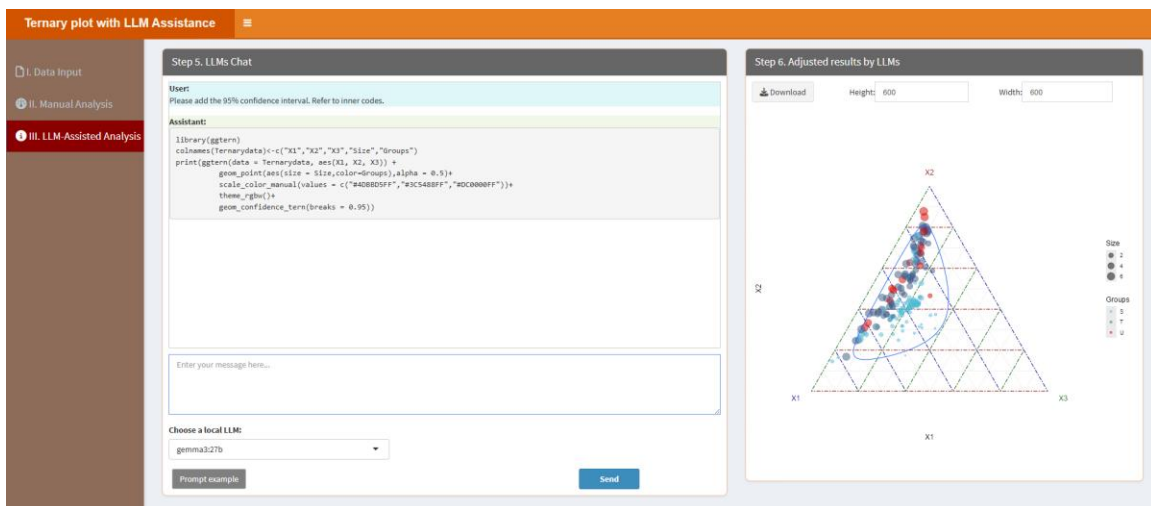
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please add the 95% confidence interval. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please add the 95% confidence interval. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.30. UpSet plot

UpSet plots visualize intersections between sets. Users can adjust parameters in the classic framework and refine the visualization with local LLMs for engaging and accurate results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'UpSet plot with LLM Assistance' interface. On the left is a sidebar with three sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?' (unchecked), and a 'Which Sheet to read?' dropdown menu set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the first column and 'No data loaded. Please upload your dataset or use the example data.' in the second column. At the bottom of the table are 'Previous', '1', and 'Next' buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'UpSet plot with LLM Assistance' interface. On the left is a sidebar with three sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains five numbered settings: '3.1. Select colors for bars:' with a dropdown menu showing 'FF0000' and 'FF0000'; '3.2. Select a color for points:' with a dropdown menu showing '408080FF'; '3.3. Select a color for lines:' with a dropdown menu showing '000000'; '3.4. Figure Height:' with a text input field containing '550'; and '3.5. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

3.1. Select colors for bars: This parameter lets users assign specific colors to the horizontal bars that represent the size of each individual set.

3.2. Select a color for points: This setting determines the color of the dots in the intersection matrix, where each row represents a set and each column shows a specific intersection.

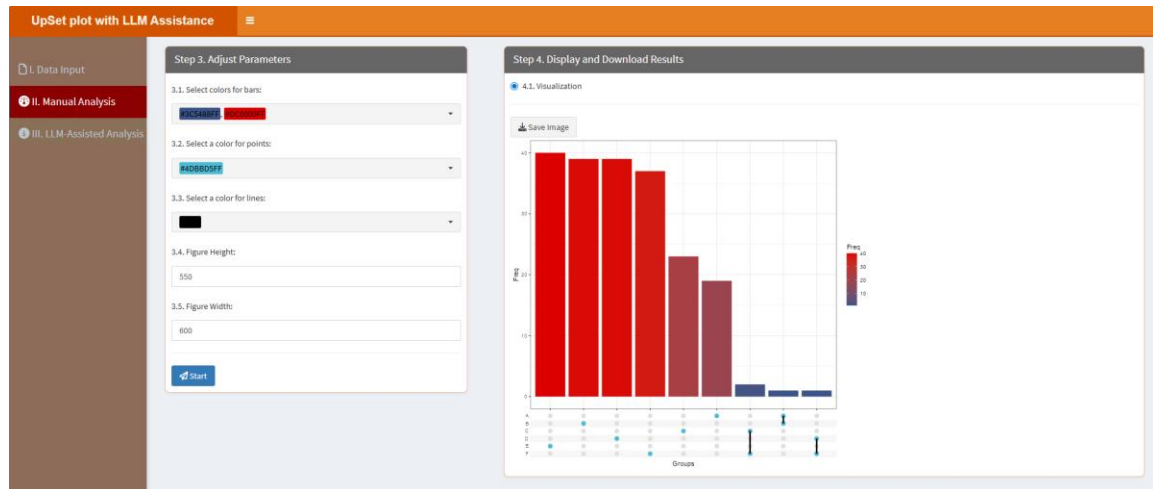
3.3. Select a color for lines: This parameter allows users to define the color of the vertical lines that connect the intersection points within each column.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

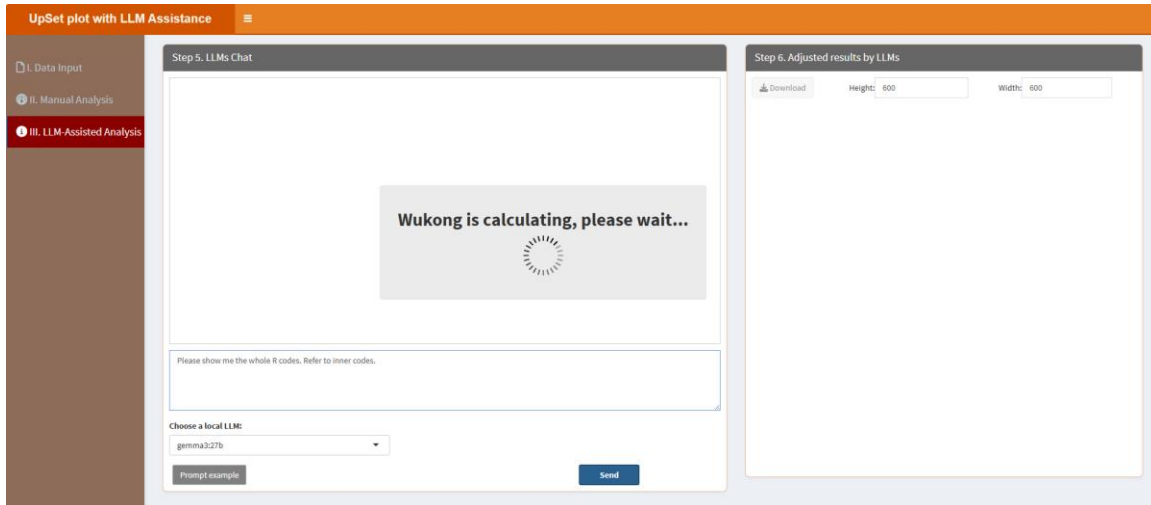
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



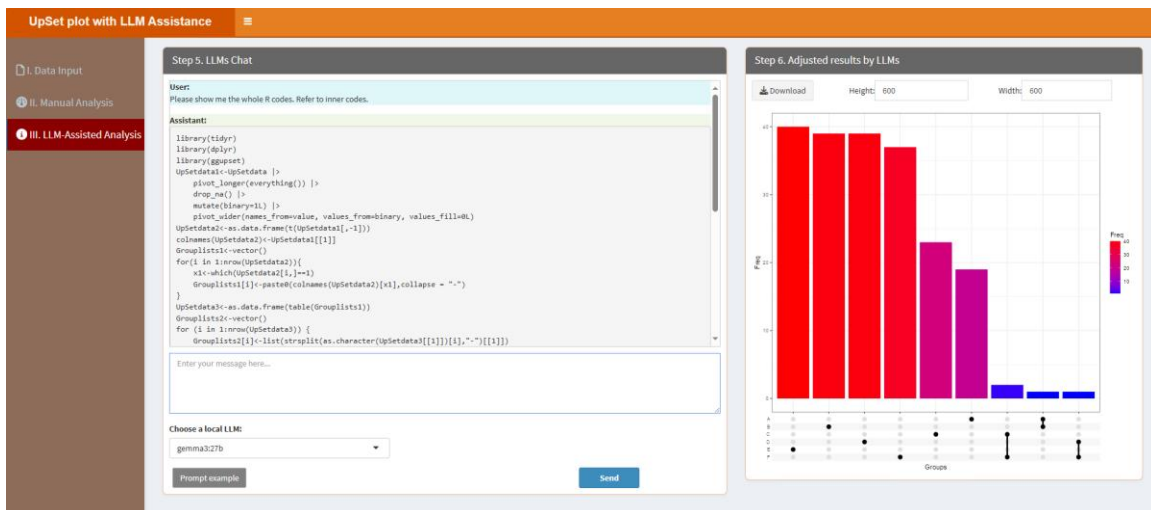
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please show me the whole R codes. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please show me the whole R codes. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.31. Venn plot

Venn plots display overlaps between sets. Users can customize the plot through the classic framework and enhance it with local LLMs for clear and informative diagrams. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Venn plot with LLM Assistance' interface. On the left is a sidebar with three sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this is a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. A 'Please import your data file:' section includes a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' (checked) and 'Is the first column names?'. A 'Which Sheet to read?' dropdown is set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the 'Description' column with the text 'No data loaded. Please upload your dataset or use the example data.' Below the table is a 'Showing 1 to 1 of 1 entries' message and 'Previous', '1', and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Venn plot with LLM Assistance' interface. The sidebar is the same as in the previous screenshot. The main area has two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three sections: '3.1. Select colors for each group:' with a dropdown menu showing 'RANDOM COLORS', 'RANDOM', and 'RANDOM'; '3.2. Figure Height:' with a text input field containing '550'; and '3.3. Figure Width:' with a text input field containing '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section '4.1. Visualization' with a 'Save Image' button.

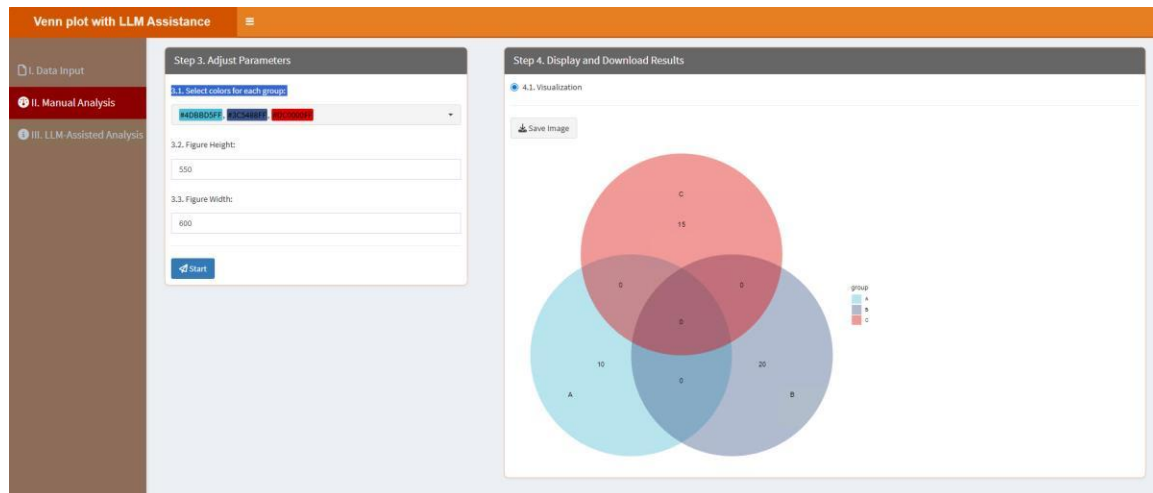
3.1. Select colors for each group: This parameter lets you choose specific colors for each group. The color number should be equal to the group number.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

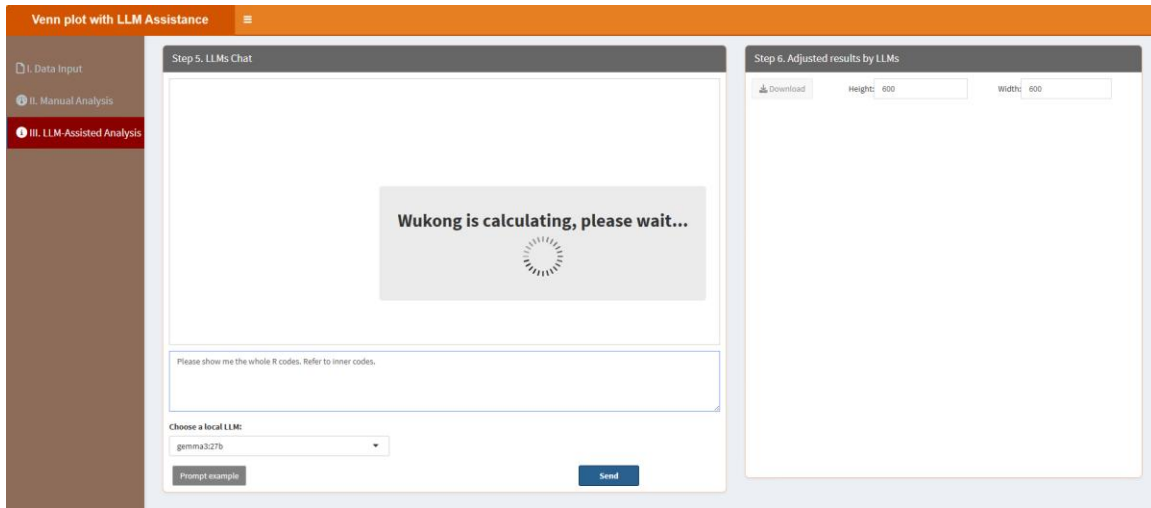
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



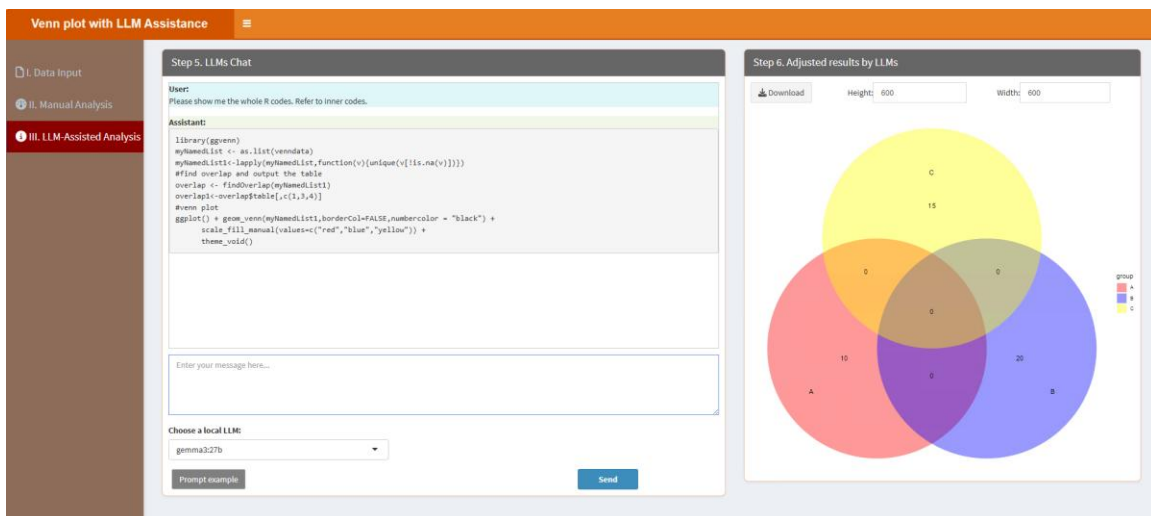
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please show me the whole R codes. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please show me the whole R codes. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.32. Violin Plot

Violin plots visualize data distributions. Users can adjust parameters in the classic framework and refine the plot with local LLMs for detailed and interpretable results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Violin Plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options to 'A. Upload' or 'B. Load example data'. Below this, it asks for the 'Select File Format' (.xlsx, .xls, or .csv/txt) and provides a 'Browse...' button. It also has checkboxes for 'Is the first row names?' and 'Is the first column names?', and a 'Which Sheet to read?' dropdown set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a table with one row and one column, indicating 'No data loaded. Please upload your dataset or use the example data.' It includes a 'Show' dropdown set to '10 entries', a 'Search' bar, and pagination controls.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Violin Plot with LLM Assistance' interface. On the left is a sidebar with three options: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains five numbered input fields: '3.1. Group and replicate number:' (with '2;4-4' entered), '3.2. Group names:' (with 'A;B' entered), '3.3. Select colors for each group:' (with a color picker showing blue and red), '3.4. Figure Height:' (with '550' entered), and '3.5. Figure Width:' (with '600' entered). There is a 'Start' button at the bottom. The right panel, titled 'Step 4. Display and Download Results', shows a '4.1. Visualization' section with a 'Save Image' button.

3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

3.3. Select colors for each group: This parameter lets you choose specific colors for each

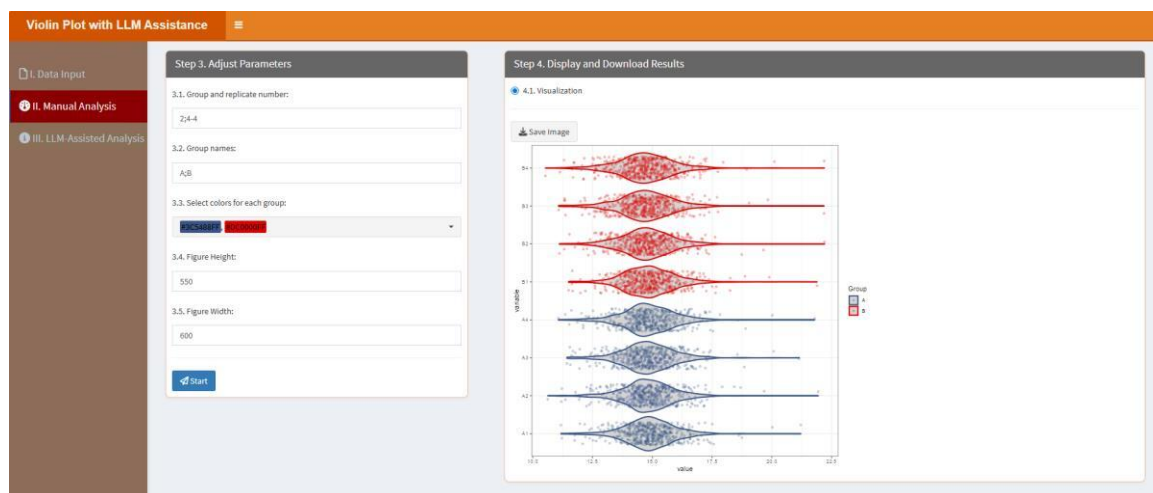
group. The color number should be equal to the group number.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

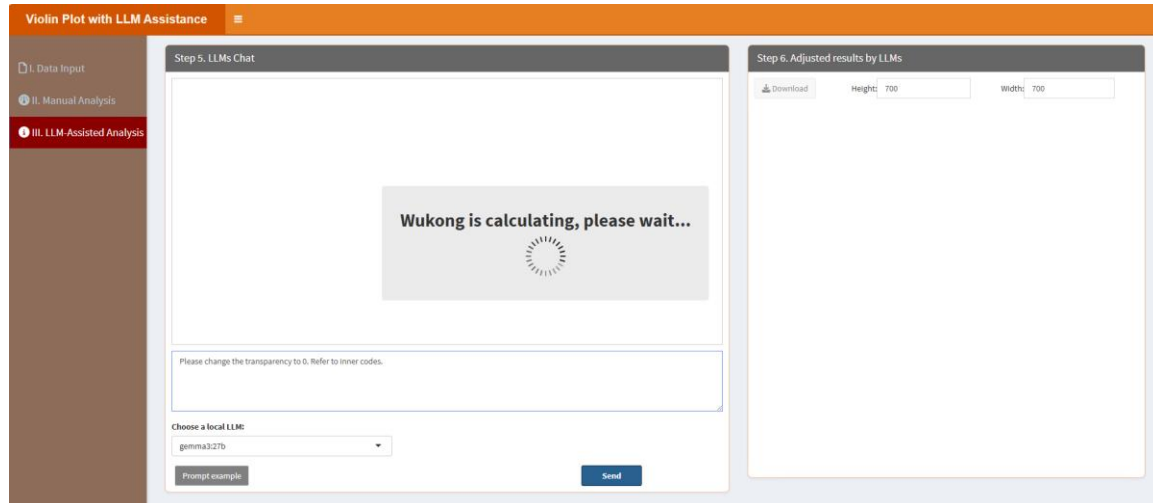


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

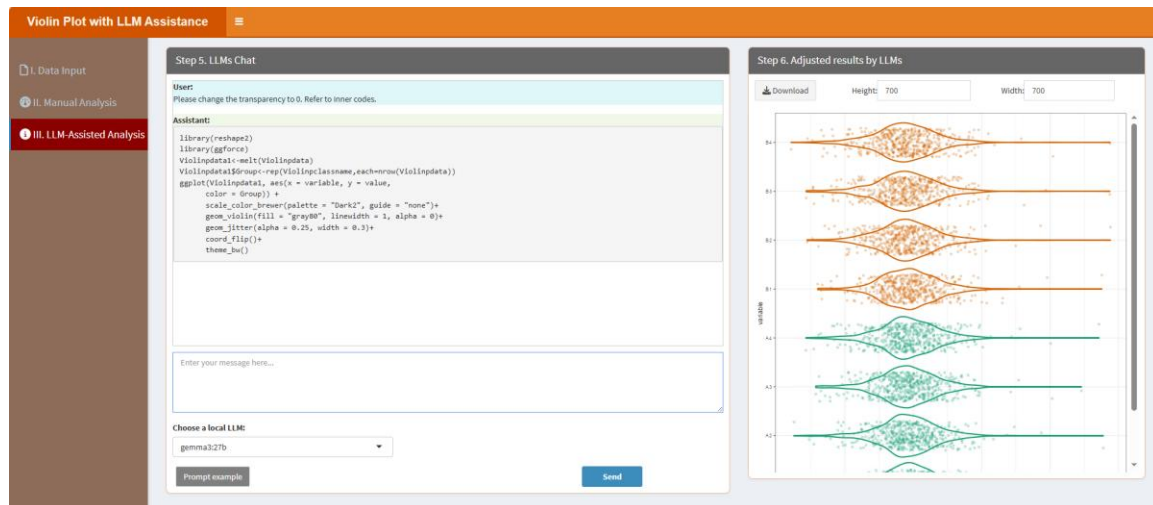
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the transparency to 0. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to

access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the transparency to 0. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.33. Volcano plot

Volcano plots identify significant changes in data. Users can customize the plot through the classic framework and refine it with local LLMs for clear and visually appealing results. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Volcano plot with LLM Assistance' interface. On the left is a sidebar with three main sections: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The 'I. Data Input' section is active. The main panel is divided into two steps. Step 1, 'Upload Data/Example Data', has two tabs: 'A. Upload' (selected) and 'B. Load example data'. Under 'A. Upload', there are options to 'Select File Format' (.xlsx, .xls, .csv/txt) and a 'Please import your data file:' section with a 'Browse...' button and 'No file selected' text. Below this are two checked checkboxes: 'Is the first row names?' and 'Is the first column names?'. At the bottom, there is a 'Which Sheet to read?' dropdown menu with '1' selected. Step 2, 'Display Uploaded Data/Example Data', is partially visible on the right, showing a 'Show 10 entries' dropdown, a search bar, and a table with one row: '1' in the 'Description' column with the text 'No data loaded. Please upload your dataset or use the example data.'.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Volcano plot with LLM Assistance' interface, Step 3: Adjust Parameters. The sidebar is the same as in the previous screenshot, but now 'II. Manual Analysis' is active. The main panel shows Step 3 with five numbered parameters: 3.1. Threshold for P value: 0.05; 3.2. Threshold for Fold Change: 1.5; 3.3. Select colors for each group: a dropdown menu showing 'grey80' and a color picker; 3.4. Figure Height: 600; 3.5. Figure Width: 600. At the bottom of the parameter list is a blue 'Start' button. Step 4, 'Display and Download Results', is partially visible on the right, showing tabs for '4.1. Visualization' (selected) and '4.2. Tables', and a 'Save Image' button.

3.1. Threshold for P value: This parameter allows users to define the cutoff for statistical significance. Variables with a p-value below this threshold will be considered significantly different between groups. Common values include 0.05.

3.2. Threshold for Fold Change: This parameter sets the minimum absolute fold change required for a variable to be considered significantly upregulated or downregulated.

3.3. Select colors for each group: This setting allows users to define distinct colors for each category of data point, typically "UP", "DOWN" and "NoChange".

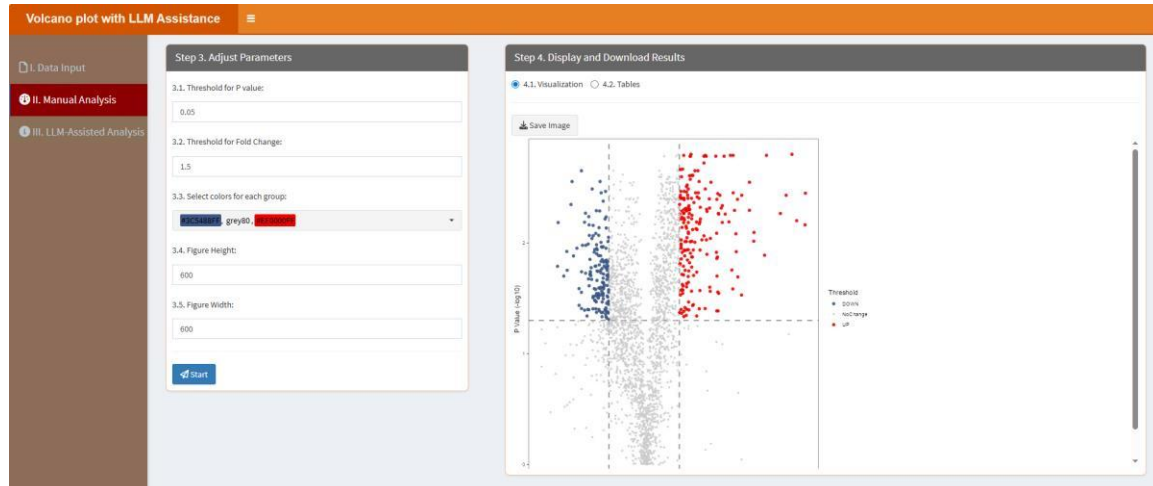
3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your

presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

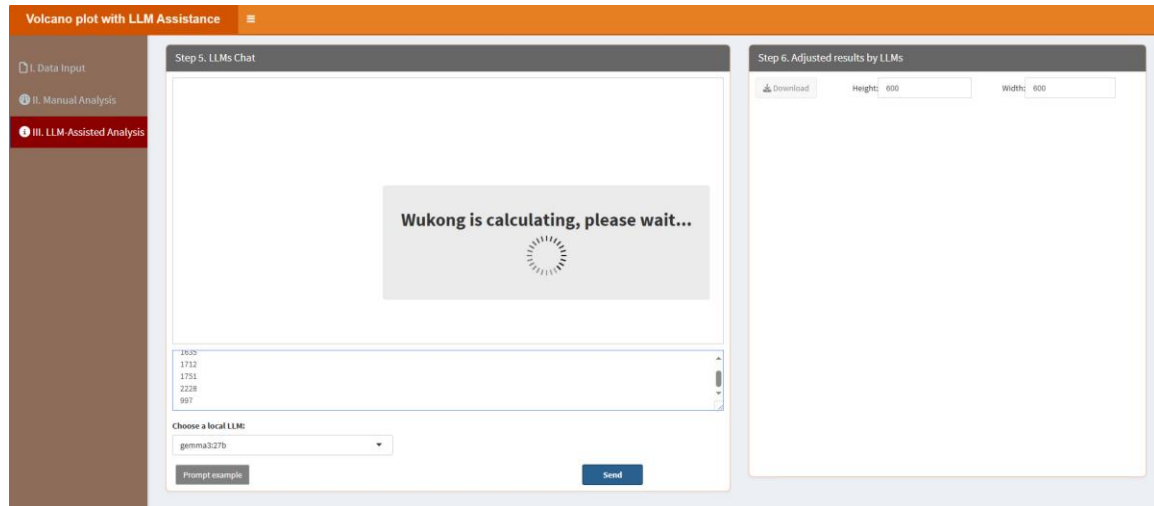
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



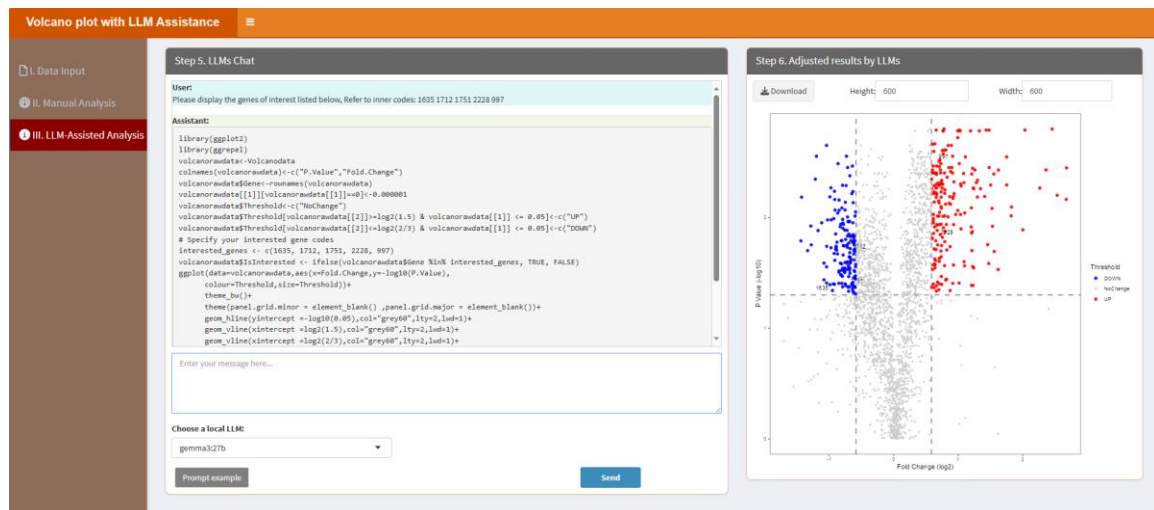
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please display the genes of interest listed below, Refer to inner codes:1635, 1712, 1751, 2228, 997.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please display the genes of interest listed below, Refer to inner codes:1635, 1712, 1751, 2228, 997. =>”, it will similarly prompt

LLMs to learn the internal R codes and produce the result table.
After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



7.34. Word cloud

Word clouds visualize text data. Users can adjust parameters in the classic framework and rely on local LLMs to generate aesthetically pleasing and informative visualizations. After clicking the “Start” button, it will be started in a new window like below:

The screenshot shows the 'Word cloud plot with LLM Assistance' interface. On the left is a sidebar with three sections: 'I. Data Input' (selected), 'II. Manual Analysis', and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 1. Upload Data/Example Data', contains options for 'A. Upload' (selected) and 'B. Load example data'. It includes a 'Select File Format' section with radio buttons for '.xlsx', '.xls', and '.csv/txt'. Below this is a 'Please import your data file:' section with a 'Browse...' button and a 'No file selected' message. There are also checkboxes for 'Is the first row names?' and 'Is the first column names?', both of which are checked. A 'Which Sheet to read?' dropdown menu is set to '1'. The right panel, titled 'Step 2. Display Uploaded Data/Example Data', shows a 'Show 10 entries' dropdown, a search bar, and a table with one row containing the text 'No data loaded. Please upload your dataset or use the example data.' Below the table, it says 'Showing 1 to 1 of 1 entries' and has 'Previous', '1', and 'Next' navigation buttons.

The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

The screenshot shows the 'Word cloud plot with LLM Assistance' interface. On the left is a sidebar with three sections: 'I. Data Input', 'II. Manual Analysis' (selected), and 'III. LLM-Assisted Analysis'. The main area is divided into two panels. The left panel, titled 'Step 3. Adjust Parameters', contains three numbered sections: '3.1. Select the shape of the cloud:' with a dropdown menu set to 'star'; '3.2. Figure Height:' with a text input field set to '600'; and '3.3. Figure Width:' with a text input field set to '600'. At the bottom of this panel is a blue 'Start' button. The right panel, titled 'Step 4. Display and Download Results', contains a section '4.1. Visualization' with a 'Save Image' button.

3.1. Select the shape of the clouds: This parameter allows users to choose the geometric layout of the word cloud. Options include star, circle, cardioid, diamond, and square, each offering a different aesthetic and spatial organization. For example, "circle" produces a balanced radial design, while "cardioid" creates a heart-shaped layout.

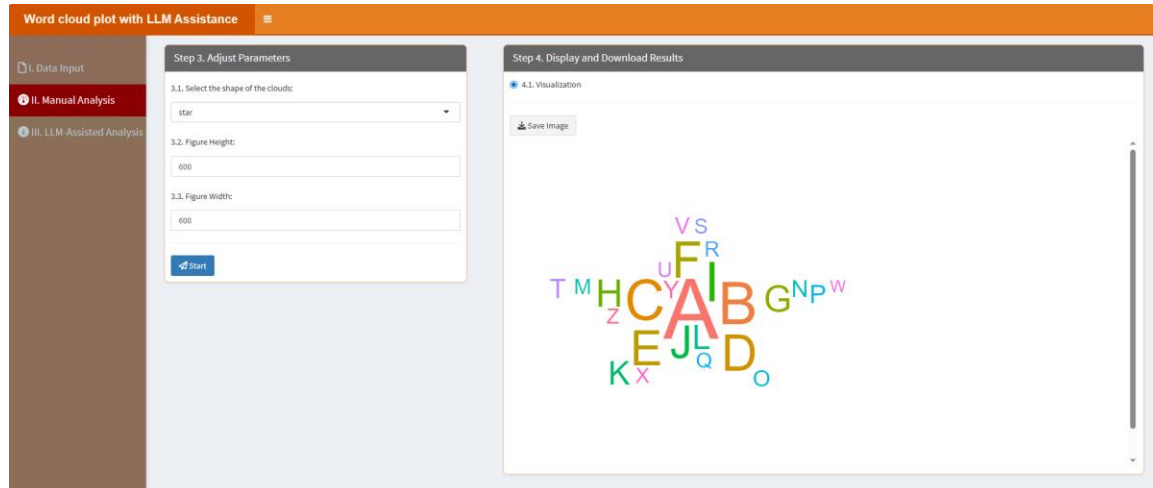
3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. *Figure Width*: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization,

ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

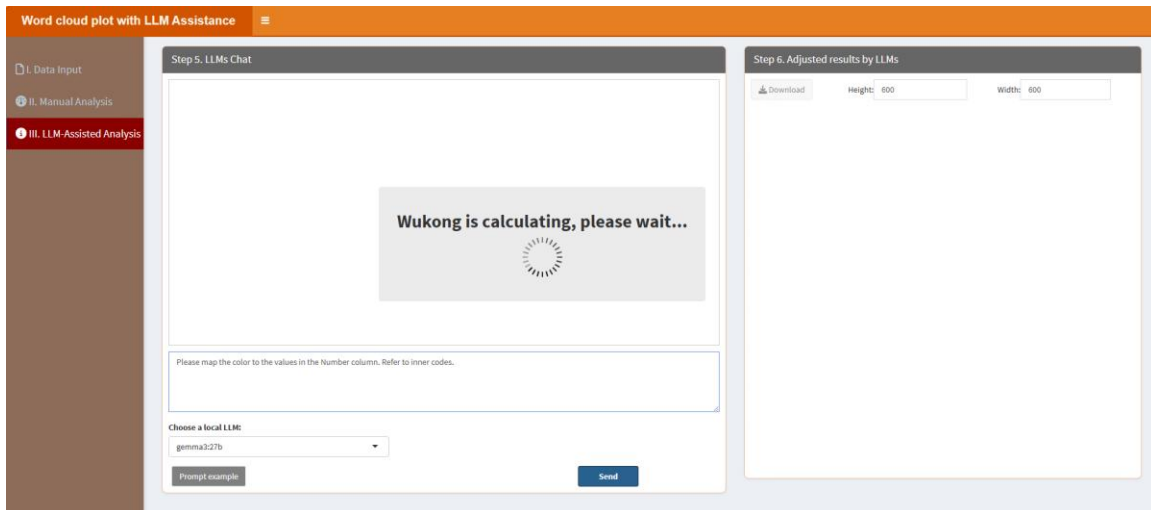
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



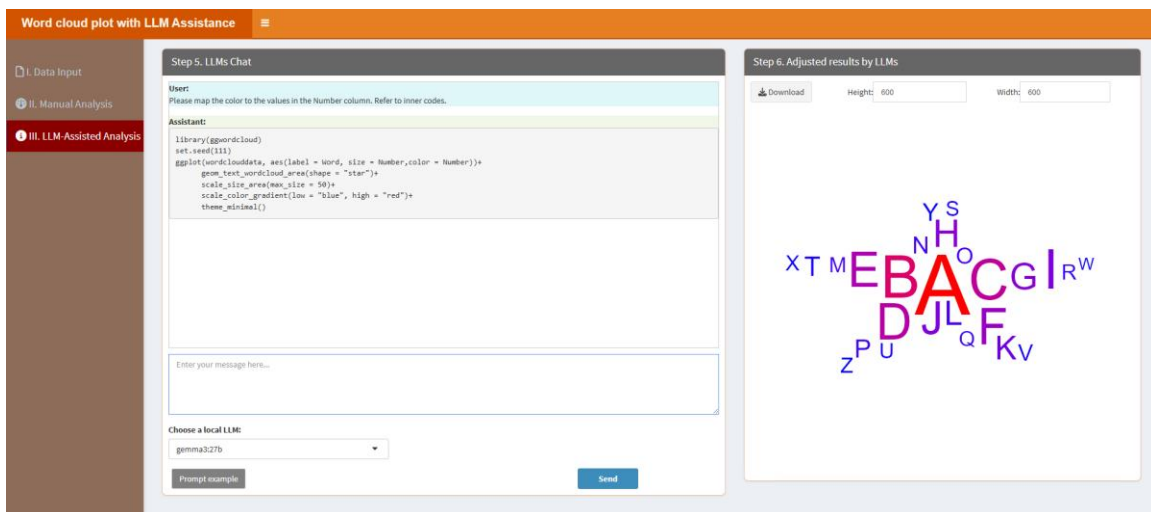
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section “Step 5. LLMs Chat” displays the conversation history, while “Enter your message here...” enables users to type messages or questions in the provided text box. Adjacent to the text box, the “Choose a local LLM” option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section “Step 6. Adjust results by LLMs” means that this panel shows the results (figures or tables) based on the conversation in the “Step 5. LLMs Chat”.

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please map the color to the values in the Number column. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please map the color to the values in the Number column. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



8. WuKongmini

The "WuKongmini" in the WuKong platform represents a significant leap forward in the way researchers approach complex proteomics data analysis. Unlike traditional and rigid analysis pipelines, this module provides a highly interactive, modular environment where users can construct, customize and execute multi-step analytical workflows tailored to their researches. The WuKongmini is intuitively organized into a stepwise interface: users begin by uploading their own dataset or selecting from example data, and then input sample information, including biological replications, group numbers, names and species. Once data is loaded, users are presented with a categorized gallery of analytical modules, spanning data preprocessing, statistical analysis, functional annotation and data visualization. By simply clicking on the desired modules, users assemble a sequential workflow, the order of which can be easily adjusted to reflect the logical progression of their analysis. Next, users could type a brief introduction about the assembled workflow for following LLMs to learn and restrict data analysis based on the brief description. For added convenience, we have built several example pipelines for users to learn how to build their own data analysis workflow and choose directly to analyze their data.

Once a workflow is constructed, users can select from a range of installed LLMs (such as gemma3:27b, gemma2:27b) and invoke the "Run Workflow" function. Please note, here we recommend users to install large-scale LLMs. The platform then automatically chains together the selected modules, passing outputs from one step as inputs to the next, and generates refined, executable R code that adheres strictly to best practices in scientific computing. The LLMs not only ensure code clarity and reproducibility but also provide stepwise explanations of input objects and workflow logic, making the process accessible to users with varying levels of programming expertise. Those without any programming background can ignore these R codes and only check the results. Results are presented in an organized fashion, with each step's outputs, including tables, plots and statistical summaries. All results can be downloaded into a zip file.

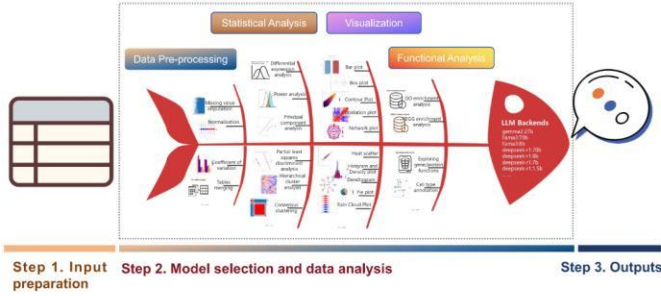
Below are the step-by-step instructions for using this module.

After installing the WuKong package (please refer to section 1 “How to run this tool locally?”), users can open the WuKong platform as shown below.


Home
Functions
Help

Welcome to WuKong Cloud Platform!

WuKong is a cutting-edge proteomics platform integrating over 72 analysis modules for data processing, visualization, and annotation. It combines classic frameworks with customizable parameters and intuitive workflows. By incorporating local large language models (LLMs), WuKong also enables conversational interactions, simplifying complex analyses and offering real-time guidance for researchers of all expertise levels.



Step 1. Input preparation **Step 2. Model selection and data analysis** **Step 3. Outputs**

Function Modules



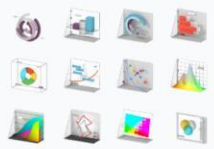
Conversation
Chat with local large language models that are deployed locally.

[Learn More](#)



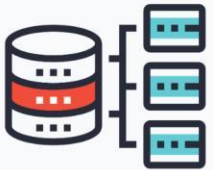
Data Pre-processing
Normalize data, handle missing values, calculate coefficients, and more.

[Learn More](#)



Statistical Analysis
Perform PCA, clustering, regression, PLS-DA, OPLS-DA, and more.

[Learn More](#)



Functional Analysis
ID conversion, enrichment analysis, custom library enrichment, and more.

[Learn More](#)

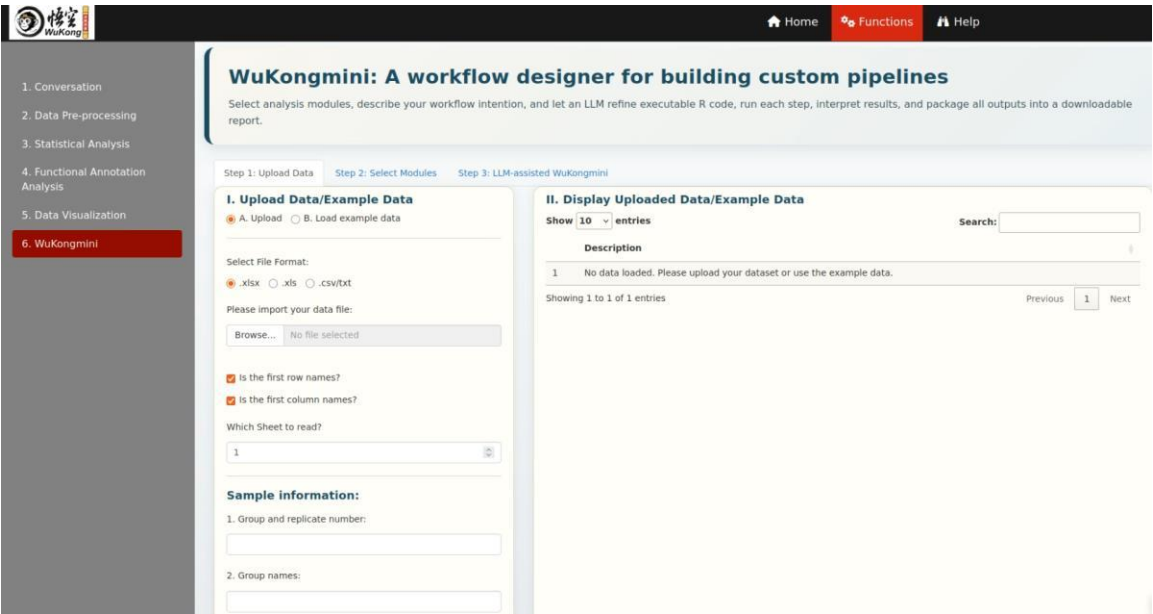


Data Visualization
Create diverse plots with ease. Click for details.

[Learn More](#)

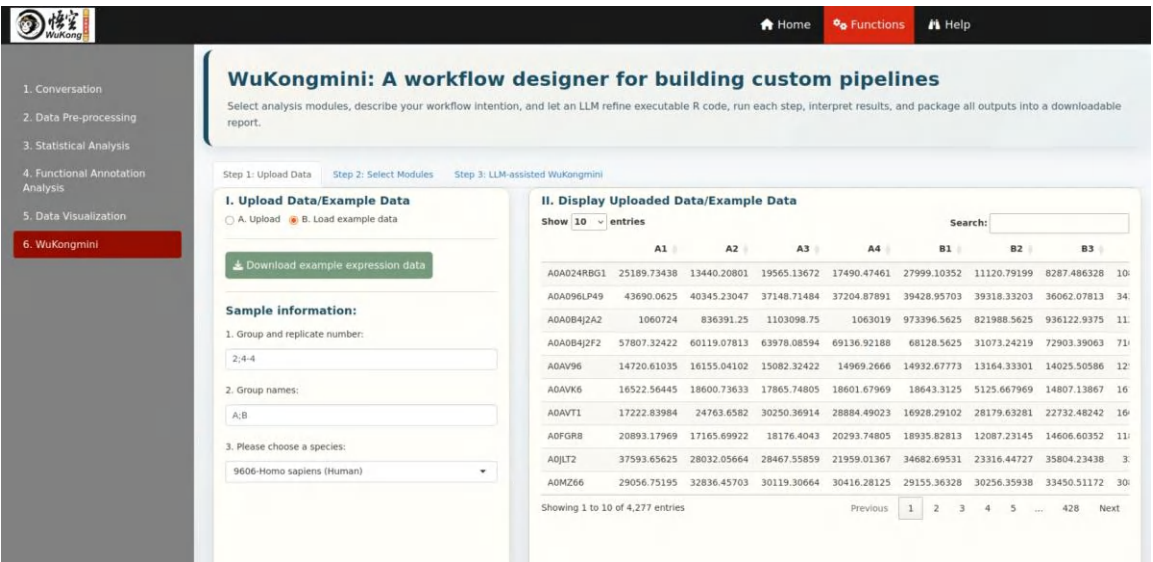
© 2025 WuKong Platform

Next, users can click “Functions” and locate the “6. WuKongmini” module. This module includes three main steps:



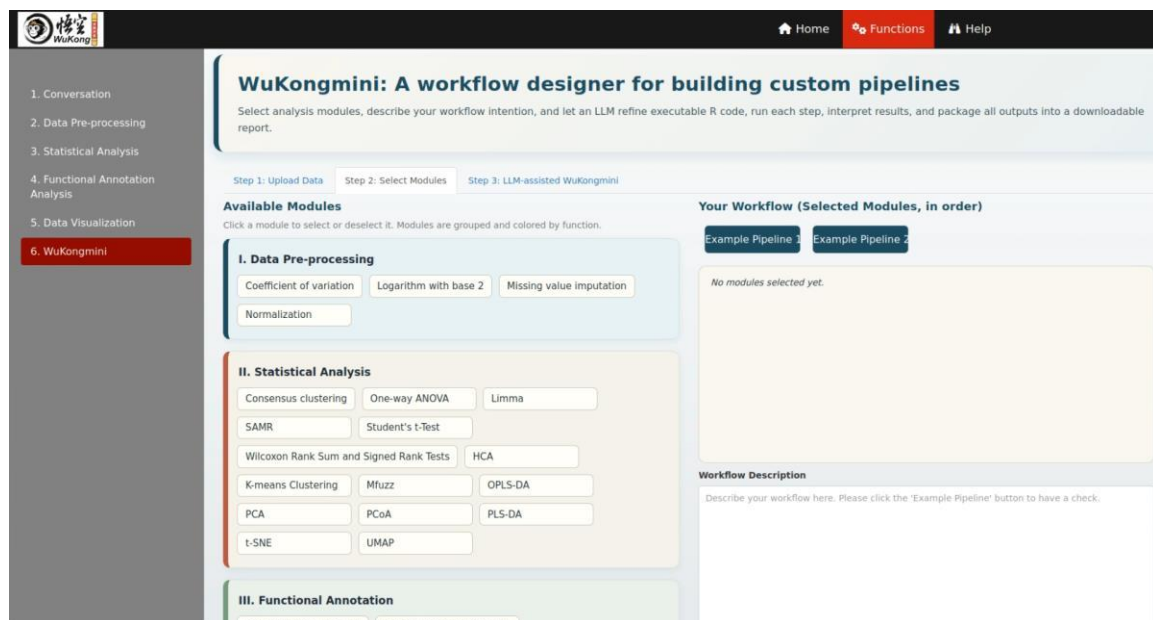
Step 1: Upload Data

In this step, users can upload their own data or load an example dataset. Please refer to section “2. Data Preparation” for details on how to format your data. If users choose “B. Load example data”, the platform displays an example dataset and corresponding sample information: this includes 4,277 proteins identified by UniProt IDs and 8 samples, divided into two groups (A and B), each with 4 biological replicates.

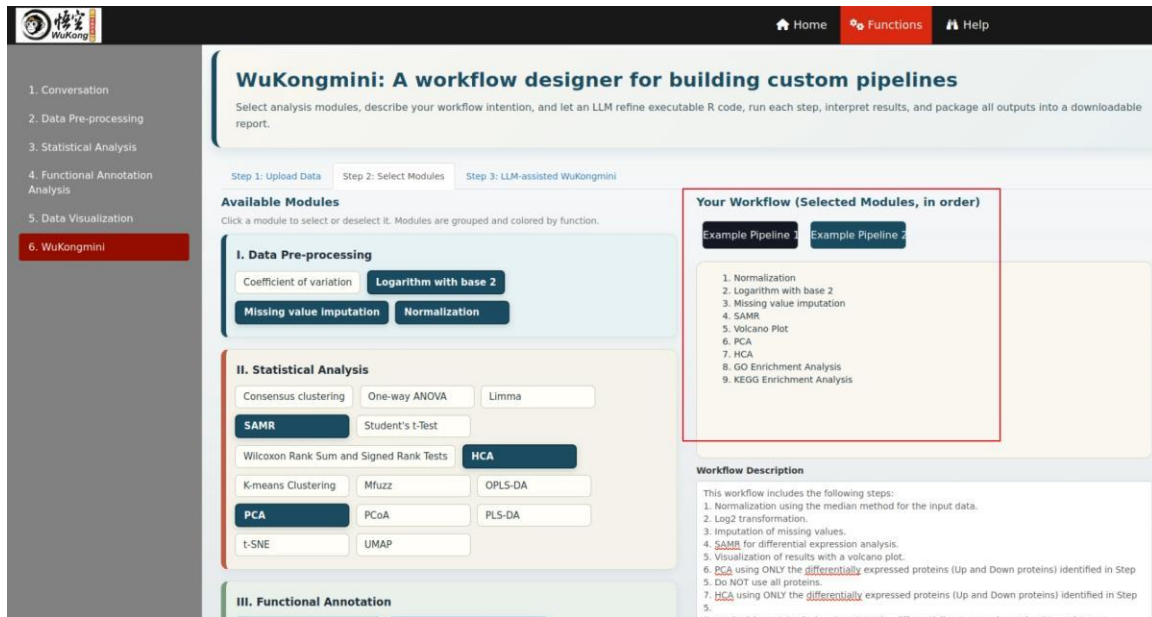


Step 2: Select Modules

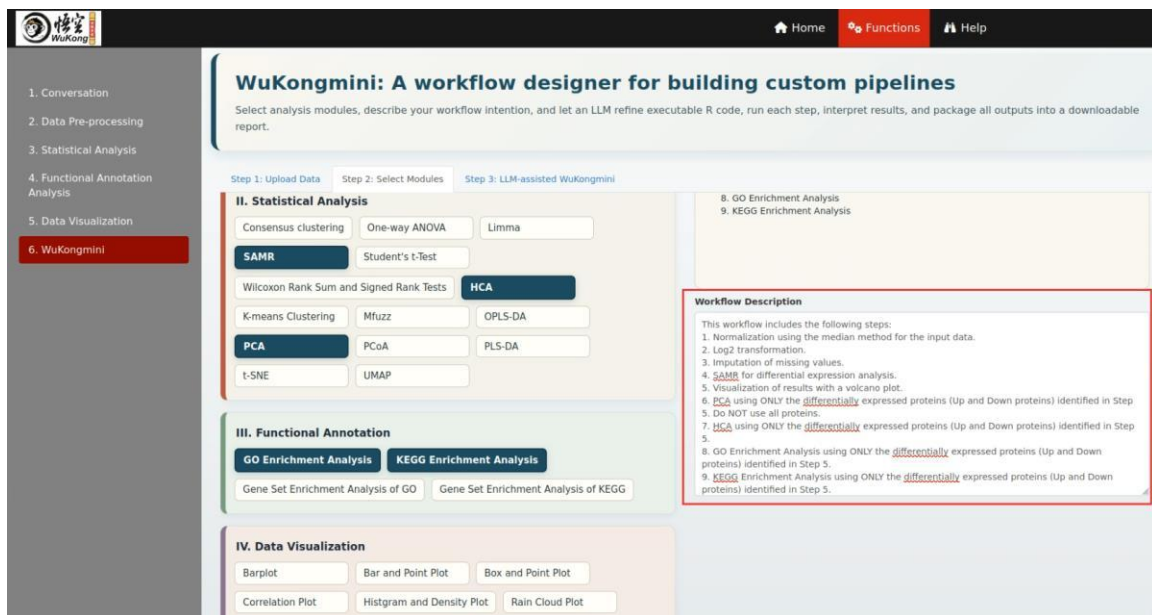
In this step, users can customize and combine various modules to process their data. These modules are categorized into four core areas as previously described: data preprocessing, statistical analysis, functional analysis and visualization. As shown below, the left panel displays all available modules, grouped and color-coded by category. Users can select any modules of interest, which will then appear in the right panel under “Your Workflow (Selected Modules, in order)”. In this section, users can also provide a brief description for each selected module in the “Workflow Description” field. These descriptions help guide the selected LLM in the next step; if left blank, the LLM will instead rely on the default R code associated with each module. Providing custom descriptions allows users to fine-tune the LLM’s behavior for more tailored analysis.



When a user clicks on a module, it is added to the “Your Workflow (Selected Modules, in order)” list on the right. Clicking the same module again will remove it from the list, allowing users to easily modify the workflow sequence.



After selecting the desired modules, users are encouraged to write descriptions for each one to help the LLM better understand the context and goals of the analysis.



For added convenience, users can click the “Example Pipeline 1” or “Example Pipeline 2” buttons to load pre-built, classical proteomics data analysis workflows. These examples serve as templates that users can run directly or study to learn how to construct their own workflows.

Clicking the “Example Pipeline 1” button loads a predefined 9-step workflow that includes Normalization, Logarithm with base 2, Missing value imputation, SAMR, Volcano Plot, PCA,

HCA, GO Enrichment Analysis, KEGG Enrichment Analysis. This pipeline comes with a sample workflow description, which helps the selected LLM in Step 3 understand and execute the analysis more effectively.

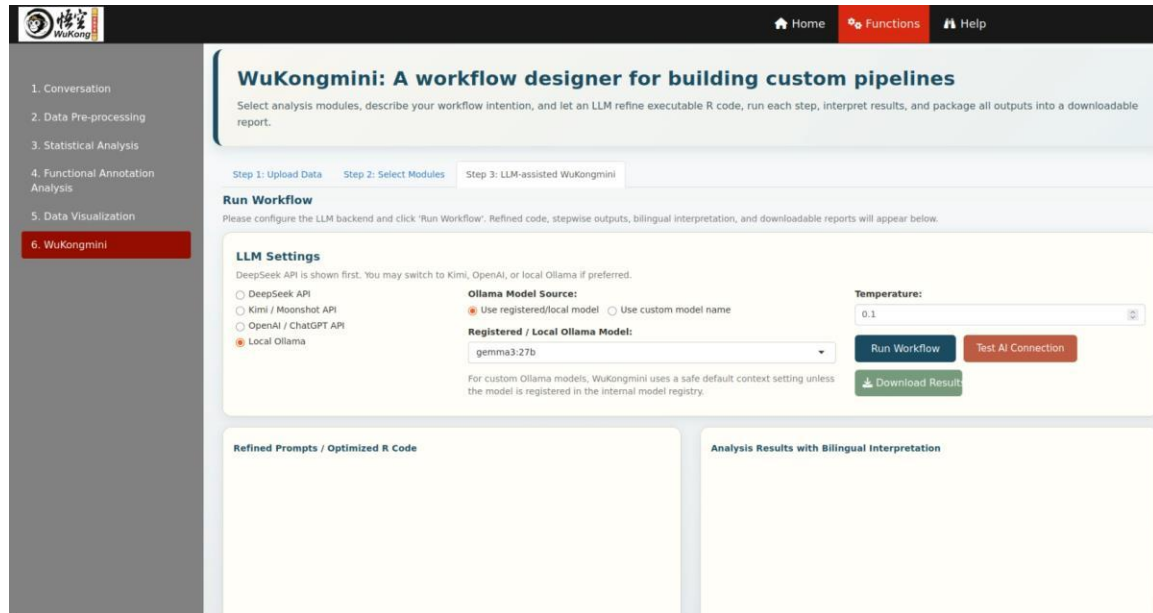
The screenshot shows the WuKongmini web interface, a workflow designer for building custom pipelines. The interface is divided into three main sections: I. Data Pre-processing, II. Statistical Analysis, and III. Functional Annotation. The left sidebar contains a navigation menu with six items: 1. Conversation, 2. Data Pre-processing, 3. Statistical Analysis, 4. Functional Annotation Analysis, 5. Data Visualization, and 6. WuKongmini (highlighted in red). The top navigation bar includes Home, Functions, and Help. The main content area displays the workflow configuration steps: Step 1: Upload Data, Step 2: Select Modules, and Step 3: LLM-assisted WuKongmini. The workflow is titled "WuKongmini: A workflow designer for building custom pipelines" and includes a description: "Select analysis modules, describe your workflow intention, and let an LLM refine executable R code, run each step, interpret results, and package all outputs into a downloadable report." The workflow is divided into three sections: I. Data Pre-processing, II. Statistical Analysis, and III. Functional Annotation. The "Example Pipeline 2" button is highlighted. The workflow description on the right lists the steps: 1. Normalization, 2. Logarithm with base 2, 3. Missing value imputation, 4. SAMR, 5. Volcano Plot, 6. PCA, 7. HCA, 8. GO Enrichment Analysis, 9. KEGG Enrichment Analysis. The workflow description also includes a detailed description of each step.

Clicking the “Example Pipeline 2” button loads a similar 9-step workflow: Normalization, Logarithm with base 2, Missing value imputation, SAMR, Volcano Plot, PCA, HCA, Gene Set Enrichment Analysis of GO, Gene Set Enrichment Analysis of KEGG. This version uses gene set enrichment analysis methods to explore protein functions. Like Pipeline 1, it includes a pre-written description that will be used by the selected LLM in Step 3 to perform the analysis.

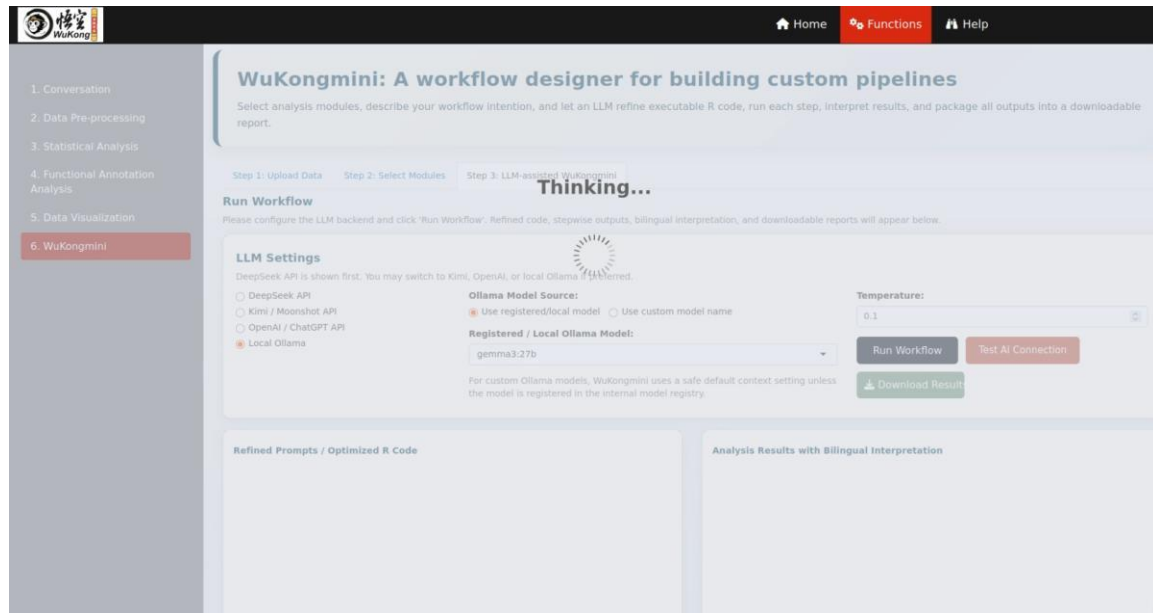
The screenshot shows the WuKongmini web interface, a workflow designer for building custom pipelines. The interface is divided into three main sections: I. Data Pre-processing, II. Statistical Analysis, and III. Functional Annotation. The left sidebar contains a navigation menu with six items: 1. Conversation, 2. Data Pre-processing, 3. Statistical Analysis, 4. Functional Annotation Analysis, 5. Data Visualization, and 6. WuKongmini (highlighted in red). The top navigation bar includes Home, Functions, and Help. The main content area displays the workflow configuration steps: Step 1: Upload Data, Step 2: Select Modules, and Step 3: LLM-assisted WuKongmini. The workflow is titled "WuKongmini: A workflow designer for building custom pipelines" and includes a description: "Select analysis modules, describe your workflow intention, and let an LLM refine executable R code, run each step, interpret results, and package all outputs into a downloadable report." The workflow is divided into three sections: I. Data Pre-processing, II. Statistical Analysis, and III. Functional Annotation. The "Example Pipeline 2" button is highlighted. The workflow description on the right lists the steps: 1. Normalization, 2. Logarithm with base 2, 3. Missing value imputation, 4. SAMR, 5. Volcano Plot, 6. PCA, 7. HCA, 8. Gene Set Enrichment Analysis of GO, 9. Gene Set Enrichment Analysis of KEGG. The workflow description also includes a detailed description of each step.

Step 3: LLM-assisted WuKongmini

In this step, the workflow constructed in Step 2 is executed with the support of a LLM, and the platform displays both the “Refined Prompts (LLM-generated R code or instructions)” and the “Analysis Results”. Users can also download all output results (e.g. tables, figures) in a ZIP file.



After selecting a local LLM (e.g., gemma3:27b), users simply click the “Run Workflow” button to begin execution. We strongly recommend using a large-scale local LLM for optimal understanding and handling of complex workflows.



Once the workflow execution is complete, results are displayed within two main sections:

“Refined Prompts” and “Analysis Results”.

Run Workflow

Please configure the LLM backend and click 'Run Workflow'. Refined code, stepwise outputs, bilingual interpretation, and downloadable reports will appear below.

LLM Settings

DeepSeek API is shown first. You may switch to Kimi, OpenAI, or local Ollama if preferred.

☐ DeepSeek API ☐ Kimi / Moonshot API ☐ OpenAI / ChatGPT API ☒ Local Ollama

Ollama Model Source:

☒ Use registered/local model ☐ Use custom model name

Registered / Local Ollama Model:

gemini3:27b

For custom Ollama models, Wukongmini uses a safe default context setting unless the model is registered in the internal model registry.

Temperature:

0.1

Run Workflow **Test AI Connection** **Download Result**

Refined Prompts / Optimized R Code

```
1 # R
2 # Input objects:
3 # - inputdata: primary expression matrix or data frame.
4 # - grname1: vector of group names.
5 # - grnum1: total number of groups.
6 # - grnum2: vector of replicate numbers per group.
7 # - grname2: sample-level group vector.
8 # - species: species id/name selected by user.
9
10 suppressPackageStartupMessages(library(input))
11 suppressPackageStartupMessages(library(samr))
12 suppressPackageStartupMessages(library(ggplot2))
13 suppressPackageStartupMessages(library(factoextra))
14 suppressPackageStartupMessages(library(factotone))
15 suppressPackageStartupMessages(library(phenoplot))
16 suppressPackageStartupMessages(library(clusterProfiler))
17 suppressPackageStartupMessages(library(dplyr))
18 suppressPackageStartupMessages(library(tidy))
19 suppressPackageStartupMessages(library(scales))
20
21 # Step 1: Normalization
22 # Input: inputdata
23 # Output: normdata1
24 normmethod <- "Median"
25 if (normmethod == "Median") {
26   normdata1 <- sweep(inputdata, 2, apply(inputdata, 2, median, na.rm = TRUE), FUN = "/")
27 } else if (normmethod == "Mean") {
28   normdata1 <- sweep(inputdata, 2, apply(inputdata, 2, mean, na.rm = TRUE), FUN = "/")
29 } else if (normmethod == "Total Intensity") {
30   normdata1 <- sweep(inputdata, 2, apply(inputdata, 2, sum, na.rm = TRUE), FUN = "/")
31 }
32 normdata1 <- as.matrix(normdata1)
33 normmethod <- "Median"
34 if (normmethod == "Median") {
35   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, median, na.rm = TRUE), FUN = "/")
36 } else if (normmethod == "Mean") {
37   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, mean, na.rm = TRUE), FUN = "/")
38 } else if (normmethod == "Total Intensity") {
39   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, sum, na.rm = TRUE), FUN = "/")
40 }
41 normdata1 <- as.matrix(normdata1)
42 normmethod <- "Centering"
43 if (normmethod == "Centering") {
44   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, mean, na.rm = TRUE), FUN = "-")
45 } else if (normmethod == "Range Normalization") {
46   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, range, na.rm = TRUE), FUN = "/")
47 } else if (normmethod == "Z-score Normalization") {
48   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, sd, na.rm = TRUE), FUN = "/")
49 } else if (normmethod == "Quantile Normalization") {
50   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, quantile, na.rm = TRUE), FUN = "/")
51 }
```

Analysis Results with Bilingual interpretation

LLM-Generated Bilingual Scientific Interpretation

Results Section - Multi-Omics Workflow for Differential Protein Expression Analysis

This study employed a multi-omics workflow to identify differentially expressed proteins and explore associated biological pathways. The workflow consisted of normalization, log transformation, missing value imputation, differential expression analysis using SAMR, visualization via volcano plot, Principal Component Analysis (PCA), Hierarchical Clustering Analysis (HCA), Gene Ontology (GO) enrichment analysis, and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis.

1 Normalization

The “Refined Prompts” section contains step-by-step R code automatically generated by the LLM, aligned with the structure of the workflow built in Step 2. Users with R programming knowledge can examine these scripts to understand how the data was processed at each stage, or reuse the code for independent analysis. For users unfamiliar with coding, this section can be skipped without impacting their ability to interpret results. The “Analysis Results” section presents the full output of the workflow, beginning with a brief, LLM-generated summary of the overall analysis. Notably, this section includes a dedicated “Differentially Expressed Protein Functional Categorization”, which provides insights into the biological roles and significance of differentially expressed proteins identified during the analysis.

Run Workflow

Please configure the LLM backend and click 'Run Workflow'. Refined code, stepwise outputs, bilingual interpretation, and downloadable reports will appear below.

LLM Settings

DeepSeek API is shown first. You may switch to Kimi, OpenAI, or local Ollama if preferred.

☐ Kimi / Moonshot API ☐ OpenAI / ChatGPT API ☒ Local Ollama

DeepSeek Model:

DeepSeek V4 Pro

DeepSeek requests use reasoning_effort='high' and thinking enabled by default.

Temperature:

0.1

Run Workflow **Test AI Connection** **Download Result**

Refined Prompts / Optimized R Code

```
1 # R
2 # Input objects: rows are proteins/genes, columns are
3 # - grname1: vector of group names (e.g., c("Control", "Treat")).
4 # - grnum1: total number of groups; for this SAM workflow this should be '2'.
5 # - grnum2: vector of replicate numbers per group (e.g., c(3, 3)), used to build the SAM
6 # - grname2: sample-level group vector matching the columns of 'inputdata'.
7 # - species: species id/name selected by the user; used for GO file lookup and KEGG organ
8
9 # Global package loading
10 suppressPackageStartupMessages(library(input))
11 suppressPackageStartupMessages(library(samr))
12 suppressPackageStartupMessages(library(ggplot2))
13 suppressPackageStartupMessages(library(factoextra))
14 suppressPackageStartupMessages(library(factotone))
15 suppressPackageStartupMessages(library(phenoplot))
16 suppressPackageStartupMessages(library(clusterProfiler))
17 suppressPackageStartupMessages(library(dplyr))
18 suppressPackageStartupMessages(library(tidy))
19
20 # Step 1: Normalization
21 # Input: inputdata (primary expression matrix or data frame)
22 # Output: normdata1 (median-normalized expression matrix)
23 normdata1 <- as.matrix(inputdata)
24 normmethod <- "Median"
25 if (normmethod == "Median") {
26   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, median, na.rm = TRUE), FUN = "/")
27 } else if (normmethod == "Mean") {
28   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, mean, na.rm = TRUE), FUN = "/")
29 } else if (normmethod == "Total Intensity") {
30   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, sum, na.rm = TRUE), FUN = "/")
31 } else if (normmethod == "Centering") {
32   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, mean, na.rm = TRUE), FUN = "-")
33 } else if (normmethod == "Range Normalization") {
34   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, range, na.rm = TRUE), FUN = "/")
35 } else if (normmethod == "Z-score Normalization") {
36   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, sd, na.rm = TRUE), FUN = "/")
37 } else if (normmethod == "Quantile Normalization") {
38   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, quantile, na.rm = TRUE), FUN = "/")
39 }
40 normdata1 <- as.matrix(normdata1)
41 normmethod <- "Quantile Normalization"
42 normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, quantile, na.rm = TRUE), FUN = "/")
43 normdata1 <- as.matrix(normdata1)
44 normmethod <- "Centering"
45 if (normmethod == "Centering") {
46   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, mean, na.rm = TRUE), FUN = "-")
47 } else if (normmethod == "Range Normalization") {
48   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, range, na.rm = TRUE), FUN = "/")
49 } else if (normmethod == "Z-score Normalization") {
50   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, sd, na.rm = TRUE), FUN = "/")
51 } else if (normmethod == "Quantile Normalization") {
52   normdata1 <- sweep(normdata1, 2, apply(normdata1, 2, quantile, na.rm = TRUE), FUN = "/")
53 }
```

Analysis Results with Bilingual interpretation

Step 8: GO Enrichment Analysis Using DEPs Only / 仅使用差异表达蛋白的 GO 富集分析

English explanation:

GO enrichment analysis is performed using `clusterProfiler::enricher` with a custom `TERM2GENE` mapping derived from a UniProt-to-Go annotation file and a GO metadata file. Only the `de.proteins` from the volcano plot are used as the input gene list. The result is merged with GO descriptions and sorted by `Count`.

Chinese explanation:

GO 富集分析使用 `clusterProfiler::enricher`，基于来自 UniProt-to-Go 注释文件和 GO 元数据文件的定制 `TERM2GENE` 映射进行。仅使用火山图筛选出的 `de.proteins` 作为输入基因列表。结果与 GO 描述合并，并按 `Count` 排序。

Result and downstream relevance:

The output table has 1,096 x 9 dimensions. The top displayed GO terms, ordered by Count, include:

GO ID	Ontology	Term	Count	GeneRatio	p.adjust
GO:0005634	CC	nucleus	58	58/140	3.04e-19
GO:0005737	CC	cytoplasm	50	50/140	6.16e-17
GO:0070062	CC	extracellular exosome	43	43/140	5.08e-29
GO:0005829	CC	cytosol	40	40/140	3.83e-22

Following this summary, the module displays the results of each individual workflow step in sequence, including figures, tables and visualizations as applicable.

The screenshot displays the DeepSeek V4 Pro interface, which is designed for running workflows and analyzing results. The interface is divided into several sections:

- Top Bar:** Contains the API selection (Kimi / Moonshot API, OpenAI / ChatGPT API, Local Ollama), the DeepSeek Model (DeepSeek V4 Pro), and buttons for "Run Workflow", "Test AI Connection", and "Download Results".
- Refined Prompts / Optimized R Code:** A section on the left showing the R code used for the analysis. The code includes comments and functions for data loading, normalization, and analysis.
- Analysis Results with Bilingual Interpretation:** A section on the right showing the results of the analysis. It includes a table of results for Step 1: Normalization, Step 2: Logarithm with base 2, and Step 5: Volcano Plot.

The "Analysis Results with Bilingual Interpretation" section is further divided into steps:

- Step 1: Normalization:** Shows a table of results for the normalization step. The table has columns for "Show", "entries", and "Search". The results are displayed in a table with columns A1, A2, and A3.
- Step 2: Logarithm with base 2:** Shows a table of results for the logarithm step. The table has columns for "Show", "entries", and "Search". The results are displayed in a table with columns A1, A2, and A3.
- Step 5: Volcano Plot:** A scatter plot showing the results of the analysis. The x-axis is labeled "Fold Change (log2)" and the y-axis is labeled "P Value (-log10)". The plot shows a distribution of points, with a legend indicating "Down", "Up", and "No Change".

Finally, users can click the “Download Results” button to export all output data in ZIP format for offline access, reporting, or further analysis.

9. Case Study

To assess the functional versatility of the WuKong platform, we utilized a publicly available proteomics dataset on lung squamous cell carcinoma from the Clinical Proteomic Tumor Analysis Consortium (CPTAC), originally published by Shankha et al. The dataset is available through ProteomeXchange with the identifier PDC000234 and includes quantitative proteomic profiles from 212 samples-comprising 110 tumor tissues and 102 matched adjacent normal tissues-with expression data for 12,759 proteins. The dataset in a .csv format can be accessed and downloaded at: <https://github.com/wangshisheng/WuKong>.

When opened in Excel, the dataset is displayed with protein names as rows and sample names as columns. Sample names containing “.N” indicate adjacent normal tissues, while those without “.N” represent tumor tissues. The data has been normalized and transformed with binary logarithm (i.e. log₂).

#	A	B	C	D	E	F	G	H	I	J	K	L
1		C3L-00081.N	C3L-00415.N	C3L-00445.N	C3L-00568.N	C3L-00603.N	C3L-00904.N	C3L-00923.N	C3L-00927.N	C3L-00993.N	C3L-01000.N	C3L-01285.N
2	ARF5	-0.04178705	0.06166895	0.11882040	-0.14546018	-0.376528442	-0.382265481	0.05380911	-0.349116468	-0.030397907	-0.19696926	-0.073082033
3	M6PR	-0.220777197	-0.13408	0.00436975	-0.204541608	-0.228215198	-0.260806008	-0.235954	0.08008223106	0.22412219226	-0.198817796	0.14948189664
4	ESRRA	-0.091096924	-0.379467	-0.191655	0.0088098161	1.10824884507	0.26254235692	-0.019013	-0.069992433	-0.347673071	0.5228739501	-0.556894679
5	KBP4	-0.759884574	-0.577785	-0.563748	-0.728165764	-0.710225996	-0.864701272	-0.828016	-0.881461878	-0.802232243	-0.899496602	-0.558510124
6	NDUFAF7	-0.349874899	-0.15448	-0.310628	0.1543552716	0.20906917292	0.06975061188	-0.101414	-0.361786358	-0.223416964	0.1109470782	-0.163725543
7	FUCA2	-0.553674481	0.1862242	0.1159842	-0.24162283	-0.082683665	0.15620422529	0.1568698	-0.079559975	-0.186408082	-0.090084366	0.01019899147
8	DBNDD1	NA	NA	-0.280613	NA	0.868230660103	NA	NA	NA	NA	NA	NA
9	HS3ST1	NA	NA	-0.67653	NA	NA	-0.113160194	-0.406904	NA	-1.297089973	NA	-2.113925964
10	SEMA3F	-0.393563725	-0.146183	-0.858224	-0.093513459	-0.337677642	-0.063408576	-0.345781	-0.36932998	-0.026083067	0.1126936372	-0.474228132
11	CFTR	-0.053583218	NA	1.3307767	0.4041811213	-0.214732913	-0.125850491	0.3582003	0.54323236225	0.02543266954	0.3021278403	0.62015993872
12	CYP51A1	0.6739195579	0.7891086	0.7283173	0.2331819236	0.01467669649	-0.180149013	0.3393526	0.70608794965	0.37800908650	0.1486026854	0.58898655964
13	TWEM176A	0.0128708493	0.9546589	0.54120507	NA	NA	NA	NA	NA	0.33456204695	0.7310973779	0.45023812776
14	SLC7A2	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
15	HSPB6	1.4118771445	0.6357099	0.9689269	0.7812973900	1.00700386140	1.21539311198	1.5841134	0.85856087856	1.24527796560	0.7407476312	0.95266155033
16	ZNF195	NA	NA	-0.730388	-0.17861356	-1.0160149	-0.250939902	NA	-0.09584301	NA	-0.796936344	NA
17	PK4	0.7687912299	NA	NA	NA	NA	0.48109176286	NA	1.05368894589	0.18285760730	1.3859576438	0.70184542196
18	RALA	0.2278686984	-0.223819	-0.052331	0.0096377815	0.29430836910	0.04090250772	0.1105418	0.35275462652	-0.165688395	0.6528871343	0.42574948654
19	BAIAP2L1	-0.613249719	-0.650435	-0.714799	-0.921410487	-0.710317018	-0.814561505	-0.859123	-0.683198408	-0.772054303	-0.688990346	-0.698446011
20	TWEM132A	-0.69609939	-0.29862	-0.389651	-0.480100615	-0.078622296	-0.229093712	-0.349371	-0.144116345	-0.144975472	-0.190470966	-0.187426633
21	DVL2	0.2392112220	-0.012549	-0.195581	-0.107080018	0.31533805291	0.12655139382	0.2788104	-0.120739981	0.09706727987	0.043119898	0.23327189770
22	RPAF3	-0.312145677	-0.106894	-0.2704	-0.135310288	-0.328286379	-0.17418853	-0.360384	-0.391965697	-0.374163516	-0.406566984	-0.447510409
23	IFRD1	0.0086317974	-0.420603	0.0428456	-0.231566167	-0.105507857	-1.059716106	-0.144047	-0.310327378	-0.367867212	-0.357068619	0.11727234062
24	SKAP2	0.4613810073	0.1515485	0.4921903	0.2785908945	-0.097887607	0.15128248631	-0.040312	0.43369400927	0.39052354794	0.1598845185	0.31338417653
25	PRSS21	-0.734084935	-0.838965	-0.715417	NA	NA	0.55160443295	-0.837499	NA	-0.147712175	NA	0.07651270913
26	CX3CL1	-0.601423723	NA	-0.843388	-0.351379827	NA	0.06873286739	-0.629065	0.23122681332	-0.126179839	-0.389659171	0.41690615991
27	PRR5	-0.294196205	-0.56282	NA	-0.109954728	-0.434026138	0.04197309017	NA	-0.523889957	-0.263962429	-0.381890214	-0.13469143
28	TRAPP6A	0.3936360317	0.1779579	0.1965846	0.2097825399	0.87053947160	0.22817369057	-0.266156	0.25107019372	0.25787227081	-0.084131333	0.15246793839
29	WDR54	0.6013513600	0.3888504	0.3657398	0.1390314186	0.52865987417	0.65543679043	0.64431417	0.07871937068	0.40467761546	0.2602207399	0.44281436471
30	SPATA20	0.1652403397	0.16517161	-0.448471	0.3127104579	0.14341974216	0.23810281721	0.0872535	-0.406106447	-0.130547405	-0.169396349	0.31830644225
31	CEACAM7	-0.285723541	-1.155254	-0.983657	NA	NA	NA	-0.637069	NA	0.44297729592	NA	-1.07952306
32	CD79B	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
33	RHBD2	-0.013992102	-0.180187	-0.021669	0.1180826883	-0.80270776	-0.396479524	0.0763964	0.13470477537	-0.542830635	0.0378378009	-0.125432203
34	RFUSD1	NA	-0.14138	0.2061062	NA	0.60796624915	NA	-0.838217	-0.455636181	-0.478324966	0.5706292682	-1.315617855
35	TSR3	-0.611176357	0.0676637	-0.090061	0.057761581	-1.090861126	-0.594102806	0.0461915	0.07850683203	0.04225285802	-0.717432068	-0.355245743

9.1. Preparation

Prior to commencing data analysis, users are required to ensure the following:

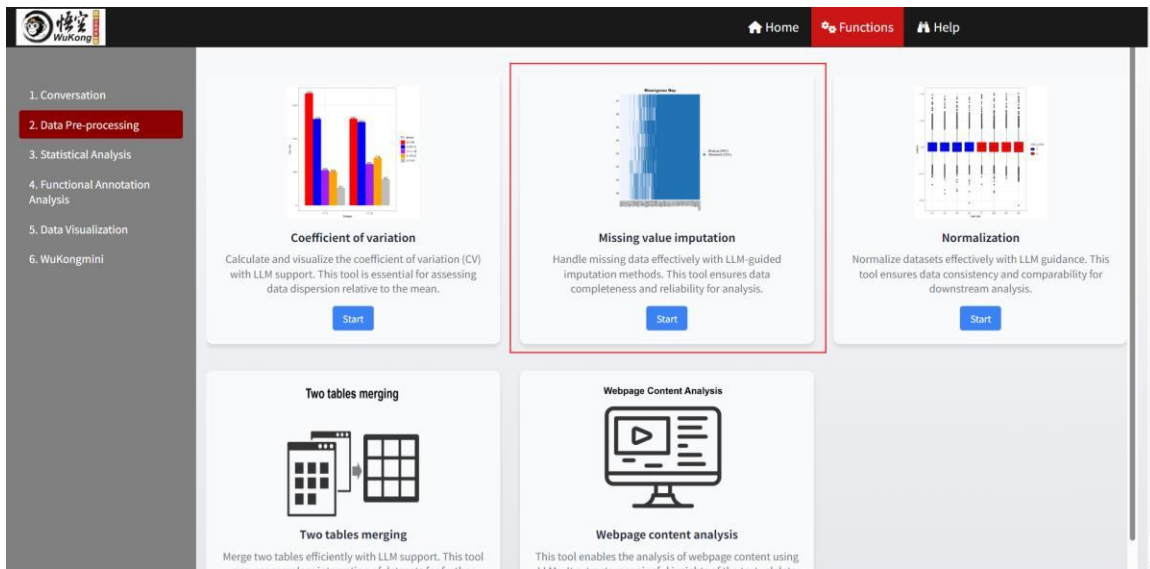
- The WuKong package has been correctly installed on their computers (Refer to the section “1. How to run this tool locally?”).
- At least one Large Language Model (LLM) has been installed from Ollama (Refer to the “How to install LLMs locally?” subsection within “3. Conversation”).
- The data has been prepared in the appropriate format as previously described (Also refer to the “2. Data Preparation” section).

9.2. Data analysis

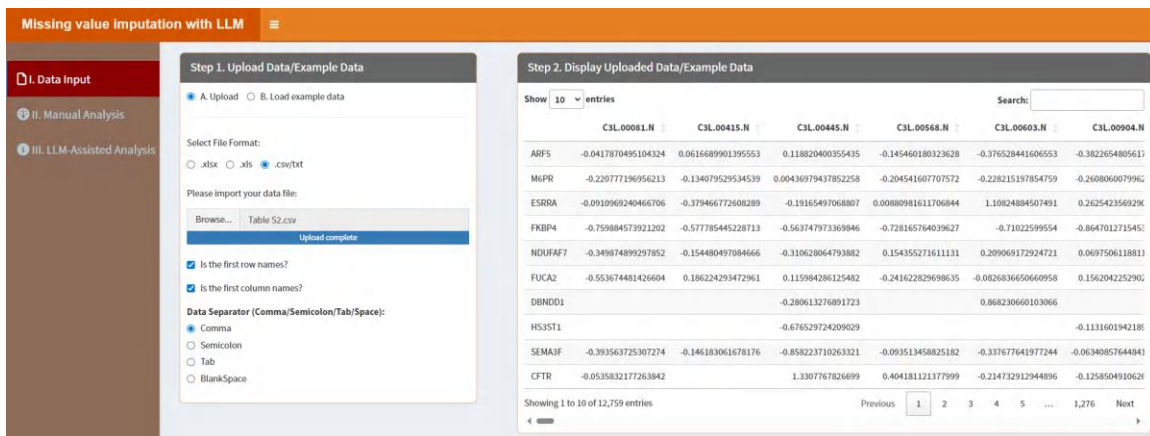
Herein, we analyzed the data above using a classical proteomics analysis pipeline, which included missing value imputation, differential protein expression analysis, volcano plot, PCA, HCA, GO and KEGG pathway enrichment analyses and post-enrichment result mining.

a. Missing value imputation

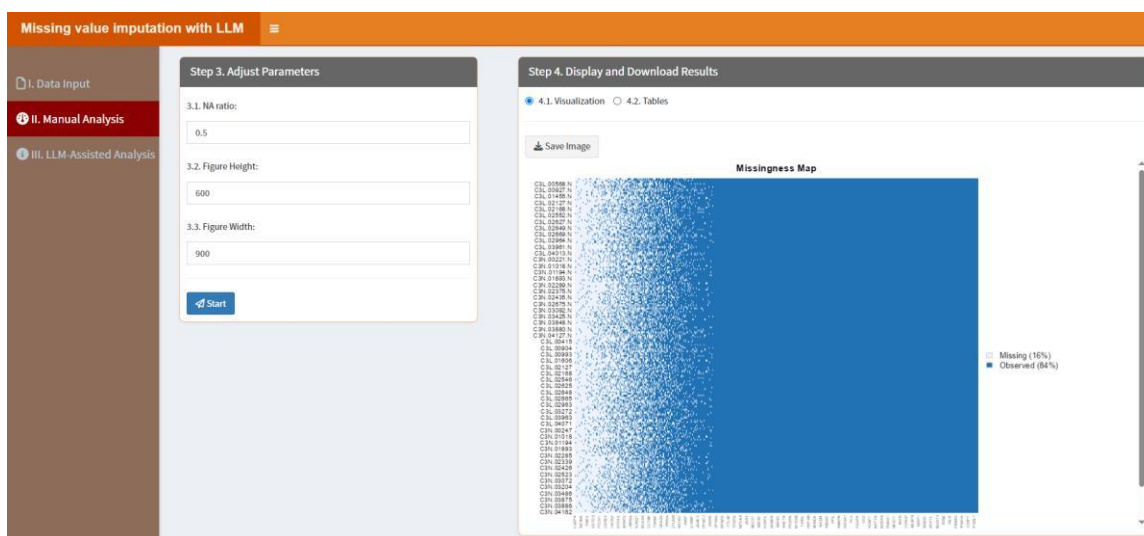
Users can find this module as below:



Then click “Start” button and upload the data, shown as below.



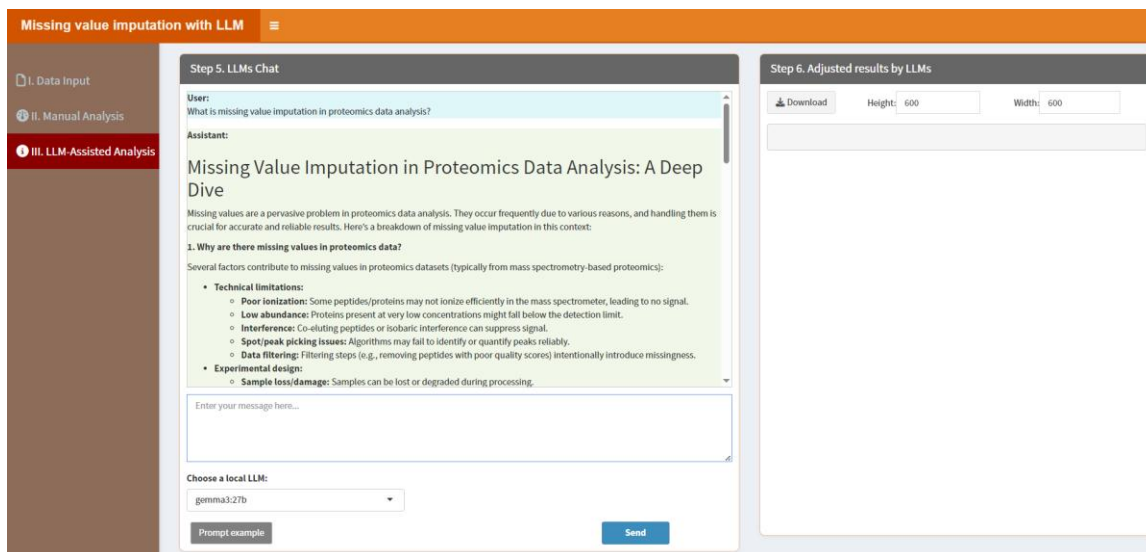
Next, click “II. Manual Analysis”. The parameters here can be kept as default, and click “Start” button. The result in “4.1. Visualization” shows the missing value distribution.



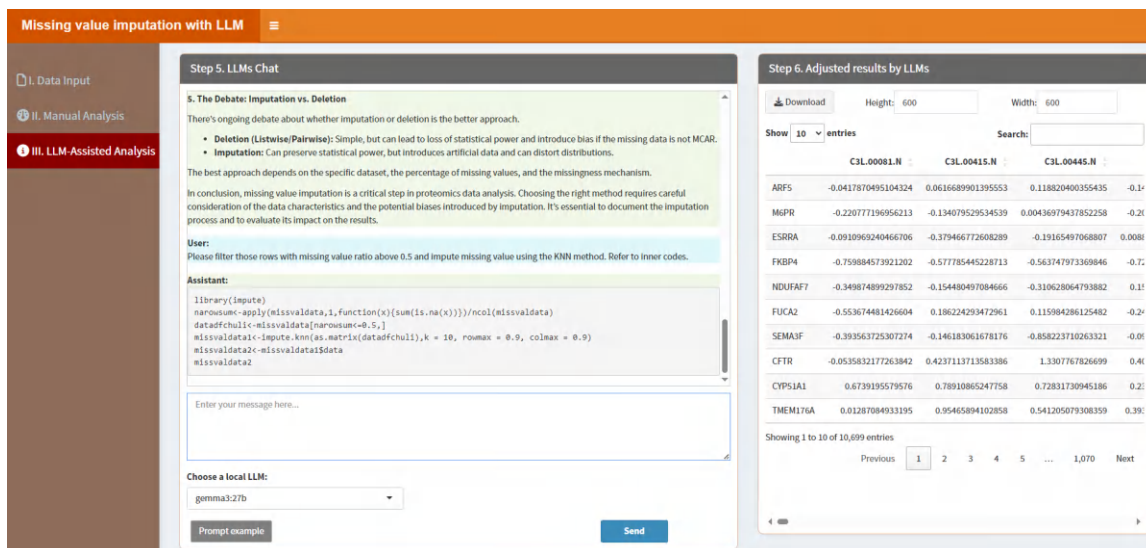
The outcomes presented in “4.2. Tables” display the imputed data generated by the K Nearest Neighbors (KNN) method as the default setting. Users are able to click the “Download” button to obtain this result for subsequent analysis.

	C3L00081.N	C3L00415.N	C3L00445.N	C3L00568.N	C3L00603.N	C3L00904.N
ARF5	-0.0417870495104324	0.0616689901395553	0.118820400355435	-0.145460180323628	-0.376528441666553	-0.38226548056
M6PR	-0.220777136956213	-0.134079529534539	0.00436979437852258	-0.204541607707572	-0.228215197854759	-0.26080600799
ESRRB	-0.0910969240466706	-0.379466772608289	-0.19165497068807	0.00880981611706944	1.10824884507491	0.26254235692
FKBP4	-0.759884573921202	-0.577785445228713	-0.563747973369846	-0.728165764039627	-0.71022599554	-0.86470127154
NDUFAT7	-0.349874899297852	-0.154408497084666	-0.310628064793882	0.154355271611131	0.209069172924721	0.06975061188
FUCAT2	-0.553674481426604	0.186224293472961	0.115984286125482	-0.241622829698635	-0.082683665060958	0.15620422529
SEMA3F	-0.393563725307274	-0.146183061678176	-0.858223710263321	-0.093513458825182	-0.337677641977244	-0.063408576448
CFTR	-0.0535832177263842	0.4237113713583386	1.3307767826699	0.404181121377999	-0.214732912944896	-0.12585049106
CYP11A1	0.6739195579576	0.78910865247758	0.72831730945186	0.233181923667317	0.0146766964916196	-0.18014901313
TMEM176A	0.012870849331395	0.95465894102858	0.541205079308359	0.3936496871454295	0.4854972024718644	0.297503970632

Additionally, users can click “III. LLM-Assisted Analysis” to perform missing value imputation with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “What is missing value imputation in proteomics data analysis?”, as below, it will give a detailed description about missing value imputation in proteomics data analysis for users.

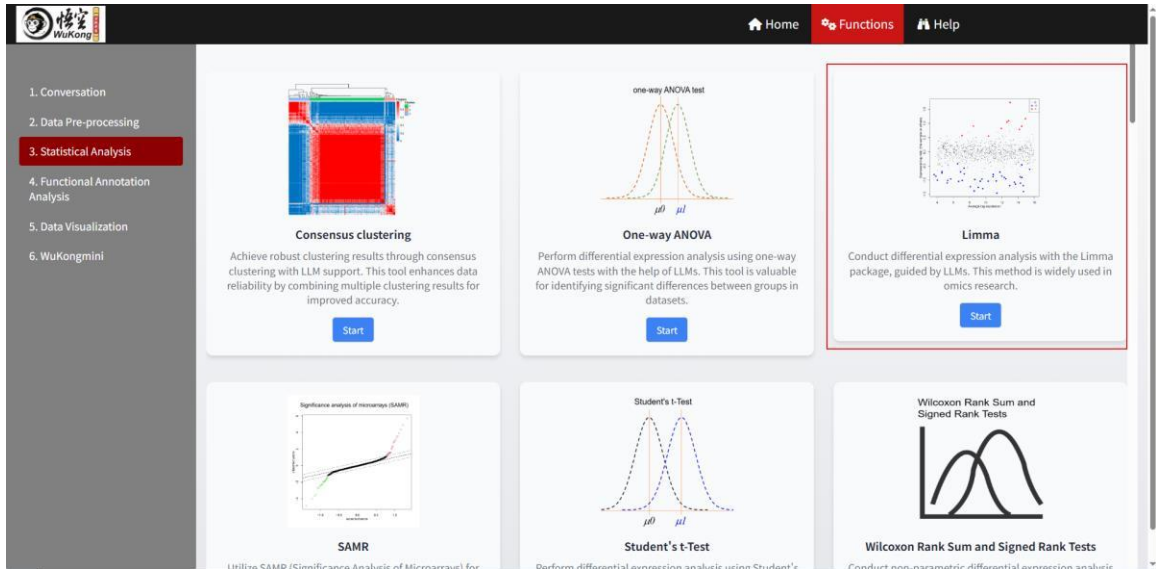


Users can input a prompt such as, “Please filter those rows with missing value ratio above 0.5 and impute missing value using the KNN method. Refer to inner codes”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate imputed results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

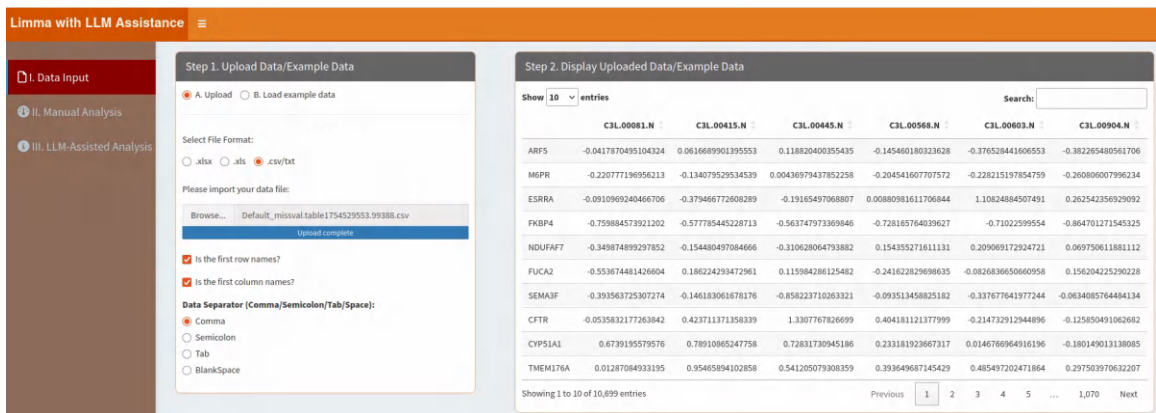


b. Differential protein expression analysis

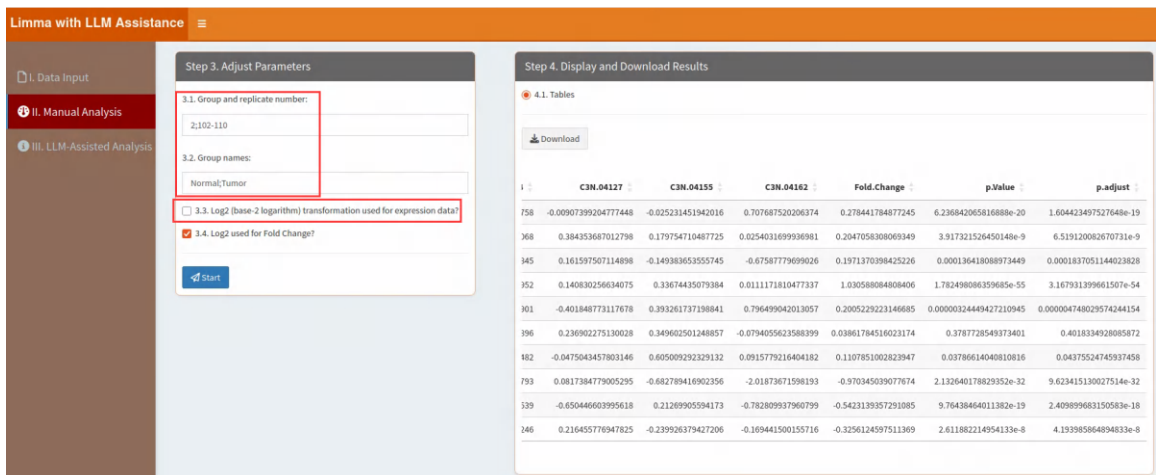
Herein, we choose Limma to process the differential protein expression analysis.



Then click “Start” button and upload the result data from the missing value imputation above, shown as below.



Next, click “II. Manual Analysis”. We should type right parameters based on the data information, and click “Start” button. The result can be downloaded in “4.1. Tables”.



Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “Please give me the differential expression analysis result table. Refer to inner codes.”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

For instance, as shown below, the internal code automatically applies log2 transformation to the uploaded data. If the data has already been log2-transformed beforehand, applying the transformation again would lead to incorrect results.

The screenshot displays the 'Limma with LLM Assistance' interface. On the left, a sidebar shows three options: 'I. Data Input', 'II. Manual Analysis', and 'III. LLM-Assisted Analysis', with the third option selected. The main area is divided into two panels. The left panel, titled 'Step 5. LLMs Chat', shows a conversation where the user asks for a differential expression analysis result table, and the assistant provides R code. The right panel, titled 'Step 6. Adjusted results by LLMs', shows a table of results. A red box highlights the first two columns of the table, which are labeled 'C3L.00081.N' and 'C3L.00415.N'.

Step 5. LLMs Chat

User: Please give me the differential expression analysis result table. Refer to inner codes.

Assistant:

```
library(limma)
DEPdata<-log2(DEPdata)
designx <- model.matrix(~factor(rep(1:DEP_leibie_num, DEP_fenzu_num)))
colnames(designx)<-c("Intercept", paste0("gr", 1:(ncol(designx)-1)))
wFit <- lmFit(DEPdata, designx)
wTEBfit <- eBayes(wFit)
wTEBfitable<-as.data.frame(topTable(wTEBfit, coef=2, number = Inf, sort.by="none"))
p.adjust<-p.adjust(wTEBfitable$p.Value, method = "BH")
FC<-rowMeans(DEPdata[,c(1:(DEP_fenzu_num[1]-1):ncol(DEPdata))], na.rm = T)-rowMeans(DEPdata[,c(1:DEP_fenzu_num[1]), na.rm = T])
wTEBfitabledf<-cbind(DEPdata, Fold.Change=FC, p.Value=wTEBfitable$p.Value, p.adjust=p.adjust)
wTEBfitabledf
```

Enter your message here...

Choose a local LLM: gemma3:27b

Step 6. Adjusted results by LLMs

Download Height: 700 Width: 700

Show 10 entries Search:

	C3L.00081.N	C3L.00415.N	C3L.00445.N	C3L
ARF5	-4.019310968115587	-3.073145540188773		
M6PR		-7.838218889619303		
ESRRA			-6.82667	
FKBP4				-2.69567
NDUFA7				-2.69567
FUCA2	-2.424886806164617	-3.107998736642192		
SEMA3F				
CFTR	-1.238846247322822	0.4122686016466228	-1.30692	
CYP51A1	-0.5693516997141984	-0.3417041360954465	-0.4573609617211082	-2.10047
TMEM176A	-6.279748931354998	-0.06694268317295407	-0.8857527155533376	-1.34501

Showing 1 to 10 of 10,699 entries

Previous 1 2 3 4 5 ... 1,070 Next

In such cases, users must revise their prompts to explicitly indicate that log2 should not be applied, ensuring the LLM interprets the data context properly.

Limma with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 5. LLMs Chat

```

FCX<-rowMeans(DEPdata[,c((DEP_fenzu_num[1]+1):ncol(DEPdata))],na.rm = T)-rowMeans(DEPdata[,c(1:DEP_fenzu_num[1]),na.rm = T])
wtfEtfitabled<-cbind(DEPdata, Fold.Change=FCX,p.Value=wtfEtfitableSP.Value,p.adjust=p.adjust)
wtfEtfitabled

```

User:

The uploaded data has already been log2-transformed, so applying log2 again is unnecessary. Refer to inner codes.

Assistant:

```

library(limma)
# DEPdata<-log2(DEPdata) # Removed this line as data is already log2 transformed
design<- model.matrix(~factor(rep(1:DEP_leible_num,DEP_fenzu_num)))
colnames(design)<-c("Intercept",paste0("gr",1:(ncol(design)-1)))
wtfFit <- lmFit(DEPdata, design)
wtfEtf <- eBayes(wtfFit)
wtfEtfitable<-as.data.frame(topTable(wtfEtf,coef=2,number = Inf,sort.by="none"))
p.adjust<-p.adjust(wtfEtfitableSP.Value,method = "BH")
FCX<-rowMeans(DEPdata[,c((DEP_fenzu_num[1]+1):ncol(DEPdata))],na.rm = T)-rowMeans(DEPdata[,c(1:DEP_fenzu_num[1]),na.rm = T])
wtfEtfitabled<-cbind(DEPdata, Fold.Change=FCX,p.Value=wtfEtfitableSP.Value,p.adjust=p.adjust)
wtfEtfitabled

```

Enter your message here...

Choose a local LLM:

gemma3.27b

Prompt example

Send

Step 6. Adjusted results by LLMs

Download

Height: 700

Width: 700

Show 10 entries

Search:

	C3L_00081.N	C3L_00415.N	C3L_00445.N	C3L
ARFS	-0.0417870495104324	0.0616689901395553	0.118620400355435	-0.14546
M6PR	-0.220777196956213	-0.134079529534539	0.00436979437852258	-0.20454
ESRRA	-0.0910969240466706	-0.379466772608289	-0.19165497068807	0.0088098
FKBP4	-0.759884573921202	-0.577785445228713	-0.563747973369846	-0.72816
NDUFAT7	-0.349874899297852	-0.154480497084666	-0.310628064793882	0.15435
FUCA2	-0.553674481426604	0.186224293472961	0.115984286125482	-0.24162
SEMA3F	-0.393563725307274	-0.146183061678176	-0.858223710263321	-0.09351
CFTR	-0.053583217263842	0.423711371358339	1.3307767826699	0.40418
CYP51A1	0.6739195579576	0.78910865247758	0.72831730945186	0.23318
TMEM176A	0.01287084933195	0.95465894102858	0.541205079308359	0.39364

Showing 1 to 10 of 10,699 entries

Previous

1

2

3

4

5

...

1,070

Next

c. volcano plot

We can find this module here:

WuKong

Home

Functions

Help

1. Conversation

2. Data Pre-processing

3. Statistical Analysis

4. Functional Annotation Analysis

5. Data Visualization

6. WuKongmini

Venn Plot

Generate Venn plots to display overlaps between sets. LLMs assist in creating clear and informative diagrams.

Start

Violin Plot

Create violin plots to visualize data distributions. LLMs guide users in generating detailed and interpretable plots.

Start

Volcano Plot

Generate volcano plots to identify significant changes in data. LLMs simplify the process, ensuring accurate and visually appealing results.

Start

Word Cloud Plot

Then click “Start” button and upload the result data from above. Herein, we extracted three columns (Protein names, p.adjust and Fold.Change), shown as below.

286

Volcano plot with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 1. Upload Data/Example Data

A. Upload

B. Load example data

Select File Format:

.xlsx

.xls

.csv/txt

Please import your data file:

Browse...

Adjusted by LLM_DEPLImma.table1754540210.13666.csv

Upload complete

☒ Is the first row names?

☒ Is the first column names?

Data Separator (Comma/Semicolon/Tab/Space):

Comma

Semicolon

Tab

BlankSpace

Step 2. Display Uploaded Data/Example Data

Show 10 entries

Search:

	p.adjust	Fold.Change
ARF5	1.60442349752765e-19	0.278441784877245
MEPR	6.51912006267073e-9	0.204705830806935
ESRRA	0.000183705114402383	0.197137039842523
FKBP4	3.16793139966151e-54	1.03058808480841
NDUFA7	0.00000474802957424415	0.200522922314669
FUCA2	0.401833492808587	0.0386178451602317
SEMA3F	0.0437552474593746	0.110785100282395
CFTR	9.62341513002751e-32	-0.970345039077674
CYP51A1	2.40989968315058e-18	-0.542313935729108
TMEM176A	4.19398586489483e-8	-0.325612459751137

Showing 1 to 10 of 10,699 entries

Previous 1 2 3 4 5 ... 1,070 Next

Next, click “II. Manual Analysis”. We should type right parameters based on the data information, and click “Start” button. The result in “4.1. Visualization” shows the volcano plot.

Volcano plot with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 3. Adjust Parameters

3.1. Threshold for P value:

0.05

3.2. Threshold for Fold Change:

1.5

3.3. Select colors for each group:

Down

grey0

Up

3.4. Figure Height:

600

3.5. Figure Width:

600

Start

Step 4. Display and Download Results

4.1. Visualization

4.2. Tables

Save Image

Threshold

Down

grey0

Up

In the “4.2. Tables”, it shows the differentially expressed protein results (UP, DOWN and NoChange):

Volcano plot with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 3. Adjust Parameters

3.1. Threshold for P value:

0.05

3.2. Threshold for Fold Change:

1.5

3.3. Select colors for each group:

Down

grey0

Up

3.4. Figure Height:

600

3.5. Figure Width:

600

Start

Step 4. Display and Download Results

4.1. Visualization

4.2. Tables

Download

Show 10 entries

Search:

	p.adjust	Fold.Change	Threshold
ARF5	1.60442349752765e-19	0.278441784877245	NoChange
MEPR	6.51912006267073e-9	0.204705830806935	NoChange
ESRRA	0.000183705114402383	0.197137039842523	NoChange
FKBP4	3.16793139966151e-54	1.03058808480841	UP
NDUFA7	0.00000474802957424415	0.200522922314669	NoChange
FUCA2	0.401833492808587	0.0386178451602317	NoChange
SEMA3F	0.0437552474593746	0.110785100282395	NoChange
CFTR	9.62341513002751e-32	-0.970345039077674	DOWN
CYP51A1	2.40989968315058e-18	-0.542313935729108	NoChange
TMEM176A	4.19398586489483e-8	-0.325612459751137	NoChange

Showing 1 to 10 of 10,699 entries

Previous 1 2 3 4 5 ... 1,070 Next

So we can click the “Download” button to save the results.

Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like,

“Please change interested genes to below genes. Refer to inner codes:

CFTR

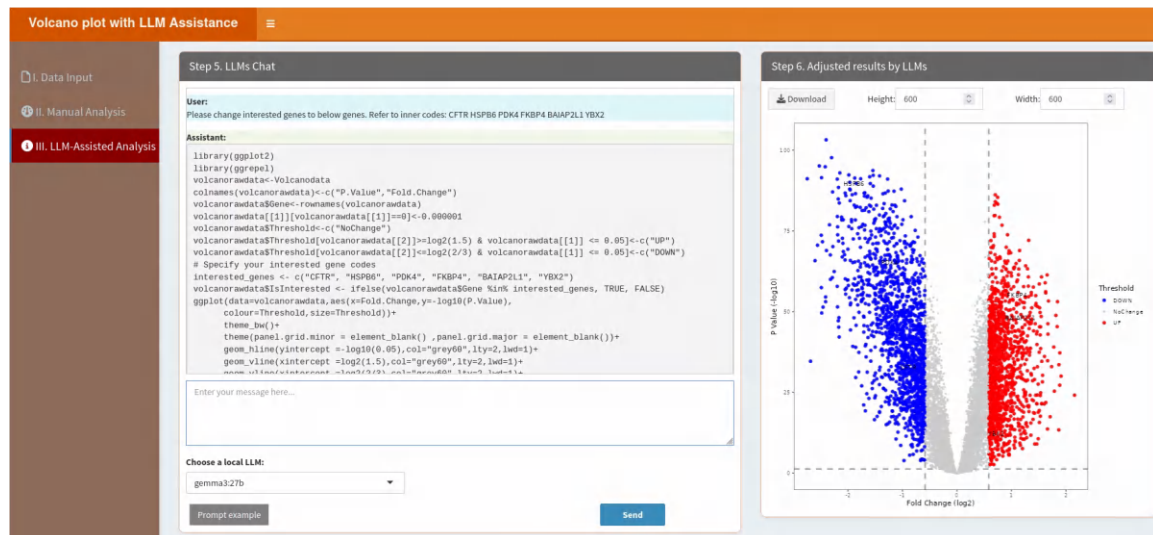
HSPB6

PDK4

FKBP4

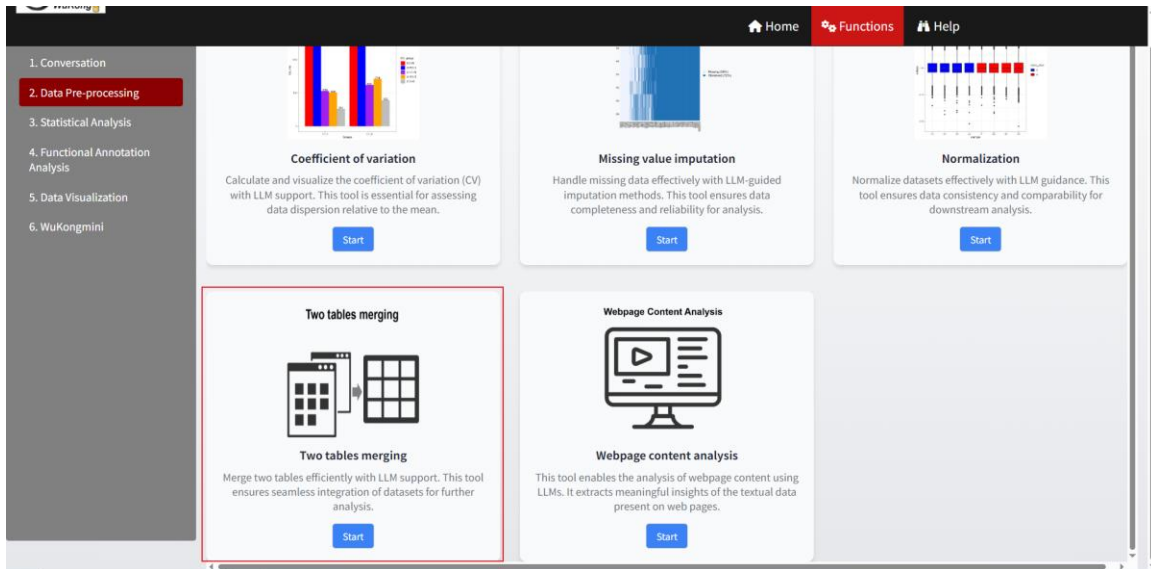
BAIAP2L1

YBX2”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.



d. Two table merging

From above, we can obtain the differentially expressed proteins (UP and DOWN). So we can use this module to extract the expression matrix of these proteins. Alternatively, users may also achieve this using other analysis tools outside the platform.



Then click “Start” button and upload two data from above: 1. Data from missing value imputation; 2. The expression matrix of differentially expressed proteins.

Two tables merging with the assistance of LLMs

1. Upload expression data:
☒ I. Upload ☐ II. Load example data

2. Conversation:

3. Results:
3.1. Adjusted results by LLMs:
[Download](#)

1.1. Import table 1:

Browse... Default_missval.table175s
Upload complete

1.2. Import table 2:

Browse... Default_volcanotable175s
Upload complete

[Review table 1](#)

[Review table 2](#)

Choose a Model:
gemma3.27b

We can click the “Review table 1” button to check the first data:

Two tables merging with the assistant

1. Upload expression data:

☒ I. Upload ☐ II. Load example data

2. Conversation

1.1. Import table 1:

Browse... Default_missval.table175

Upload complete

1.2. Import table 2:

Browse... Default_Volcanotable175

Upload complete

Review table 1

Review table 2

Choose a Model:

gemma3:27b

Uploaded table 1

Show 10 entries

Search:

	C3L.00081.N	C3L.00415.N	C3L.00445.N	C3L.00568.N	C3L.00603.N
ARF5	-0.0417870495104324	0.0616689901395553	0.118820400355435	-0.145460180323628	-0.376528441606
M6PR	-0.220777196956213	-0.134079529534539	0.00436979437852258	-0.204541607707572	-0.228215197854
ESRRA	-0.0910969240466706	-0.379466772608289	-0.19165497068807	0.00880981611706844	1.10824884507
FKBP4	-0.759884573921202	-0.577785445228713	-0.563747973369846	-0.728165764039627	-0.71022599
NDUFAF7	-0.349874899297852	-0.154480497084666	-0.310628064793882	0.154355271611131	0.209069172924
FUCA2	-0.553674481426604	0.186224293472961	0.115984286125482	-0.241622829698635	-0.0826836650660
SEMA3F	-0.393563725307274	-0.146183061678176	-0.858223710263321	-0.093513458825182	-0.337677641977
CFTR	-0.0535832177263842	0.423711371358339	1.3307767826699	0.404181121377999	-0.214732912944
CYP51A1	0.6739195579576	0.78910865247758	0.72831730945186	0.233181923667317	0.0146766964916
TMEM176A	0.01287084933195	0.95465894102858	0.541205079308359	0.393649687145429	0.485497202471

Showing 1 to 10 of 10,699 entries

Previous 1 2 3 4 5 ... 1,070 Next

Cancel

Click the “Review table 2” button to check the Second data:

Two tables merging with the assistant

1. Upload expression data:

☒ I. Upload ☐ II. Load example data

2. Conversation

1.1. Import table 1:

Browse... Default_missval.table175

Upload complete

1.2. Import table 2:

Browse... Default_Volcanotable175

Upload complete

Review table 1

Review table 2

Choose a Model:

gemma3:27b

Uploaded table 2

Show 10 entries

Search:

	p.adjust	Fold.Change	Threshold
FKBP4	3.17e-54	1.03058808480841	UP
CFTR	9.62e-32	-0.970345039077674	DOWN
HSPB6	1.17e-88	-1.96753862428871	DOWN
PDK4	2.29e-64	-1.30913384004645	DOWN
BAIAP2L1	3e-47	1.08733219138274	UP
YBX2	1.41e-14	0.801076005689295	UP
LRRC23	4.16e-14	-0.691104803791415	DOWN
CD74	9.95e-22	-0.622585929964037	DOWN
TSPAN9	4.14e-45	-0.864644400793722	DOWN
NME1	3.4e-37	0.721052191817554	UP

Showing 1 to 10 of 2,752 entries

Previous 1 2 3 4 5 ... 276 Next

Cancel

Next, users can click “Prompt example” and the “Send” button to perform this module with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “Please merge the two uploaded tables by rownames. Refer to inner codes.”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

Two tables merging with the assistance of LLMs

1. Upload expression data:
☒ I. Upload ☐ II. Load example data

1.1. Import table 1:
 Browse... Default_missval.table175
 Upload complete

1.2. Import table 2:
 Browse... Default_Volcanotable175
 Upload complete

[Review table 1](#)

[Review table 2](#)

Choose a Model:
 gemma3.27b

2. Conversation:

User:
 Please merge the two uploaded tables by rownames. Refer to inner codes.

Assistant:

```
mergedata1$ID<-rownames(mergedata1)
mergedata2$ID<-rownames(mergedata2)
mergedata<-base::merge(mergedata1,mergedata2,by.x="ID",by.y="ID",all=FALSE,all.x=FALSE,all.y=FALSE,sort=F)
mergedata
```

3. Results:

3.1. Adjusted results by LLMs:

[Download](#)

Show 10 entries

ID	C3L00081.N	C3L00415.N	C3L00445.N	C3L005
1 FKBP4	-0.759884573921202	-0.577785445228713	-0.563747973369646	-0.728165764
2 CFTR	-0.0535832177263842	0.423711371358339	1.3307767826699	0.404181121
3 HSPB6	1.41187714454355	0.635709961286391	0.968926919360946	0.781297390
4 PDK4	0.76879122996354	0.415129568459868	0.665183006035176	0.45642376
5 BAIAP2L1	-0.613249719442923	-0.650434755675018	-0.714798761879639	-0.921410486
6 YBX2	-2.11397478606034	-0.334983648572457	-0.773489472743645	-0.466448395
7 LRRC23	0.997660196155017	0.748359212064887	0.717830145027005	0.100249122
8 CD74	0.218705477730364	0.387891214421998	0.696805052451243	0.590511762
9 TSPAN9	0.652419956316173	0.139775355285359	0.160388786935676	0.499537072
10 NME1	-0.343970553382384	0.20340127009257	-0.332182813551469	-0.530414691

Showing 1 to 10 of 2,752 entries

Previous 1 2 3 4 5 ... 276 Next

Type your message here... [Prompt example](#) [Send](#)

In the “3. Results” part, it shows the merged results and click the “Download” button to save the result.

e. PCA

We can find this module here:

Home Functions Help

1. Conversation
 2. Data Pre-processing
 3. Statistical Analysis
 4. Functional Annotation Analysis
 5. Data Visualization
 6. WuKongmini

Network Degree Matrix
 Analyze network structures using the Network Degree Matrix (NDM). LLMs assist in interpreting network connectivity and relationships.
[Start](#)

OPLS-DA
 Perform Orthogonal Partial Least Squares Discriminant Analysis (OPLS-DA) with LLM support. This tool is widely used for classification and biomarker discovery.
[Start](#)

PCA
 Conduct Principal Component Analysis (PCA) with LLM guidance. This tool reduces data dimensionality while preserving important information.
[Start](#)

PCoA
 Perform Correspondence Analysis (PCoA) with LLM support. This tool is used for analyzing the relationship between objects based on the matrix of dissimilarity or distance.
[Start](#)

PLS-DA
 Perform Partial Least Squares Discriminant Analysis (PLS-DA) with LLM support. This tool is used for classification and biomarker discovery.
[Start](#)

Power Analysis
 Perform Power Analysis with LLM support. This tool is used for determining the sample size required for a study.
[Start](#)

Then click “Start” button and upload the result data from above. Please note that users should remove the last three columns (p.adjust, Fold.Change, Threshold).

Principal Component Analysis (PCA) with LLM Assistance

I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis

Step 1. Upload Data/Example Data

☒ A. Upload ☐ B. Load example data

Select File Format:

☐ .xlsx ☐ .xls ☒ .csv/txt

Please import your data file:

Browse... Adjusted by LLM_merge.table1754551280.09979.csv

Upload complete

☒ Is the first row names?

☒ Is the first column names?

Data Separator (Comma/Semicolon/Tab/Space):

☒ Comma ☐ Semicolon ☐ Tab ☐ BlankSpace

Step 2. Display Uploaded Data/Example Data

Show: 10 entries Search:

	C3L.00081.N	C3L.00415.N	C3L.00445.N	C3L.00568.N	C3L.00603.N	C3L.00904.N
FKBP4	-0.759884573921202	-0.577785445228713	-0.563747973369846	-0.728165764039627	-0.71022599554	-0.864701271545325
CFTR	-0.0535832177263842	0.423711371358339	1.3307767826699	0.404181121377999	-0.214732912944896	-0.125850491062682
HSPB6	1.41187714454355	0.635709961286391	0.968926919360946	0.781297390041477	1.00700386140999	1.2153931119819
PDK4	0.76879122996354	0.415129568459868	0.665183006035176	0.456423780960551	0.685909107068108	0.481091762863645
BAIAP2L1	-0.613249719442923	-0.650434755675018	-0.714798761879639	-0.921410486743699	-0.710317018116189	-0.814561504953272
YBX2	-2.11397478606034	-0.334983648572457	-0.773489472743645	-0.466448395439126	-0.169381541994429	-0.420246281833476
LRRC23	0.997660196155017	0.748359212064887	0.717830145027005	0.100249122546571	0.210927088333916	0.586999344656502
CD74	0.218705477730364	0.387891214421998	0.696805052451243	0.590511762057318	-0.22388168123727	0.080423289828626
TSPAN9	0.652419956316173	0.139775355285359	0.160368786935676	0.499537072850487	0.383839633937253	0.0639503544841524
NME1	-0.343970553382384	0.20340127009257	-0.332182813551469	-0.530414691926005	-0.603012720421447	0.00962433248620442

Showing 1 to 10 of 2,752 entries

Previous 1 2 3 4 5 ... 276 Next

Next, click “II. Manual Analysis”. We should type right parameters based on the data information, and click “Start” button. The result in “4.1. Visualization” shows the PCA plot.

Principal Component Analysis (PCA) with LLM Assistance

I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis

Step 3. Adjust Parameters

3.1. Group and replicate number:

2;102-110

3.2. Group names:

Normal;Tumor

3.3. Select colors for each group:

☒ C3L.00081.N ☒ C3L.00415.N

3.4. Point size:

2

3.5. Figure Height:

600

3.6. Figure Width:

600

Start

Step 4. Display and Download Results

☒ 4.1. Visualization ☐ 4.2. Results summary

Save Image

Groups

☒ Normal ☒ Tumor

In the “4.2. Results summary”, it shows the summary of PCA.

Principal Component Analysis (PCA) with LLM Assistance

I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis

Step 3. Adjust Parameters

3.1. Group and replicate number:

2;102-110

3.2. Group names:

Normal;Tumor

3.3. Select colors for each group:

FC4A8BFC4A8B

3.4. Point size:

2

3.5. Figure Height:

600

3.6. Figure Width:

600

Start

Step 4. Display and Download Results

4.1. Visualization 4.2. Results summary

	eigenvalue	percentage of variance	cumulative percentage of variance
comp 1	1645.0695361	59.77723605	59.77724
comp 2	90.8312334	3.27148377	63.04872
comp 3	65.6948279	2.38716671	65.43589
comp 4	55.9659087	2.06999215	67.50587
comp 5	43.6110507	1.58470388	69.09057
comp 6	38.6731343	1.40527378	70.49585
comp 7	31.4839822	1.14404905	71.63989
comp 8	22.0582152	0.80153398	72.44142
comp 9	21.4968889	0.78113695	73.22256
comp 10	18.7001012	0.67950949	73.90207
comp 11	17.1865769	0.62451224	74.52658
comp 12	15.8232728	0.57497357	75.10155
comp 13	14.8375663	0.53915575	75.64071
comp 14	14.0634555	0.51102673	76.15174
comp 15	13.7335510	0.49903892	76.65077
comp 16	12.5244287	0.45510279	77.10588
comp 17	12.3632900	0.44924528	77.55512
comp 18	11.5689547	0.42038353	77.97551
comp 19	11.3680444	0.41308301	78.38859
comp 20	10.5591753	0.38336393	78.77195
comp 21	10.3584400	0.37639680	79.14835
comp 22	9.9961042	0.36323053	79.51158
comp 23	9.8725444	0.35874971	79.87032
comp 24	9.6061312	0.34960900	80.21938
comp 25	9.4823009	0.34456035	80.56394
comp 26	9.1826634	0.33367236	80.89761
comp 27	9.1557479	0.33269433	81.23031
comp 28	8.8217396	0.32055740	81.55087
comp 29	8.6078141	0.31278394	81.86365
comp 30	8.3955919	0.30507238	82.16872

Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “I want all point size are bigger. Refer to inner codes.”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

Principal Component Analysis (PCA) with LLM Assistance

I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis

Step 5. LLMs Chat

User:

I want all point size are bigger. Refer to inner codes.

Assistant:

```
library(factoextra)
library(FactoMineR)
prdataread<-PCA(t(pcadata),scale.unit = TRUE, ncp = ncol(pcadata),graph = FALSE)
fviz_pca_ind(prdataread,
  label = "none",
  habillage = factor(pca$classname,levels = pca$textvalue1),
  palette = c("#00AFBB", "#F4A077"),
  mean.point = FALSE,
  pointshape="circle",pointsize = 10,
  addEllipses = TRUE)
theme_bw()
theme(panel.grid.major=element_blank(),panel.grid.minor=element_blank())
```

Enter your message here...

Choose a local LLM:

gemma327b

Prompt example Send

Step 6. Adjusted results by LLMs

Download Height: 600 Width: 700

Individuals - PCA

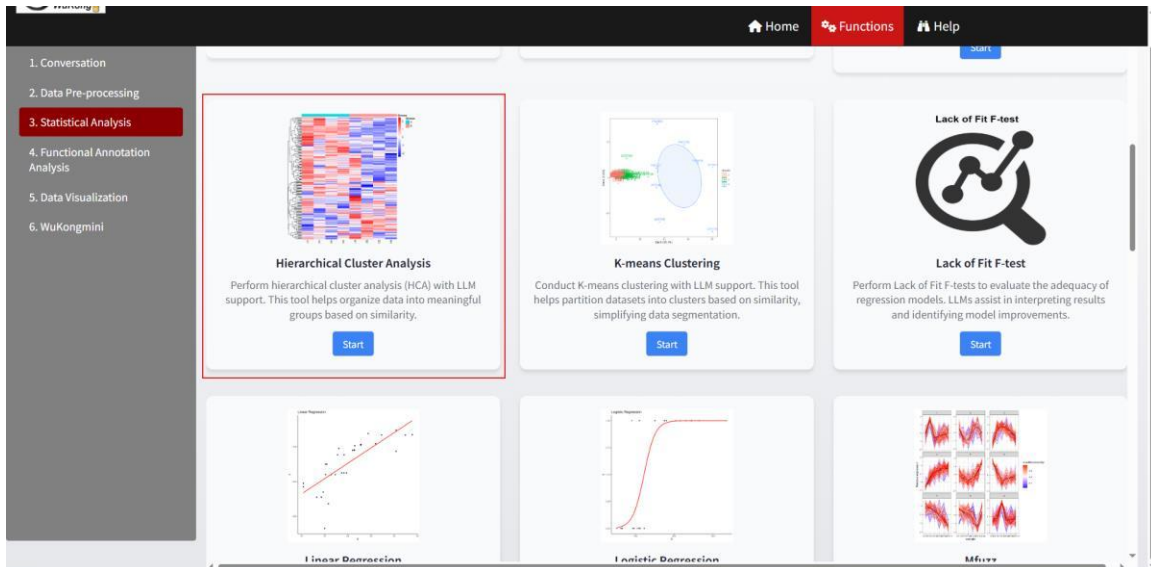
Groups

Normal

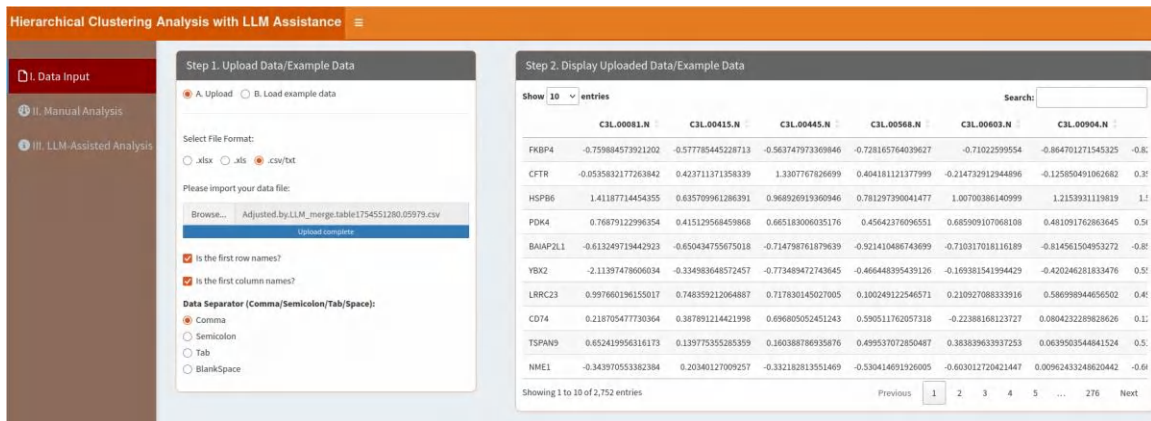
Tumor

f. HCA

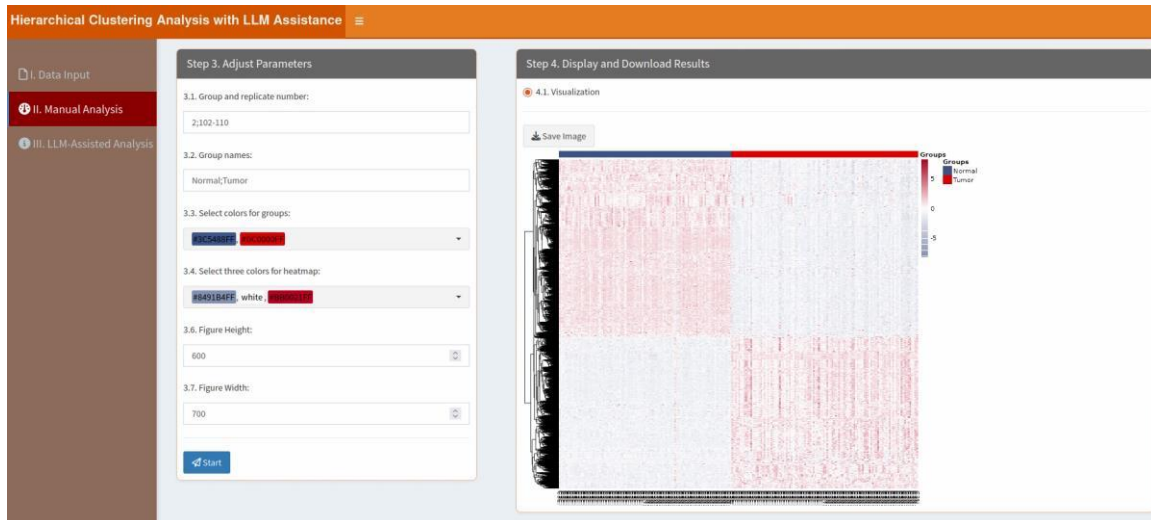
We can find this module here:



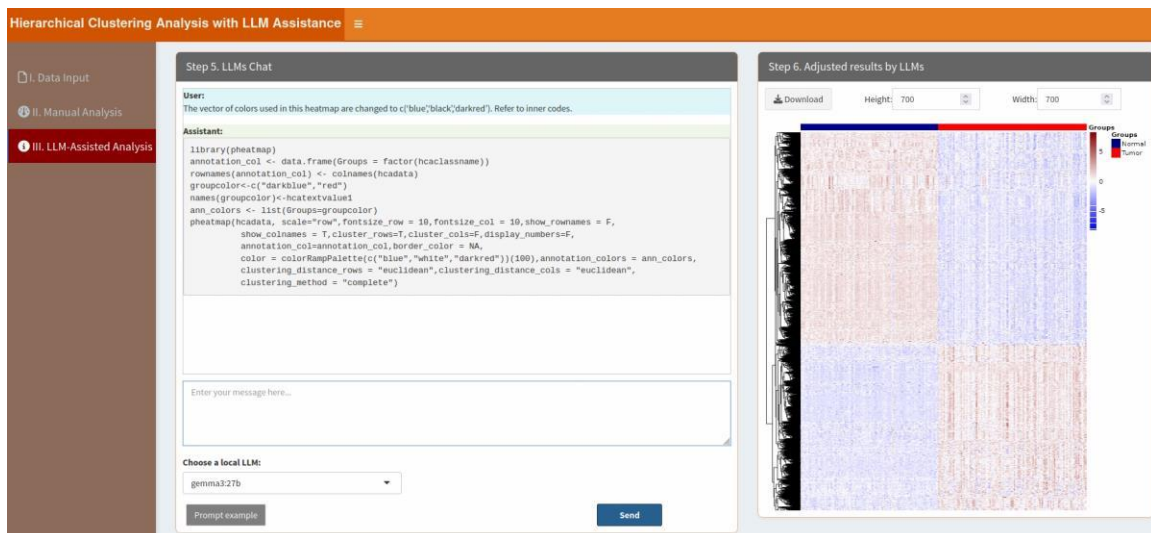
Then click “Start” button and upload the result data from the “Two table merging”. Please note that users should remove the last three columns (p.adjust, Fold.Change, Threshold).



Next, click “II. Manual Analysis”. We should type right parameters based on the data information, and click “Start” button. The result in “4.1. Visualization” shows the HCA plot.

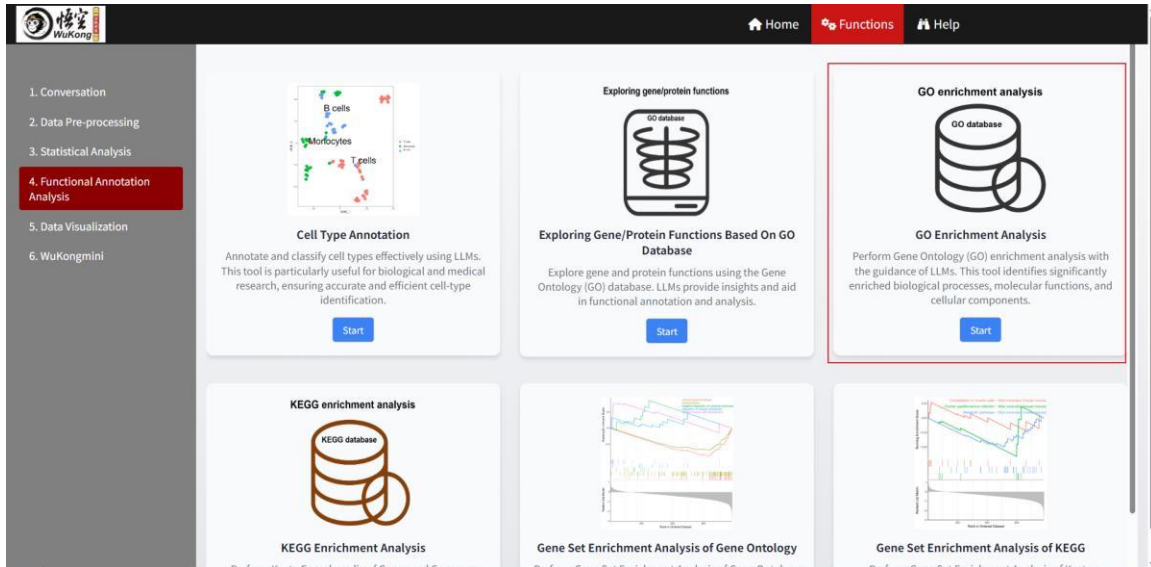


Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “The vector of colors used in this heatmap are changed to c('blue','black','darkred'). Refer to inner codes.”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.



g. GO enrichment analysis

We can find this module here:



Then click “Start” button and upload the differentially expressed proteins. This module needs one column of UniProt IDs, so we need transform these protein names to UniProt IDs using the ID mapping function of the UniProt web (<https://www.uniprot.org/tool-dash-board>).

Then we can upload these UniProt IDs:

GO enrichment analysis with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 1. Upload Data/Example Data

A. Upload B. Load example data

Select File Format:

☒ .xlsx
☐ .xls
☐ .csv/txt

Please import your data file:

Browse...

idmapping_2025_08_07.xlsx

upload complete

☒ Is the first row names?
☐ Is the first column names?

Which Sheet to read?

1

Step 2. Display Uploaded Data/Example Data

Show 10 entries

Search:

Entry
1 Q62790
2 P13569
3 Q14558
4 Q16654
5 Q8UHR4
6 Q9Y217
7 Q53EV4
8 P04233
9 O75954
10 P15531

Showing 1 to 10 of 2,942 entries

Previous 1 2 3 4 5 ... 285 Next

Next, click “II. Manual Analysis”. We should type right parameters based on the data information, and click “Start” button. The enrichment table is shown in the “4.1. Tables” part. Users can click the “Download” button to save the result.

GO enrichment analysis with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 3. Adjust Parameters

3.1. Please choose a species:

9606-Homo sapiens (Human)

3.2. Select two colors for Barplot:

blue red

3.3. Total number of GOs displayed in the Barplot:

15

3.4. Figure Height:

600

3.5. Figure Width:

600

Start

Step 4. Display and Download Results

4.1. Tables 4.2. Visualization

Download

Show 10 entries

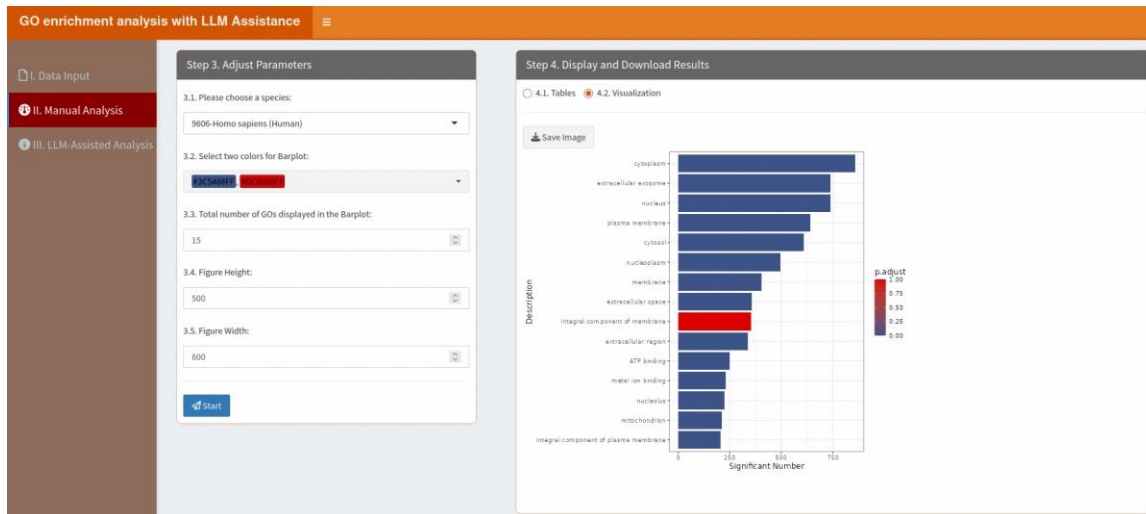
Search:

	ID	ONTOLOGY	TERM	Count	GeneRatio	BgRatio	pvalue	p.adjust	qval
3	GO:0000111	BP	vacuole inheritance	1	1/2747	1/95459	0.02877675322298771	0.05814437436324039	QH
4	GO:0000112	BP	single strand break repair	1	1/2747	25/95459	0.5181229017066751	0.5342152807168986	QH
6	GO:0000115	CC	phosphorynate hydratase complex	1	1/2747	31/95459	0.59583889176242	0.6086161929117421	PI
11	GO:000033	MF	alpha-1,3-mannosyltransferase activity	1	1/2747	3/95459	0.0838706357011848	0.1209481288740394	QH
16	GO:000054	BP	ribosomal subunit export from nucleus	1	1/2747	2/95459	0.056725695956509	0.09203034401894081	PE
20	GO:000064	MF	L-ornithine transmembrane transporter activity	1	1/2747	4/95459	0.1102346920593896	0.1469203425609343	PI
30	GO:0000104	MF	succinate dehydrogenase activity	1	1/2747	11/95459	0.2747256004053397	0.3027842417267384	OI
35	GO:0000137	CC	Golgi cis cisterna	1	1/2747	7/95459	0.1848629101721435	0.2178299097841965	QH
40	GO:0000150	MF	recombinase activity	1	1/2747	9/95459	0.2311054705585568	0.2605452563607515	OI
43	GO:0000155	MF	phosphorelay sensor kinase activity	1	1/2747	16/95459	0.3732978087354516	0.3970337211436421	QI

Showing 1 to 10 of 8,147 entries

Previous 1 2 3 4 5 ... 815 Next

In the “4.2. Visualization” part, it shows the top 15 GOs ordered by the counts.



Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. Two types of prompts are supported: prompts beginning with “Text:” are used to request interpretation or explanation of the enrichment results (e.g. summarizing pathways relevance), while prompts beginning with “Plot:” allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed pathways). For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results.”, where “Refer to inner codes” instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

GO enrichment analysis with LLM Assistance

I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis

Step 5. LLMs Chat

User:

Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results.

Assistant:

Okay, based on your focus on lung cancer and inflammation, here are 3 of the most relevant GO IDs, descriptions, and associated UniProt IDs from the provided list, prioritizing those with strong connections to both cancer and inflammatory processes:

- GO:2000097: Regulation of smooth muscle cell-matrix adhesion**
 - Description:** Regulation of the adhesion of smooth muscle cells to the extracellular matrix. This is relevant to lung cancer as the extracellular matrix plays a crucial role in tumor progression, metastasis, and the tumor microenvironment. Inflammation can significantly alter the ECM, promoting cancer development.
 - UniProt ID:** P00749
- GO:2000446: Regulation of macrophage migration inhibitory factor signaling pathway**
 - Description:** Regulation of the signaling pathway mediated by Macrophage Migration Inhibitory Factor (MIF). MIF is a key pro-inflammatory cytokine heavily implicated in cancer progression, metastasis, and angiogenesis. It's often upregulated in the tumor microenvironment and promotes immune suppression.
 - UniProt ID:** Q15389
- GO:2000514: Regulation of CD4-positive, alpha-beta T cell activation**
 - Description:** Regulation of the activation of CD4-positive, alpha-beta T cells. T cells are central to the immune response, and their

Enter your message here...

Choose a local LLM:

gemma3.27b

Prompt example 1
Prompt example 2
Send

Step 6. Adjusted results by LLMs

Download

Height: 900

Width: 700

Users can also adjust the plot by type “Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors.”

GO enrichment analysis with LLM Assistance

I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis

Step 5. LLMs Chat

User:

Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors.

Assistant:

Assuming 'allRes.df' is already defined and contains your data

```
# Order the dataframe by Count in descending order
allRes.df1.new1 <- allRes.df[order(allRes.dfsCount, decreasing = TRUE),]

# Select the top 15 rows
allRes.df1.new <- allRes.df1.new1[1:15,]

# Create the barplot with yellow as the low color and darkred as the high color
ggplot(allRes.df1.new, aes(x = reorder(TERM, Count), y = Count, fill = p.adjust)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  scale_fill_gradient(low = "yellow", high = "darkred") + # Changed colors here
  theme_bw() +
  xlab("Description") +
  ylab("Significant Number") +
  theme(text = element_text(size = 12))
```

Enter your message here...

Choose a local LLM:

gemma3.27b

Prompt example 1
Prompt example 2
Send

Step 6. Adjusted results by LLMs

Download

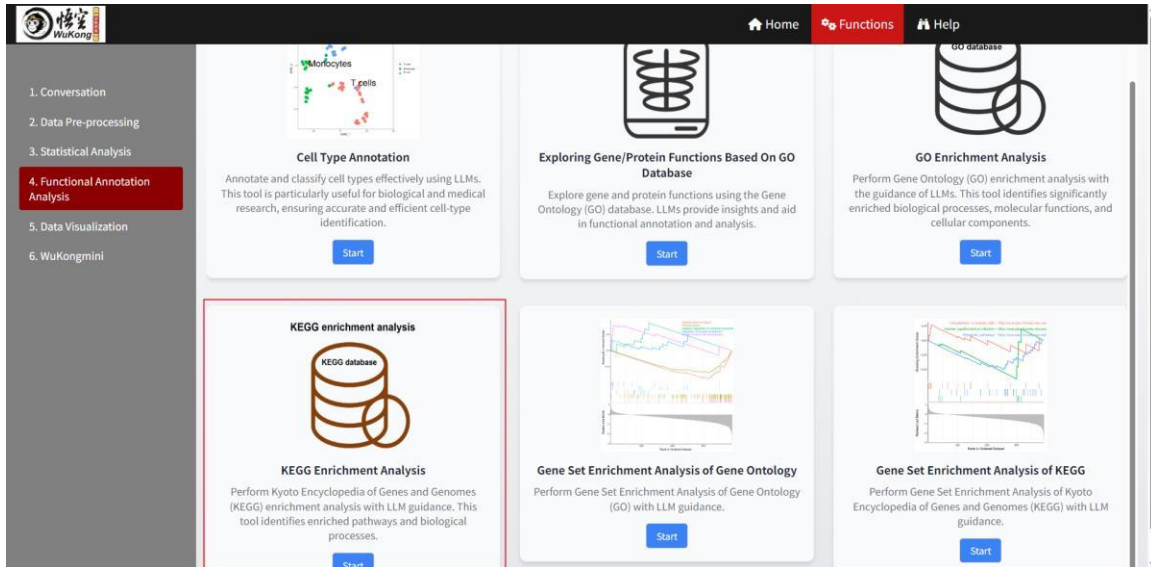
Height: 600

Width: 700

h. KEGG pathway enrichment analysis

We can find this module here:

299



Then click “Start” button and upload the differentially expressed proteins. Same as above, this module also needs one column of UniProt IDs, so we need transform these protein names to UniProt IDs using the ID mapping function of the UniProt web (<https://www.uniprot.org/tool-dashboard>).

Then we can upload these UniProt IDs:

KEGG enrichment analysis with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 1. Upload Data/Example Data

A. Upload

B. Load example data

Select File Format:

.xlsx

.xls

.csv/txt

Please import your data file:

Browse...

ldmapping_2025_08_07.xlsx

Upload complete

Is the first row names?

Is the first column names?

Which Sheet to read?

1

Step 2. Display Uploaded Data/Example Data

Show

10

entries

Search:

Entry	
1	Q02790
2	P13569
3	Q14558
4	Q16654
5	Q9UHR4
6	Q9Y217
7	Q53EV4
8	P04233
9	O75954
10	P15531

Showing 1 to 10 of 2,842 entries

Previous

1

2

3

4

5

...

285

Next

Next, click “II. Manual Analysis”. We should type right parameters based on the data information, and click “Start” button. The enrichment table is shown in the “4.1. Tables” part. Users can click the “Download” button to save the result.

KEGG enrichment analysis with LLM Assistance

I. Data Input

II. Manual Analysis

III. LLM-Assisted Analysis

Step 3. Adjust Parameters

3.1. KEGG organism:

hsa

3.2. Select two colors for Barplot:

Blue

Red

3.3. Total number of Pathways displayed in the Barplot:

15

3.4. Figure Height:

500

3.5. Figure Width:

600

Start

Step 4. Display and Download Results

4.1. Tables

4.2. Visualization

Download

Show

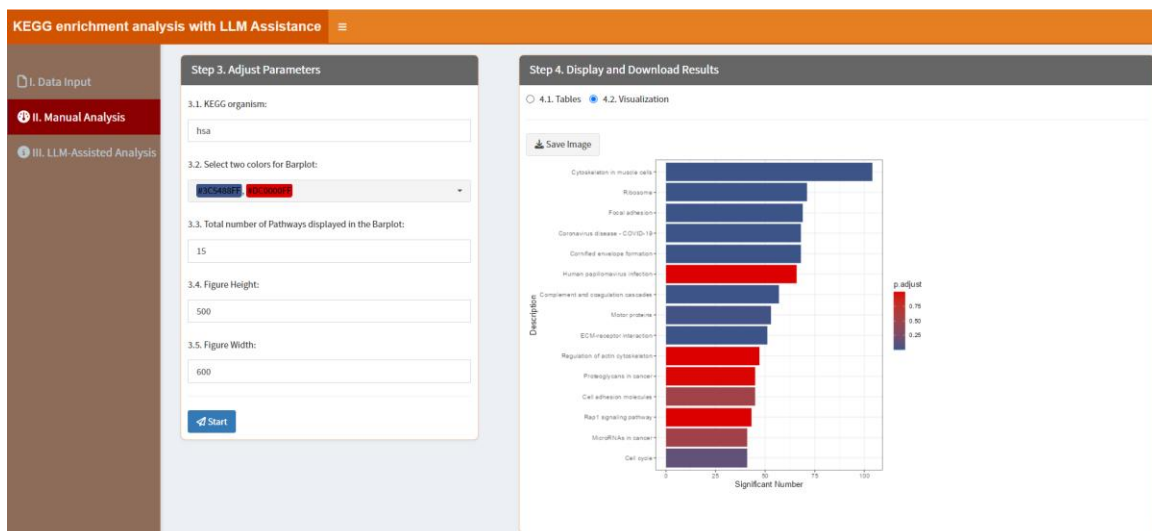
10

entries

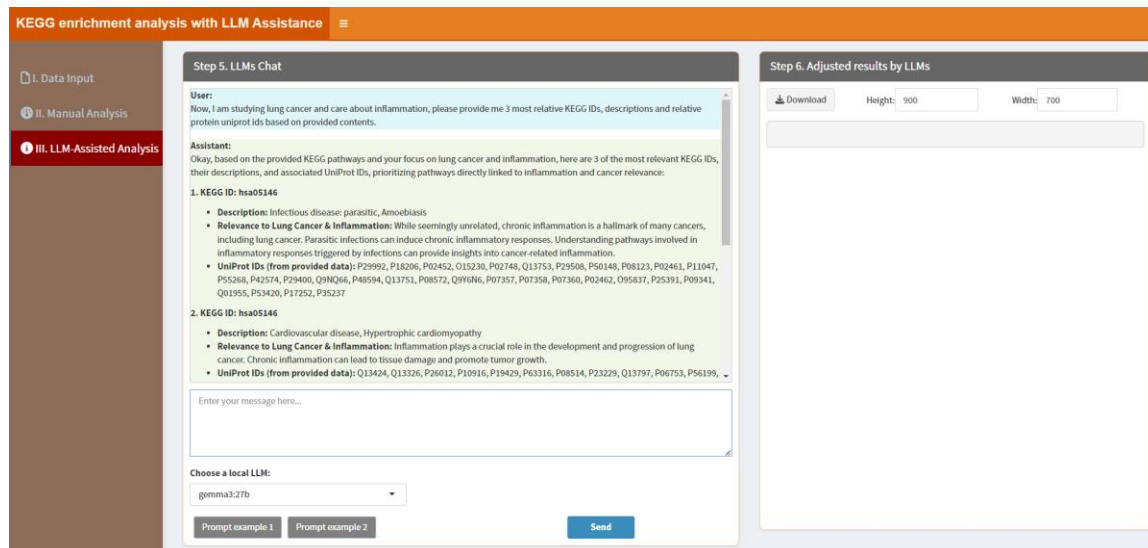
Search:

category	subcategory	ID	Description	GeneRatio	BgRatio	RichFactor	FoldEnrichment
		hsa04120	Cytoskeleton in muscle cells	104/1433	322/10978	0.3229813664596273	2.474312340749329
Genetic Information Processing	Translation	hsa03010	Ribosome	71/1433	190/10978	0.3585858585858586	2.747072962704505
Cellular Processes	Cellular community-eukaryotes	hsa04510	Focal adhesion	69/1433	307/10978	0.2247557003257329	1.721820012683807
		hsa04382	Cornified envelope formation	68/1433	258/10978	0.2635658914728682	2.019139118345532
Human Diseases	Infectious disease: viral	hsa05171	Coronavirus disease - COVID-19	68/1433	325/10978	0.2092307692307692	1.602885823178914
Human Diseases	Infectious disease: viral	hsa05165	Human papillomavirus infection	66/1433	478/10978	0.1380753138075314	1.85777445567277
Organismal Systems	Immune system	hsa04610	Complement and coagulation cascades	57/1433	112/10978	0.5089285714285714	3.898826138969195

In the “4.2. Visualization” part, it shows the top 15 pathways ordered by the counts.



Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. Two types of prompts are supported: prompts beginning with “Text:” are used to request interpretation or explanation of the enrichment results (e.g. summarizing pathways relevance), while prompts beginning with “Plot:” allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed pathways). For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results.”, where “Refer to inner codes” instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.



Users can also adjust the plot by type “Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors.”

